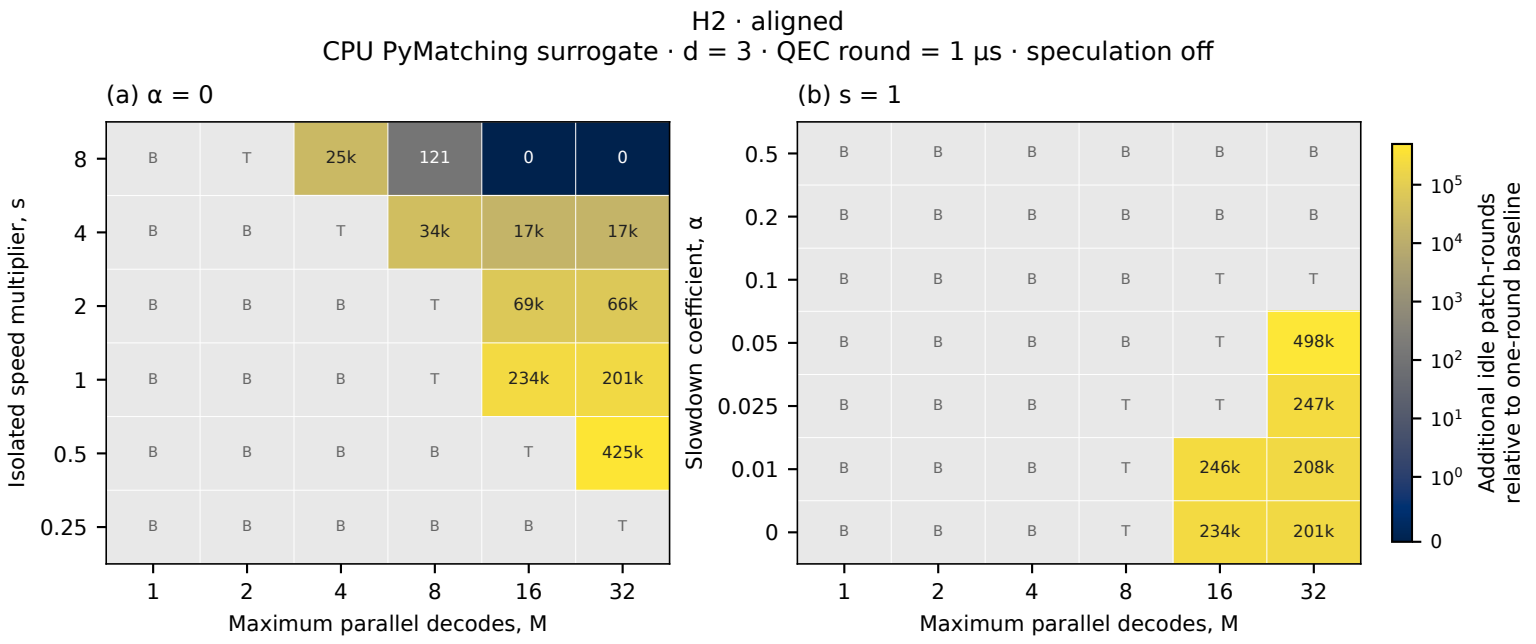
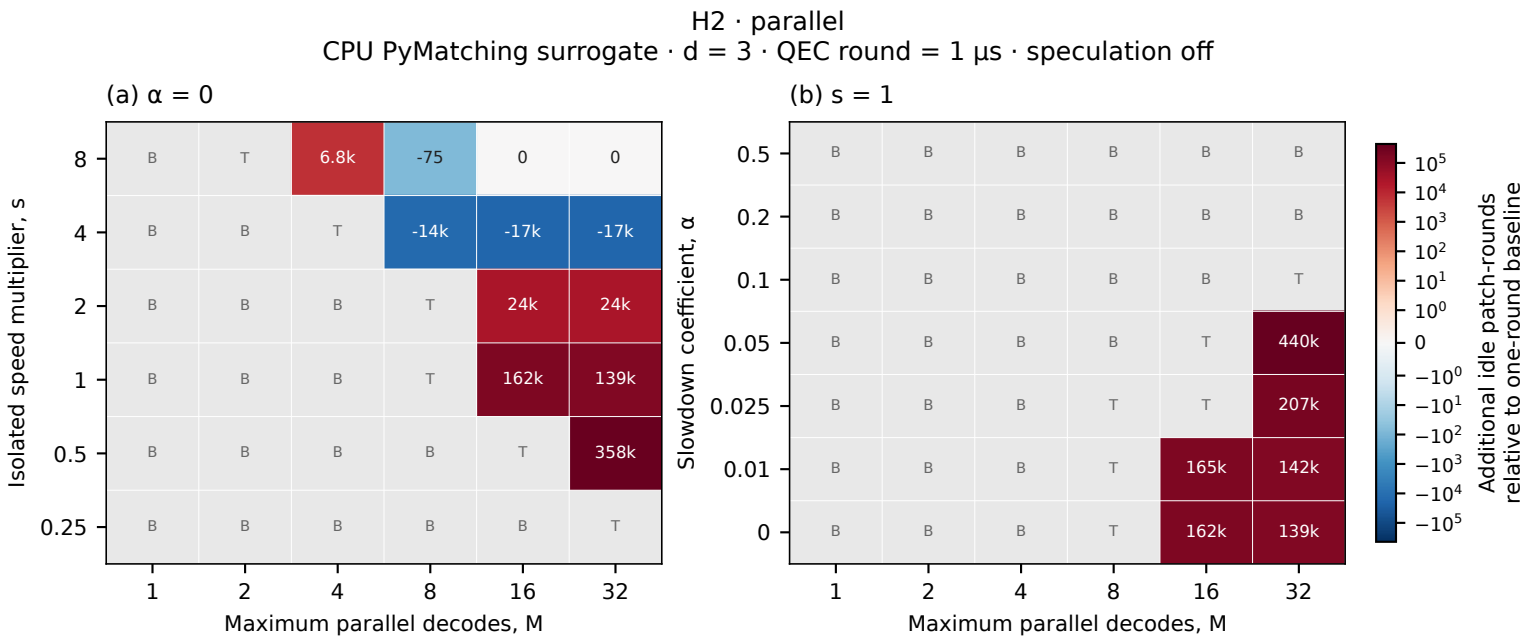
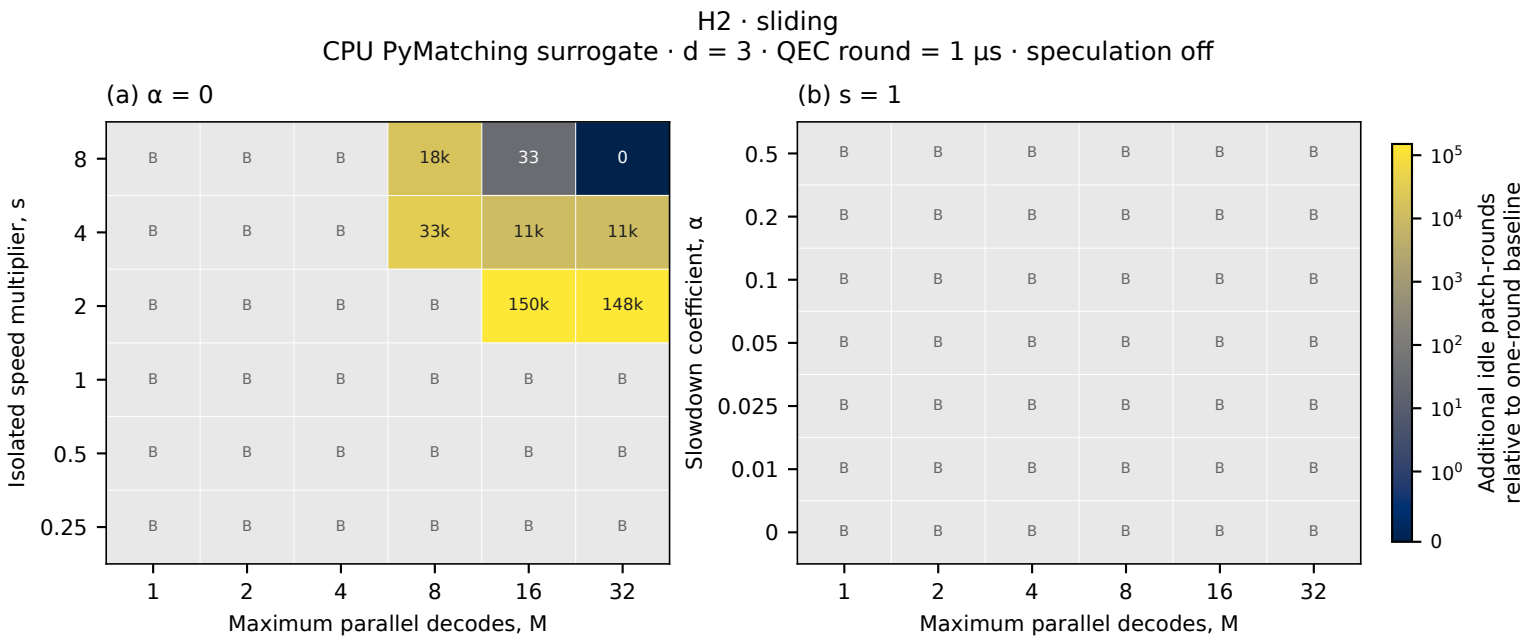
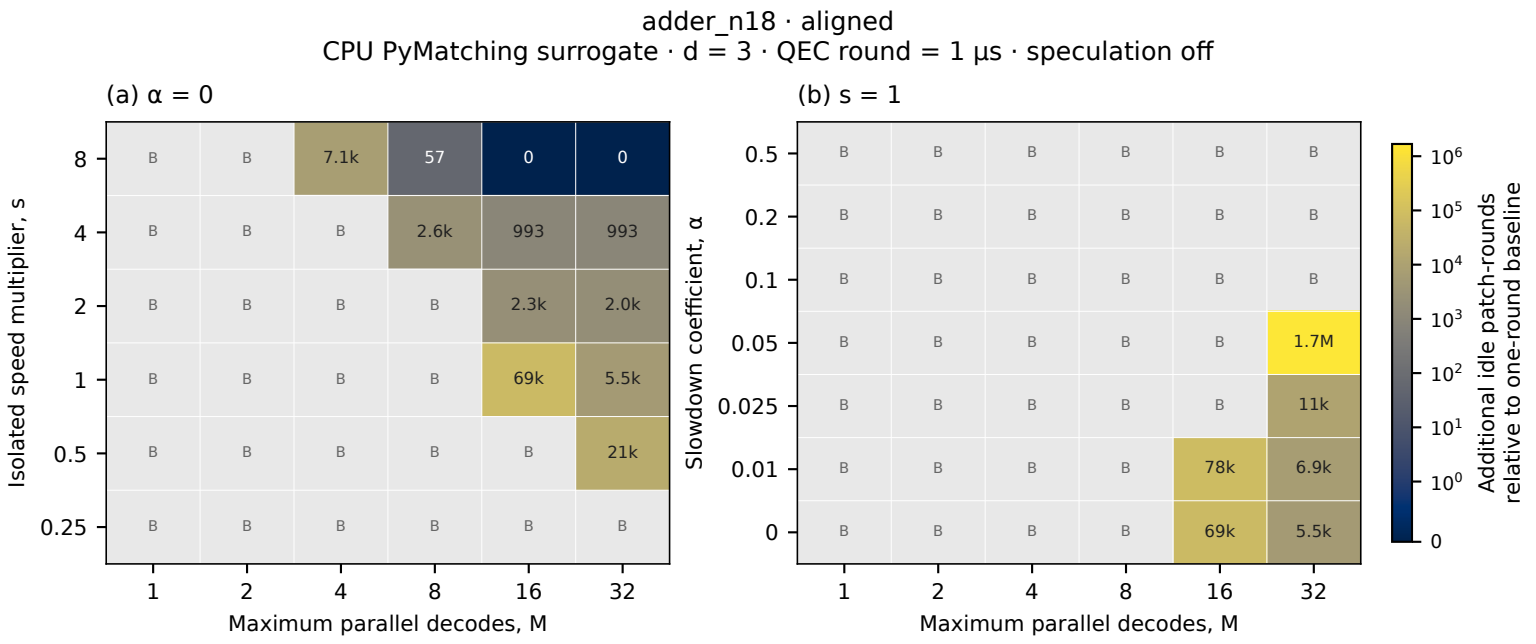
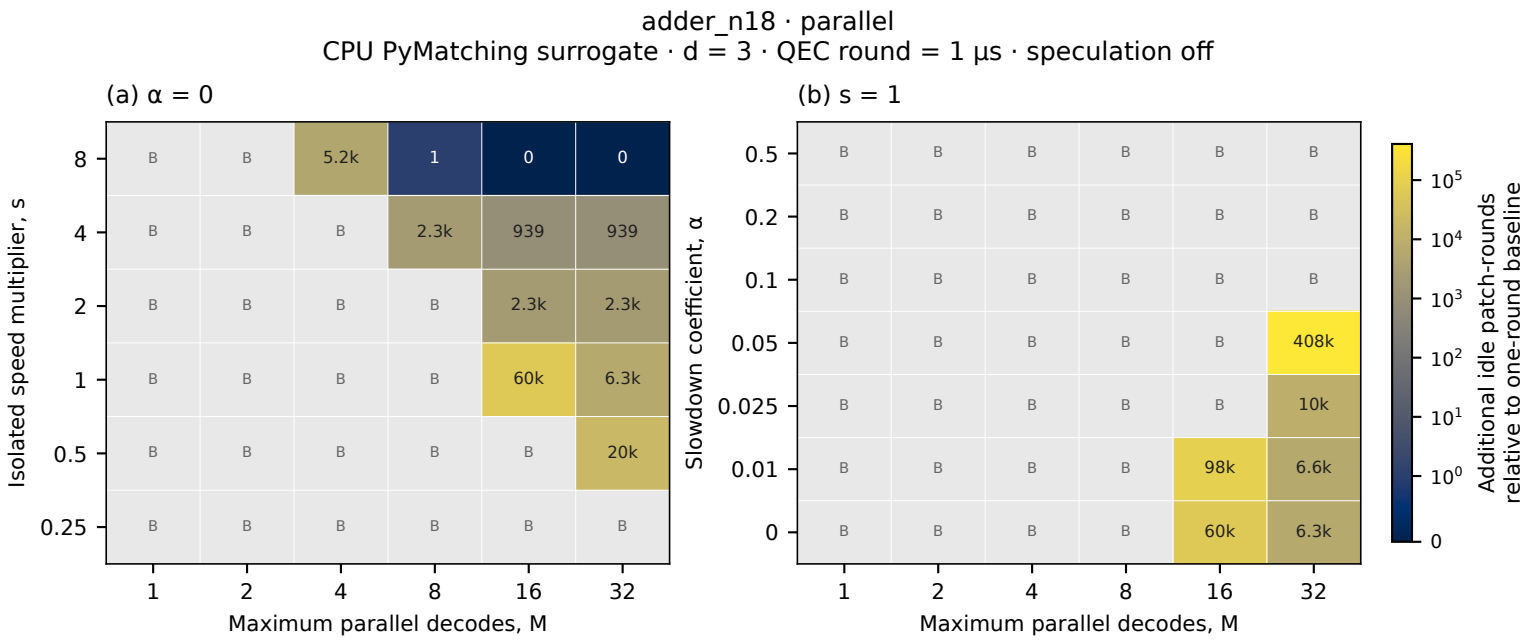
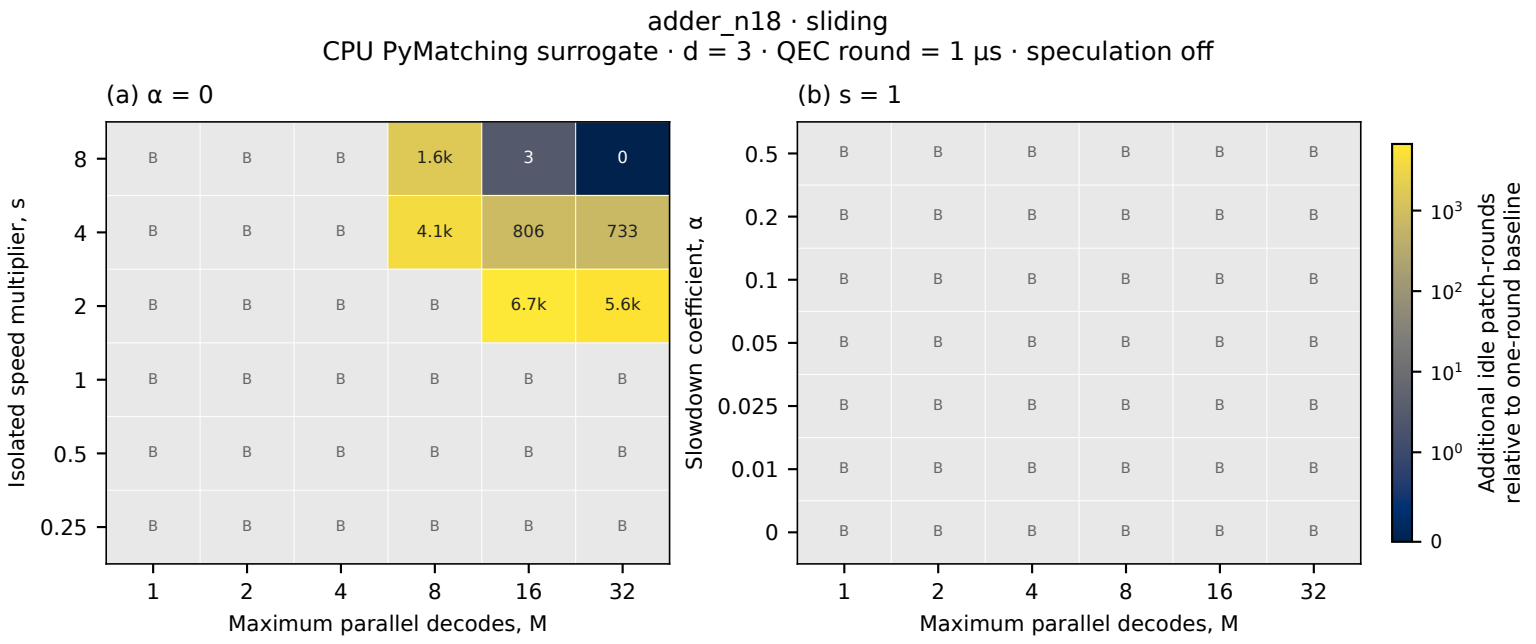

H2 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
B
T
25k
121
0
34k
17k
69k
66k
234k
201k
425k
498k
247k
246k
208k
10^5
10^4
10^3
10^2
10^1
10^0
0

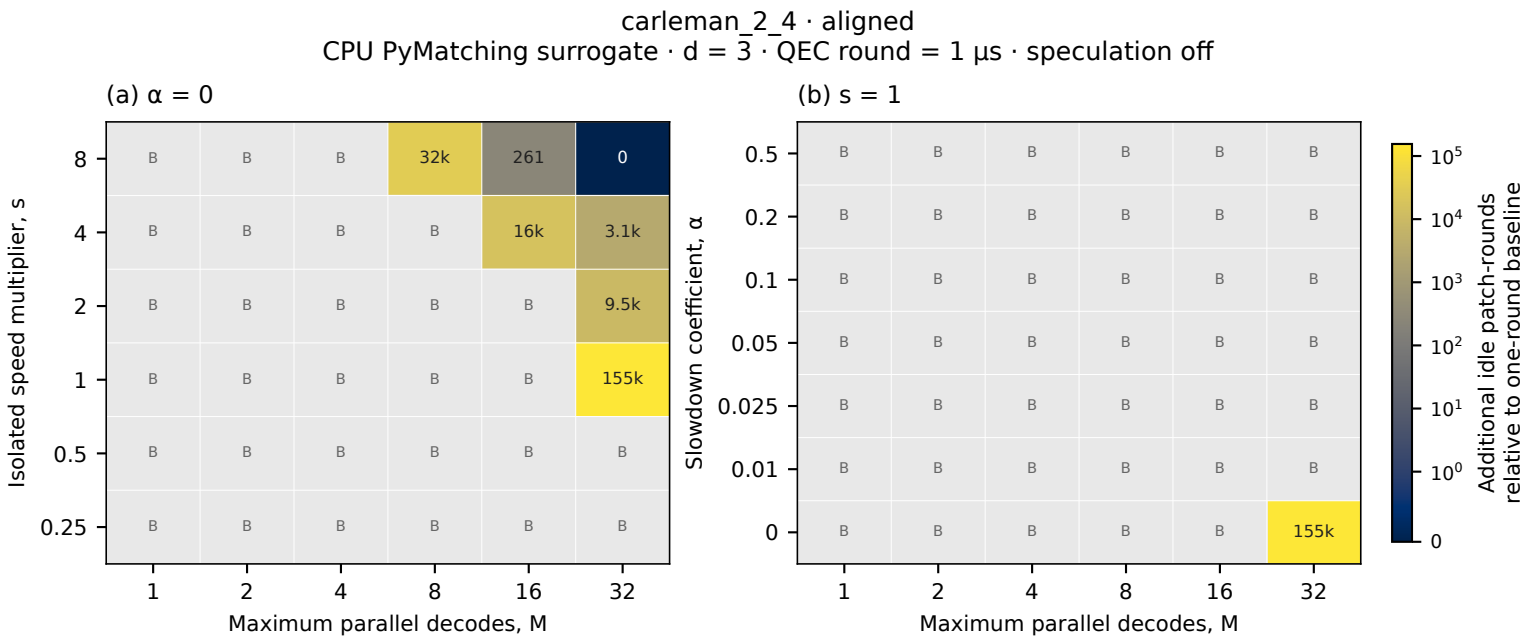

H2 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
B
T
6.8k
-75
0
-14k
-17k
24k
162k
139k
358k
440k
207k
165k
142k
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
0
-10^{0}
-10^{1}
-10^{2}
-10^{3}
-10^{4}
-10^{5}

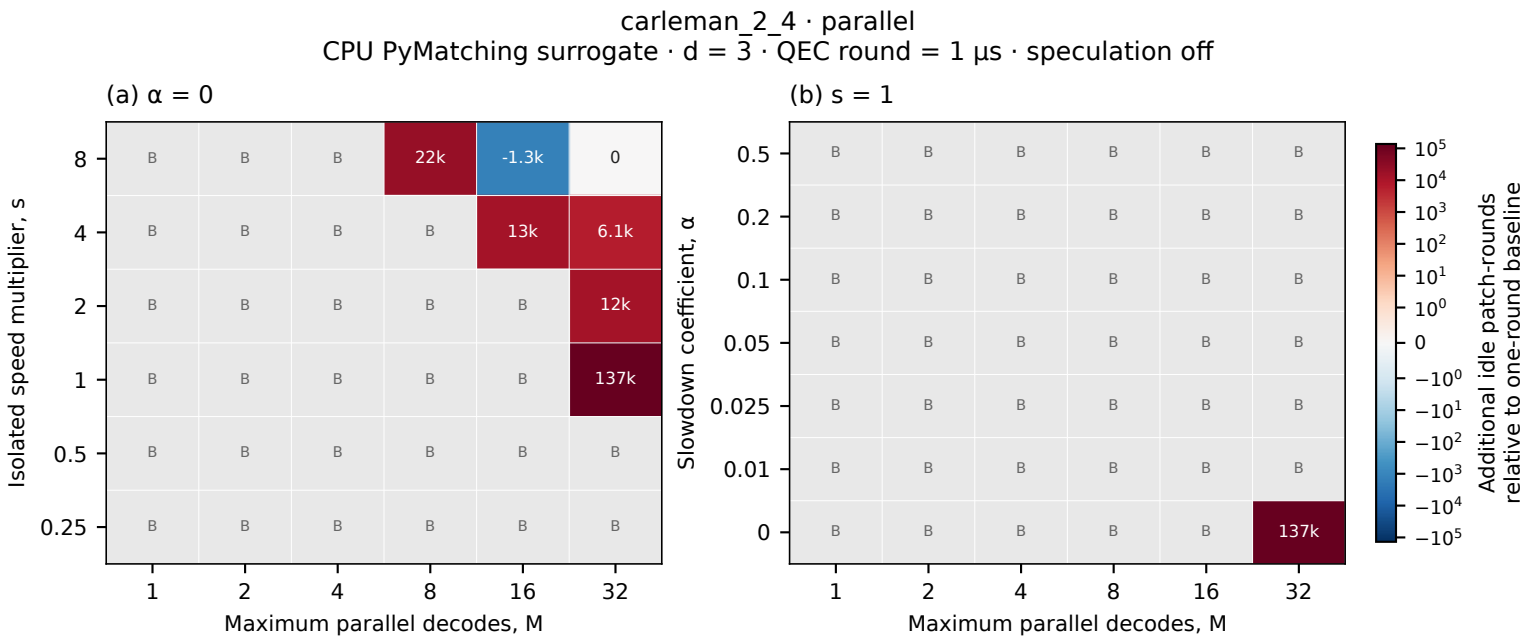

H2 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
18k
33
0
33k
11k
11k
150k
148k
B
10^5
10^4
10^3
10^2
10^1
10^0
0

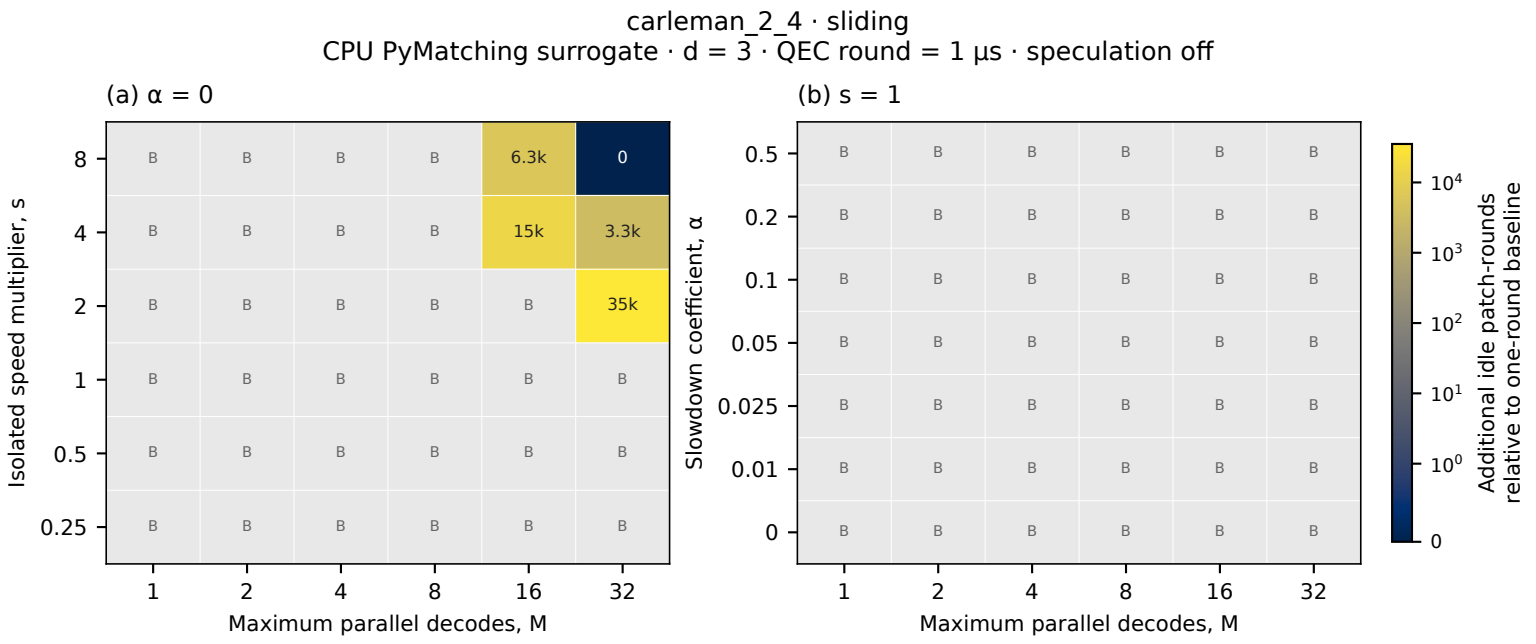

adder_n18 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
B
7.1k
57
0
2.6k
993
2.3k
2.0k
69k
5.5k
21k
1.7M
11k
78k
6.9k

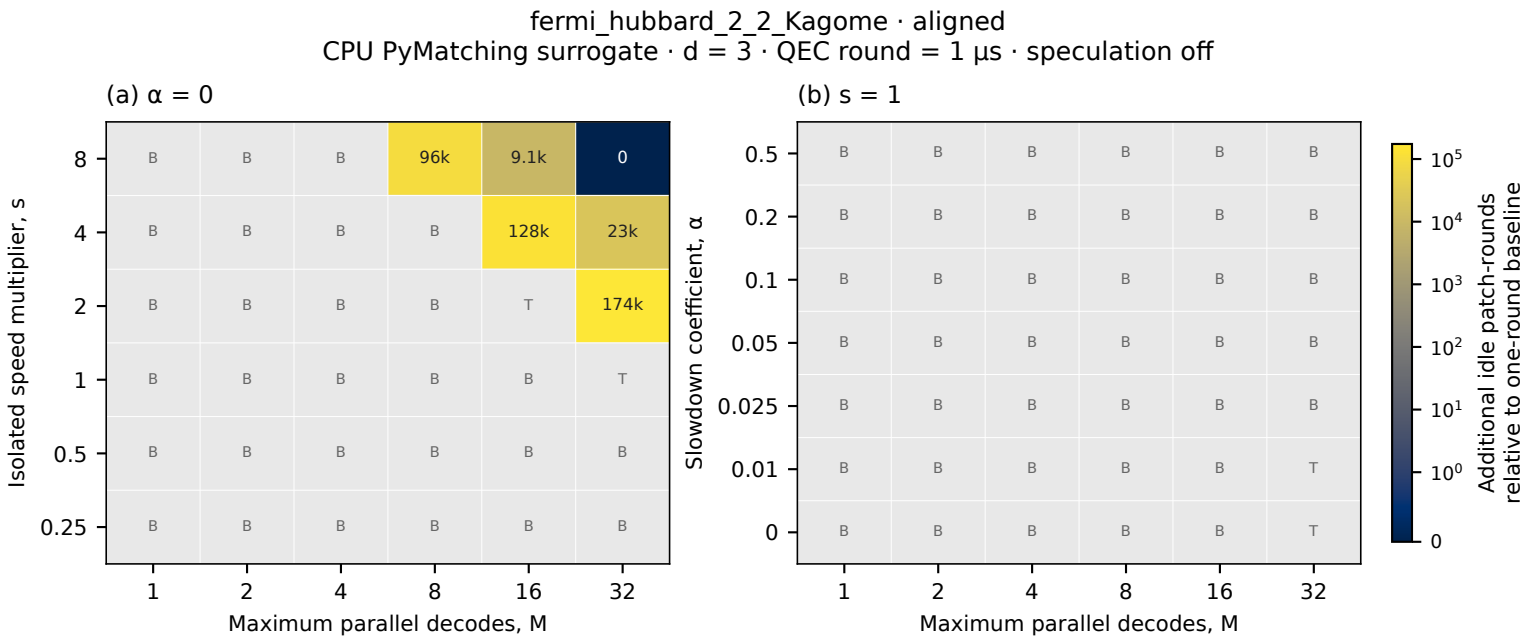

adder_n18 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
B
5.2k
1
0
2.3k
939
2.3k
60k
6.3k
20k
408k
10k
98k
6.6k
10^5
10^4
10^3
10^2
10^1
10^0
0

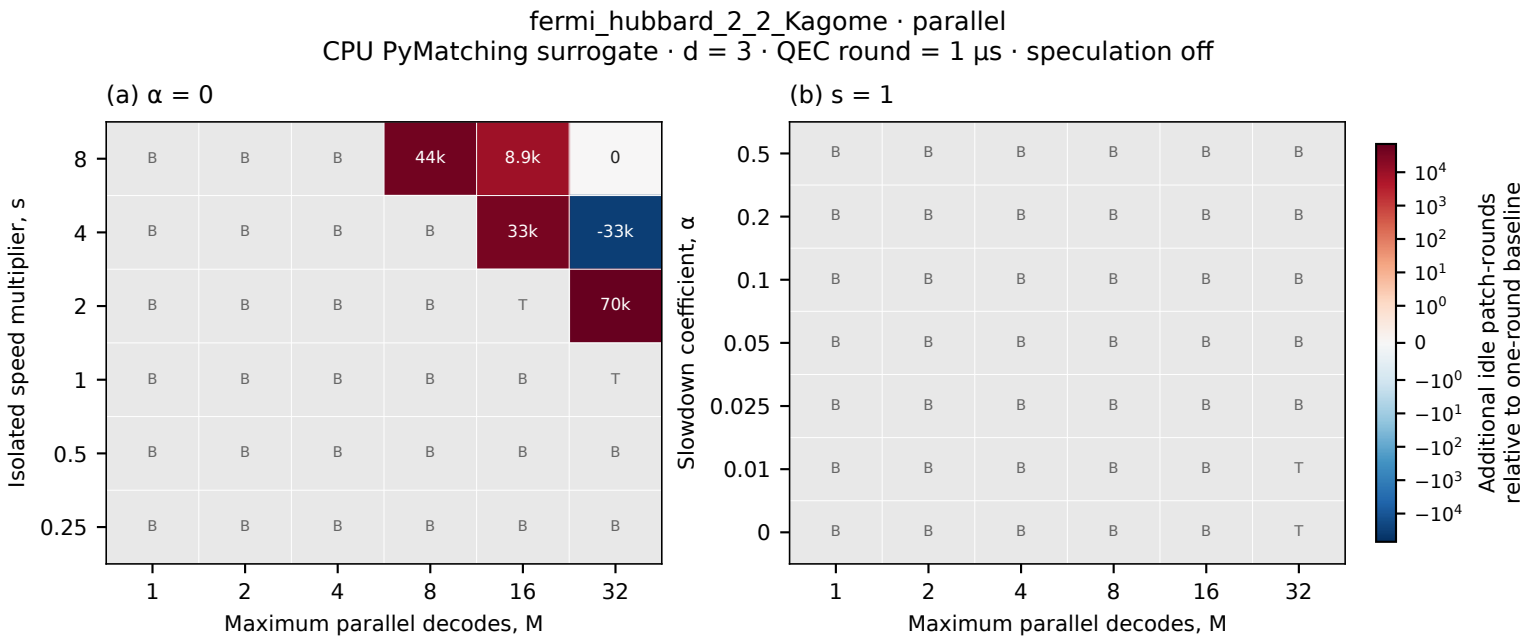

adder_n18 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
B
1.6k
3
0
4.1k
806
733
6.7k
5.6k

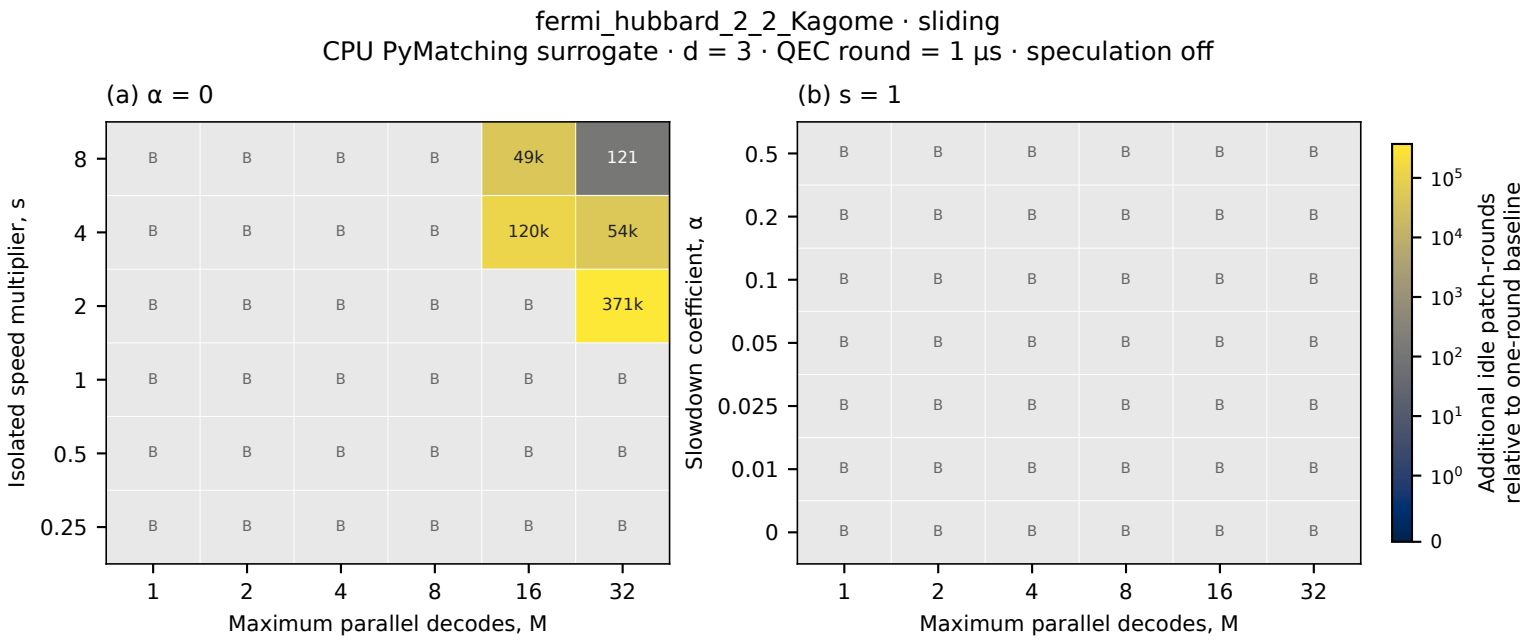

carleman_2_4 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
B
32k
261
0
16k
3.1k
9.5k
155k
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
10^5
10^4
10^3
10^2
10^1
10^0
0

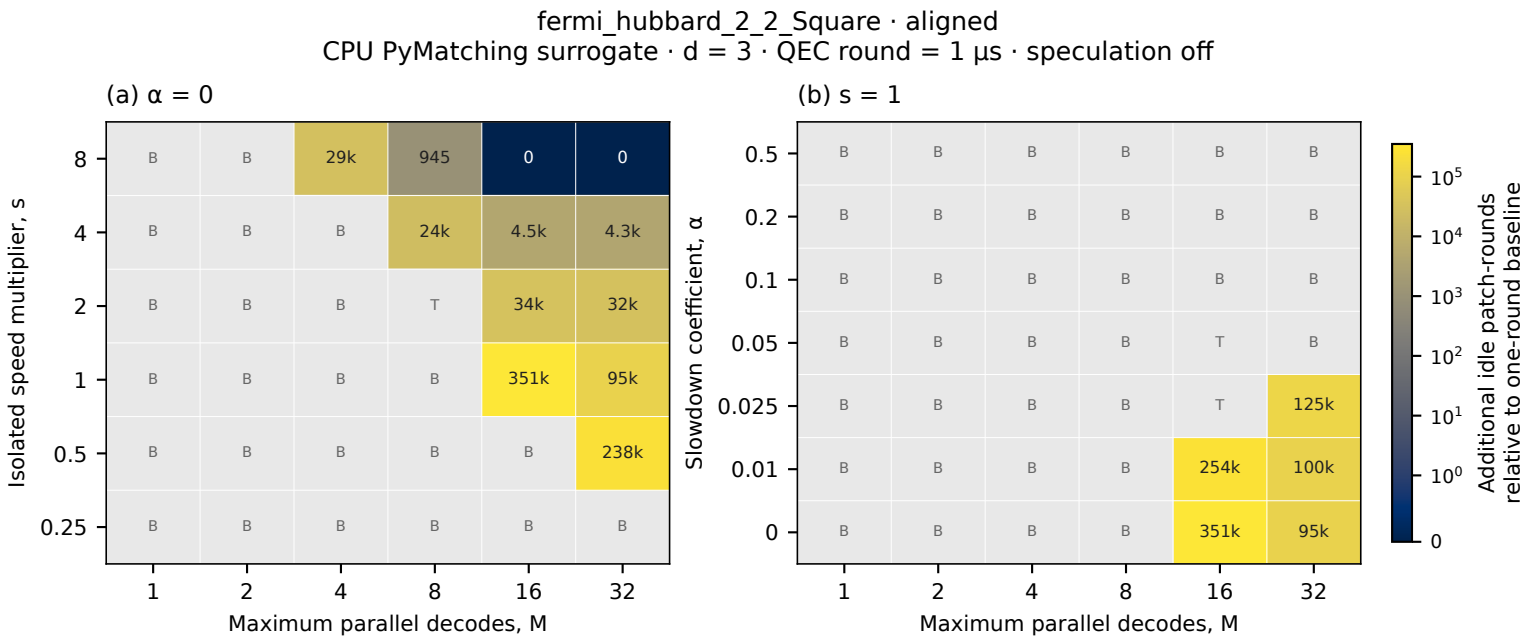

carleman_2_4 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
B
22k
-1.3k
0
13k
6.1k
12k
137k
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
1
2
4
8
16
32
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
-10^{0}
-10^{1}
-10^{2}
-10^{3}
-10^{4}
-10^{5}

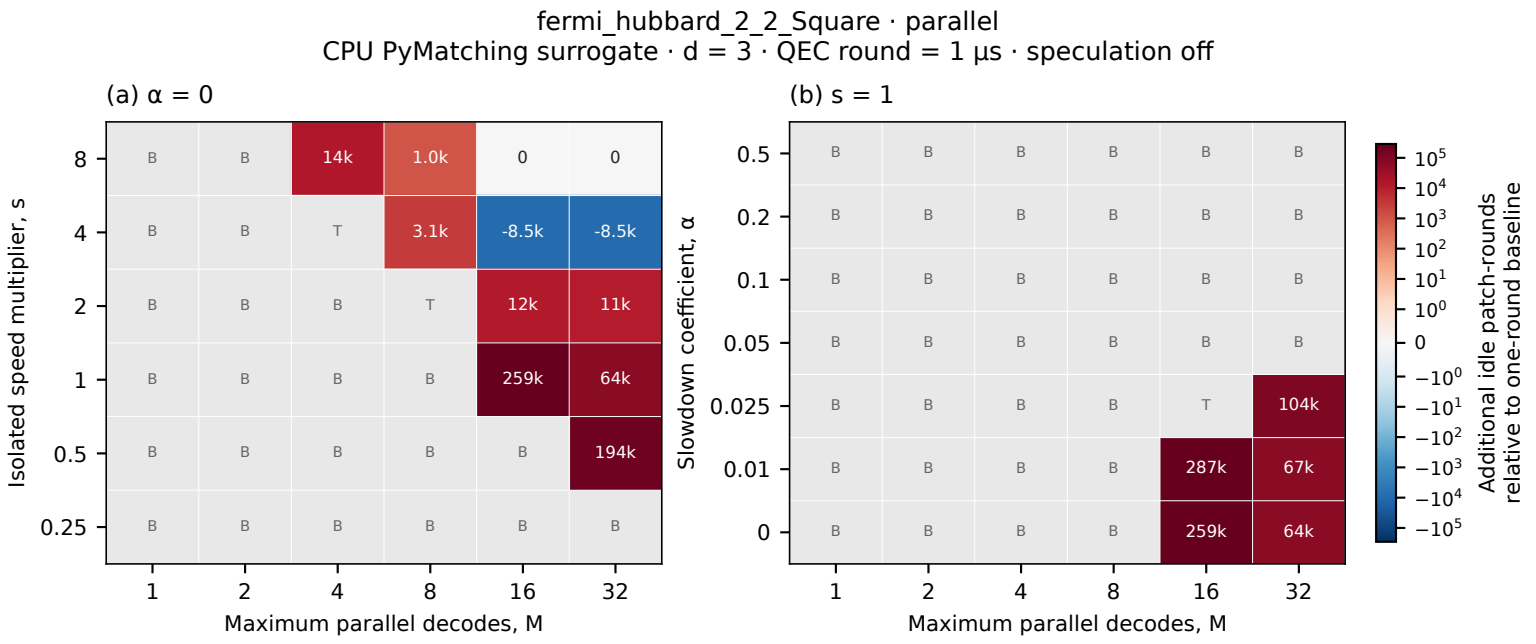

carleman_2_4 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
B
6.3k
0
15k
3.3k
35k

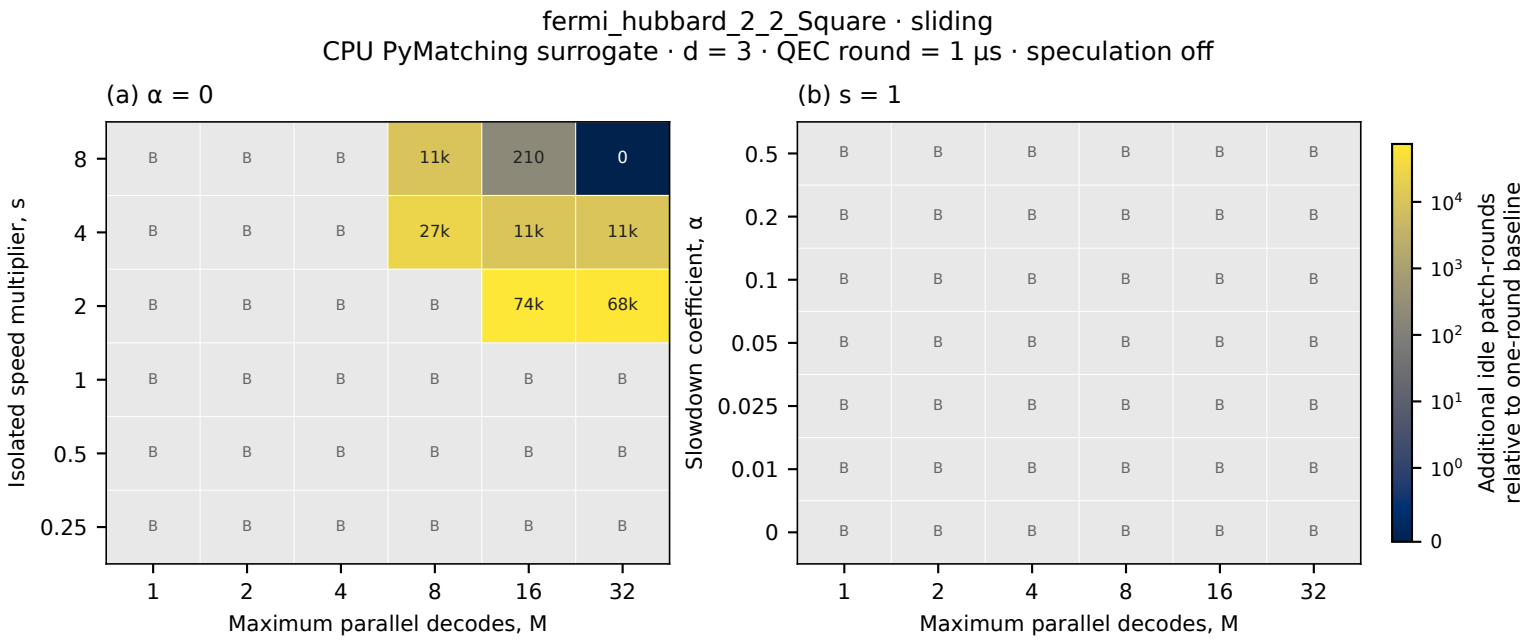

fermi_hubbard_2_2_Kagome · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
96k
9.1k
0
128k
23k
174k
B
T
10^5
10^4
10^3
10^2
10^1
10^0

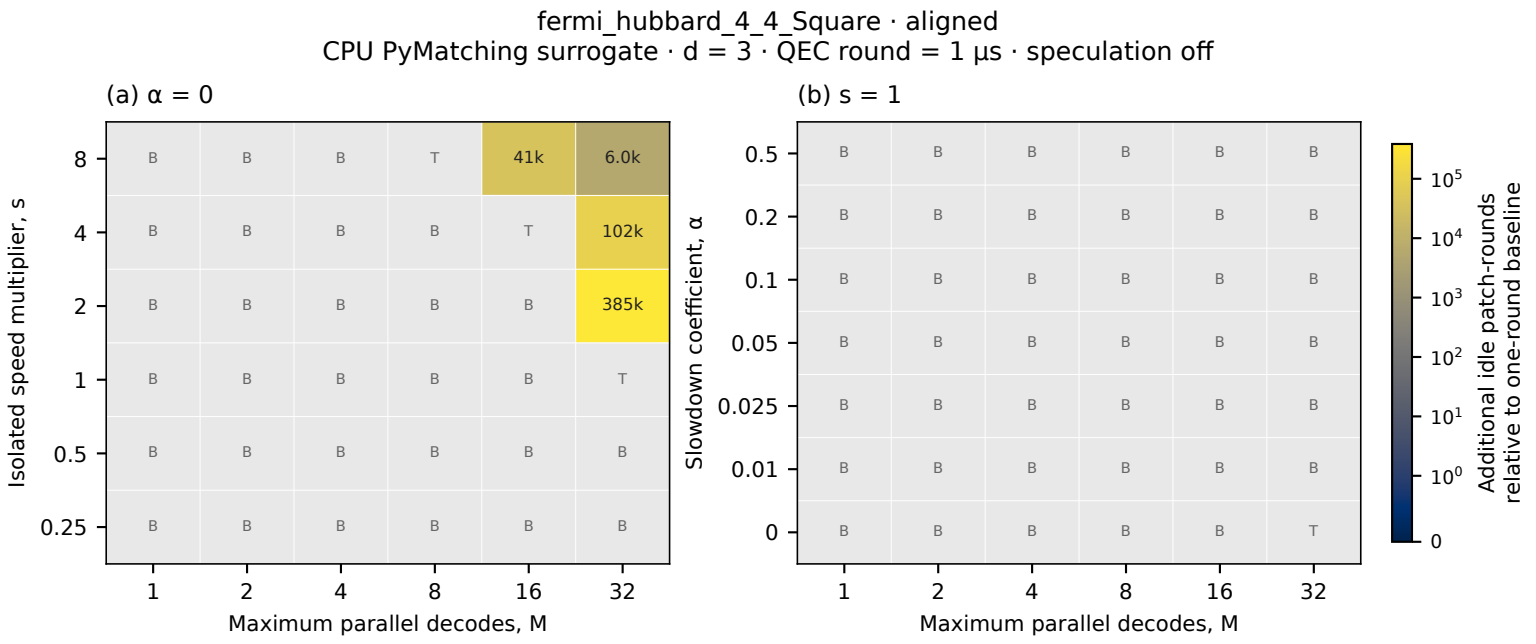

fermi_hubbard_2_2_Kagome · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
44k
8.9k
0
33k
-33k
70k
B
T

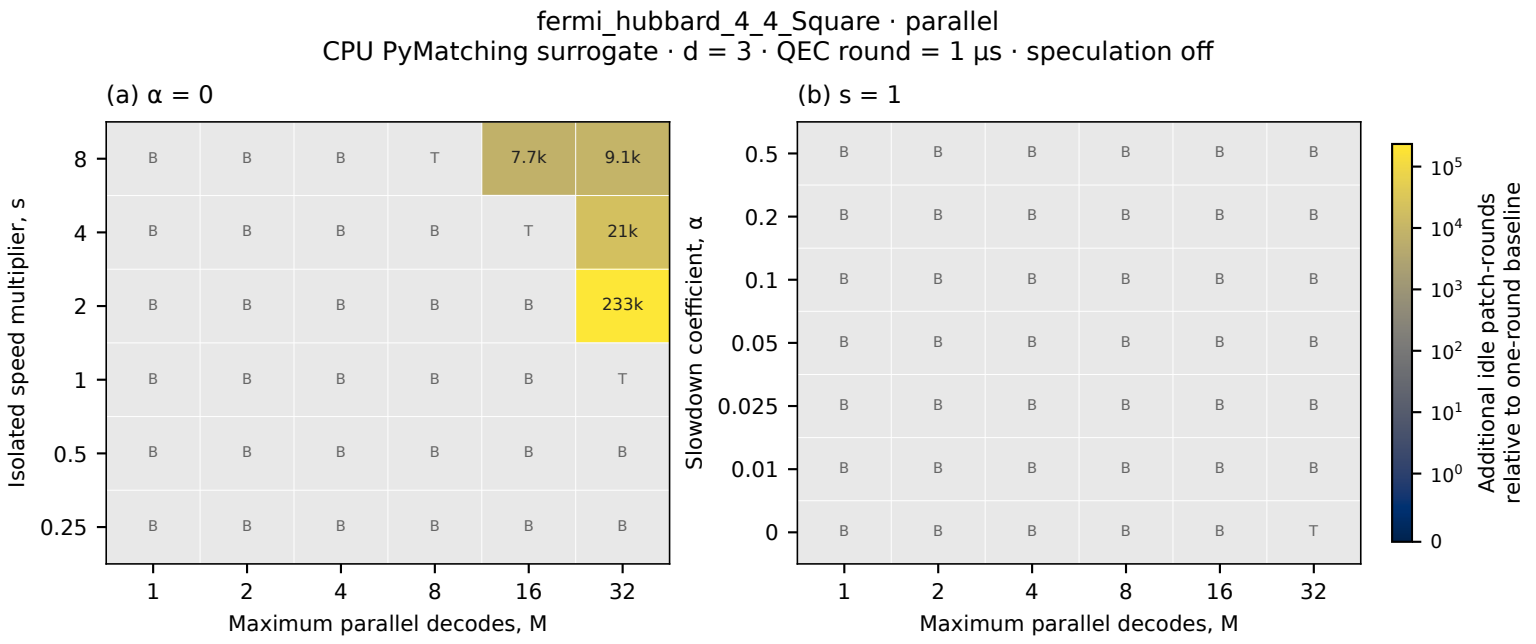

fermi_hubbard_2_2_Kagome · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
B
49k
121
120k
54k
371k

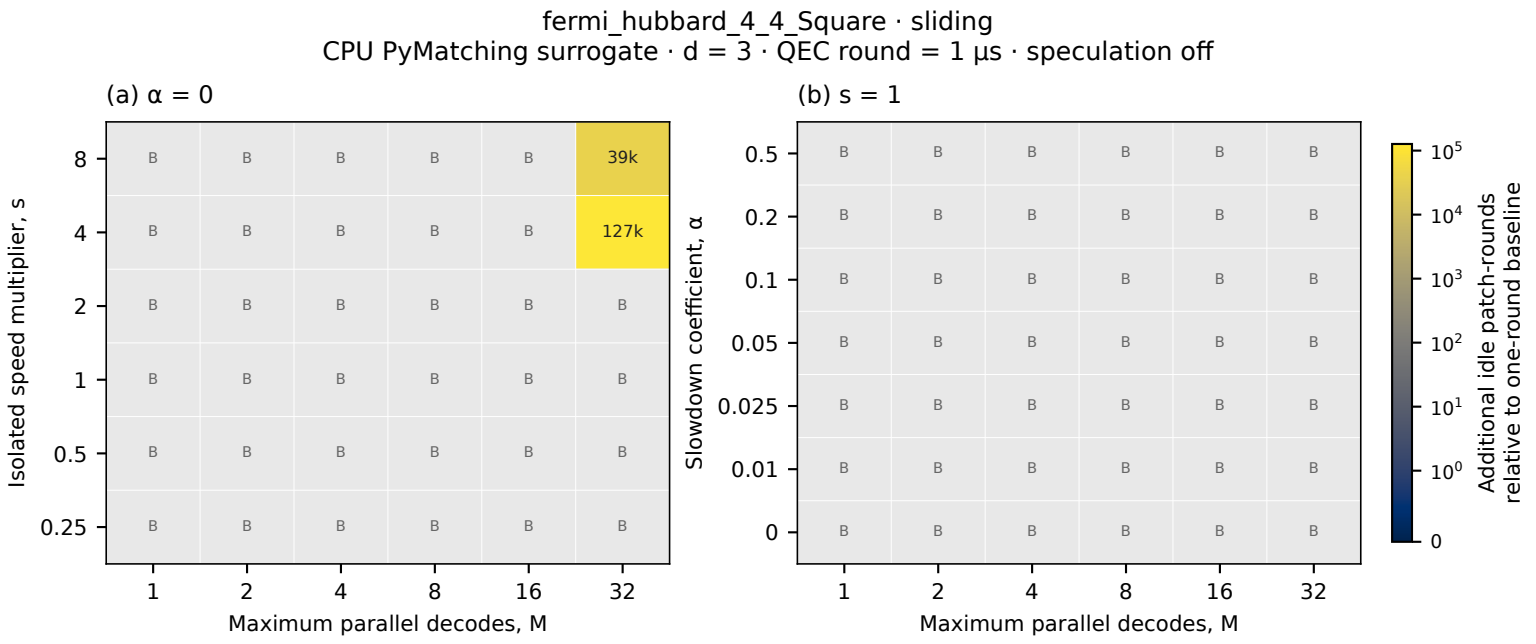

fermi_hubbard_2_2_Square · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
B
T
29k
945
0
24k
4.5k
4.3k
34k
32k
351k
95k
238k
125k
254k
100k
10^5
10^4
10^3
10^2
10^1
10^0

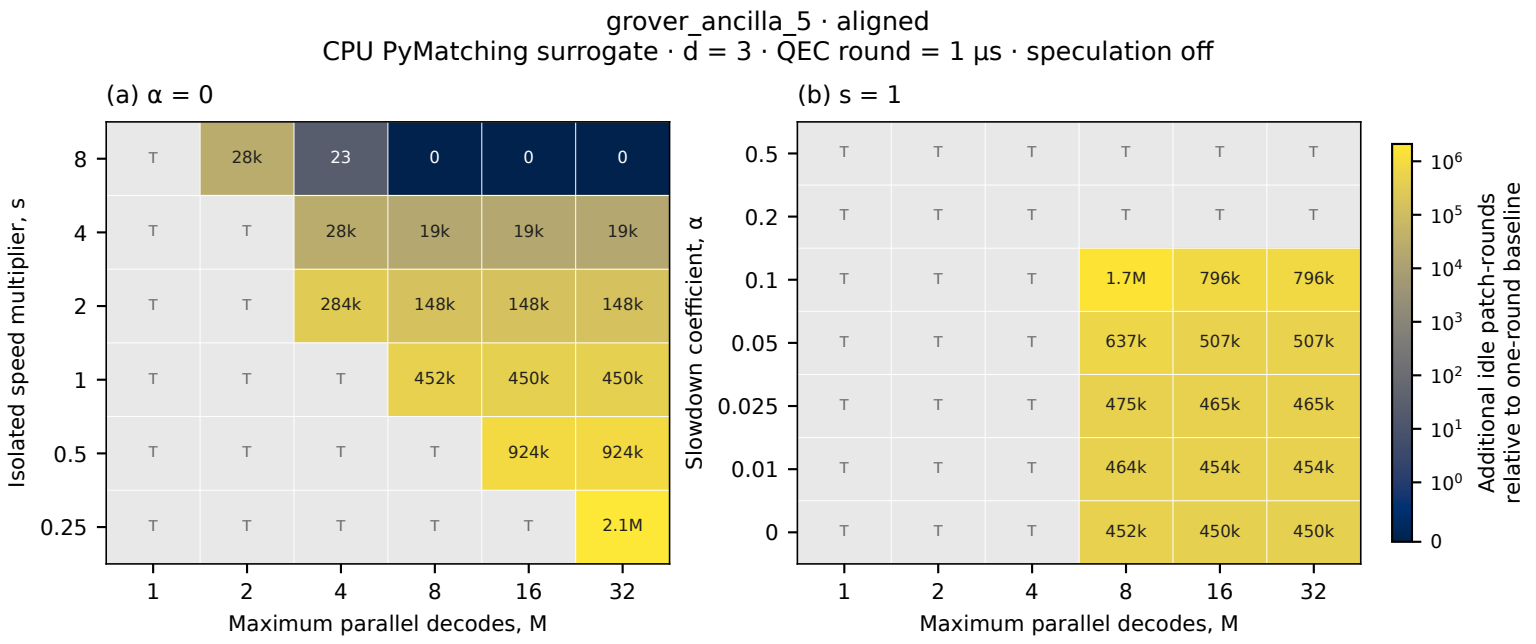

fermi_hubbard_2_2_Square · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
B
T
14k
1.0k
0
3.1k
-8.5k
12k
11k
259k
64k
194k
104k
287k
67k

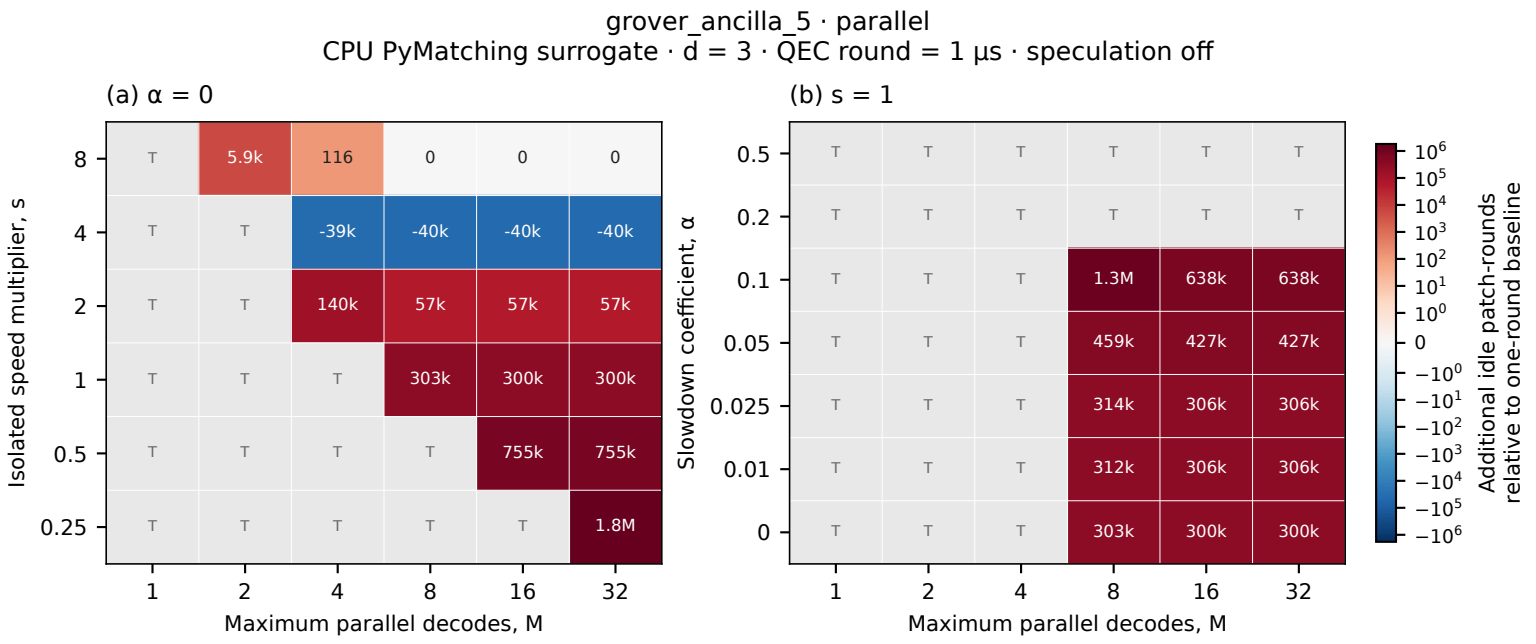

fermi_hubbard_2_2_Square · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
B
11k
210
0
27k
11k
11k
74k
68k
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
1
2
4
8
16
32
10^4
10^3
10^2
10^1
10^0

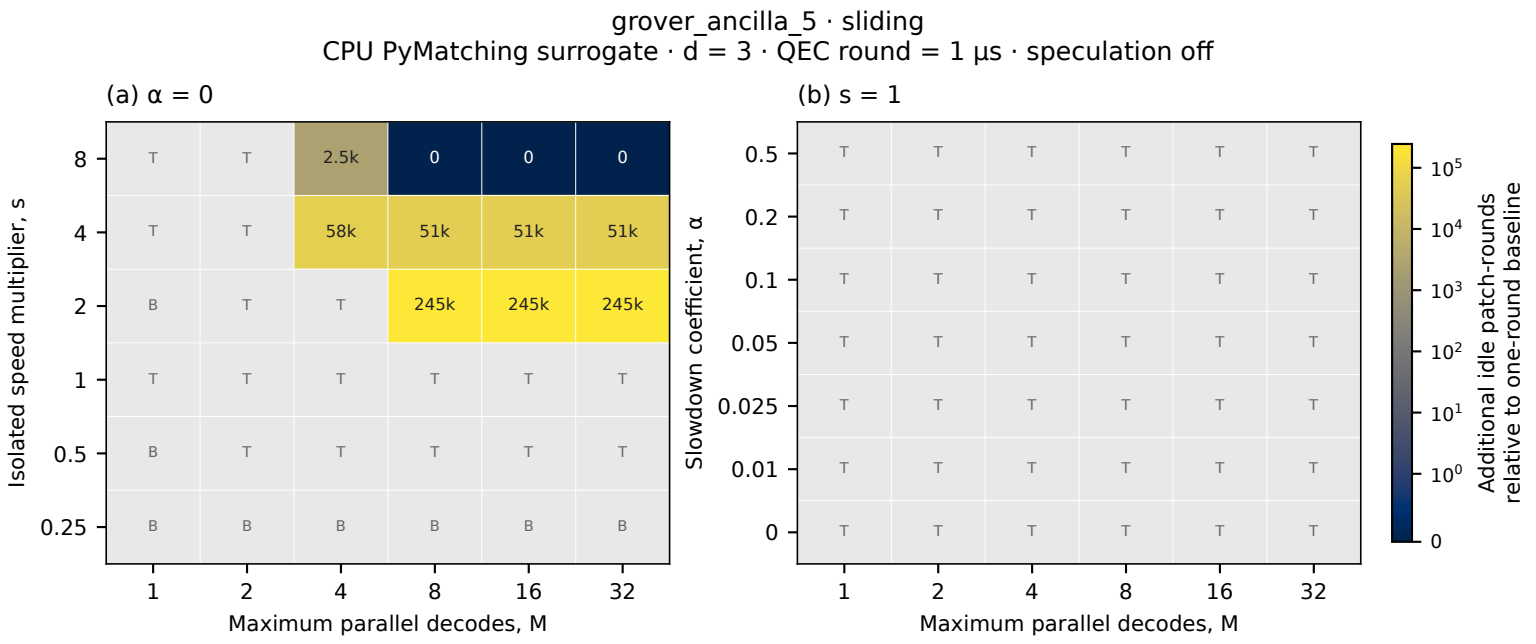

fermi_hubbard_4_4_Square · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
41k
6.0k
102k
385k
B
T
10^5
10^4
10^3
10^2
10^1
10^0
0

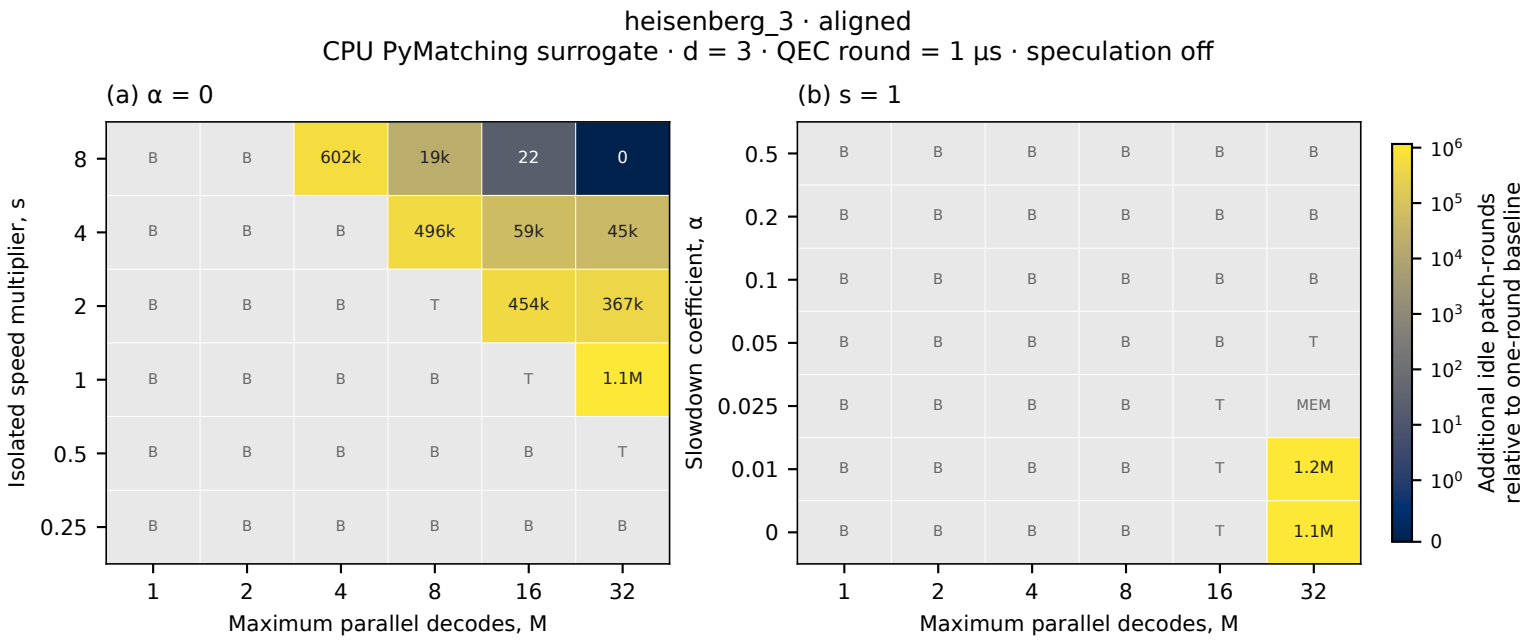

fermi_hubbard_4_4_Square · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
7.7k
9.1k
21k
233k
B
T

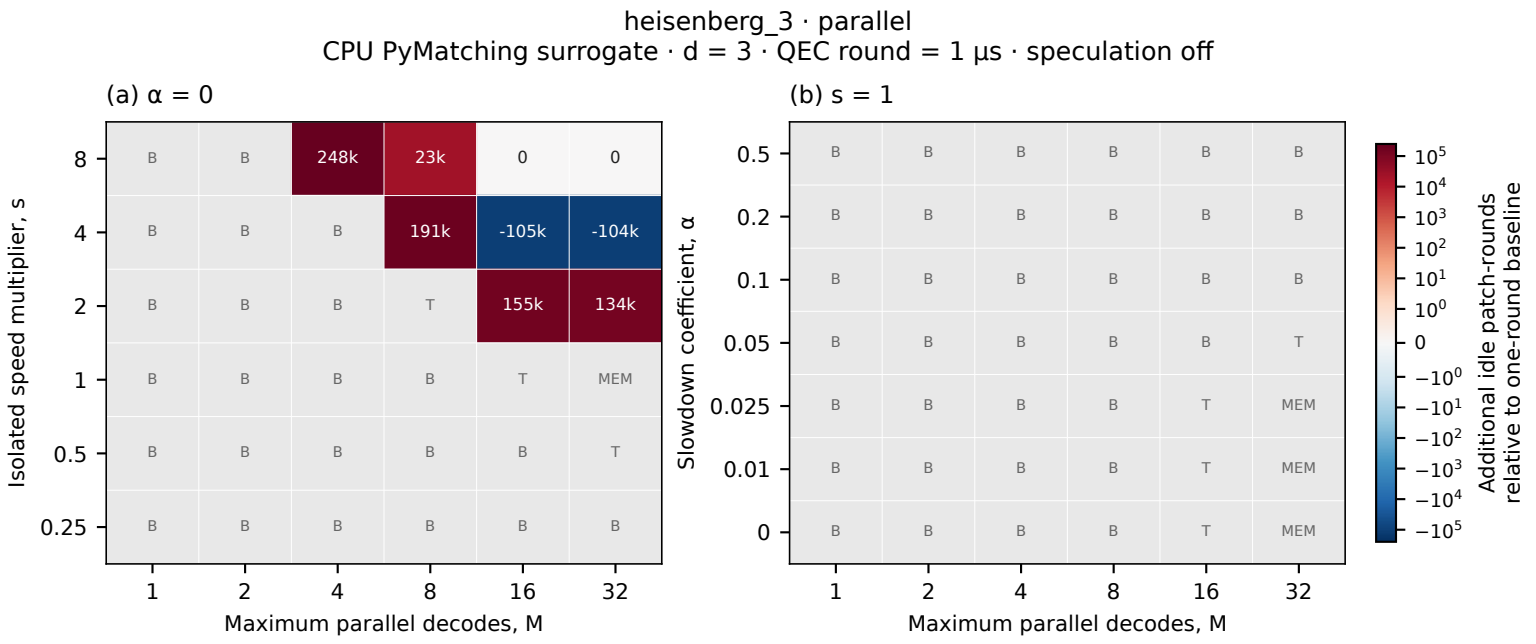

fermi_hubbard_4_4_Square · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
B
39k
127k
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
10^5
10^4
10^3
10^2
10^1
10^0

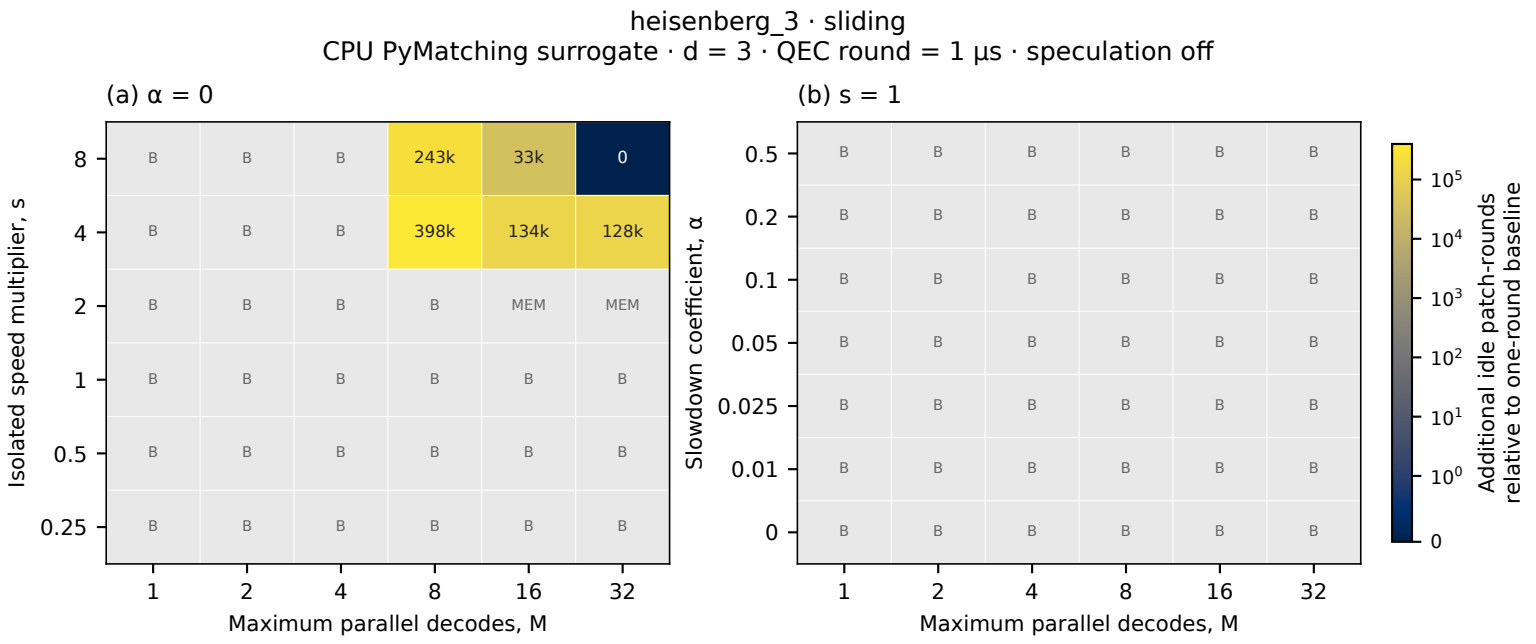

grover_ancilla_5 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
T
28k
23
0
19k
284k
148k
452k
450k
924k
2.1M
1.7M
796k
637k
507k
475k
465k
464k
454k
452k

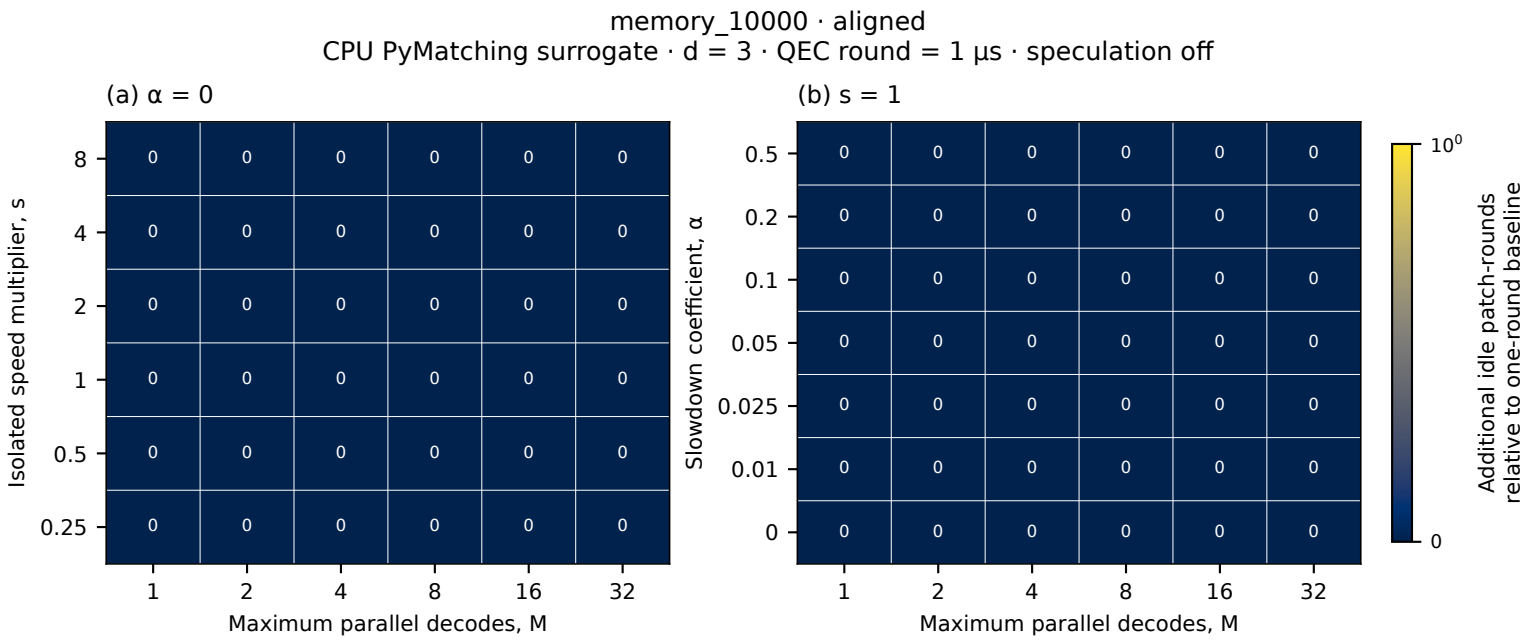

grover_ancilla_5 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
T
5.9k
116
0
-39k
-40k
140k
57k
303k
300k
755k
1.8M
1.3M
638k
459k
427k
314k
306k
312k

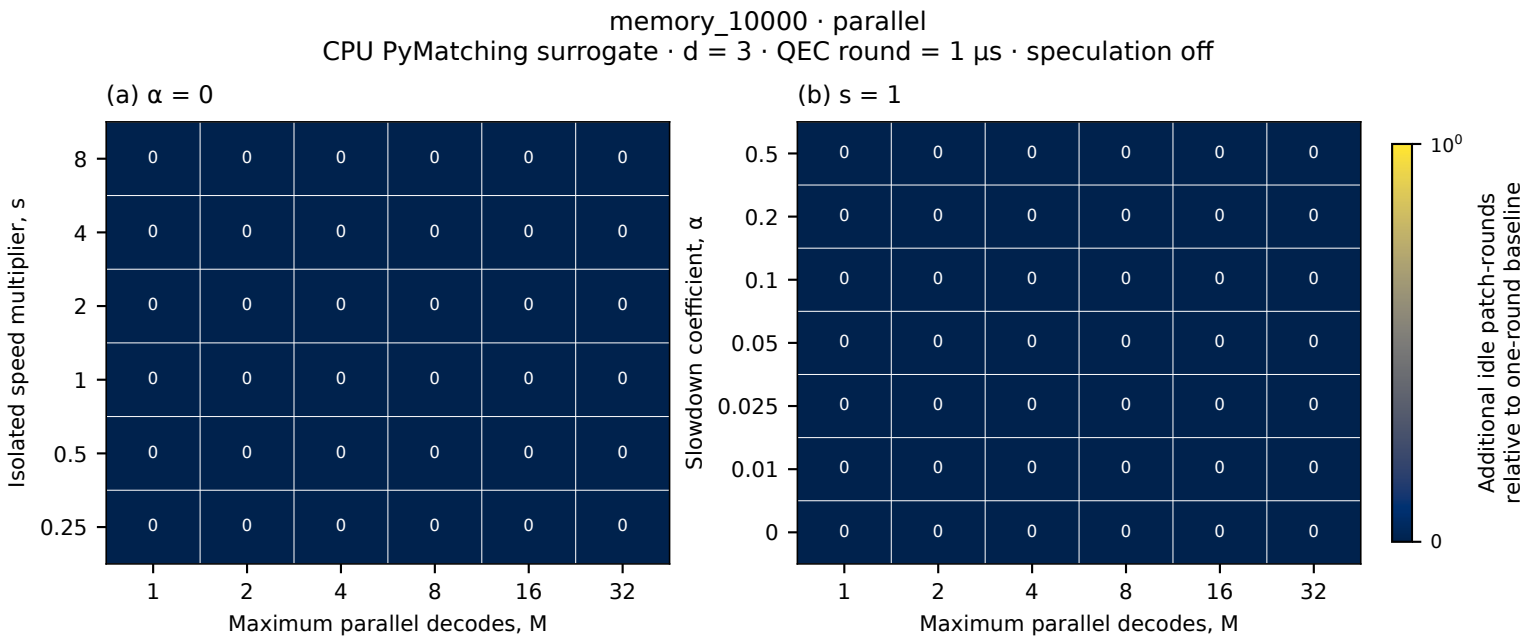

grover_ancilla_5 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
2.5k
0
58k
51k
245k
T
B

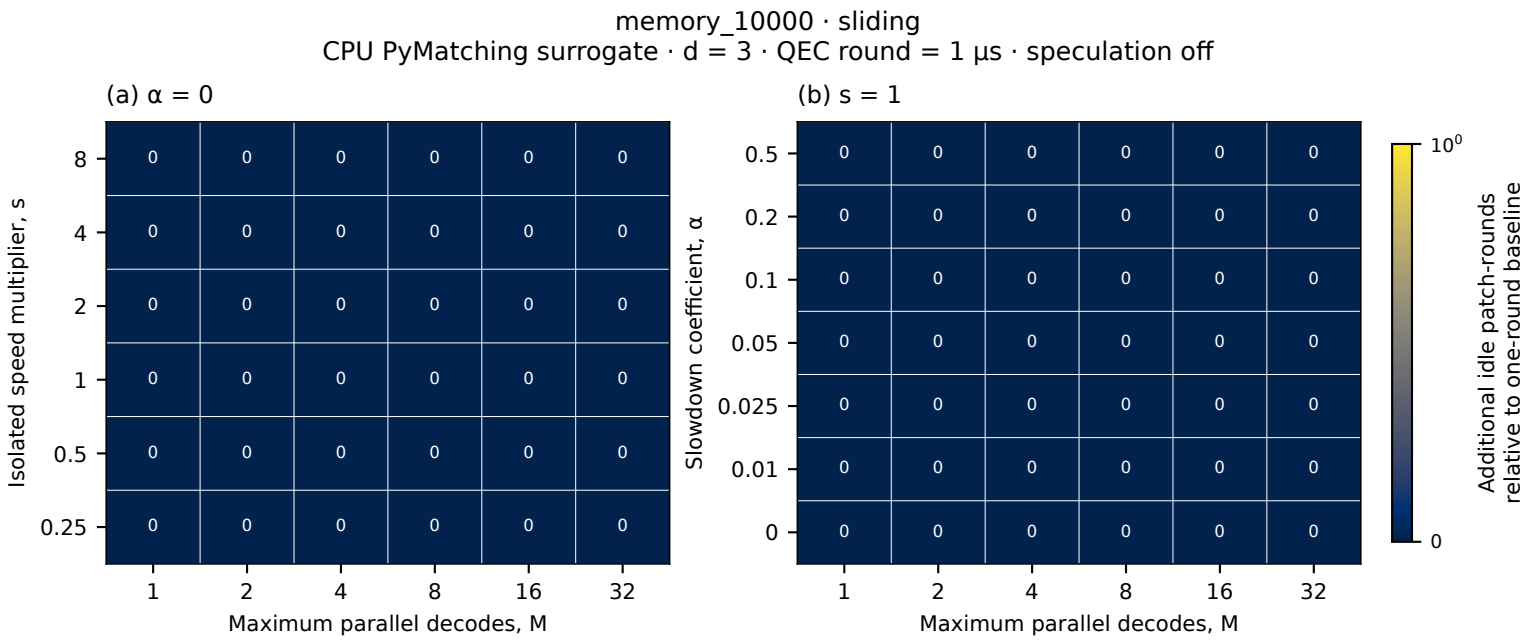

heisenberg_3 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
B
T
MEM
602k
19k
22
0
496k
59k
45k
454k
367k
1.1M
1.2M
10^6
10^5
10^4
10^3
10^2
10^1
10^0
0

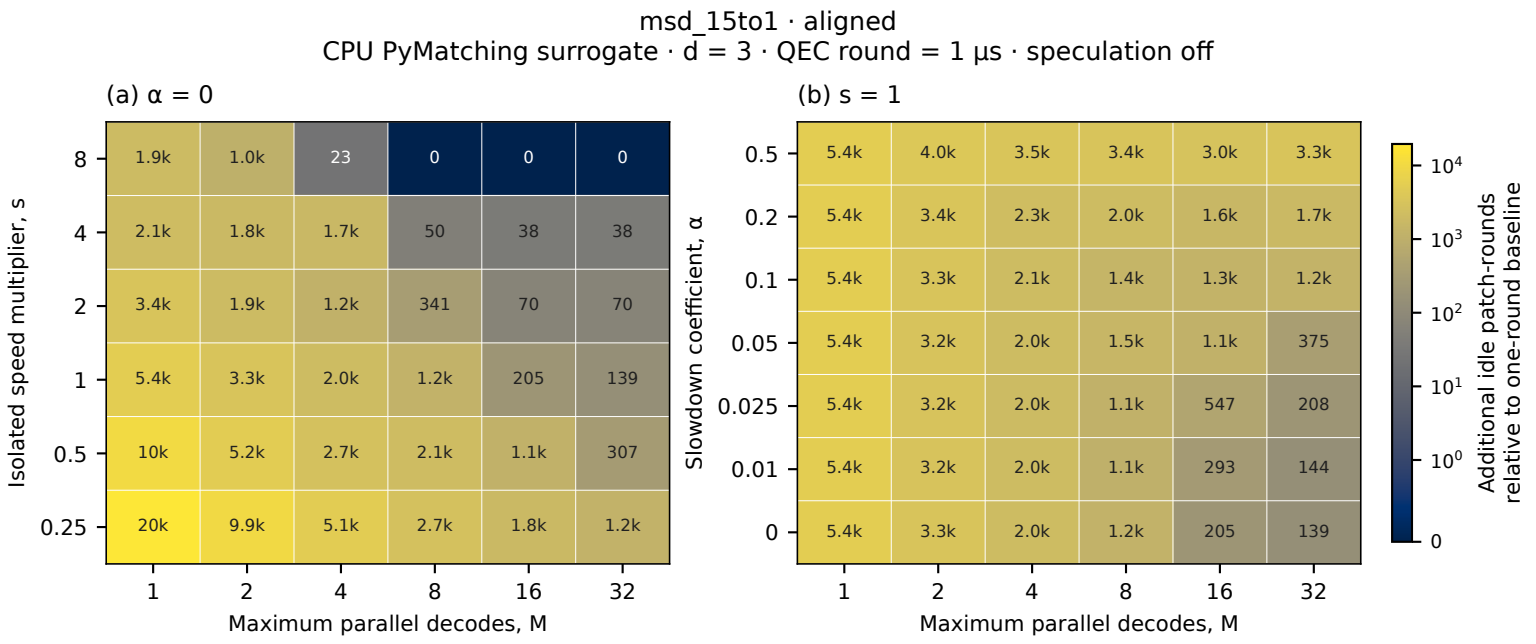

heisenberg_3 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
B
T
MEM
248k
23k
0
191k
-105k
-104k
155k
134k
10^5
10^4
10^3
10^2
10^1
10^0
-10^0
-10^1
-10^2
-10^3
-10^4
-10^5

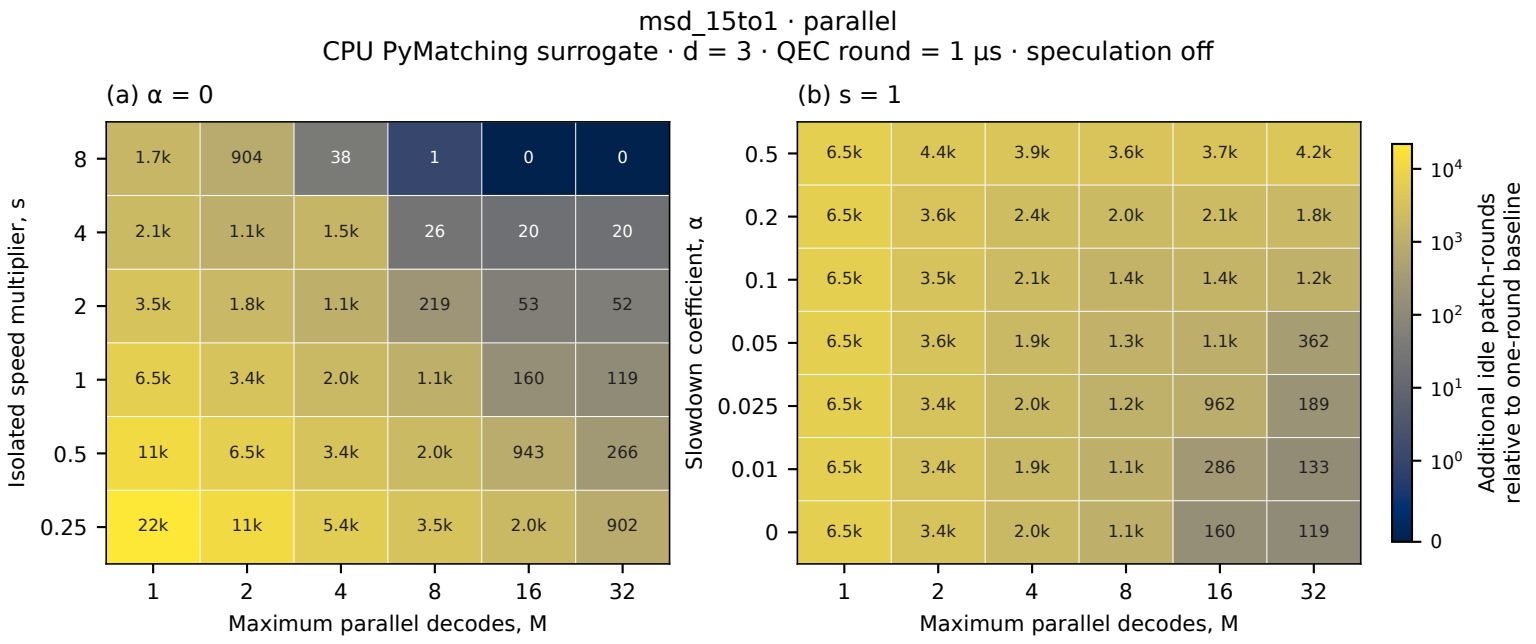

heisenberg_3 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
B
243k
33k
0
398k
134k
128k
MEM

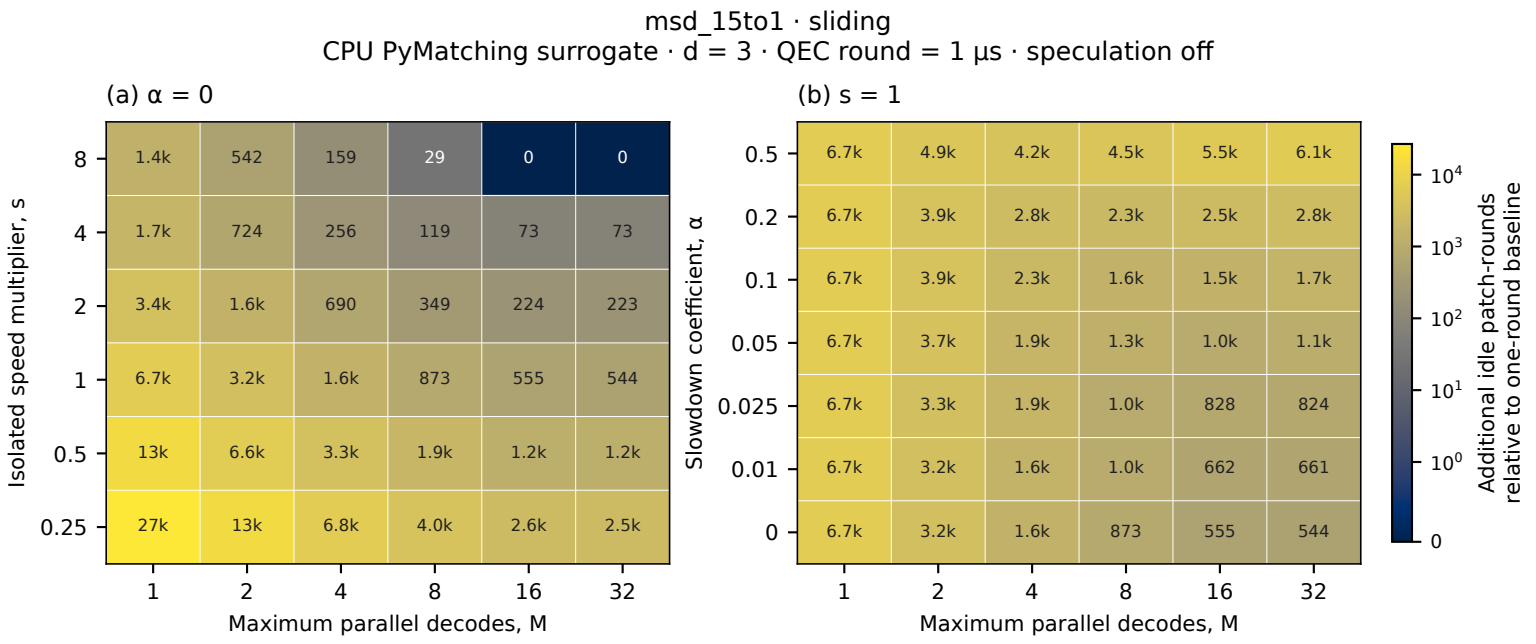

memory_10000 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^0
0
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32

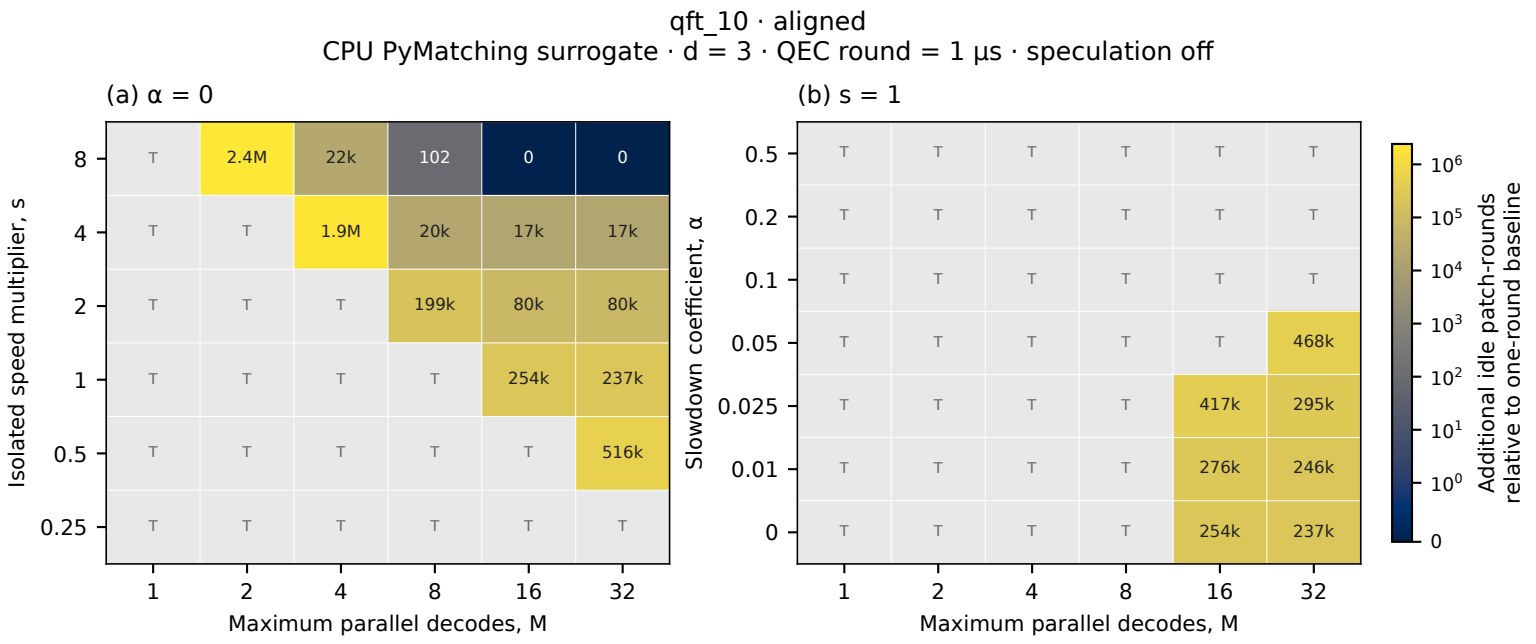

memory_10000 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^0
0
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32

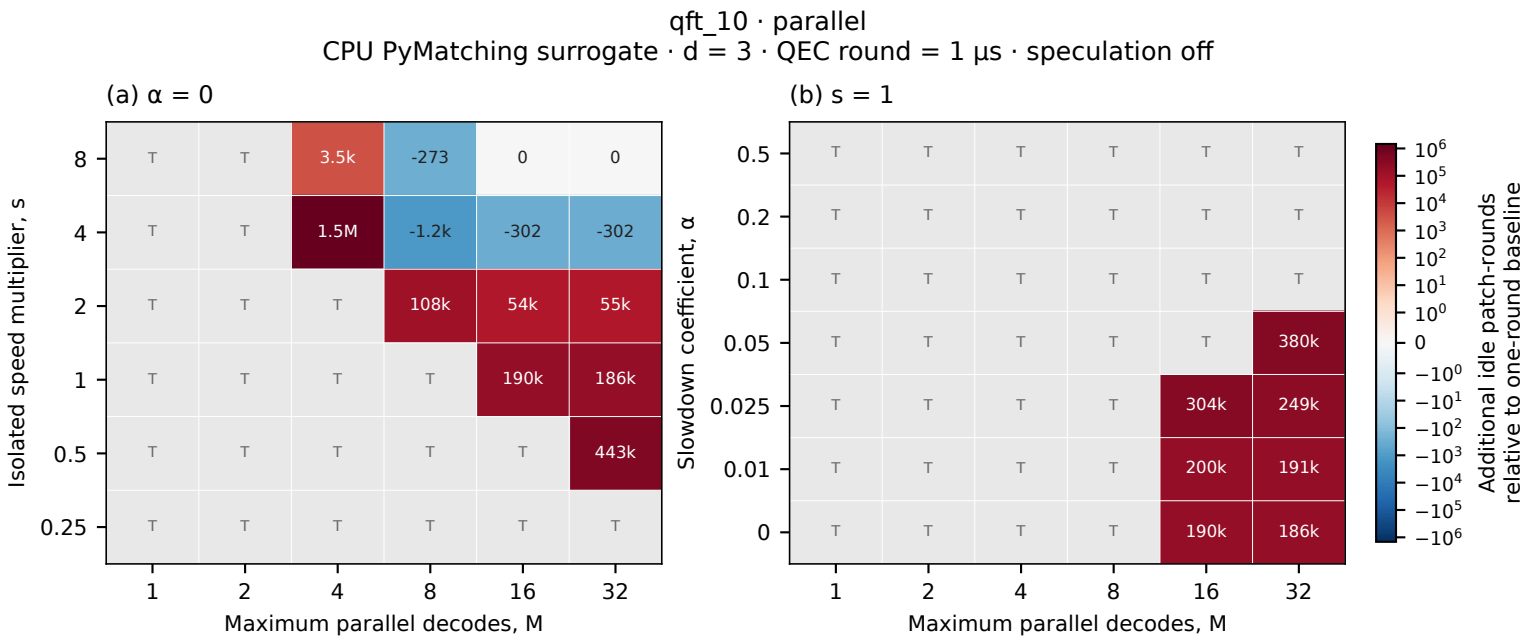

memory_10000 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^0
0
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32

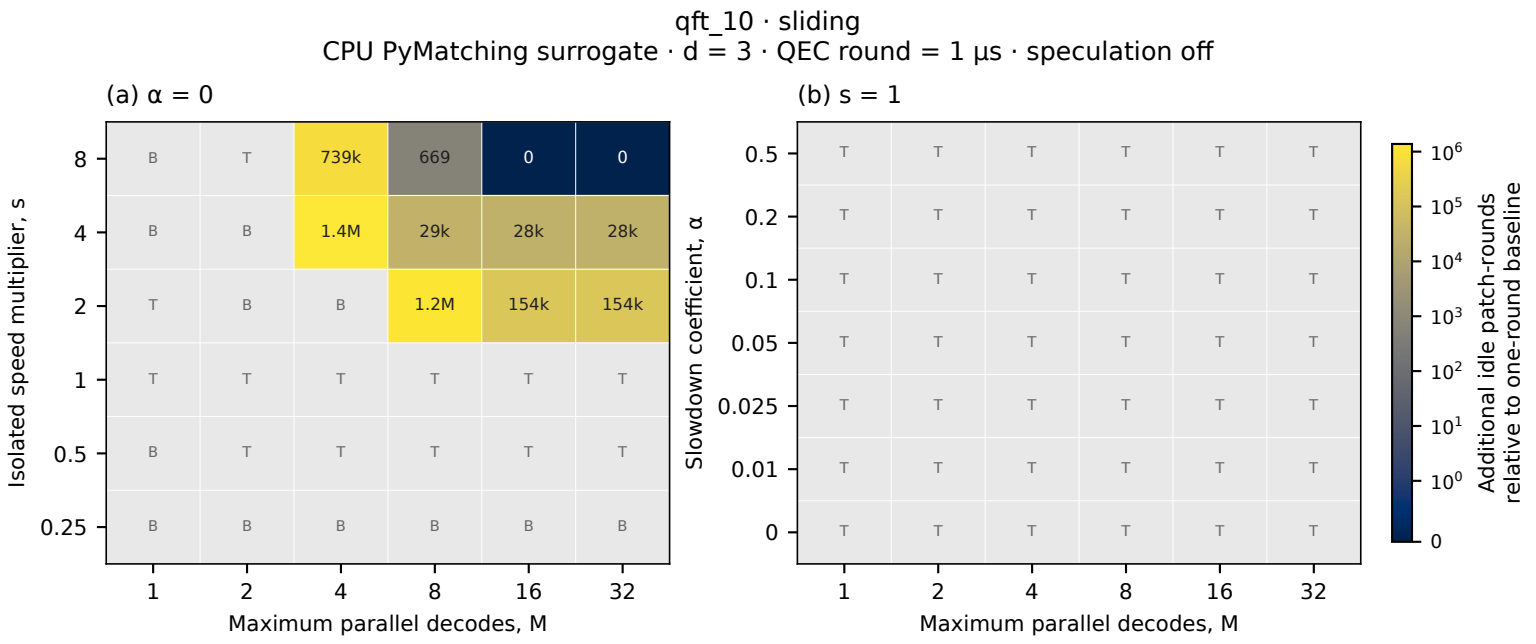

msd_15to1 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8 4 2 1 0.5 0.25
0.5 0.2 0.1 0.05 0.025 0.01 0
1 2 4 8 16 32
1.9k 1.0k 23 0 0 0
2.1k 1.8k 1.7k 50 38 38
3.4k 1.9k 1.2k 341 70 70
5.4k 3.3k 2.0k 1.2k 205 139
10k 5.2k 2.7k 2.1k 1.1k 307
20k 9.9k 5.1k 2.7k 1.8k 1.2k
5.4k 4.0k 3.5k 3.4k 3.0k 3.3k
5.4k 3.4k 2.3k 2.0k 1.6k 1.7k
5.4k 3.3k 2.1k 1.4k 1.3k 1.2k
5.4k 3.2k 2.0k 1.5k 1.1k 375
5.4k 3.2k 2.0k 1.1k 547 208
5.4k 3.2k 2.0k 1.1k 293 144
5.4k 3.3k 2.0k 1.2k 205 139
10^4 10^3 10^2 10^1 10^0 0

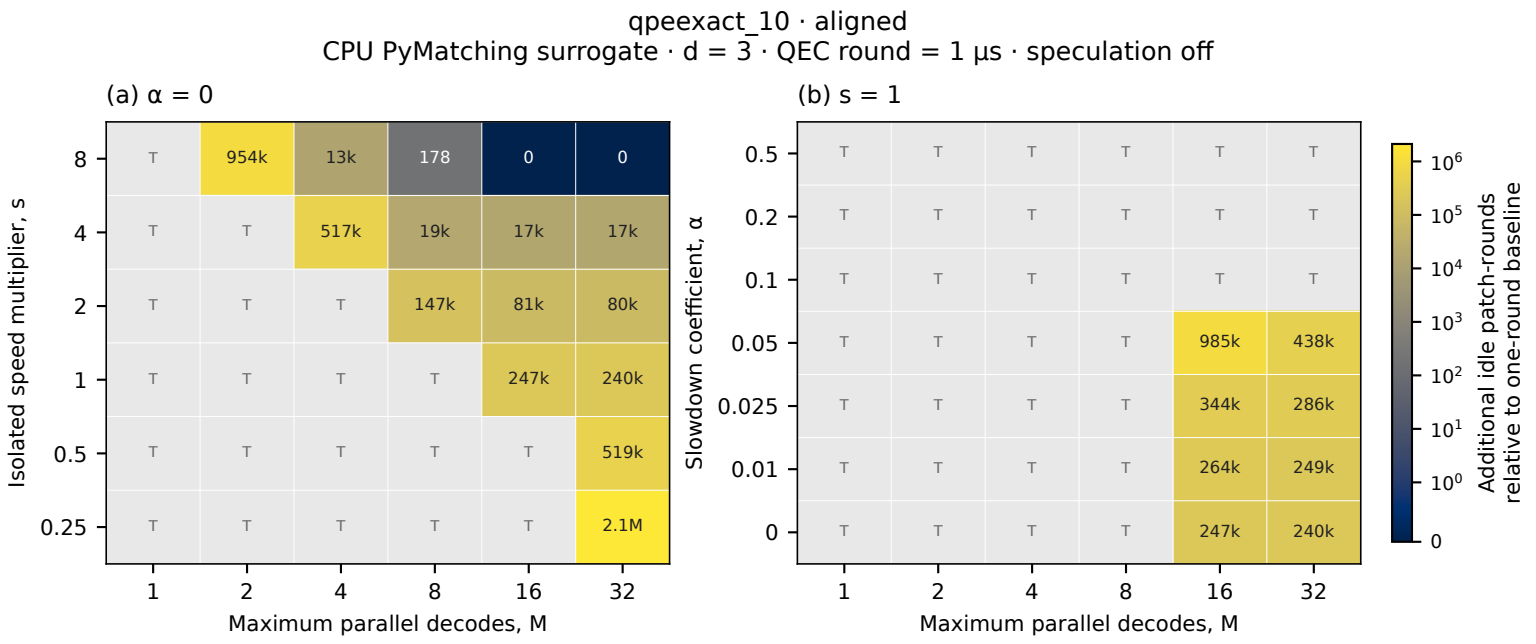

msd_15to1 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
1.7k 904 38 1 0 0
2.1k 1.1k 1.5k 26 20 20
3.5k 1.8k 1.1k 219 53 52
6.5k 3.4k 2.0k 1.1k 160 119
11k 6.5k 3.4k 2.0k 943 266
22k 11k 5.4k 3.5k 2.0k 902
1 2 4 8 16 32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
6.5k 4.4k 3.9k 3.6k 3.7k 4.2k
6.5k 3.6k 2.4k 2.0k 2.1k 1.8k
6.5k 3.5k 2.1k 1.4k 1.4k 1.2k
6.5k 3.6k 1.9k 1.3k 1.1k 362
6.5k 3.4k 2.0k 1.2k 962 189
6.5k 3.4k 1.9k 1.1k 286 133
6.5k 3.4k 2.0k 1.1k 160 119
1 2 4 8 16 32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

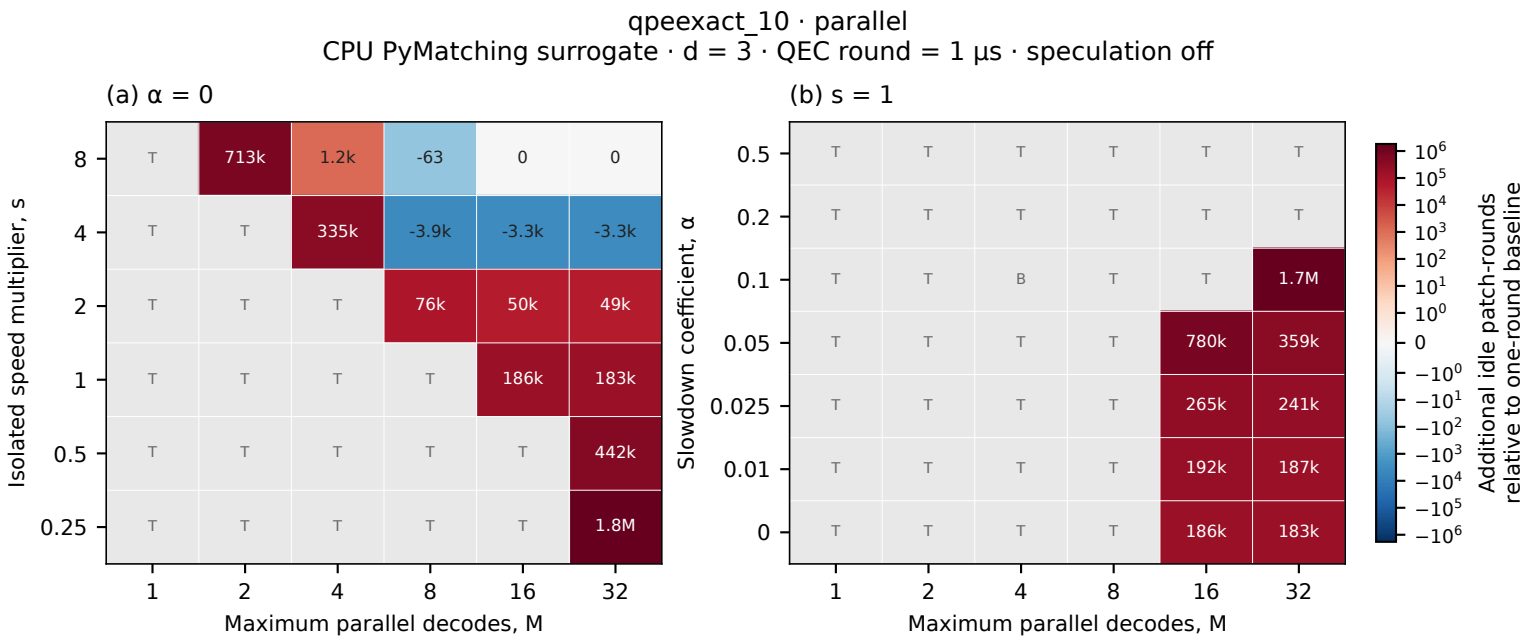

msd_15to1 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
1.4k 542 159 29 0 0
1.7k 724 256 119 73 73
3.4k 1.6k 690 349 224 223
6.7k 3.2k 1.6k 873 555 544
13k 6.6k 3.3k 1.9k 1.2k 1.2k
27k 13k 6.8k 4.0k 2.6k 2.5k
1 2 4 8 16 32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
6.7k 4.9k 4.2k 4.5k 5.5k 6.1k
6.7k 3.9k 2.8k 2.3k 2.5k 2.8k
6.7k 3.9k 2.3k 1.6k 1.5k 1.7k
6.7k 3.7k 1.9k 1.3k 1.0k 1.1k
6.7k 3.3k 1.9k 1.0k 828 824
6.7k 3.2k 1.6k 1.0k 662 661
6.7k 3.2k 1.6k 873 555 544
1 2 4 8 16 32
Maximum parallel decodes, M
10^4
10^3
10^2
10^1
10^0
0
Additional idle patch-rounds
relative to one-round baseline

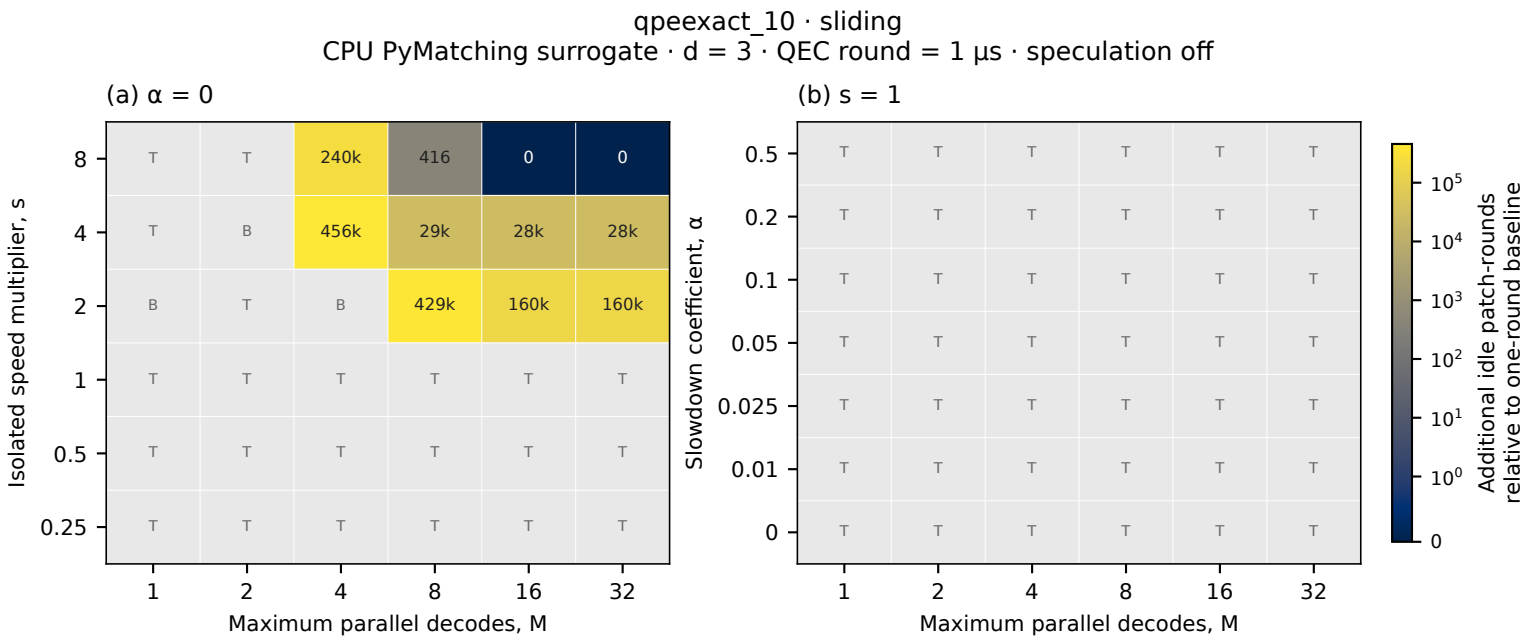

qft_10 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
T
2.4M
22k
102
0
1.9M
20k
17k
199k
80k
254k
516k
468k
417k
295k
276k
246k

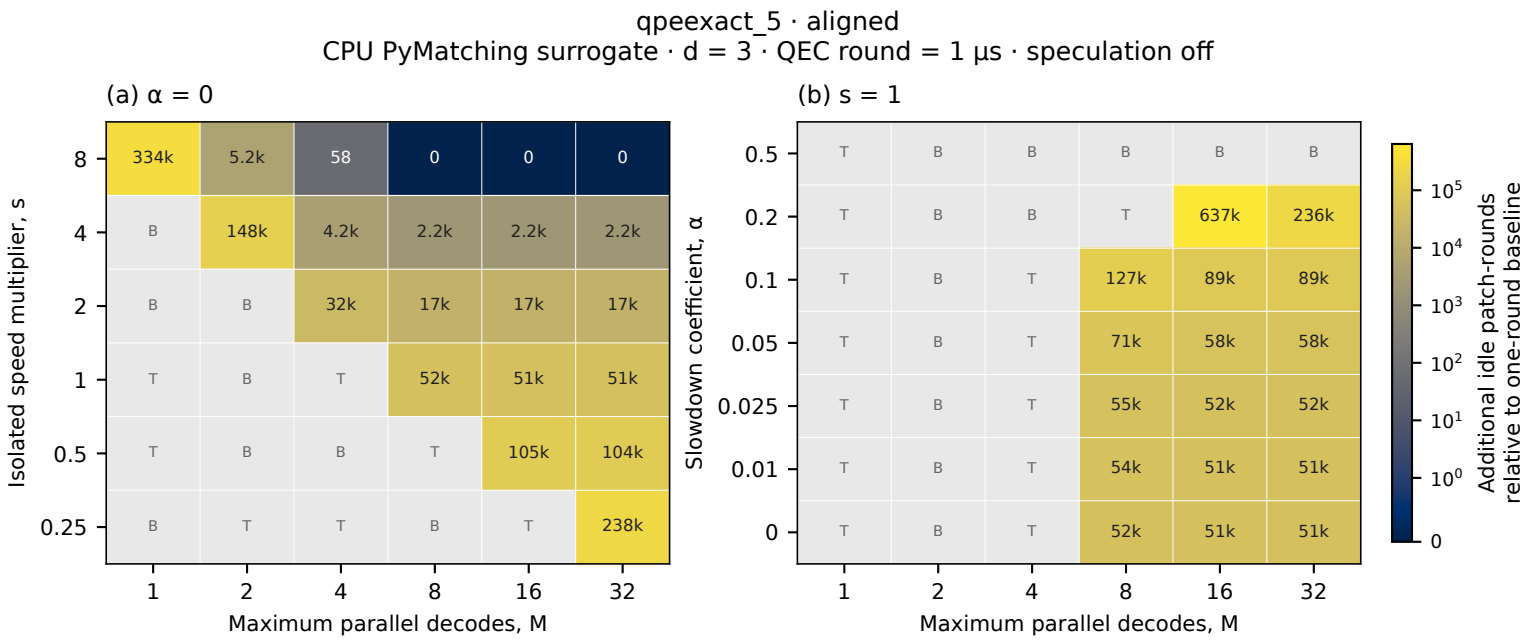

qft_10 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
T
3.5k
-273
0
0
1.5M
-1.2k
-302
-302
108k
54k
55k
190k
186k
443k
380k
304k
249k
200k
191k
190k
186k
10^{6}
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
0
-10^{0}
-10^{1}
-10^{2}
-10^{3}
-10^{4}
-10^{5}
-10^{6}

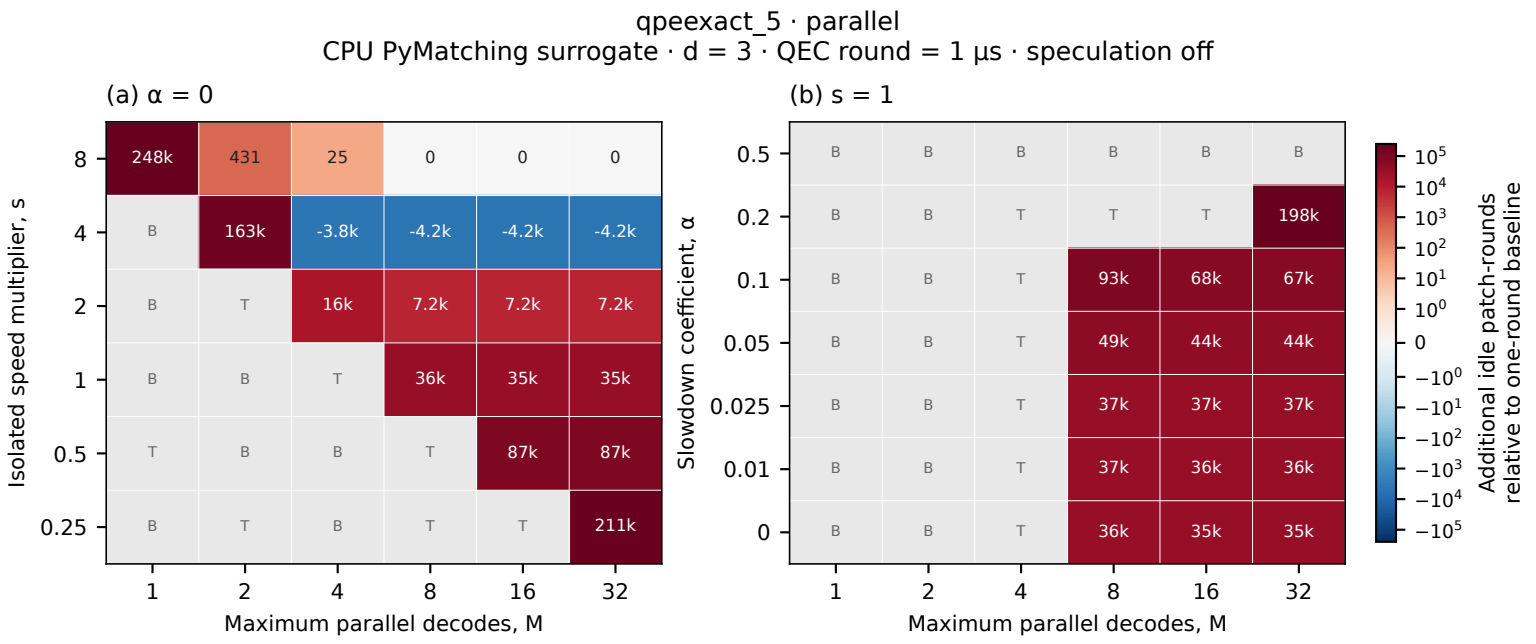

qft_10 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
B
T
739k
669
0
1.4M
29k
28k
1.2M
154k
10^6
10^5
10^4
10^3
10^2
10^1
10^0

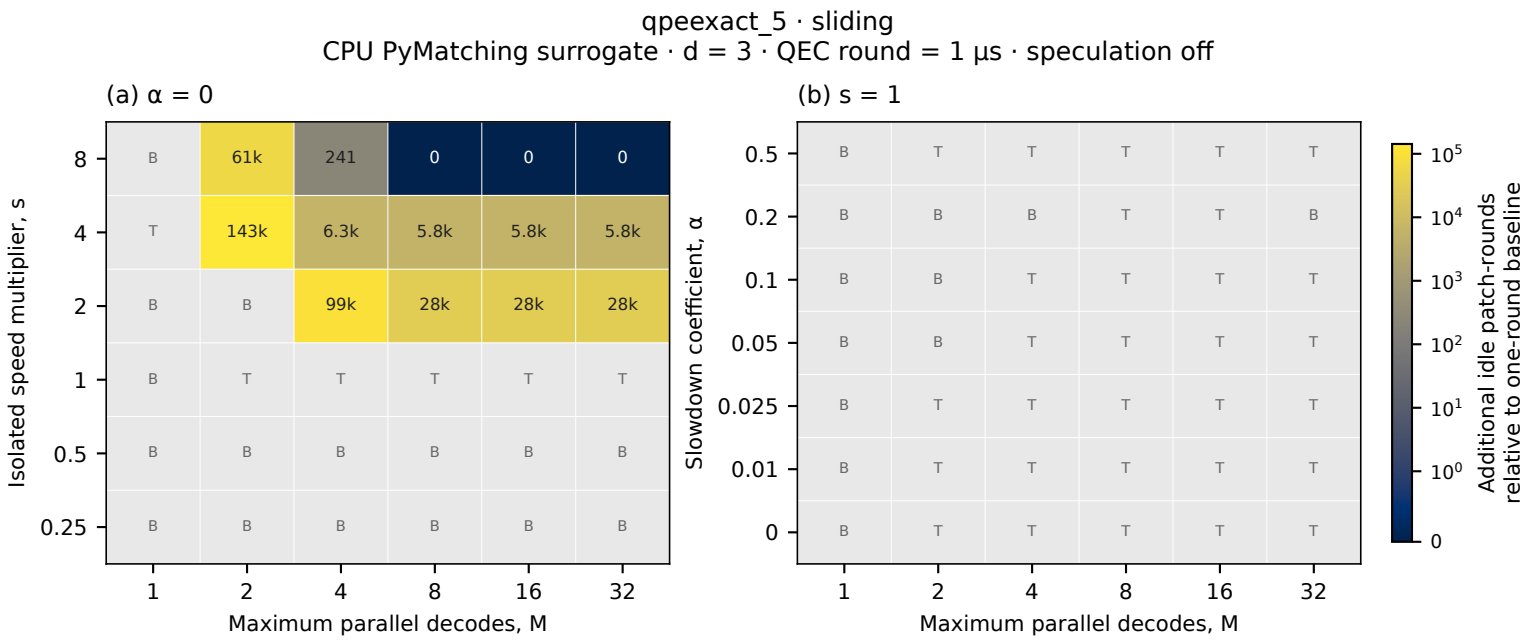

qpeexact_10 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
T
954k
13k
178
0
0
517k
19k
17k
17k
147k
81k
80k
247k
240k
519k
2.1M
985k
438k
344k
286k
264k
249k
247k
240k
10^6
10^5
10^4
10^3
10^2
10^1
10^0
0

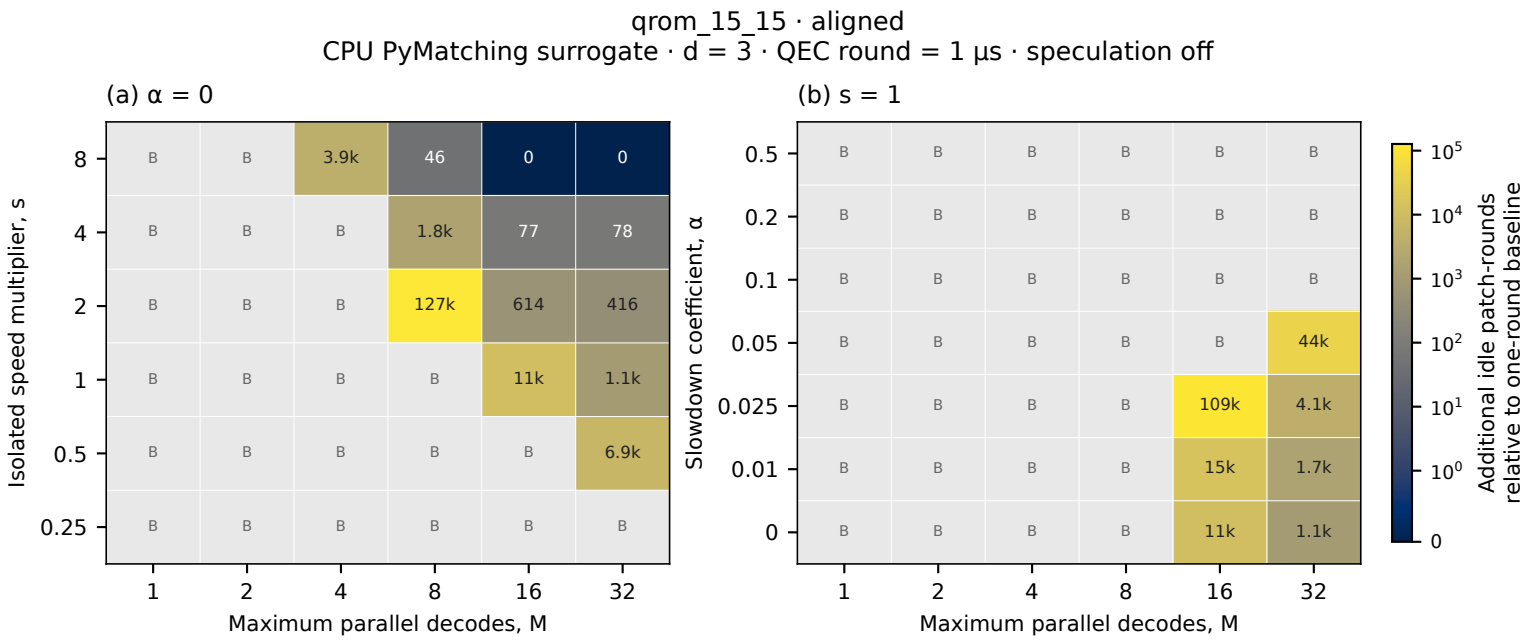

qpeexact_10 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
T
B
713k
1.2k
-63
0
335k
-3.9k
-3.3k
76k
50k
49k
186k
183k
442k
1.8M
1.7M
780k
359k
265k
241k
192k
187k

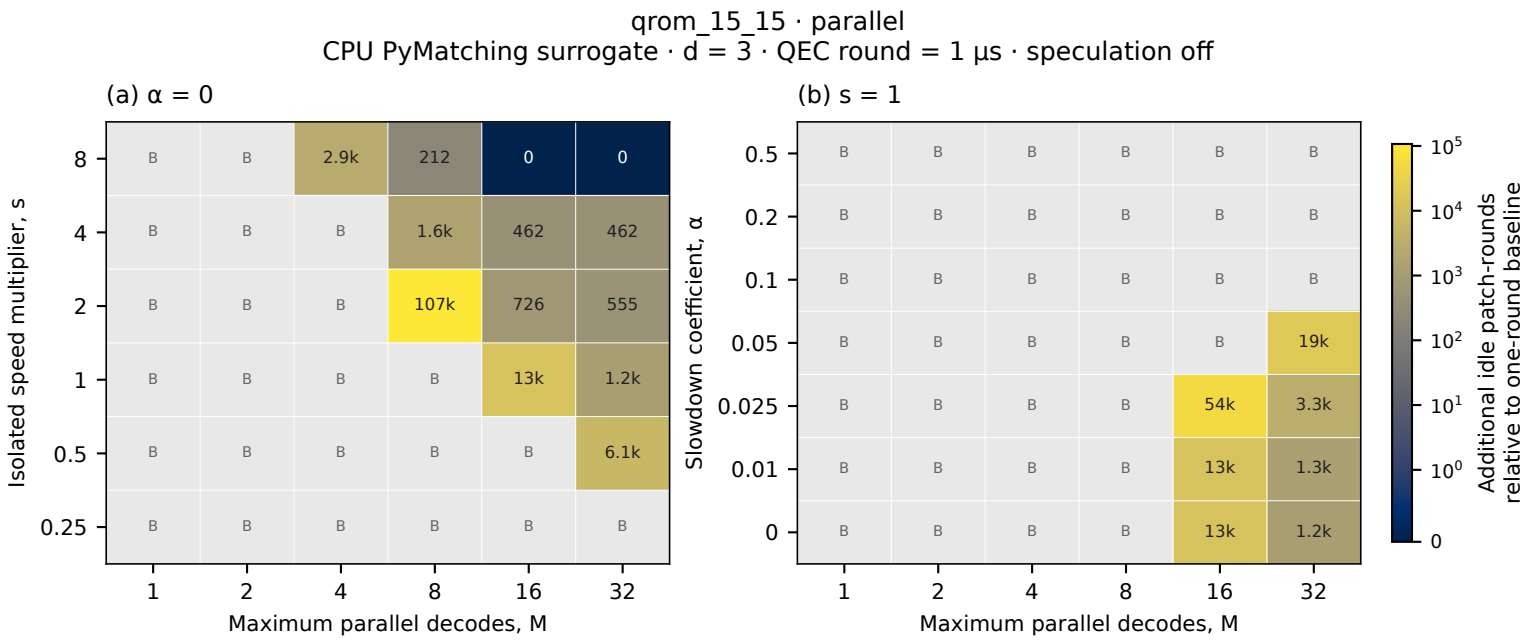

qpeexact_10 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
240k
416
0
456k
29k
28k
429k
160k

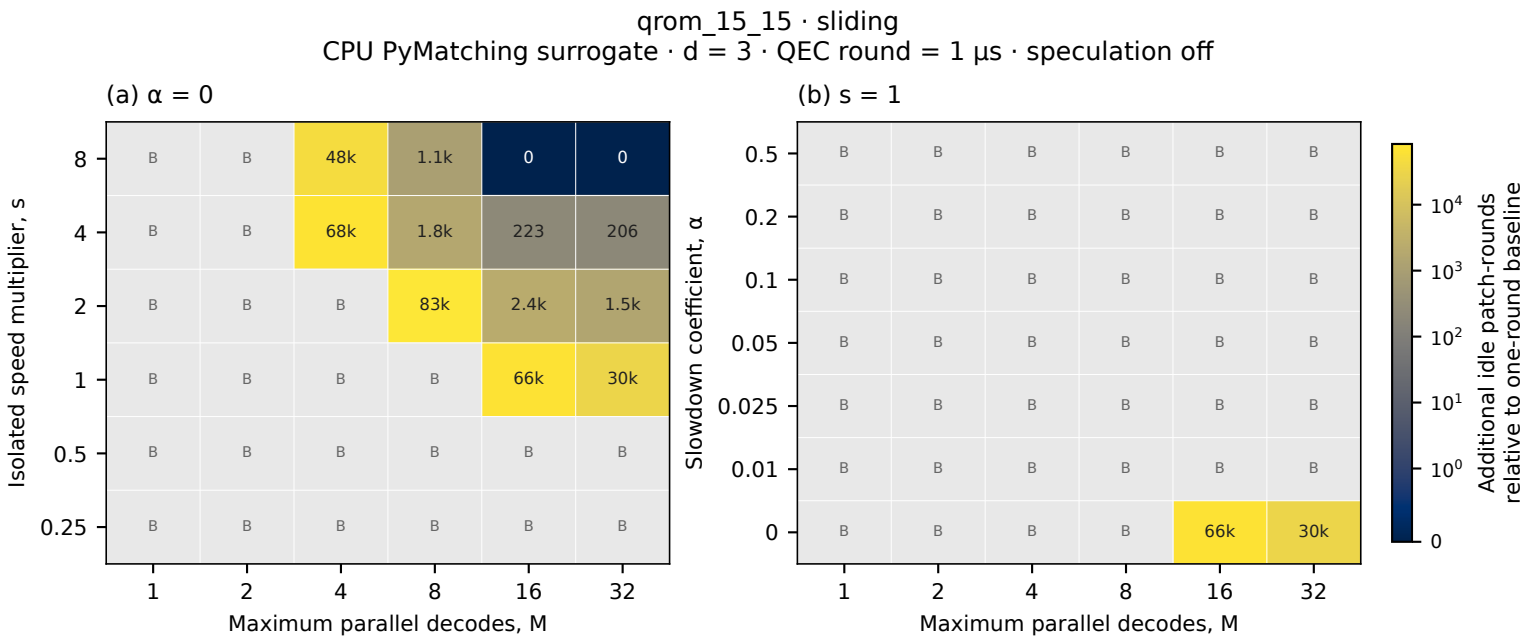

qpeexact_5 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
334k
5.2k
58
0
0
0
148k
4.2k
2.2k
2.2k
2.2k
32k
17k
17k
17k
52k
51k
51k
105k
104k
238k
637k
236k
127k
89k
89k
71k
58k
58k
55k
52k
52k
54k
51k
51k
52k
51k
51k
B
T
10^5
10^4
10^3
10^2
10^1
10^0
0

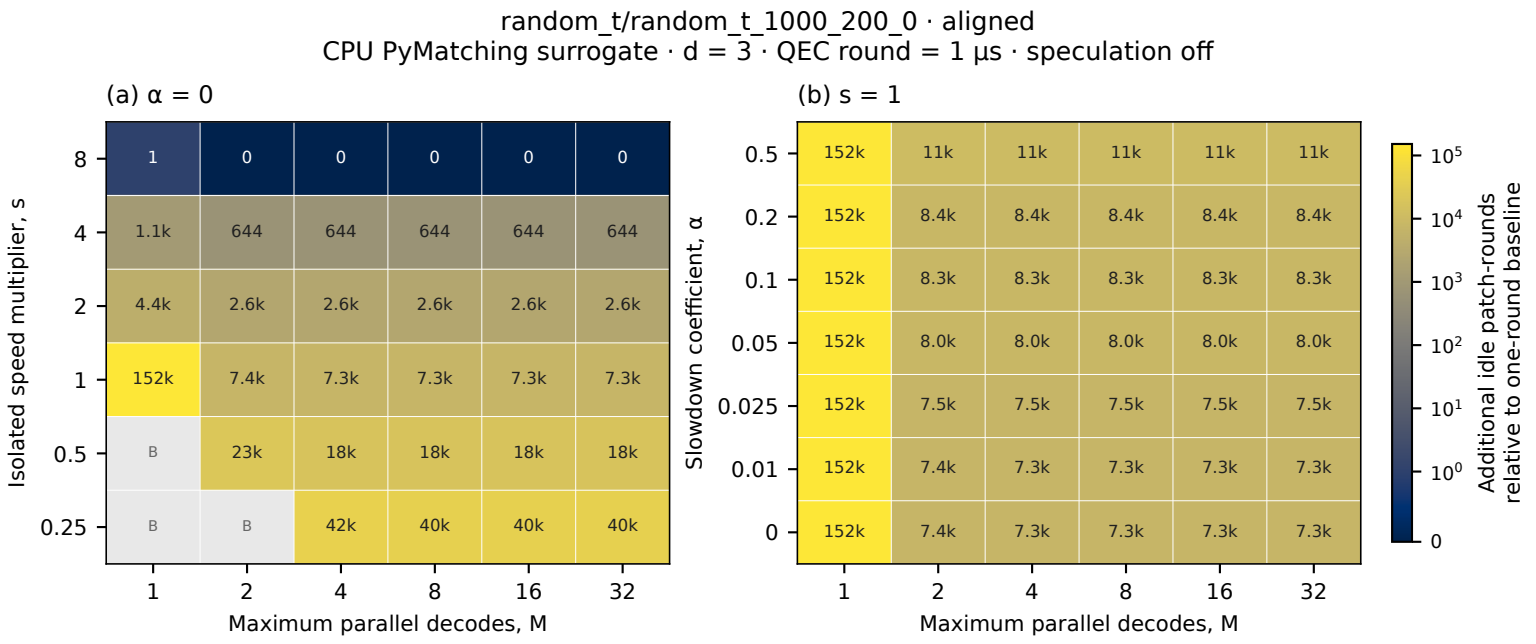

qpeexact_5 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
248k
431
25
0
0
0
163k
-3.8k
-4.2k
-4.2k
-4.2k
16k
7.2k
7.2k
7.2k
36k
35k
35k
87k
87k
211k
198k
93k
68k
67k
49k
44k
44k
37k
37k
37k
37k
36k
36k
36k
35k
35k
B
T

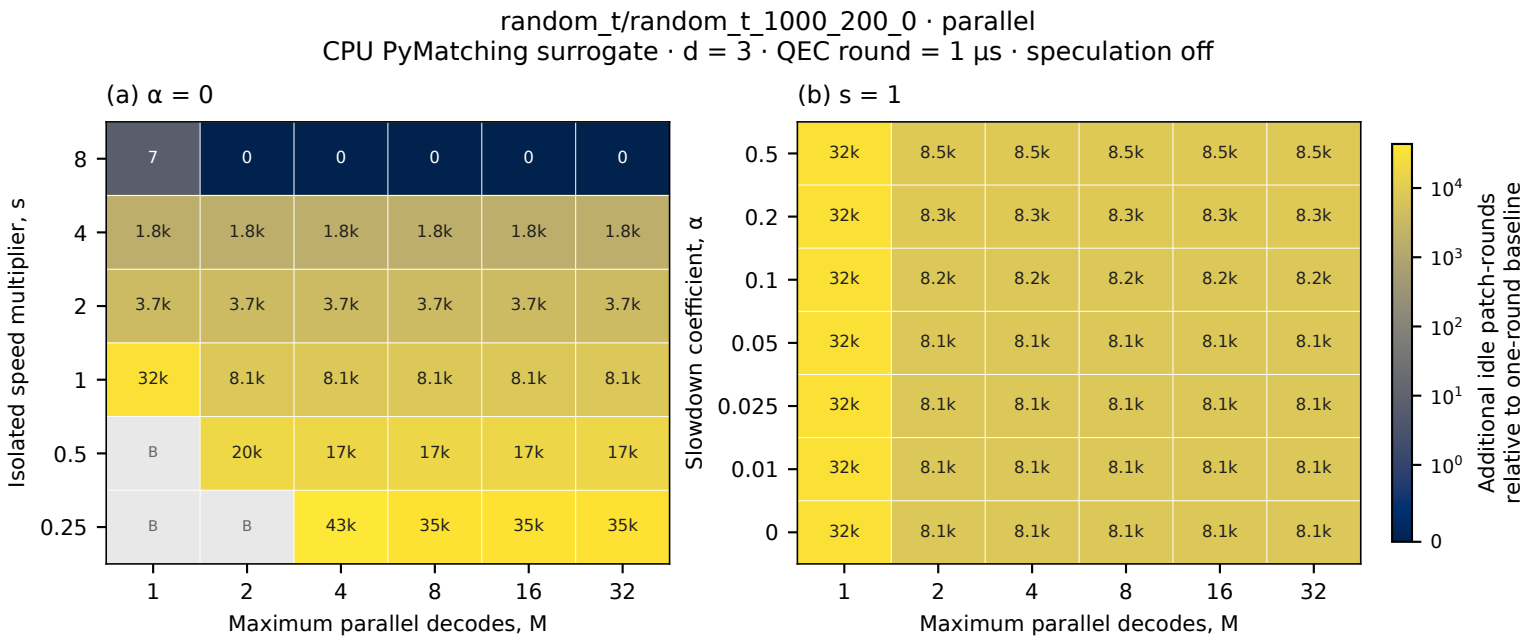

qpeexact_5 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
61k
241
143k
6.3k
5.8k
99k
28k
B
T
10^5
10^4
10^3
10^2
10^1
10^0

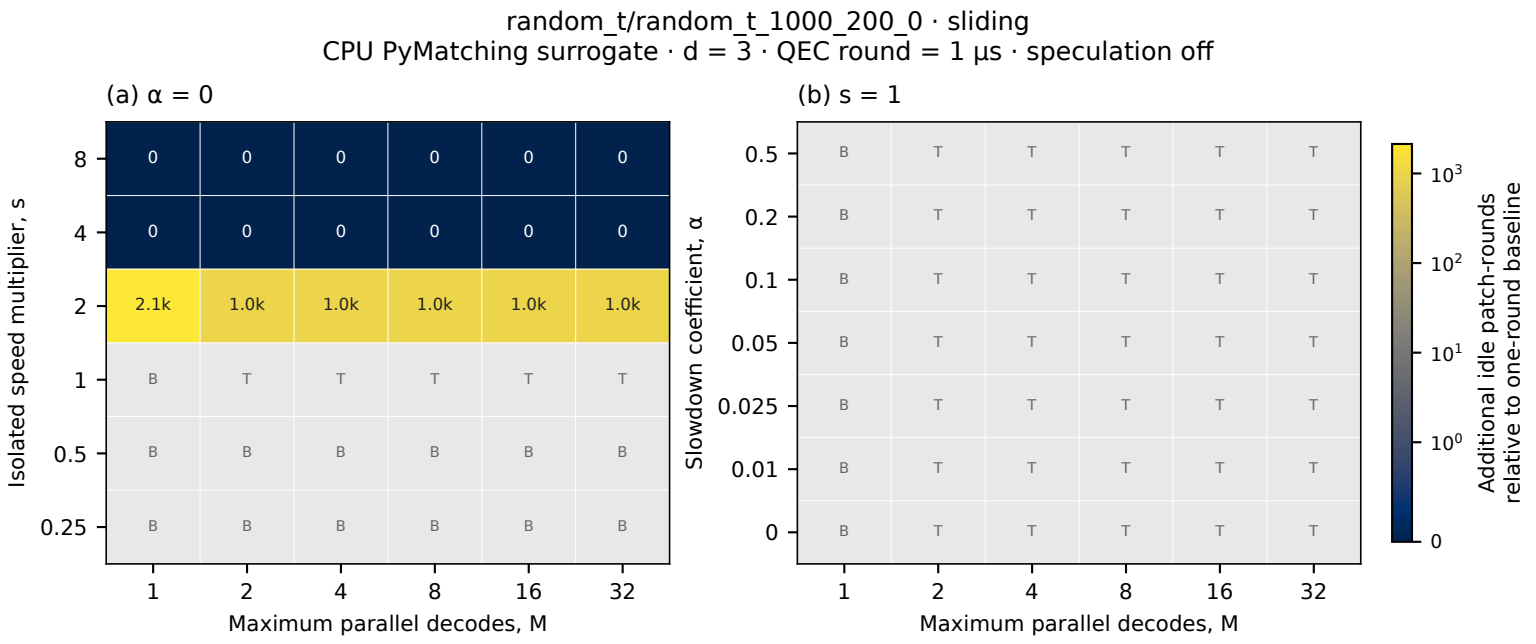

qrom_15_15 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
B
3.9k
46
0
1.8k
77
78
127k
614
416
11k
1.1k
6.9k
44k
109k
4.1k
15k
1.7k
11k
1.1k


qrom_15_15 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
B
2.9k
212
0
1.6k
462
107k
726
555
13k
1.2k
6.1k
19k
54k
3.3k
1.3k

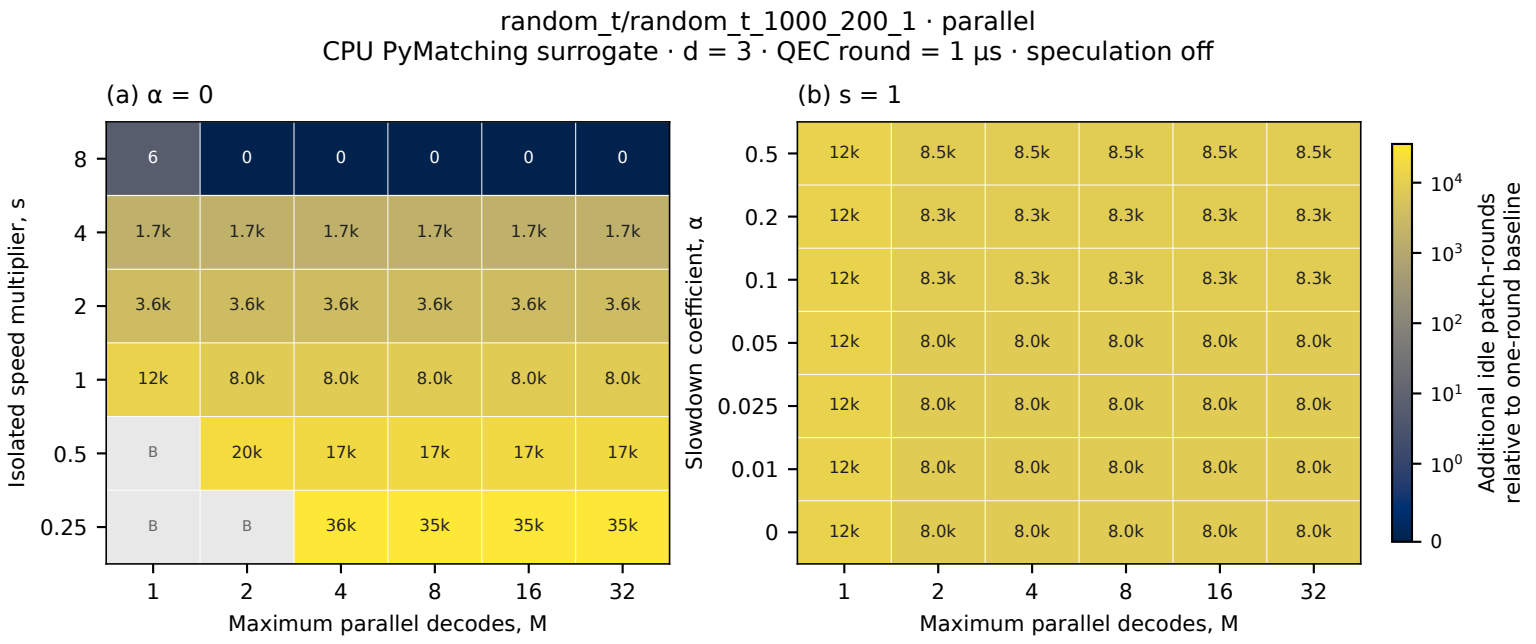

qrom_15_15 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
B
48k
1.1k
0
68k
1.8k
223
206
83k
2.4k
1.5k
66k
30k

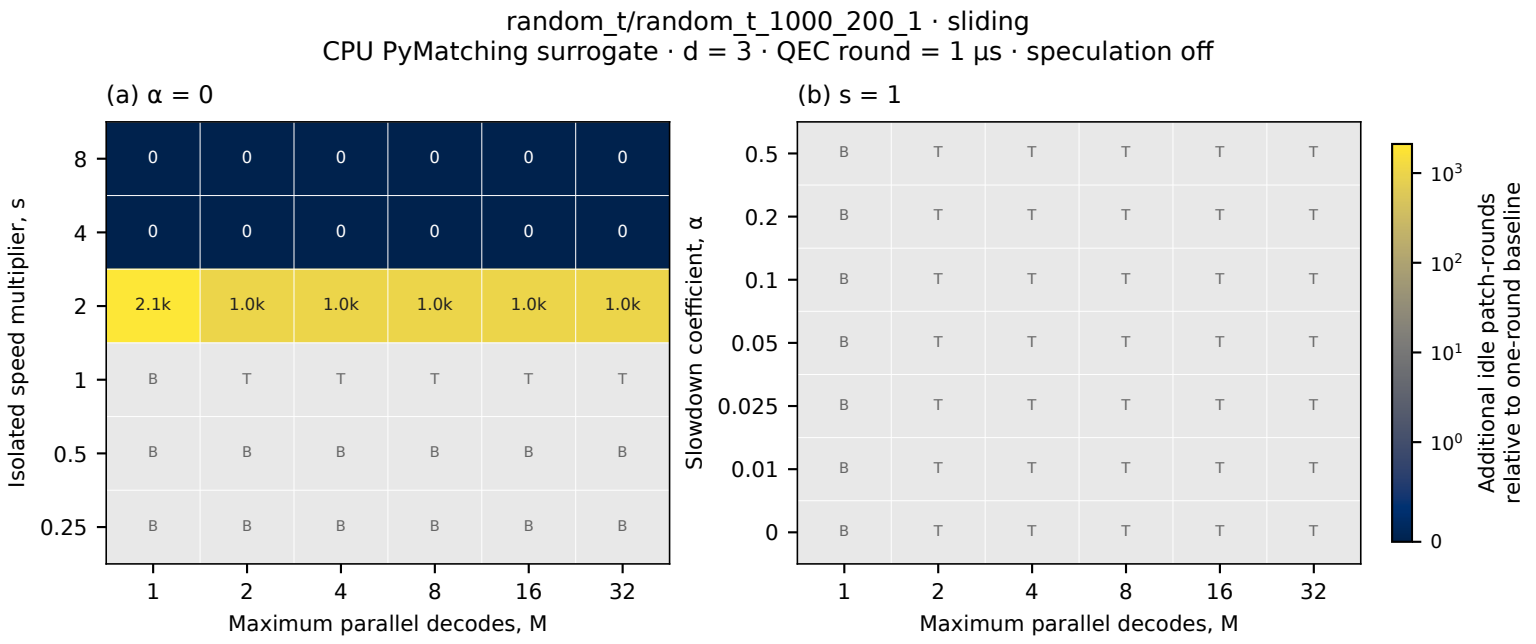

random_t/random_t_1000_200_0 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
1
0
0
0
0
0
1.1k
644
644
644
644
644
4.4k
2.6k
2.6k
2.6k
2.6k
2.6k
152k
7.4k
7.3k
7.3k
7.3k
7.3k
B
23k
18k
18k
18k
18k
B
B
42k
40k
40k
40k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
152k
11k
11k
11k
11k
11k
152k
8.4k
8.4k
8.4k
8.4k
8.4k
152k
8.3k
8.3k
8.3k
8.3k
8.3k
152k
8.0k
8.0k
8.0k
8.0k
8.0k
152k
7.5k
7.5k
7.5k
7.5k
7.5k
152k
7.4k
7.3k
7.3k
7.3k
7.3k
152k
7.4k
7.3k
7.3k
7.3k
7.3k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^5
10^4
10^3
10^2
10^1
10^0
0

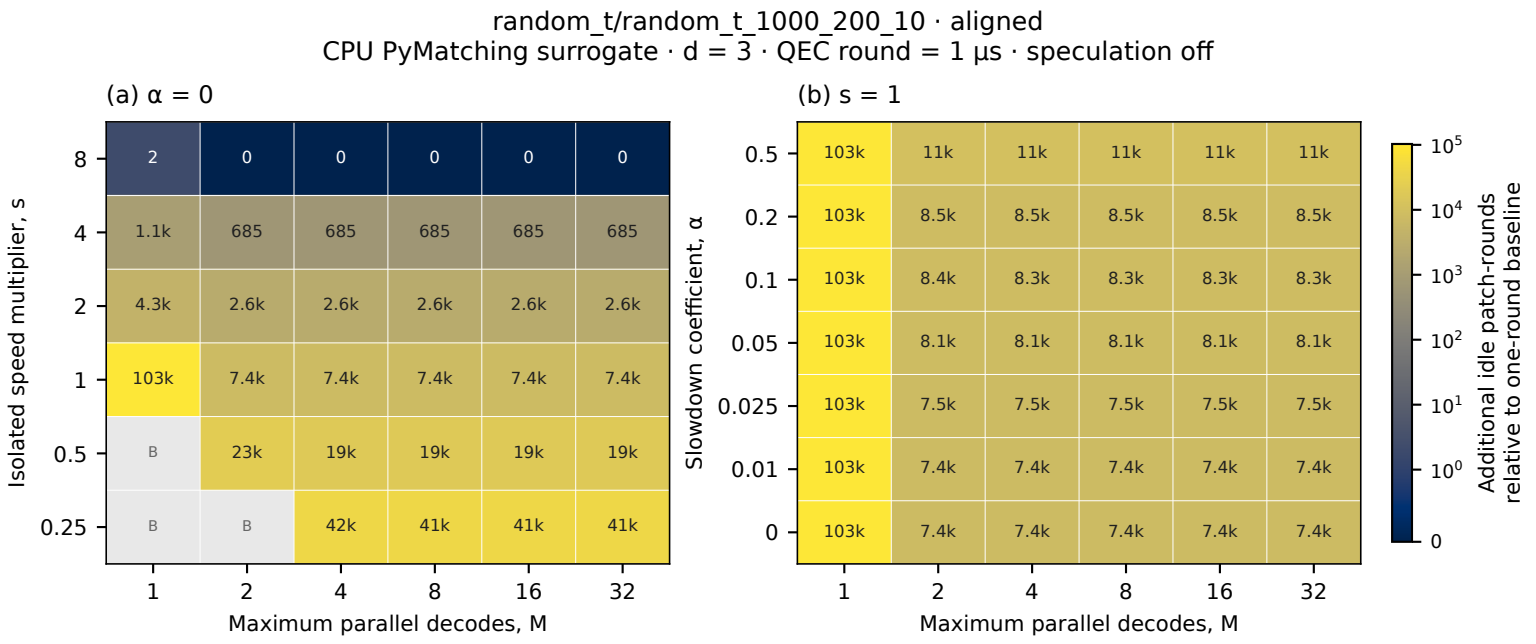

random_t/random_t_1000_200_0 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
7
0
0
0
0
0
1.8k
1.8k
1.8k
1.8k
1.8k
1.8k
3.7k
3.7k
3.7k
3.7k
3.7k
3.7k
32k
8.1k
8.1k
8.1k
8.1k
8.1k
B
20k
17k
17k
17k
17k
B
B
43k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
32k
8.5k
8.5k
8.5k
8.5k
8.5k
32k
8.3k
8.3k
8.3k
8.3k
8.3k
32k
8.2k
8.2k
8.2k
8.2k
8.2k
32k
8.1k
8.1k
8.1k
8.1k
8.1k
32k
8.1k
8.1k
8.1k
8.1k
8.1k
32k
8.1k
8.1k
8.1k
8.1k
8.1k
32k
8.1k
8.1k
8.1k
8.1k
8.1k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

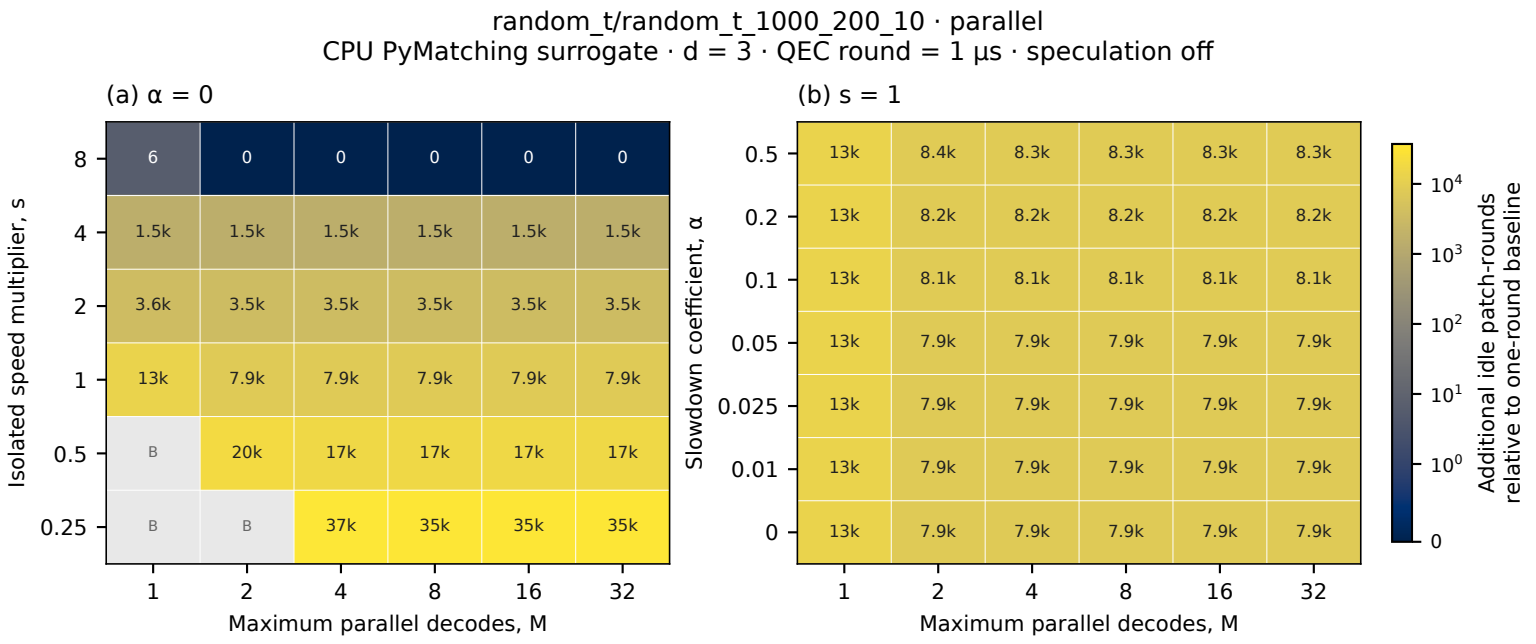

random_t/random_t_1000_200_0 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

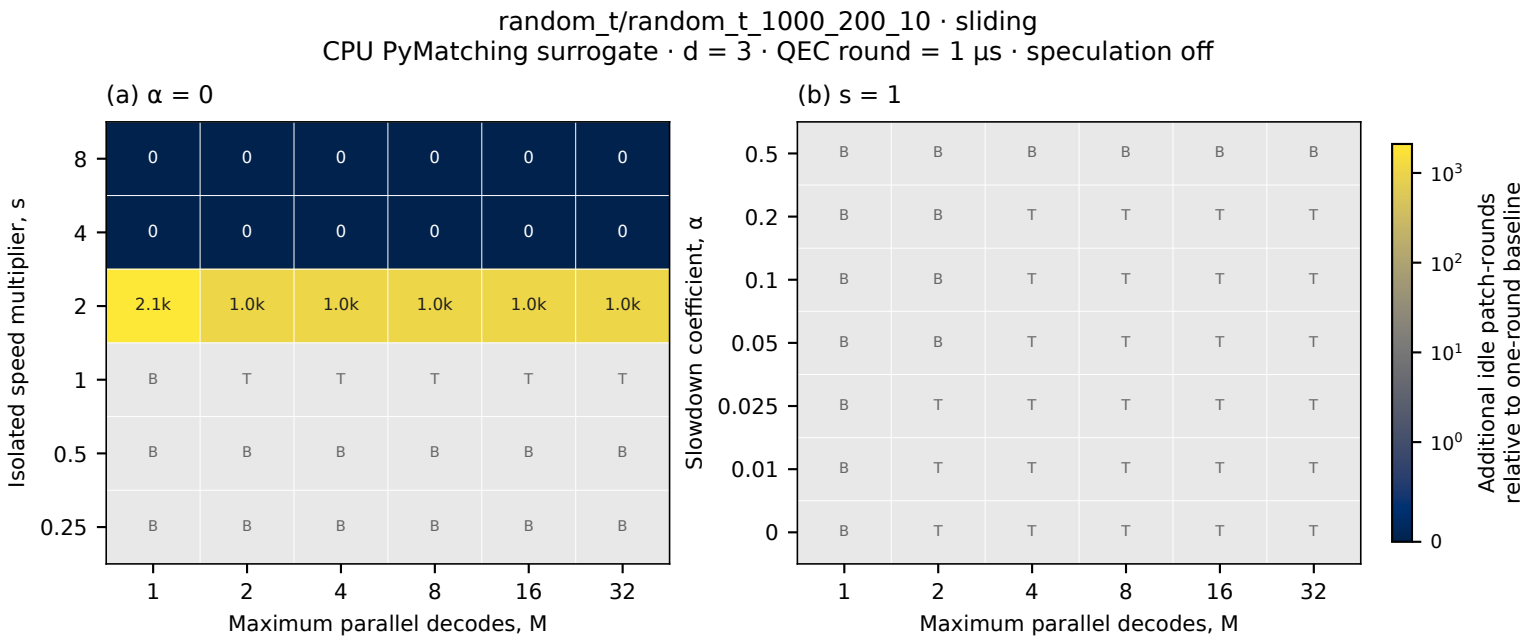

random_t/random_t_1000_200_1 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
4
0
0
0
0
0
1.1k
678
678
678
678
678
4.3k
2.6k
2.6k
2.6k
2.6k
2.6k
18k
7.3k
7.3k
7.3k
7.3k
7.3k
B
22k
18k
18k
18k
18k
B
B
41k
40k
40k
40k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
18k
11k
10k
10k
10k
10k
18k
8.5k
8.5k
8.5k
8.5k
8.5k
18k
8.4k
8.3k
8.3k
8.3k
8.3k
18k
8.1k
8.1k
8.1k
8.1k
8.1k
18k
7.6k
7.6k
7.6k
7.6k
7.6k
18k
7.3k
7.3k
7.3k
7.3k
7.3k
18k
7.3k
7.3k
7.3k
7.3k
7.3k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

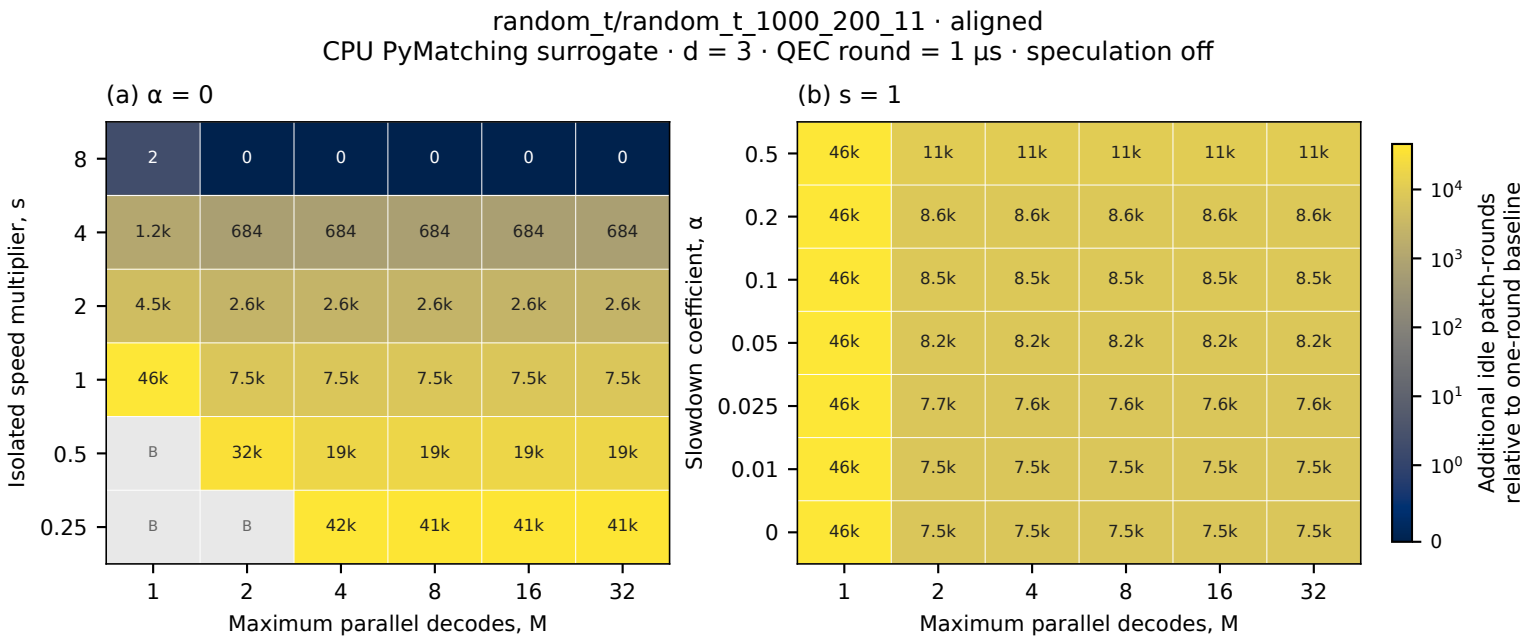

random_t/random_t_1000_200_1 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
6
0
0
0
0
0
1.7k
1.7k
1.7k
1.7k
1.7k
1.7k
3.6k
3.6k
3.6k
3.6k
3.6k
3.6k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
B
20k
17k
17k
17k
17k
B
B
36k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
12k
8.5k
8.5k
8.5k
8.5k
8.5k
12k
8.3k
8.3k
8.3k
8.3k
8.3k
12k
8.3k
8.3k
8.3k
8.3k
8.3k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

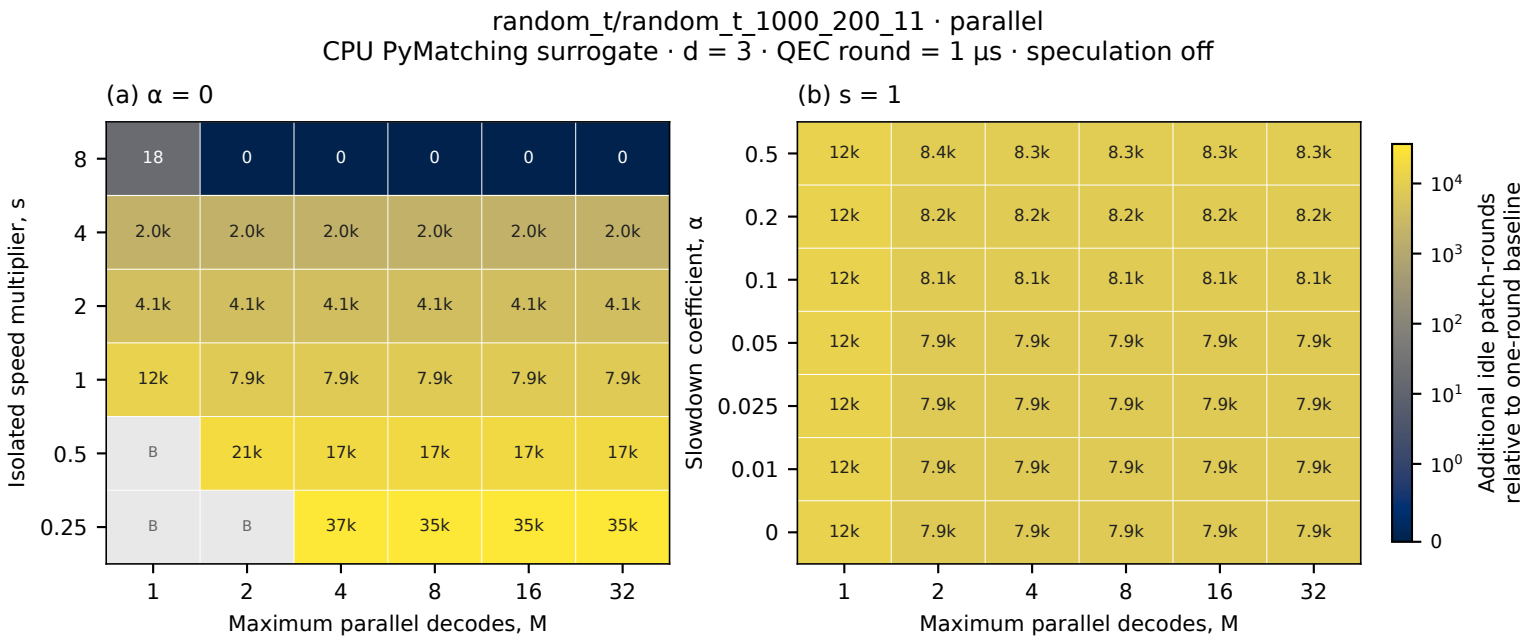

random_t/random_t_1000_200_1 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

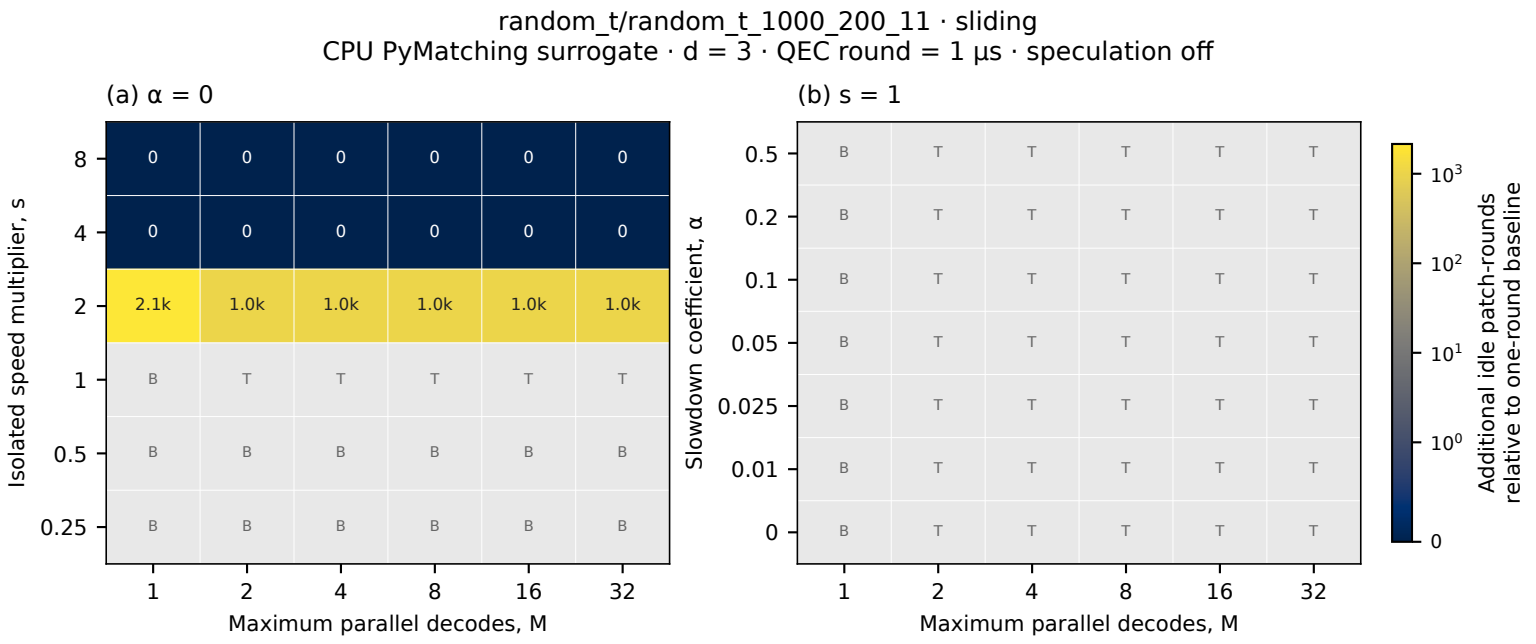

random_t/random_t_1000_200_10 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
2
0
1.1k
685
4.3k
2.6k
103k
7.4k
B
23k
19k
42k
41k
11k
8.5k
8.4k
8.3k
8.1k
7.5k
7.4k
10^5
10^4
10^3
10^2
10^1
10^0
0

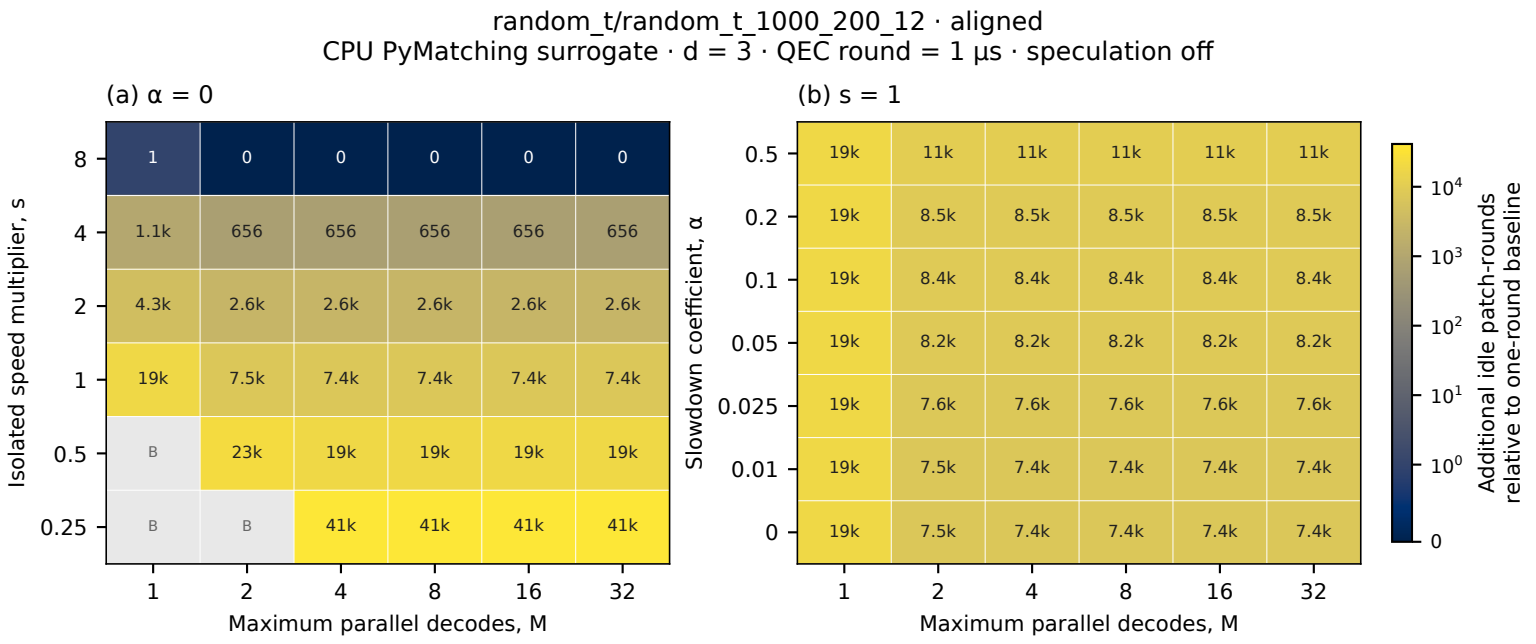

random_t/random_t_1000_200_10 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
6
0
0
0
0
0
1.5k
1.5k
1.5k
1.5k
1.5k
1.5k
3.6k
3.5k
3.5k
3.5k
3.5k
3.5k
13k
7.9k
7.9k
7.9k
7.9k
7.9k
B
20k
17k
17k
17k
17k
B
B
37k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
13k
8.4k
8.3k
8.3k
8.3k
8.3k
13k
8.2k
8.2k
8.2k
8.2k
8.2k
13k
8.1k
8.1k
8.1k
8.1k
8.1k
13k
7.9k
7.9k
7.9k
7.9k
7.9k
13k
7.9k
7.9k
7.9k
7.9k
7.9k
13k
7.9k
7.9k
7.9k
7.9k
7.9k
13k
7.9k
7.9k
7.9k
7.9k
7.9k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

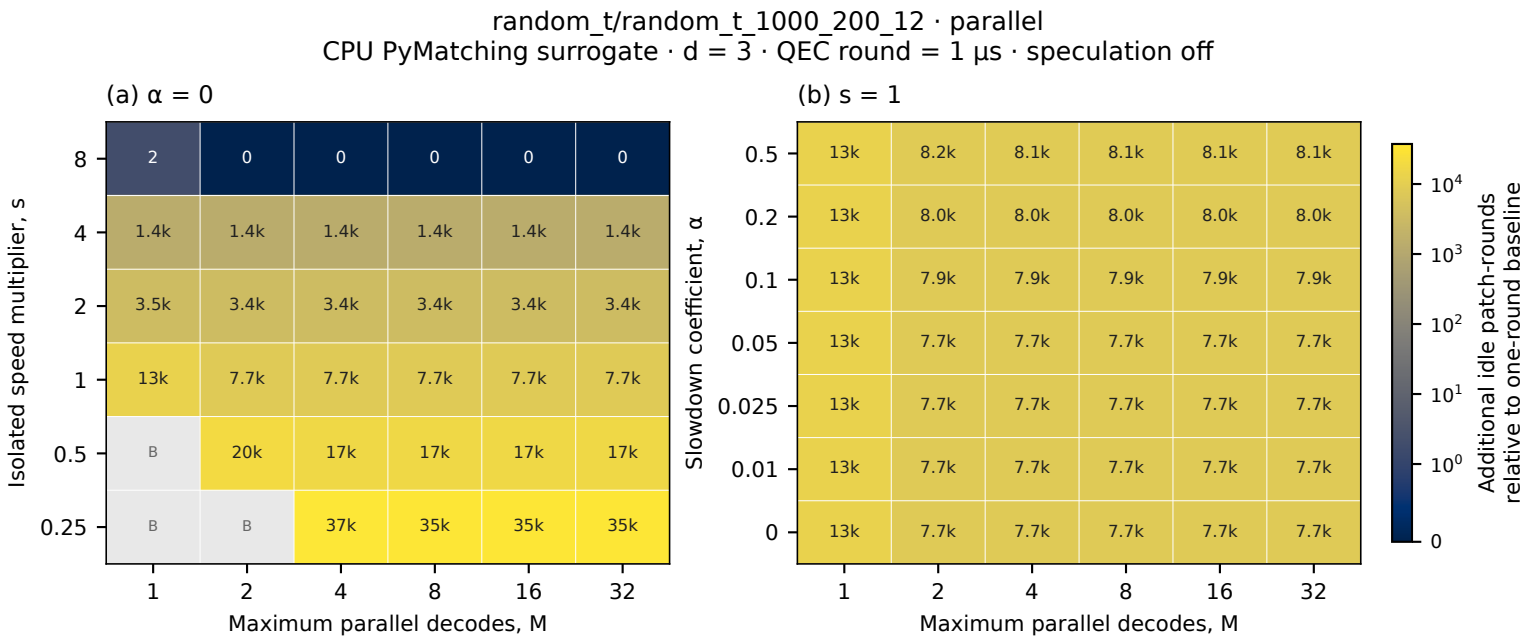

random_t/random_t_1000_200_10 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

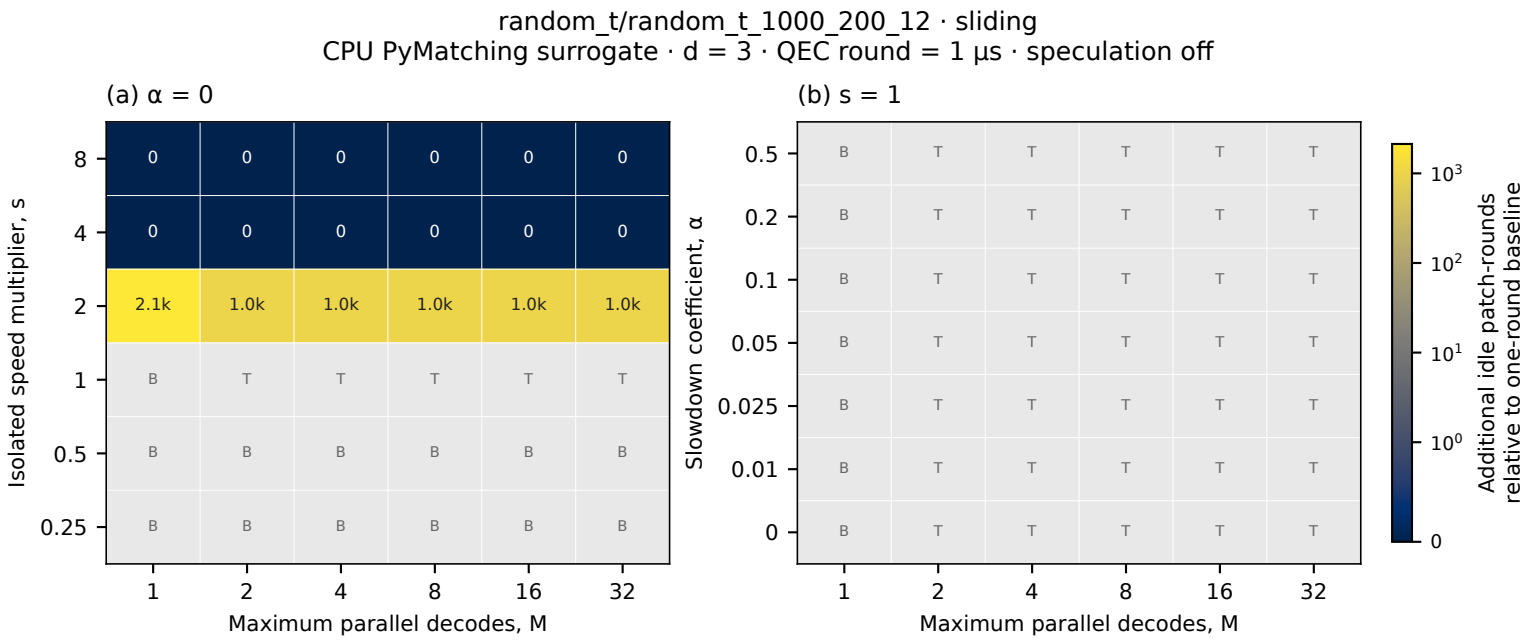

random_t/random_t_1000_200_11 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
2
0
0
0
0
0
1.2k
684
684
684
684
684
4.5k
2.6k
2.6k
2.6k
2.6k
2.6k
46k
7.5k
7.5k
7.5k
7.5k
7.5k
B
32k
19k
19k
19k
19k
B
B
42k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
46k
11k
11k
11k
11k
11k
46k
8.6k
8.6k
8.6k
8.6k
8.6k
46k
8.5k
8.5k
8.5k
8.5k
8.5k
46k
8.2k
8.2k
8.2k
8.2k
8.2k
46k
7.7k
7.6k
7.6k
7.6k
7.6k
46k
7.5k
7.5k
7.5k
7.5k
7.5k
46k
7.5k
7.5k
7.5k
7.5k
7.5k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

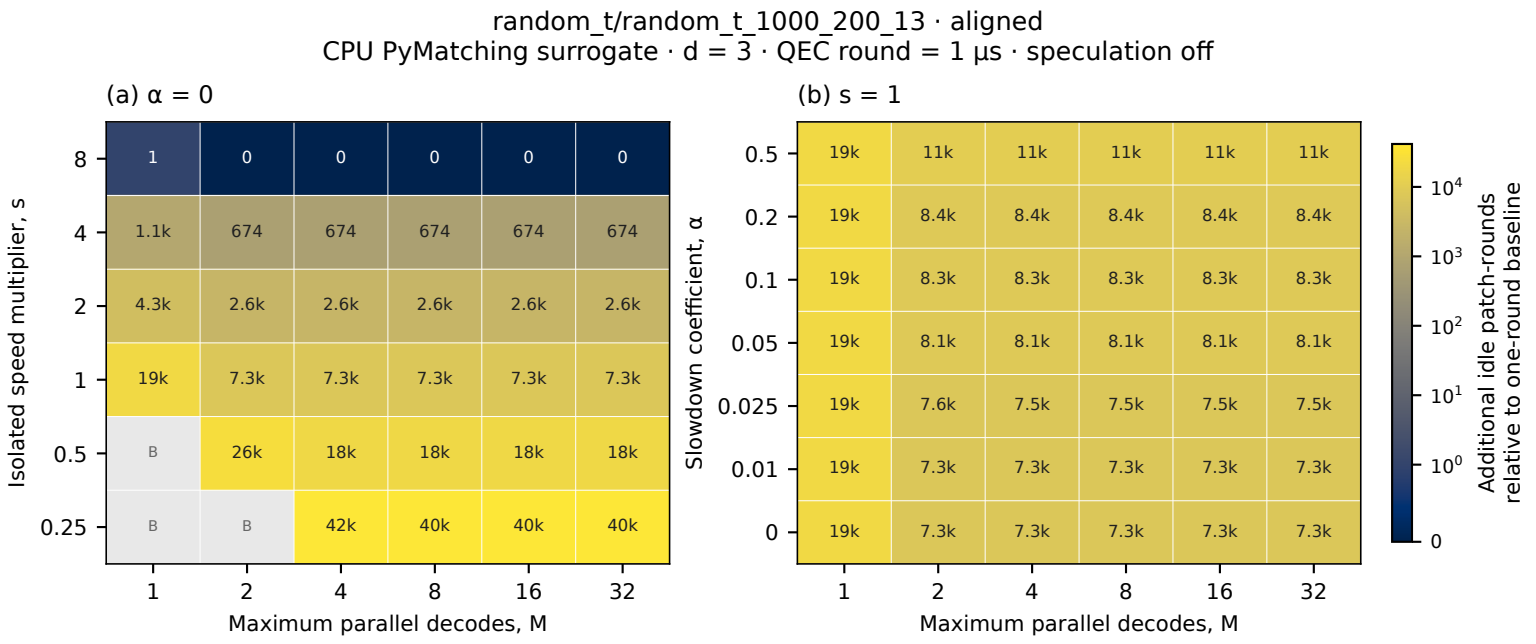

random_t/random_t_1000_200_11 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
18
0
0
0
0
0
2.0k
2.0k
2.0k
2.0k
2.0k
2.0k
4.1k
4.1k
4.1k
4.1k
4.1k
4.1k
12k
7.9k
7.9k
7.9k
7.9k
7.9k
B
21k
17k
17k
17k
17k
B
B
37k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
12k
8.4k
8.3k
8.3k
8.3k
8.3k
12k
8.2k
8.2k
8.2k
8.2k
8.2k
12k
8.1k
8.1k
8.1k
8.1k
8.1k
12k
7.9k
7.9k
7.9k
7.9k
7.9k
12k
7.9k
7.9k
7.9k
7.9k
7.9k
12k
7.9k
7.9k
7.9k
7.9k
7.9k
12k
7.9k
7.9k
7.9k
7.9k
7.9k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

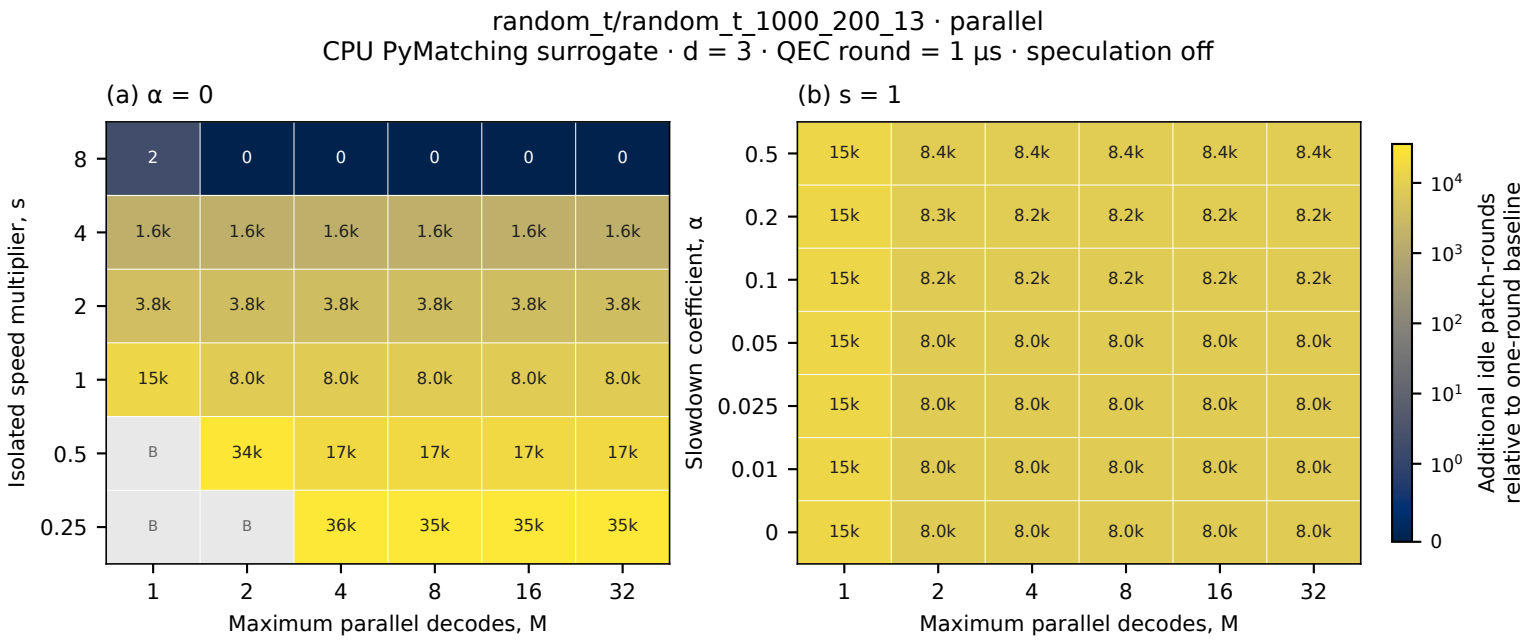

random_t/random_t_1000_200_11 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

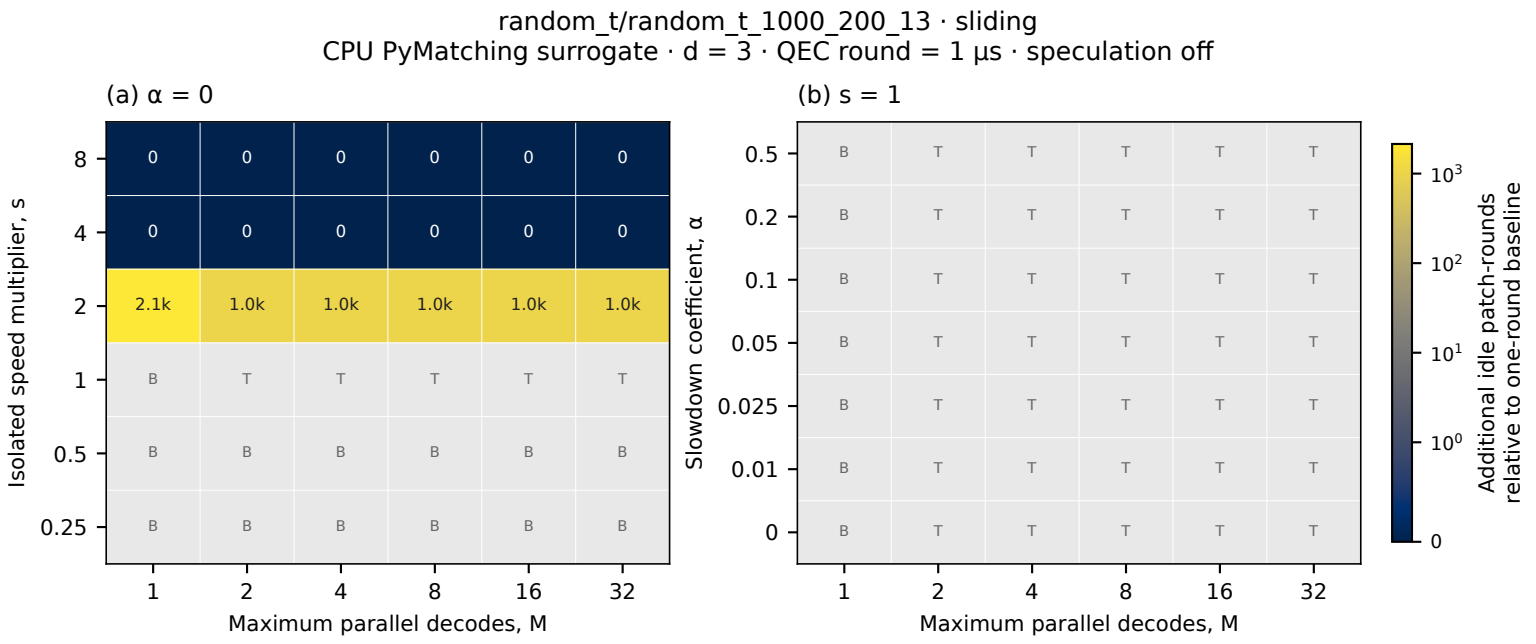

random_t/random_t_1000_200_12 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
1
0
0
0
0
0
1.1k
656
656
656
656
656
4.3k
2.6k
2.6k
2.6k
2.6k
2.6k
19k
7.5k
7.4k
7.4k
7.4k
7.4k
B
23k
19k
19k
19k
19k
B
B
41k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
19k
11k
11k
11k
11k
11k
19k
8.5k
8.5k
8.5k
8.5k
8.5k
19k
8.4k
8.4k
8.4k
8.4k
8.4k
19k
8.2k
8.2k
8.2k
8.2k
8.2k
19k
7.6k
7.6k
7.6k
7.6k
7.6k
19k
7.5k
7.4k
7.4k
7.4k
7.4k
19k
7.5k
7.4k
7.4k
7.4k
7.4k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

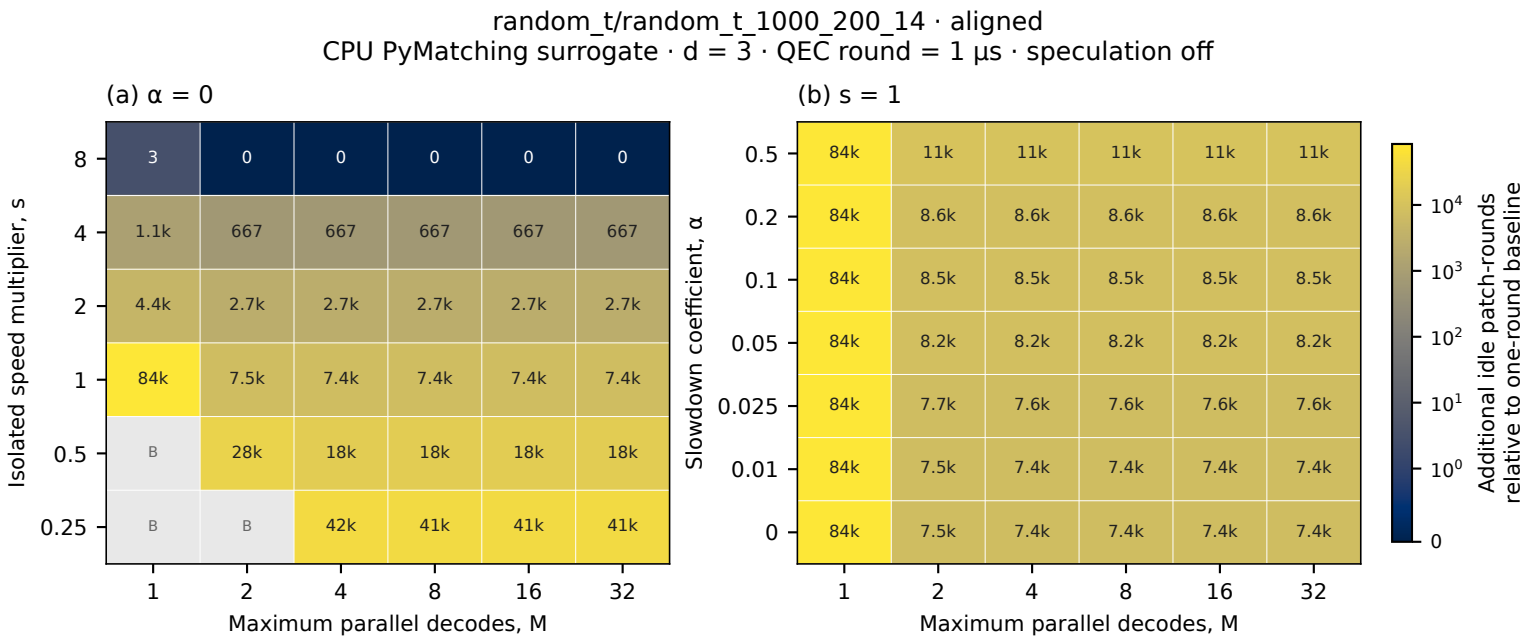

random_t/random_t_1000_200_12 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
2
0
0
0
0
0
1.4k
1.4k
1.4k
1.4k
1.4k
1.4k
3.5k
3.4k
3.4k
3.4k
3.4k
3.4k
13k
7.7k
7.7k
7.7k
7.7k
7.7k
B
20k
17k
17k
17k
17k
B
B
37k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
13k
8.2k
8.1k
8.1k
8.1k
8.1k
13k
8.0k
8.0k
8.0k
8.0k
8.0k
13k
7.9k
7.9k
7.9k
7.9k
7.9k
13k
7.7k
7.7k
7.7k
7.7k
7.7k
13k
7.7k
7.7k
7.7k
7.7k
7.7k
13k
7.7k
7.7k
7.7k
7.7k
7.7k
13k
7.7k
7.7k
7.7k
7.7k
7.7k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

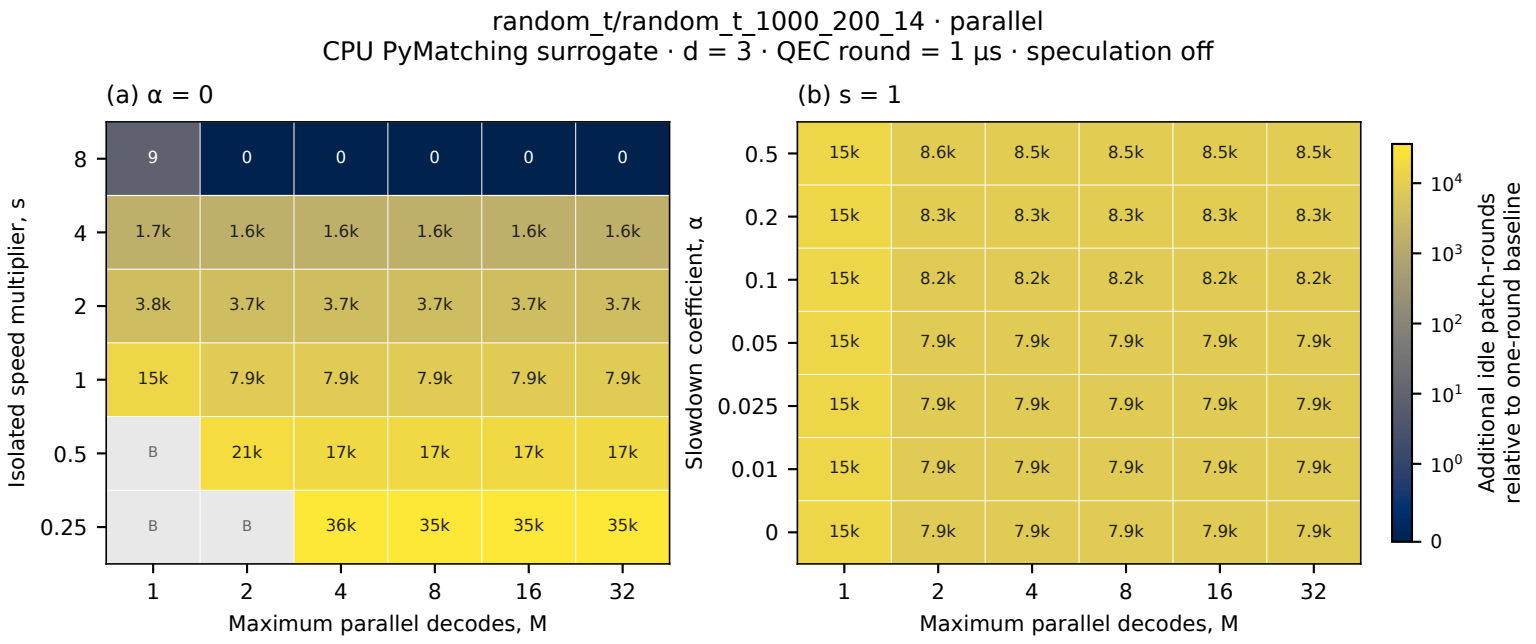

random_t/random_t_1000_200_12 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

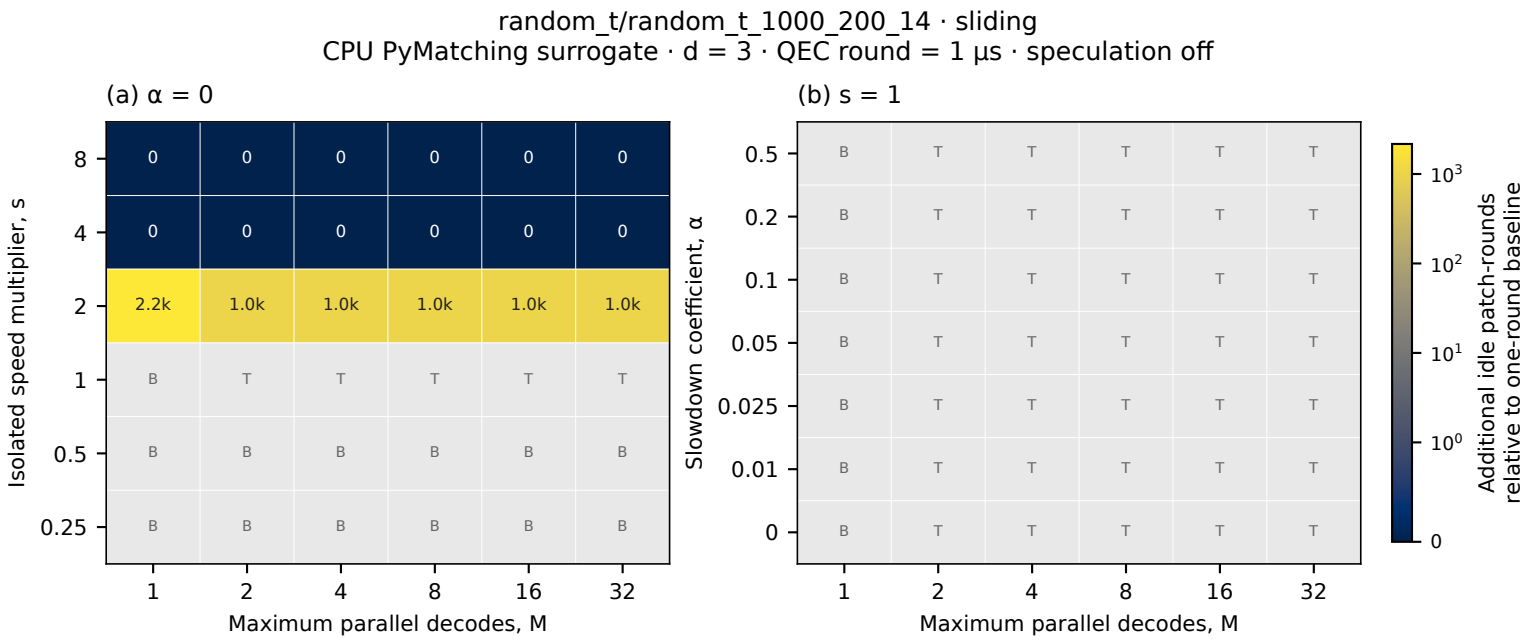

random_t/random_t_1000_200_13 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
1
0
0
0
0
0
1.1k
674
674
674
674
674
4.3k
2.6k
2.6k
2.6k
2.6k
2.6k
19k
7.3k
7.3k
7.3k
7.3k
7.3k
B
26k
18k
18k
18k
18k
B
B
42k
40k
40k
40k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
19k
11k
11k
11k
11k
11k
19k
8.4k
8.4k
8.4k
8.4k
8.4k
19k
8.3k
8.3k
8.3k
8.3k
8.3k
19k
8.1k
8.1k
8.1k
8.1k
8.1k
19k
7.6k
7.5k
7.5k
7.5k
7.5k
19k
7.3k
7.3k
7.3k
7.3k
7.3k
19k
7.3k
7.3k
7.3k
7.3k
7.3k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

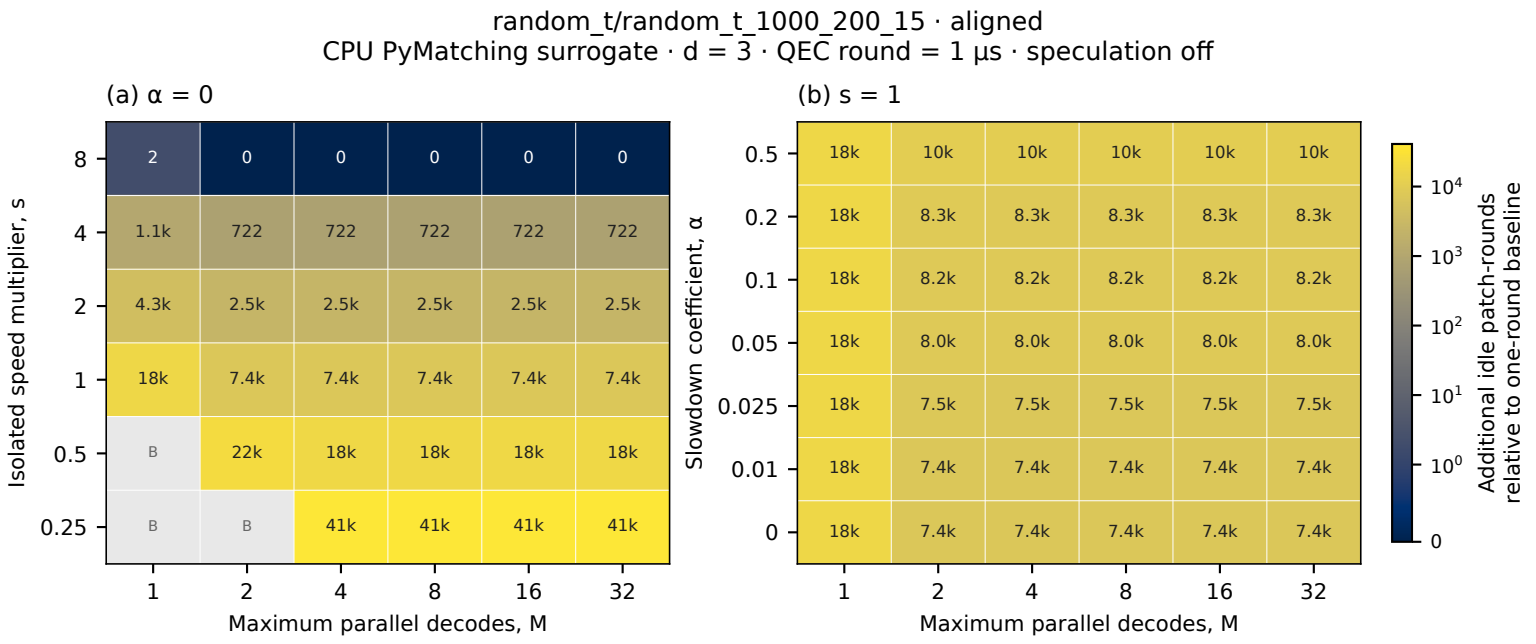

random_t/random_t_1000_200_13 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
2
0
0
0
0
0
1.6k
1.6k
1.6k
1.6k
1.6k
1.6k
3.8k
3.8k
3.8k
3.8k
3.8k
3.8k
15k
8.0k
8.0k
8.0k
8.0k
8.0k
B
34k
17k
17k
17k
17k
B
B
36k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
15k
8.4k
8.4k
8.4k
8.4k
8.4k
15k
8.3k
8.2k
8.2k
8.2k
8.2k
15k
8.2k
8.2k
8.2k
8.2k
8.2k
15k
8.0k
8.0k
8.0k
8.0k
8.0k
15k
8.0k
8.0k
8.0k
8.0k
8.0k
15k
8.0k
8.0k
8.0k
8.0k
8.0k
15k
8.0k
8.0k
8.0k
8.0k
8.0k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

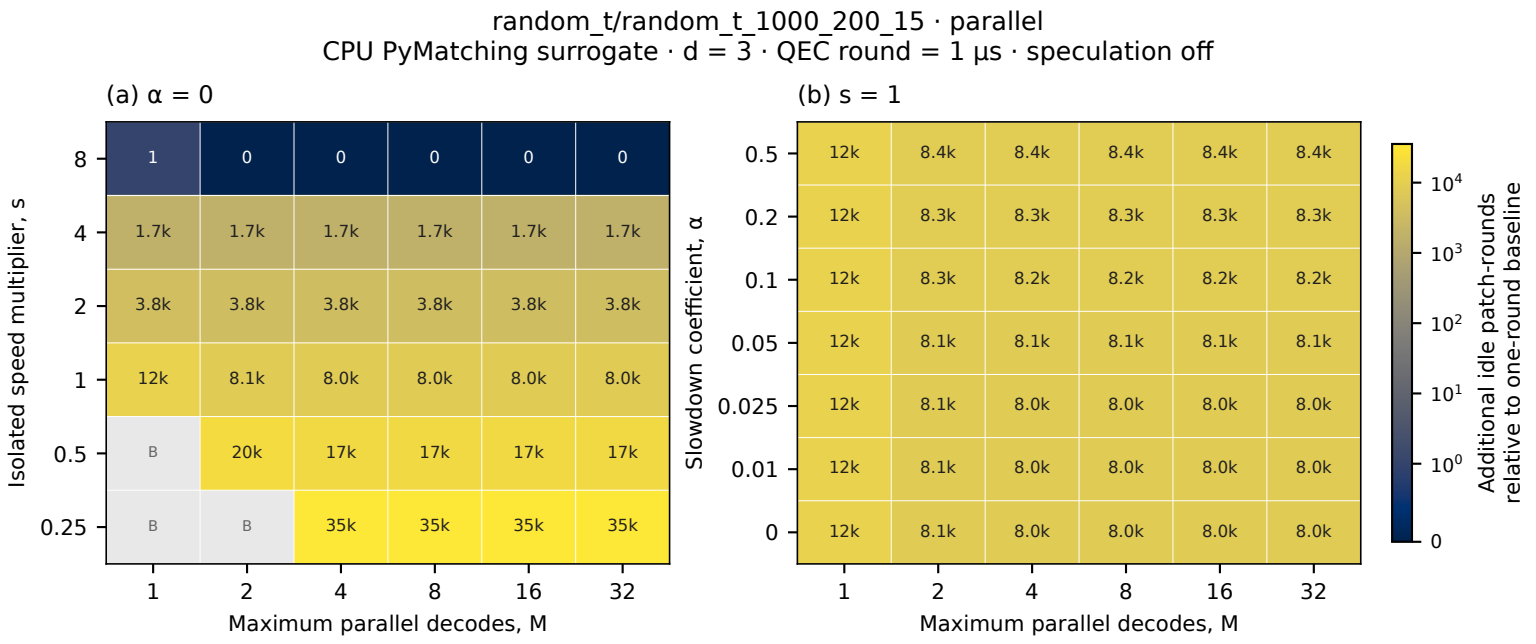

random_t/random_t_1000_200_13 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

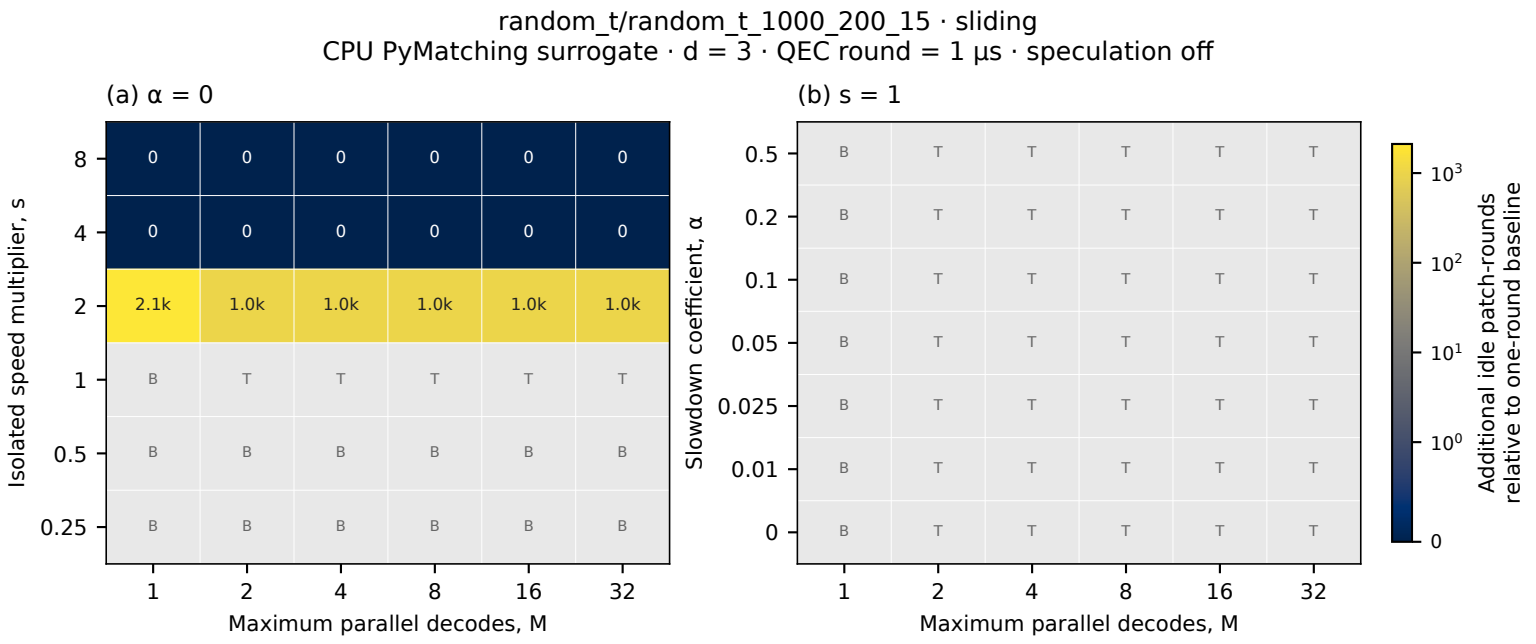

random_t/random_t_1000_200_14 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
3
0
0
0
0
0
1.1k
667
667
667
667
667
4.4k
2.7k
2.7k
2.7k
2.7k
2.7k
84k
7.5k
7.4k
7.4k
7.4k
7.4k
B
28k
18k
18k
18k
18k
B
B
42k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
84k
11k
11k
11k
11k
11k
84k
8.6k
8.6k
8.6k
8.6k
8.6k
84k
8.5k
8.5k
8.5k
8.5k
8.5k
84k
8.2k
8.2k
8.2k
8.2k
8.2k
84k
7.7k
7.6k
7.6k
7.6k
7.6k
84k
7.5k
7.4k
7.4k
7.4k
7.4k
84k
7.5k
7.4k
7.4k
7.4k
7.4k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

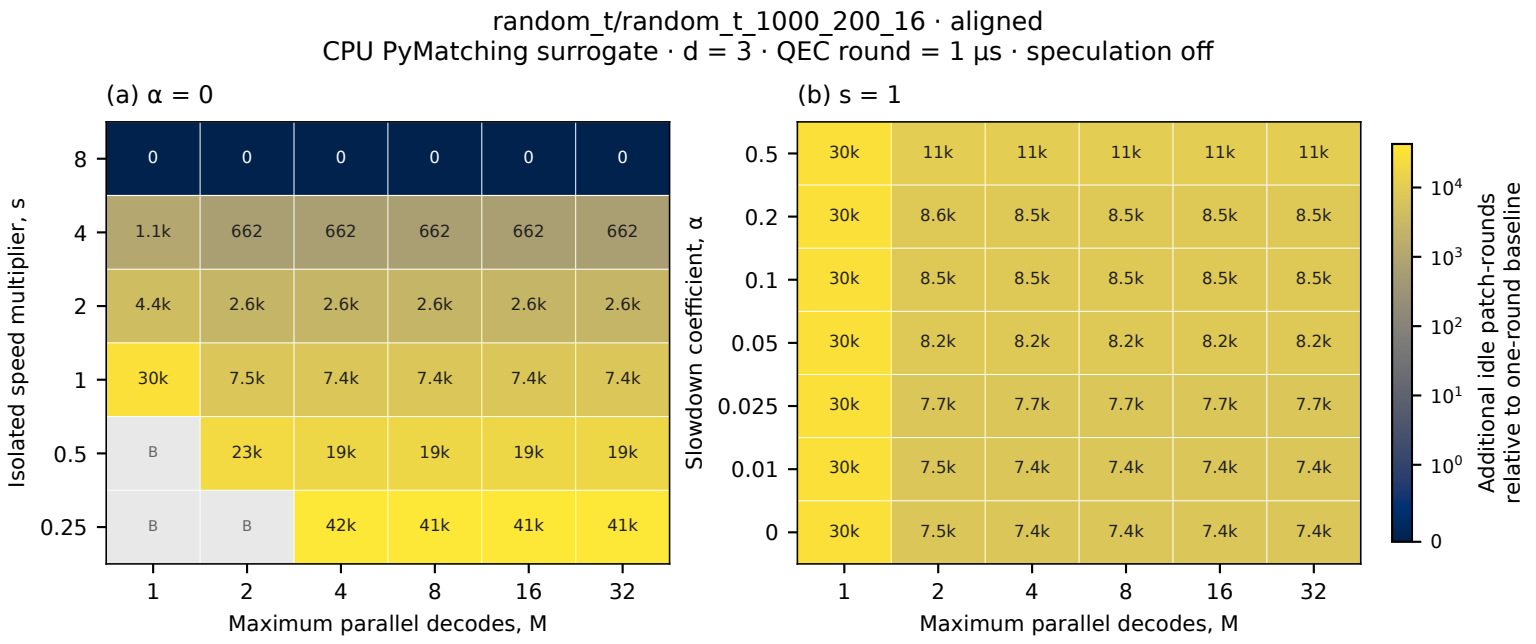

random_t/random_t_1000_200_14 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
9
0
0
0
0
0
1.7k
1.6k
1.6k
1.6k
1.6k
1.6k
3.8k
3.7k
3.7k
3.7k
3.7k
3.7k
15k
7.9k
7.9k
7.9k
7.9k
7.9k
B
21k
17k
17k
17k
17k
B
B
36k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
15k
8.6k
8.5k
8.5k
8.5k
8.5k
15k
8.3k
8.3k
8.3k
8.3k
8.3k
15k
8.2k
8.2k
8.2k
8.2k
8.2k
15k
7.9k
7.9k
7.9k
7.9k
7.9k
15k
7.9k
7.9k
7.9k
7.9k
7.9k
15k
7.9k
7.9k
7.9k
7.9k
7.9k
15k
7.9k
7.9k
7.9k
7.9k
7.9k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

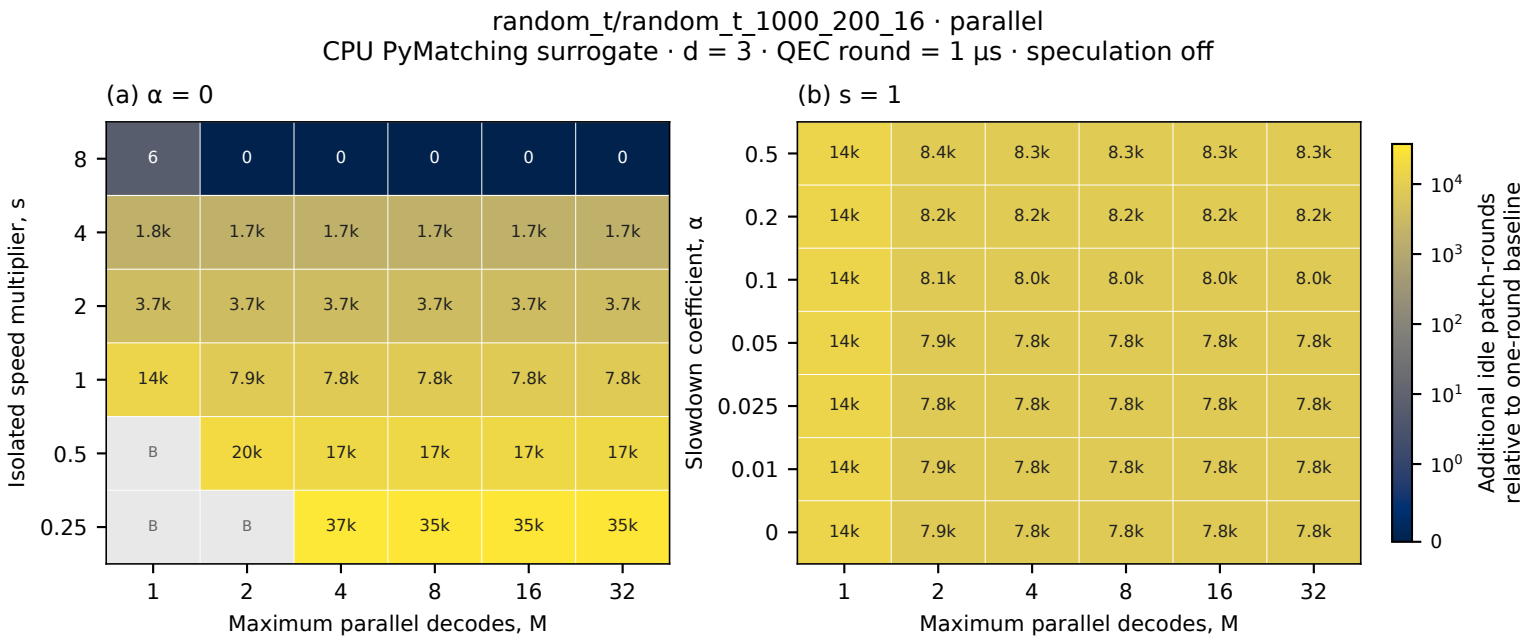

random_t/random_t_1000_200_14 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.2k
1.0k
B
T
10^3
10^2
10^1
10^0

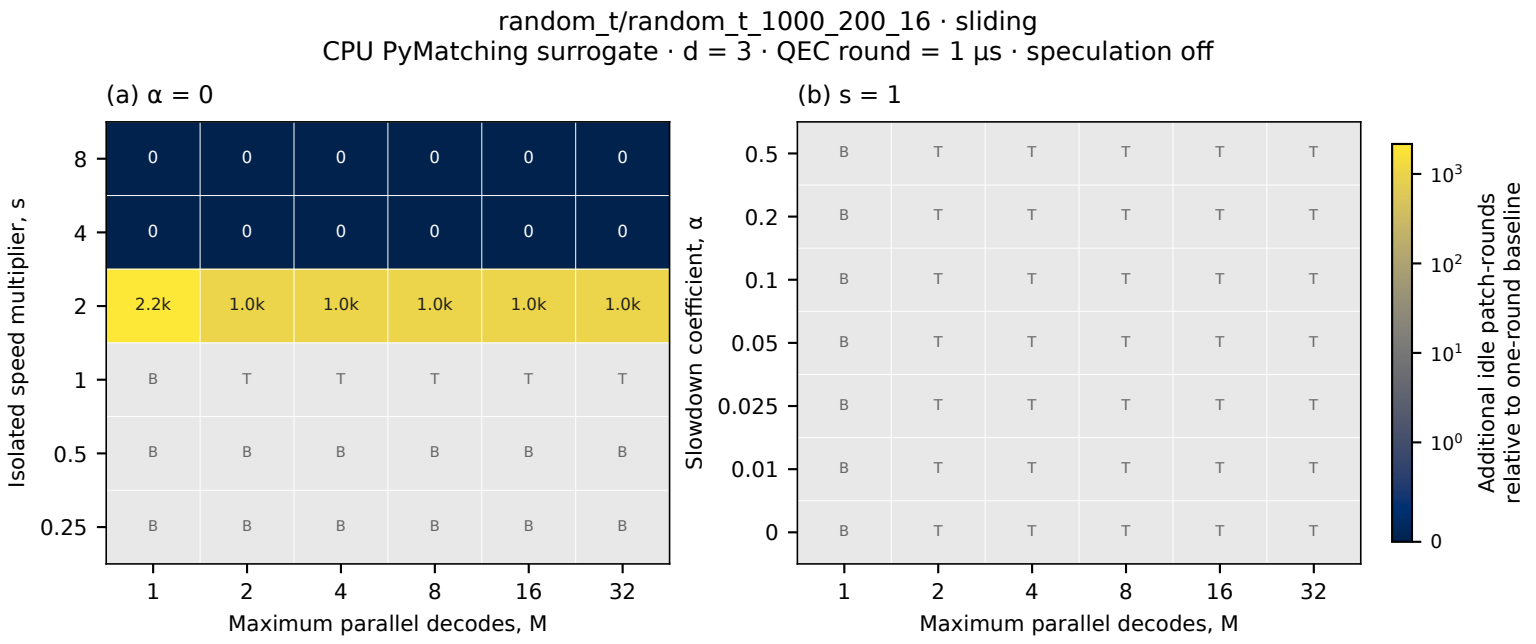

random_t/random_t_1000_200_15 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
2
0
0
0
0
0
1.1k
722
722
722
722
722
4.3k
2.5k
2.5k
2.5k
2.5k
2.5k
18k
7.4k
7.4k
7.4k
7.4k
7.4k
B
22k
18k
18k
18k
18k
B
B
41k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
18k
10k
10k
10k
10k
10k
18k
8.3k
8.3k
8.3k
8.3k
8.3k
18k
8.2k
8.2k
8.2k
8.2k
8.2k
18k
8.0k
8.0k
8.0k
8.0k
8.0k
18k
7.5k
7.5k
7.5k
7.5k
7.5k
18k
7.4k
7.4k
7.4k
7.4k
7.4k
18k
7.4k
7.4k
7.4k
7.4k
7.4k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

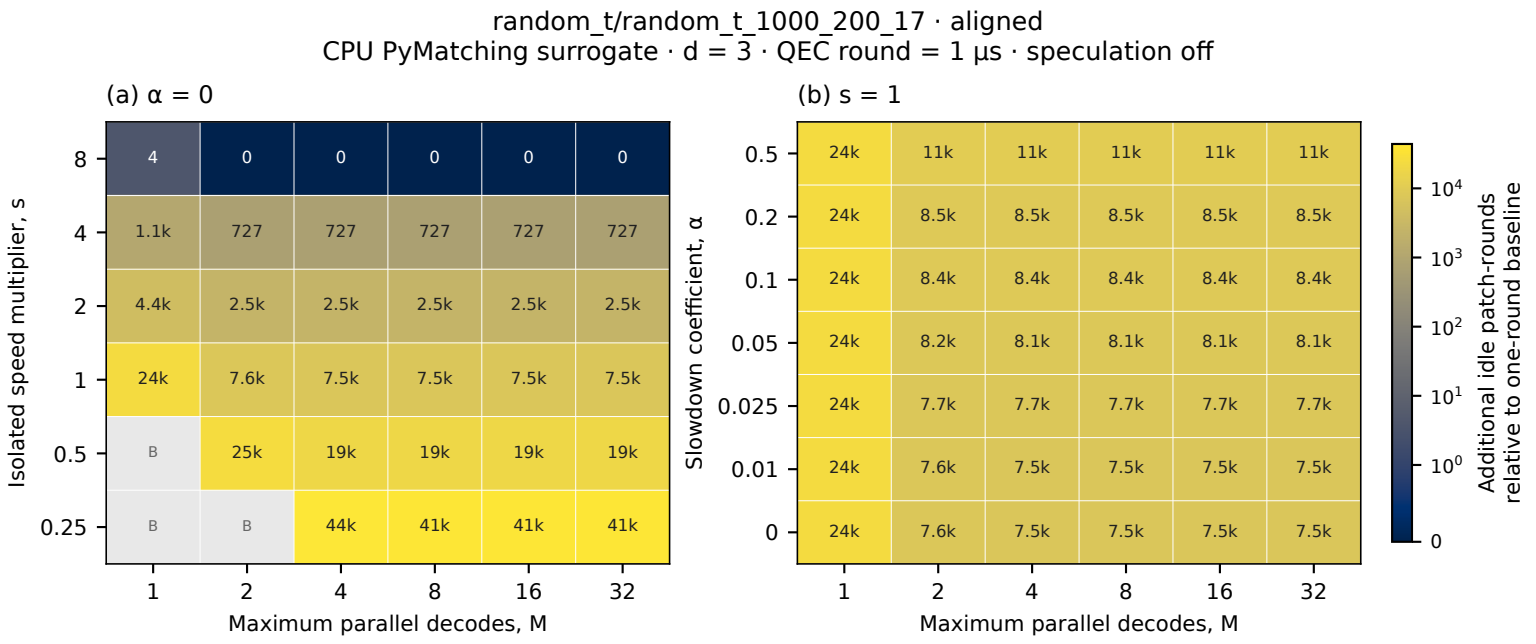

random_t/random_t_1000_200_15 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
1
0
0
0
0
0
1.7k
1.7k
1.7k
1.7k
1.7k
1.7k
3.8k
3.8k
3.8k
3.8k
3.8k
3.8k
12k
8.1k
8.0k
8.0k
8.0k
8.0k
B
20k
17k
17k
17k
17k
B
B
35k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
12k
8.4k
8.4k
8.4k
8.4k
8.4k
12k
8.3k
8.3k
8.3k
8.3k
8.3k
12k
8.3k
8.2k
8.2k
8.2k
8.2k
12k
8.1k
8.1k
8.1k
8.1k
8.1k
12k
8.1k
8.0k
8.0k
8.0k
8.0k
12k
8.1k
8.0k
8.0k
8.0k
8.0k
12k
8.1k
8.0k
8.0k
8.0k
8.0k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

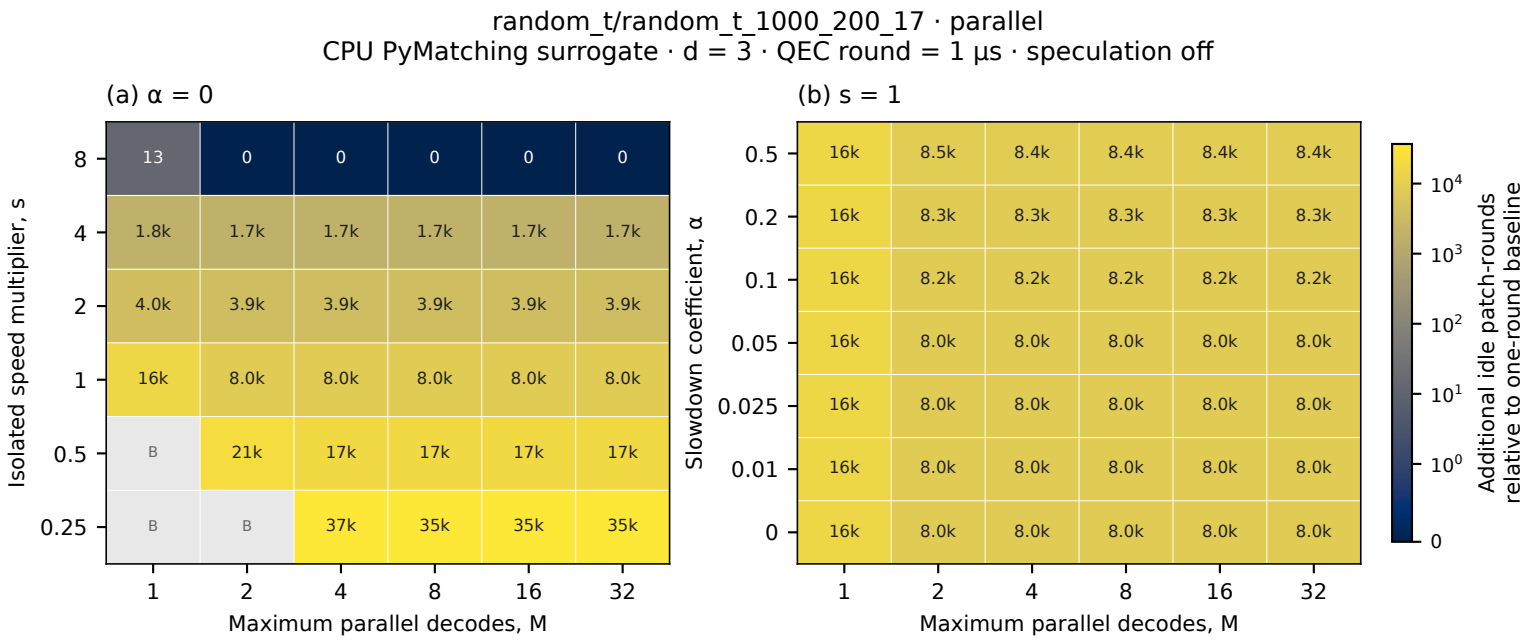

random_t/random_t_1000_200_15 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

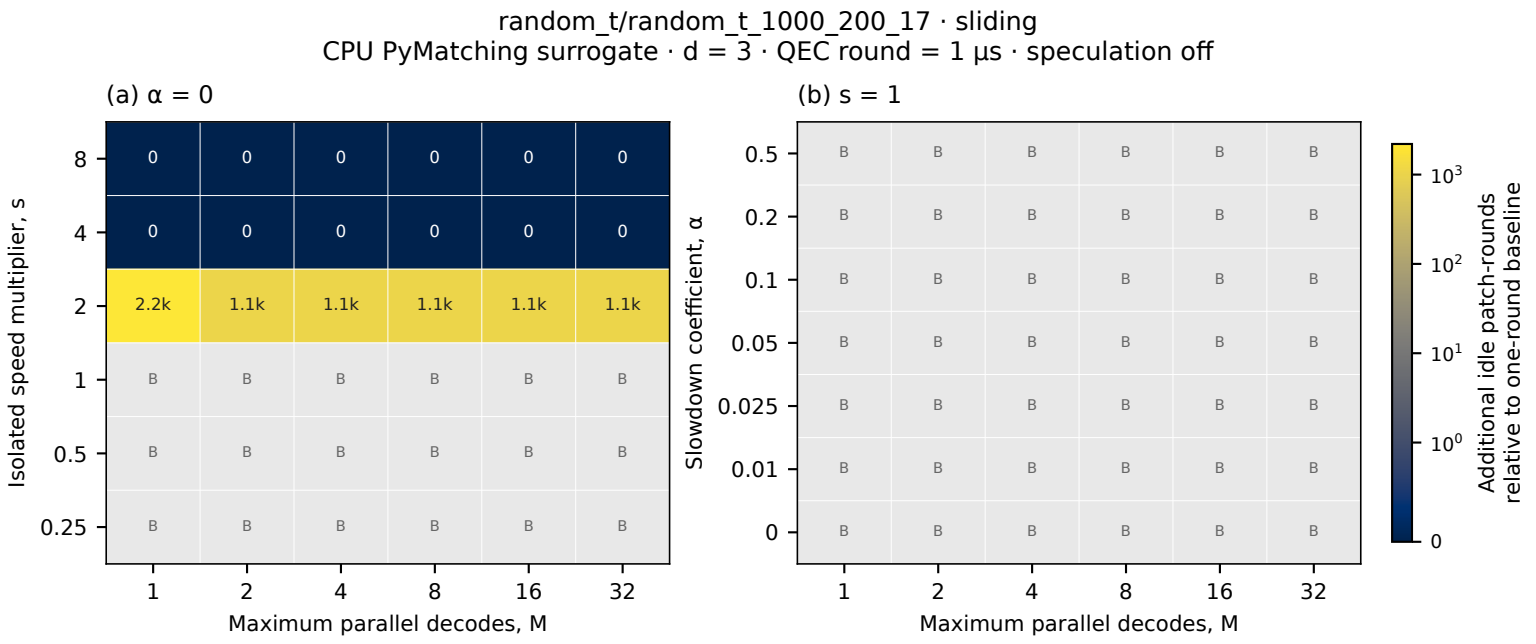

random_t/random_t_1000_200_16 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
0
1.1k
662
662
662
662
4.4k
2.6k
2.6k
2.6k
2.6k
30k
7.5k
7.4k
7.4k
7.4k
7.4k
B
23k
19k
19k
19k
B
B
42k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
30k
11k
11k
11k
11k
8.6k
8.5k
8.5k
8.5k
8.5k
8.2k
8.2k
8.2k
8.2k
7.7k
7.7k
7.7k
7.7k
7.5k
7.4k
7.4k
7.4k
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

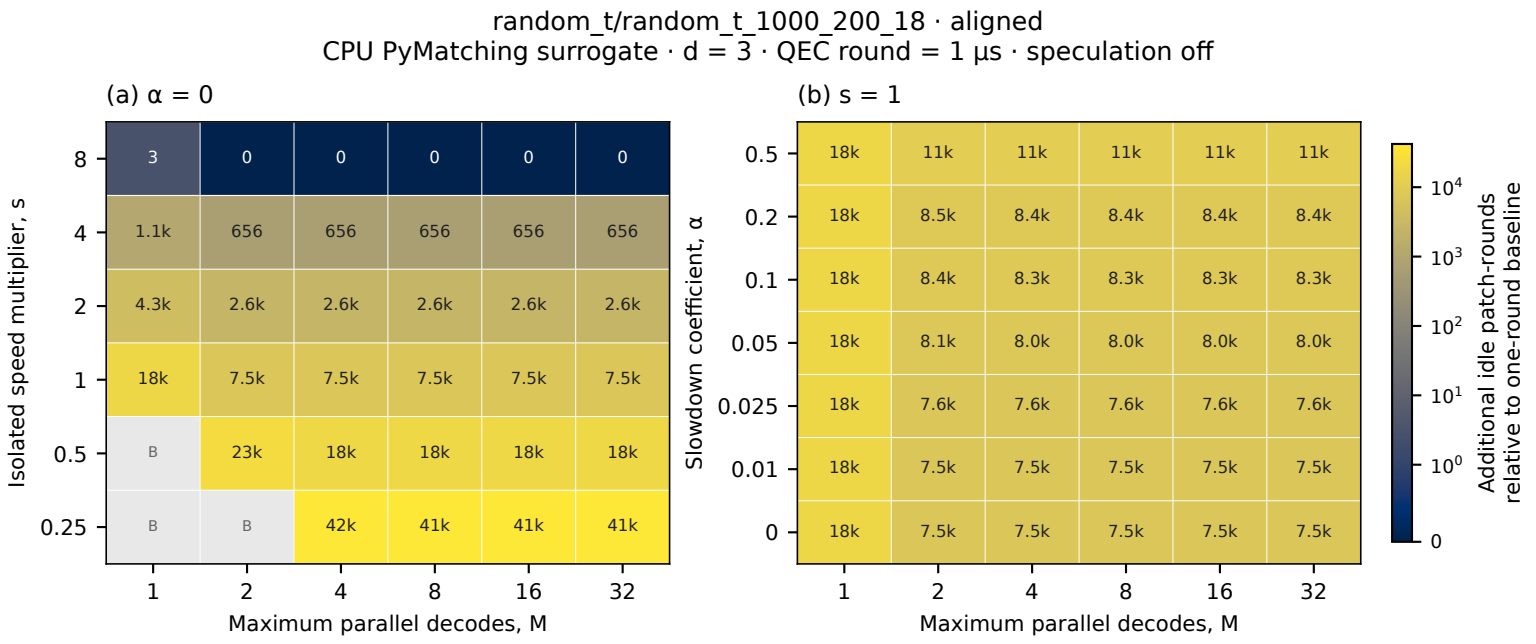

random_t/random_t_1000_200_16 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
6
0
0
0
0
0
1.8k
1.7k
1.7k
1.7k
1.7k
1.7k
3.7k
3.7k
3.7k
3.7k
3.7k
3.7k
14k
7.9k
7.8k
7.8k
7.8k
7.8k
B
20k
17k
17k
17k
17k
B
B
37k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
14k
8.4k
8.3k
8.3k
8.3k
8.3k
14k
8.2k
8.2k
8.2k
8.2k
8.2k
14k
8.1k
8.0k
8.0k
8.0k
8.0k
14k
7.9k
7.8k
7.8k
7.8k
7.8k
14k
7.8k
7.8k
7.8k
7.8k
7.8k
14k
7.9k
7.8k
7.8k
7.8k
7.8k
14k
7.9k
7.8k
7.8k
7.8k
7.8k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

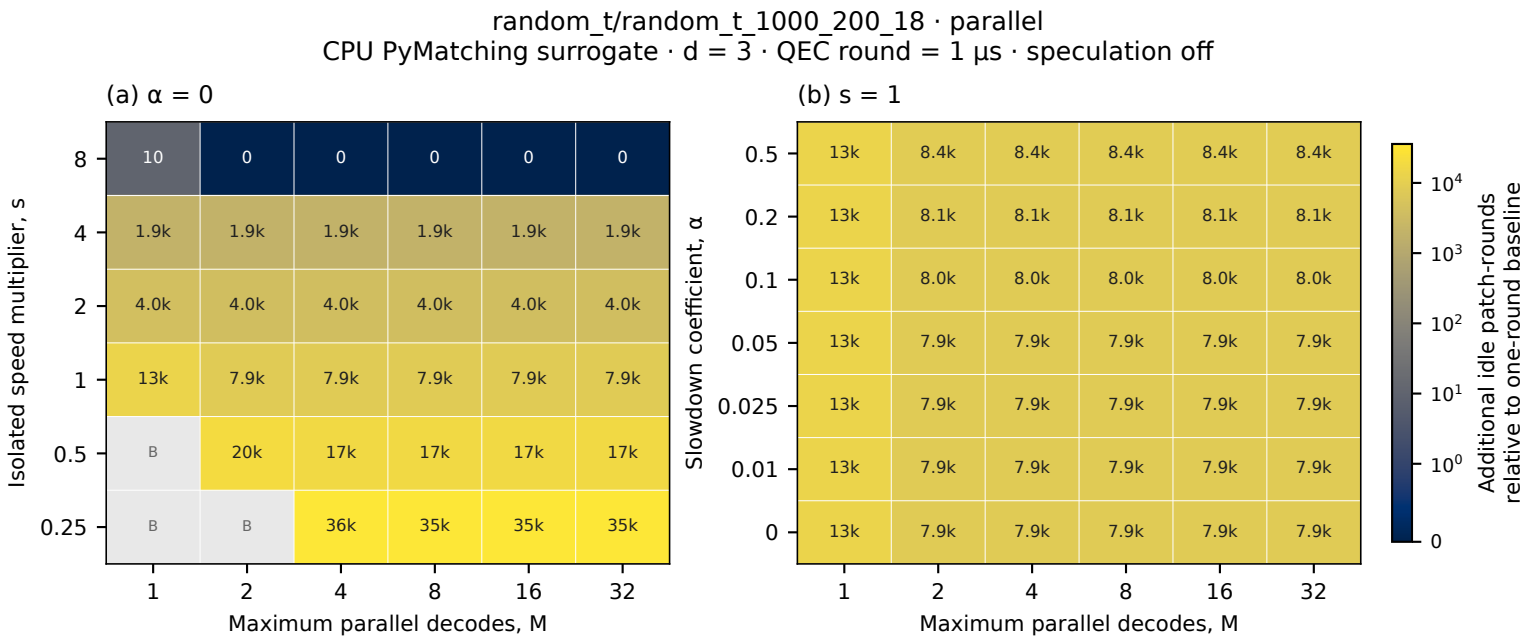

random_t/random_t_1000_200_16 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.2k
1.0k
B
T
10^3
10^2
10^1
10^0

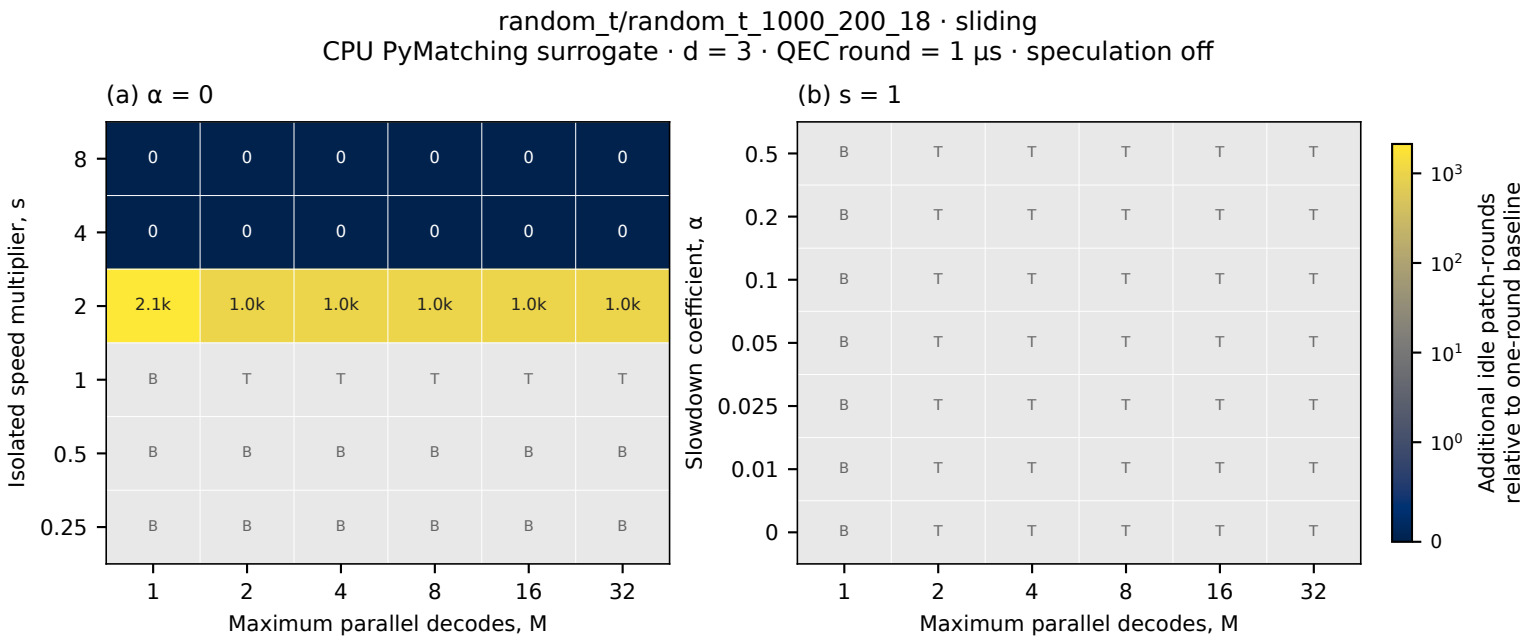

random_t/random_t_1000_200_17 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
4
0
0
0
0
0
1.1k
727
727
727
727
727
4.4k
2.5k
2.5k
2.5k
2.5k
2.5k
24k
7.6k
7.5k
7.5k
7.5k
7.5k
B
25k
19k
19k
19k
19k
B
B
44k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
24k
11k
11k
11k
11k
11k
24k
8.5k
8.5k
8.5k
8.5k
8.5k
24k
8.4k
8.4k
8.4k
8.4k
8.4k
24k
8.2k
8.1k
8.1k
8.1k
8.1k
24k
7.7k
7.7k
7.7k
7.7k
7.7k
24k
7.6k
7.5k
7.5k
7.5k
7.5k
24k
7.6k
7.5k
7.5k
7.5k
7.5k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

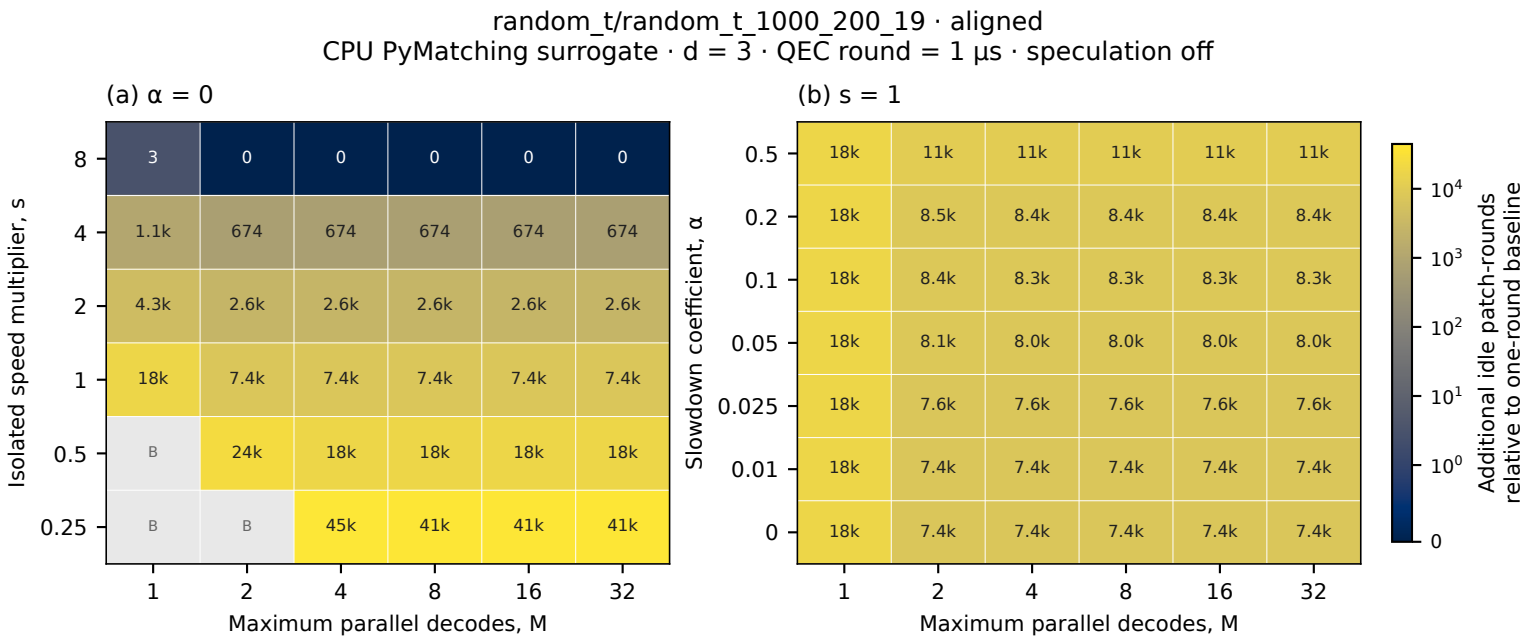

random_t/random_t_1000_200_17 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
13
0
0
0
0
0
1.8k
1.7k
1.7k
1.7k
1.7k
1.7k
4.0k
3.9k
3.9k
3.9k
3.9k
3.9k
16k
8.0k
8.0k
8.0k
8.0k
8.0k
B
21k
17k
17k
17k
17k
B
B
37k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
16k
8.5k
8.4k
8.4k
8.4k
8.4k
16k
8.3k
8.3k
8.3k
8.3k
8.3k
16k
8.2k
8.2k
8.2k
8.2k
8.2k
16k
8.0k
8.0k
8.0k
8.0k
8.0k
16k
8.0k
8.0k
8.0k
8.0k
8.0k
16k
8.0k
8.0k
8.0k
8.0k
8.0k
16k
8.0k
8.0k
8.0k
8.0k
8.0k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

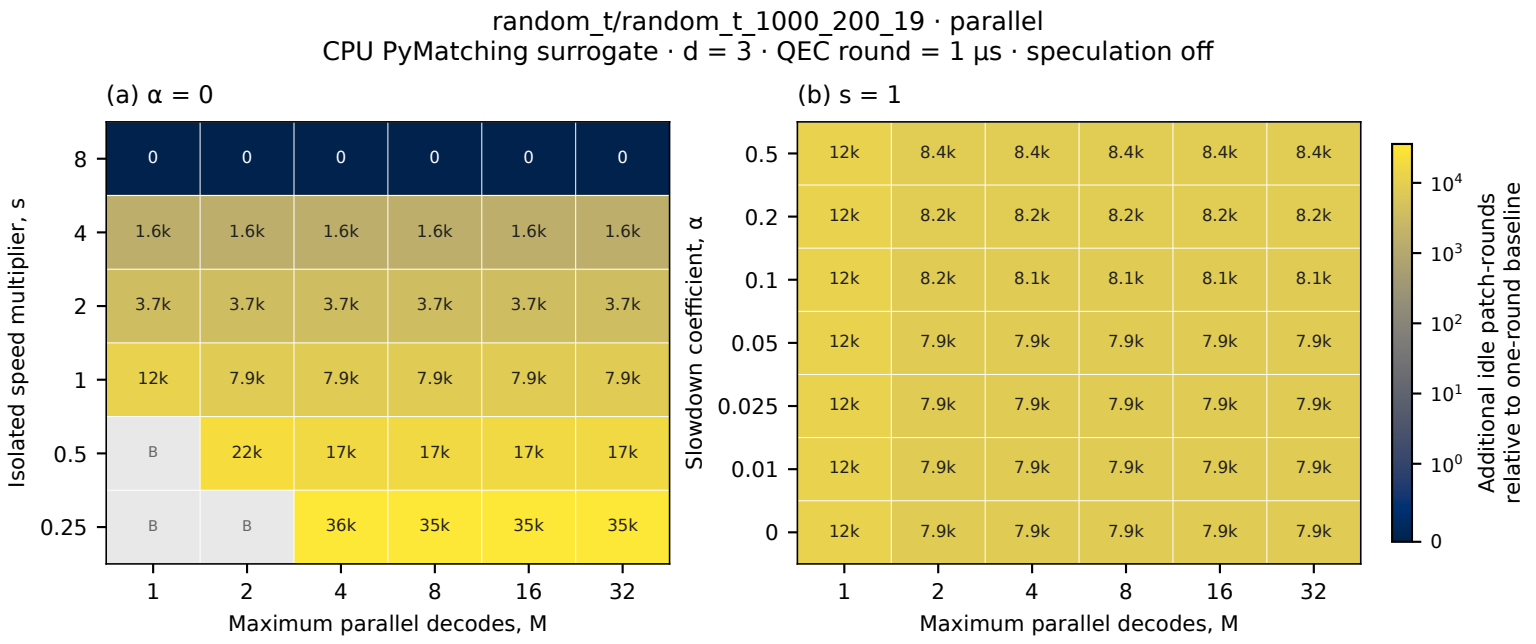

random_t/random_t_1000_200_17 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.2k
1.1k
B
10^3
10^2
10^1
10^0

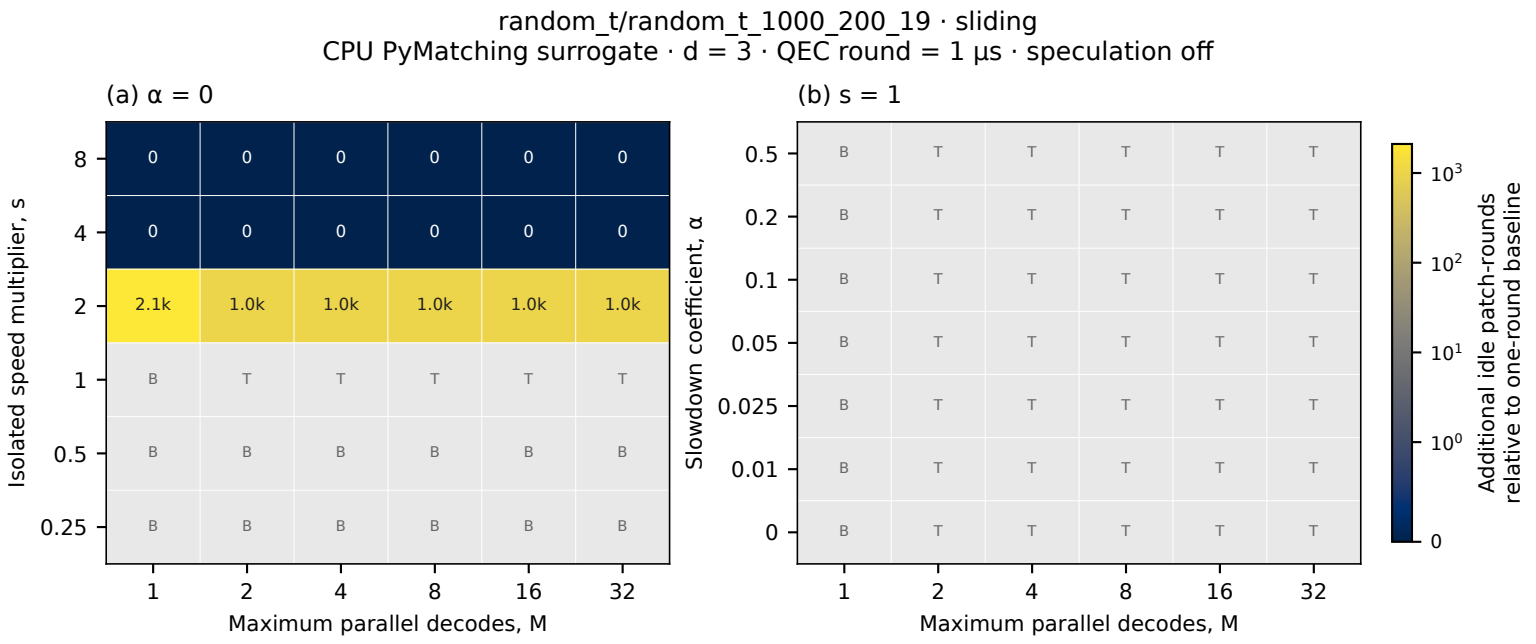

random_t/random_t_1000_200_18 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
3
0
0
0
0
0
1.1k
656
656
656
656
656
4.3k
2.6k
2.6k
2.6k
2.6k
2.6k
18k
7.5k
7.5k
7.5k
7.5k
7.5k
B
23k
18k
18k
18k
18k
B
B
42k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
18k
11k
11k
11k
11k
11k
18k
8.5k
8.4k
8.4k
8.4k
8.4k
18k
8.4k
8.3k
8.3k
8.3k
8.3k
18k
8.1k
8.0k
8.0k
8.0k
8.0k
18k
7.6k
7.6k
7.6k
7.6k
7.6k
18k
7.5k
7.5k
7.5k
7.5k
7.5k
18k
7.5k
7.5k
7.5k
7.5k
7.5k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

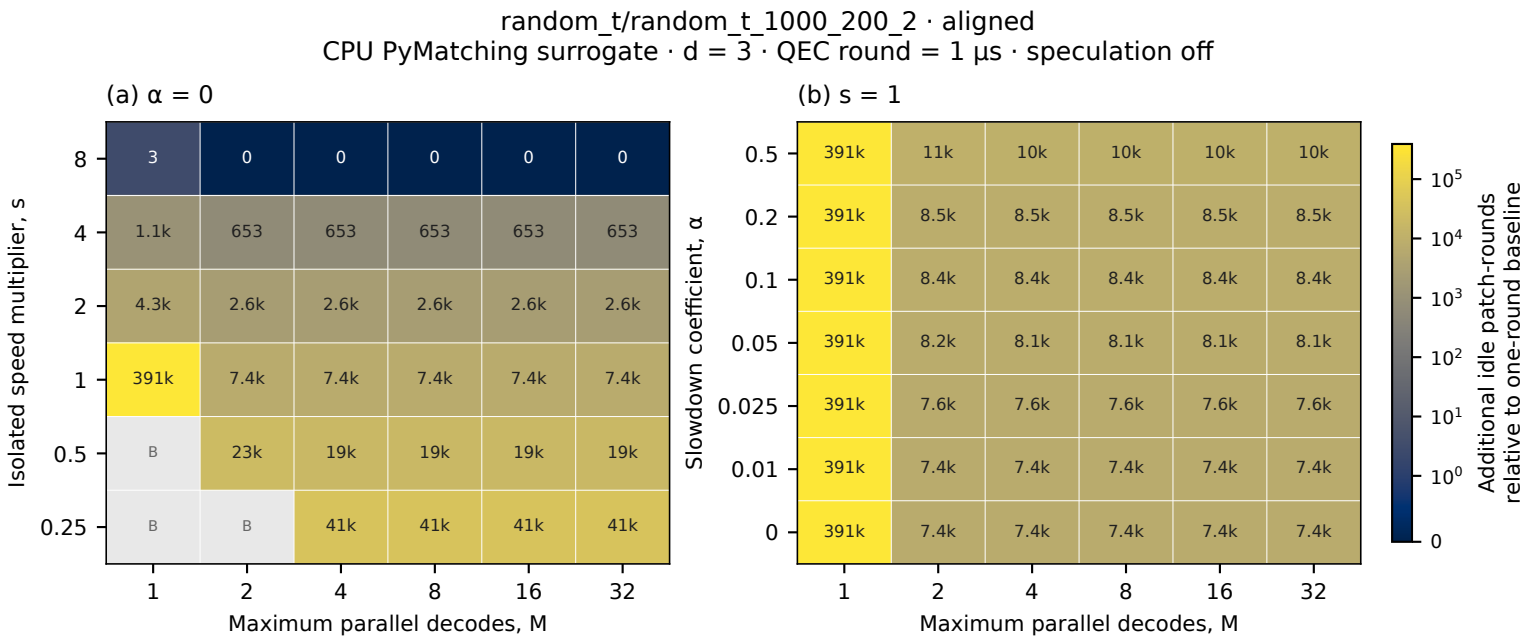

random_t/random_t_1000_200_18 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
10
0
0
0
0
0
1.9k
1.9k
1.9k
1.9k
1.9k
1.9k
4.0k
4.0k
4.0k
4.0k
4.0k
4.0k
13k
7.9k
7.9k
7.9k
7.9k
7.9k
B
20k
17k
17k
17k
17k
B
B
36k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
13k
8.4k
8.4k
8.4k
8.4k
8.4k
13k
8.1k
8.1k
8.1k
8.1k
8.1k
13k
8.0k
8.0k
8.0k
8.0k
8.0k
13k
7.9k
7.9k
7.9k
7.9k
7.9k
13k
7.9k
7.9k
7.9k
7.9k
7.9k
13k
7.9k
7.9k
7.9k
7.9k
7.9k
13k
7.9k
7.9k
7.9k
7.9k
7.9k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

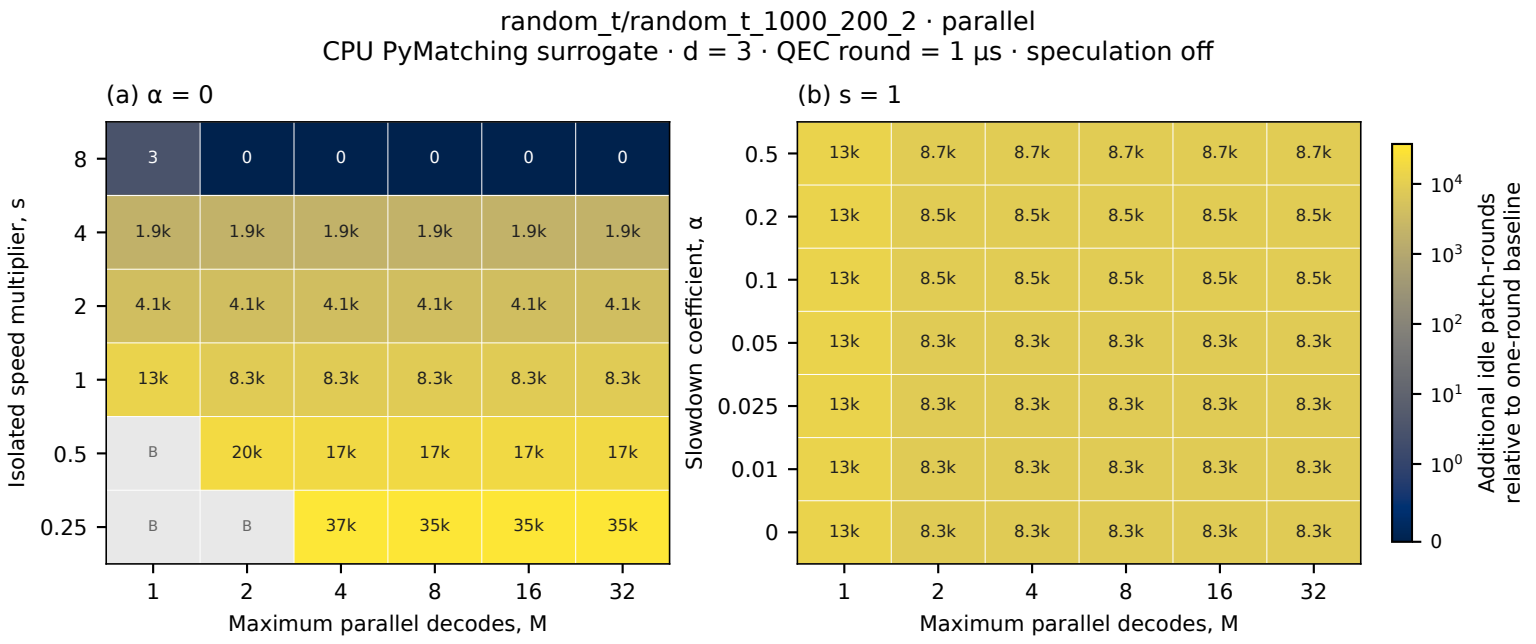

random_t/random_t_1000_200_18 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

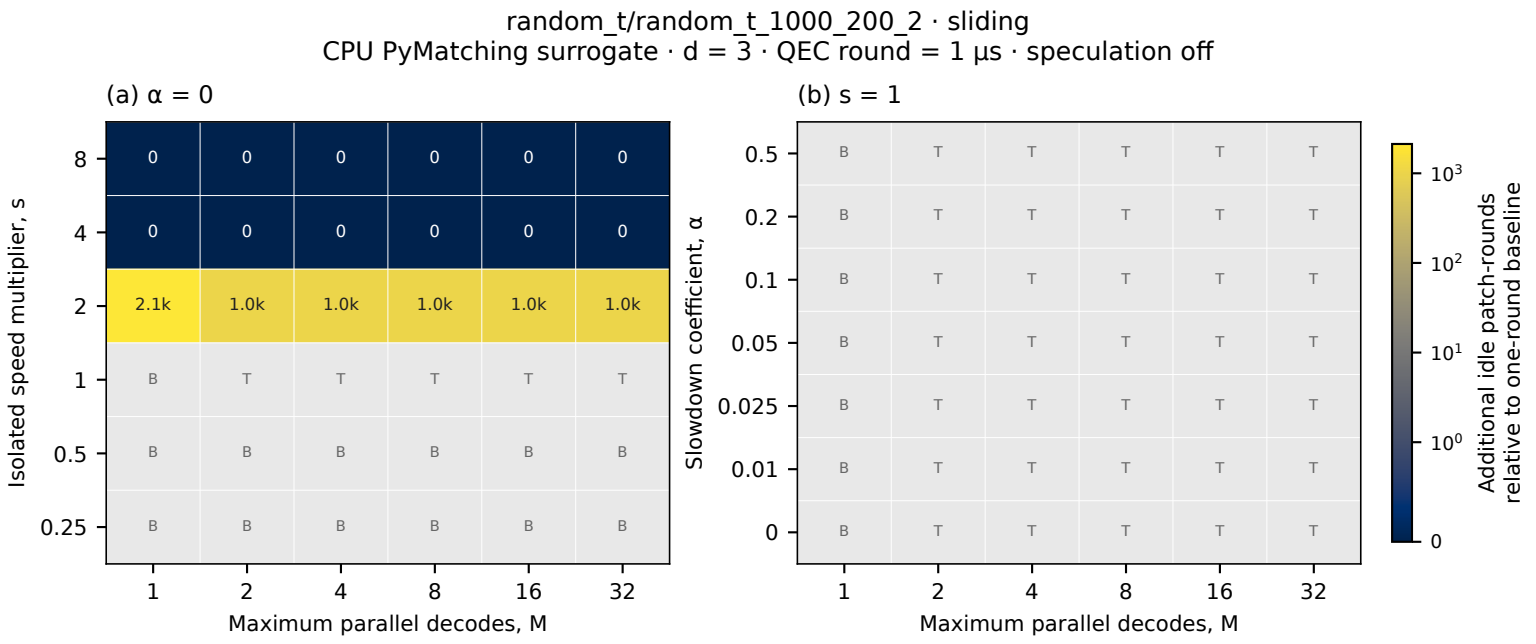

random_t/random_t_1000_200_19 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
3
0
0
0
0
0
1.1k
674
674
674
674
674
4.3k
2.6k
2.6k
2.6k
2.6k
2.6k
18k
7.4k
7.4k
7.4k
7.4k
7.4k
B
24k
18k
18k
18k
18k
B
B
45k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
18k
11k
11k
11k
11k
11k
18k
8.5k
8.4k
8.4k
8.4k
8.4k
18k
8.4k
8.3k
8.3k
8.3k
8.3k
18k
8.1k
8.0k
8.0k
8.0k
8.0k
18k
7.6k
7.6k
7.6k
7.6k
7.6k
18k
7.4k
7.4k
7.4k
7.4k
7.4k
18k
7.4k
7.4k
7.4k
7.4k
7.4k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

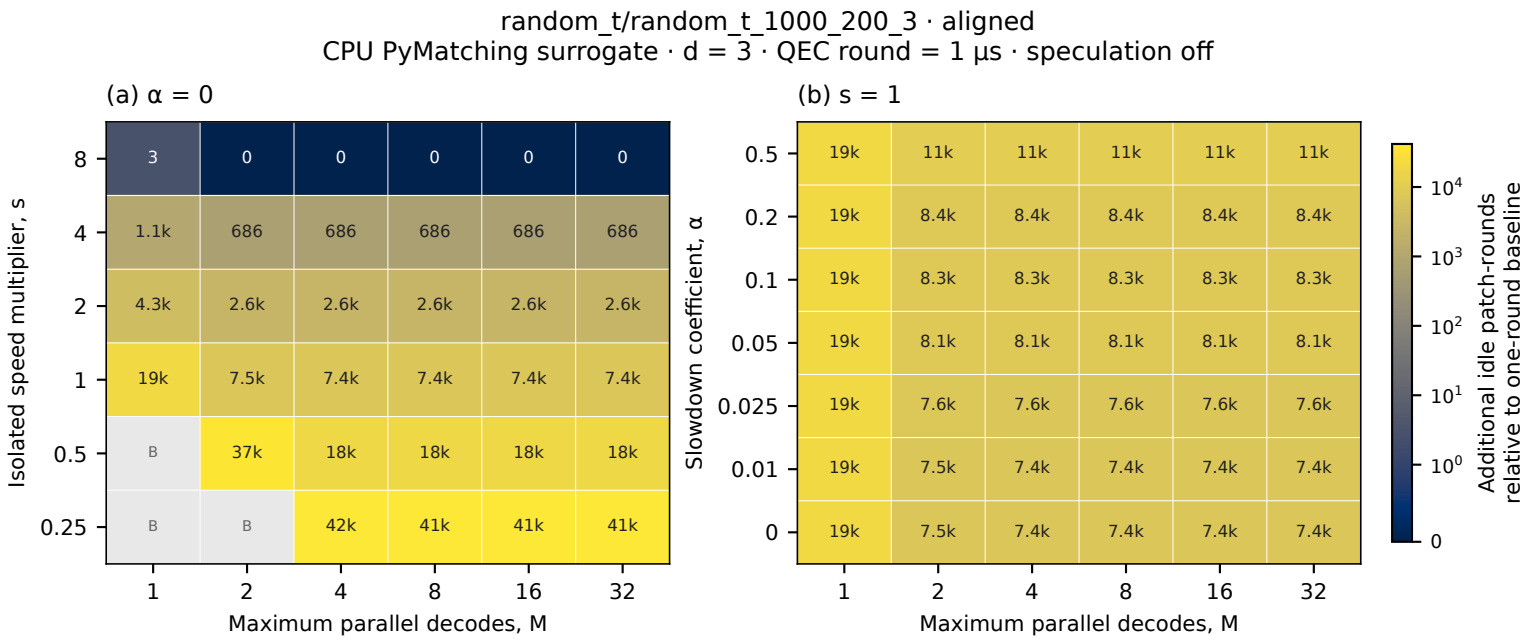

random_t/random_t_1000_200_19 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
0 0 0 0 0 0
1.6k 1.6k 1.6k 1.6k 1.6k 1.6k
3.7k 3.7k 3.7k 3.7k 3.7k 3.7k
12k 7.9k 7.9k 7.9k 7.9k 7.9k
B 22k 17k 17k 17k 17k
B B 36k 35k 35k 35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
12k 8.4k 8.4k 8.4k 8.4k 8.4k
12k 8.2k 8.2k 8.2k 8.2k 8.2k
12k 8.2k 8.1k 8.1k 8.1k 8.1k
12k 7.9k 7.9k 7.9k 7.9k 7.9k
12k 7.9k 7.9k 7.9k 7.9k 7.9k
12k 7.9k 7.9k 7.9k 7.9k 7.9k
12k 7.9k 7.9k 7.9k 7.9k 7.9k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

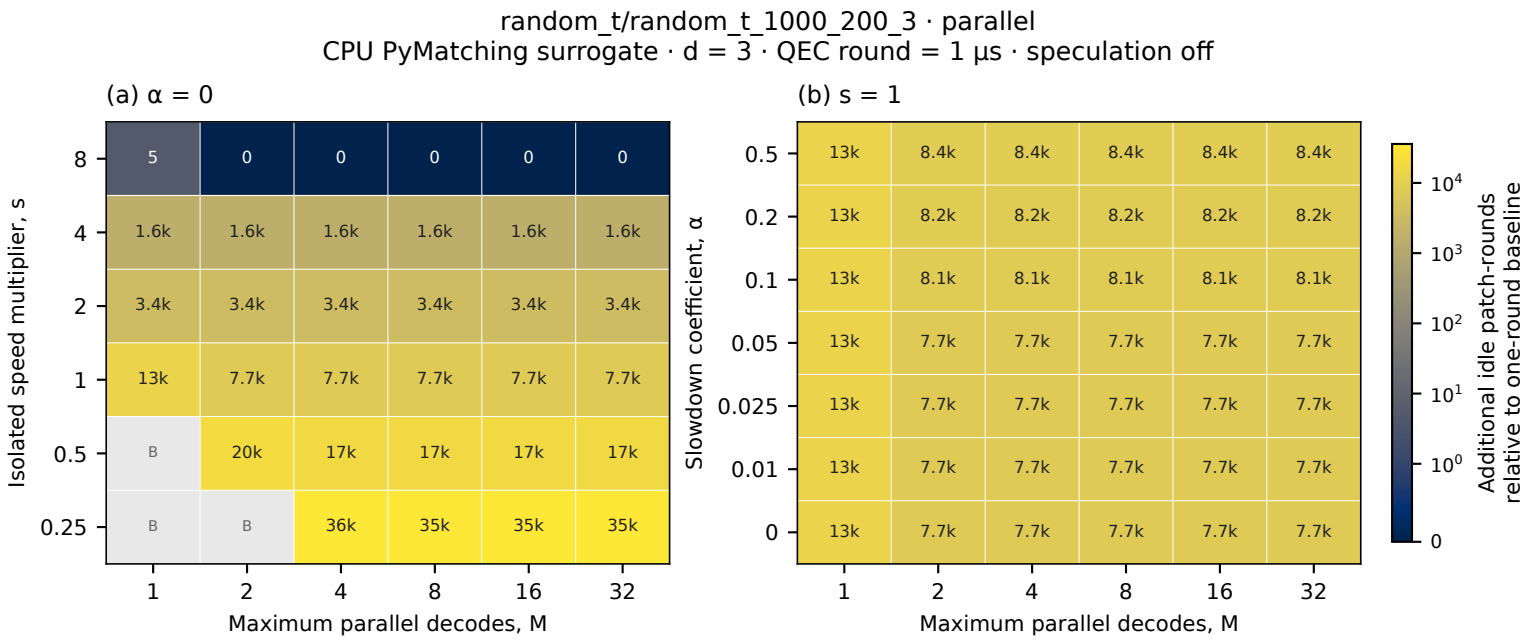

random_t/random_t_1000_200_19 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

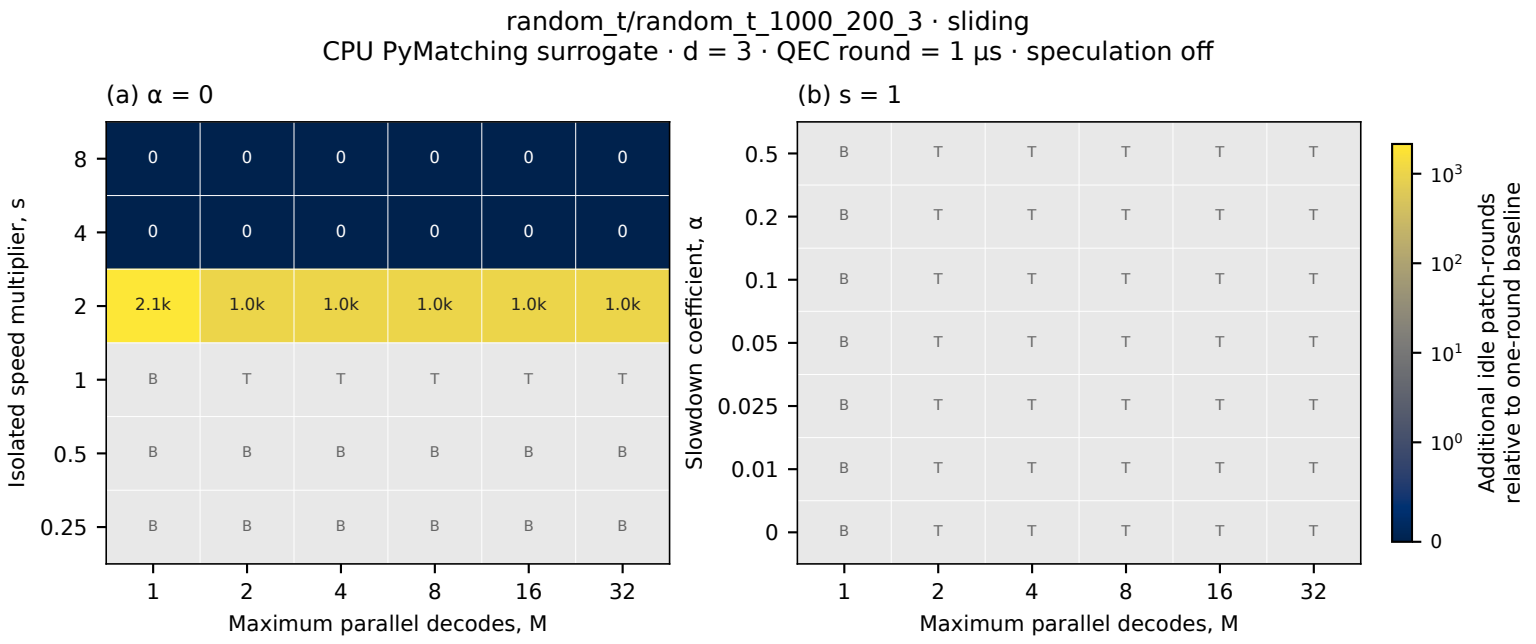

random_t/random_t_1000_200_2 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
3
0
0
0
0
0
1.1k
653
653
653
653
653
4.3k
2.6k
2.6k
2.6k
2.6k
2.6k
391k
7.4k
7.4k
7.4k
7.4k
7.4k
B
23k
19k
19k
19k
19k
B
B
41k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
391k
11k
10k
10k
10k
10k
391k
8.5k
8.5k
8.5k
8.5k
8.5k
391k
8.4k
8.4k
8.4k
8.4k
8.4k
391k
8.2k
8.1k
8.1k
8.1k
8.1k
391k
7.6k
7.6k
7.6k
7.6k
7.6k
391k
7.4k
7.4k
7.4k
7.4k
7.4k
391k
7.4k
7.4k
7.4k
7.4k
7.4k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^5
10^4
10^3
10^2
10^1
10^0
0

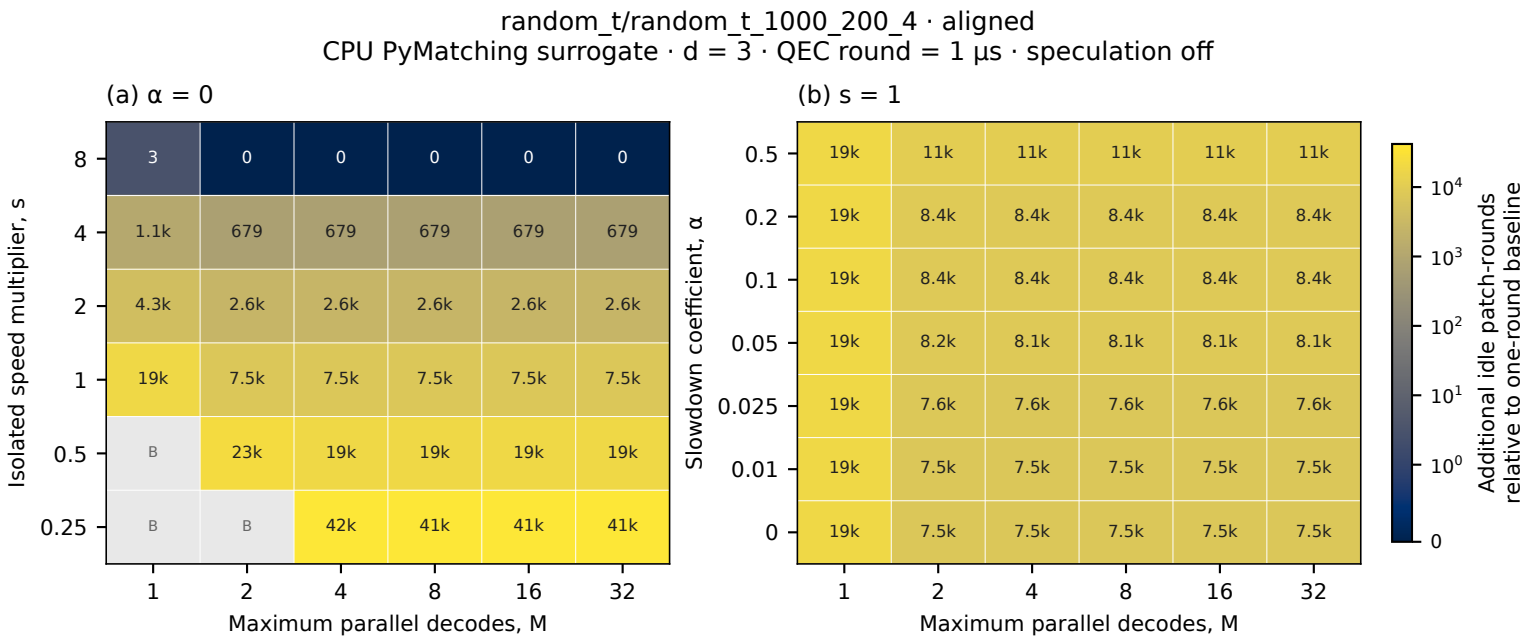

random_t/random_t_1000_200_2 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
3
0
0
0
0
0
1.9k
1.9k
1.9k
1.9k
1.9k
1.9k
4.1k
4.1k
4.1k
4.1k
4.1k
4.1k
13k
8.3k
8.3k
8.3k
8.3k
8.3k
B
20k
17k
17k
17k
17k
B
B
37k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
13k
8.7k
8.7k
8.7k
8.7k
8.7k
13k
8.5k
8.5k
8.5k
8.5k
8.5k
13k
8.5k
8.5k
8.5k
8.5k
8.5k
13k
8.3k
8.3k
8.3k
8.3k
8.3k
13k
8.3k
8.3k
8.3k
8.3k
8.3k
13k
8.3k
8.3k
8.3k
8.3k
8.3k
13k
8.3k
8.3k
8.3k
8.3k
8.3k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

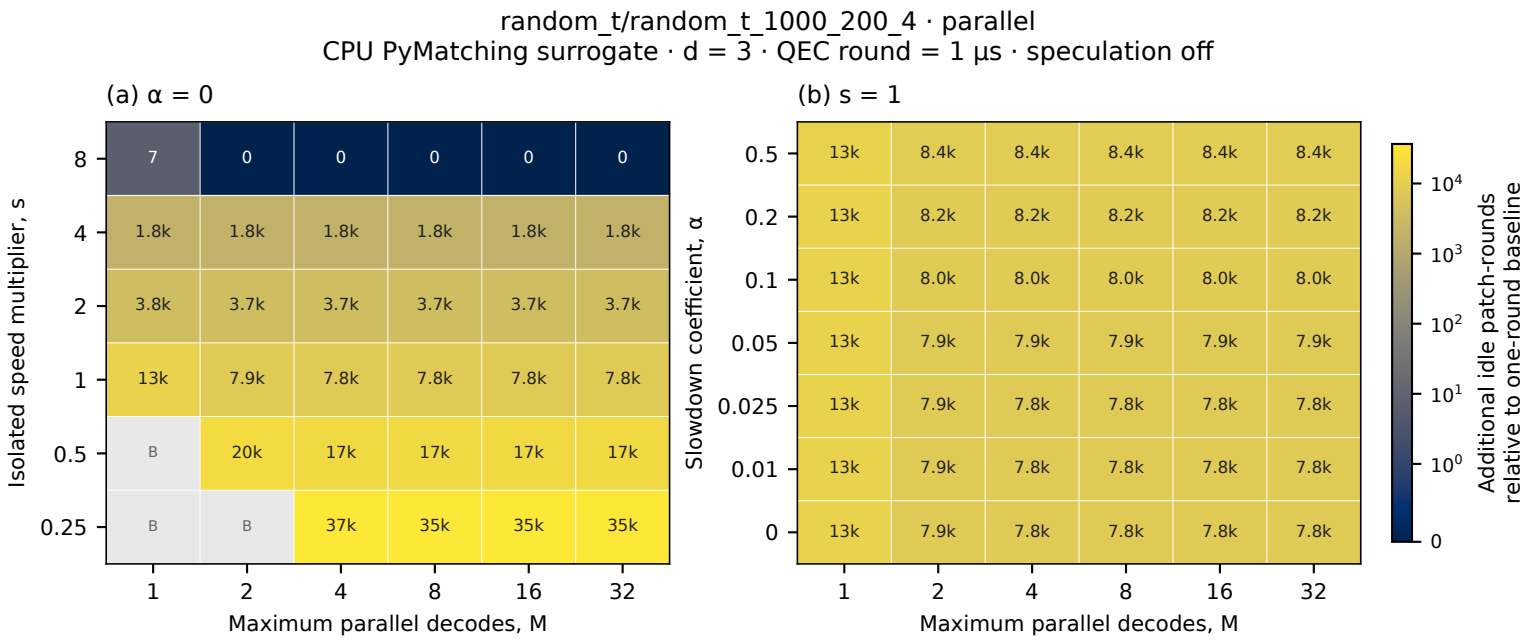

random_t/random_t_1000_200_2 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

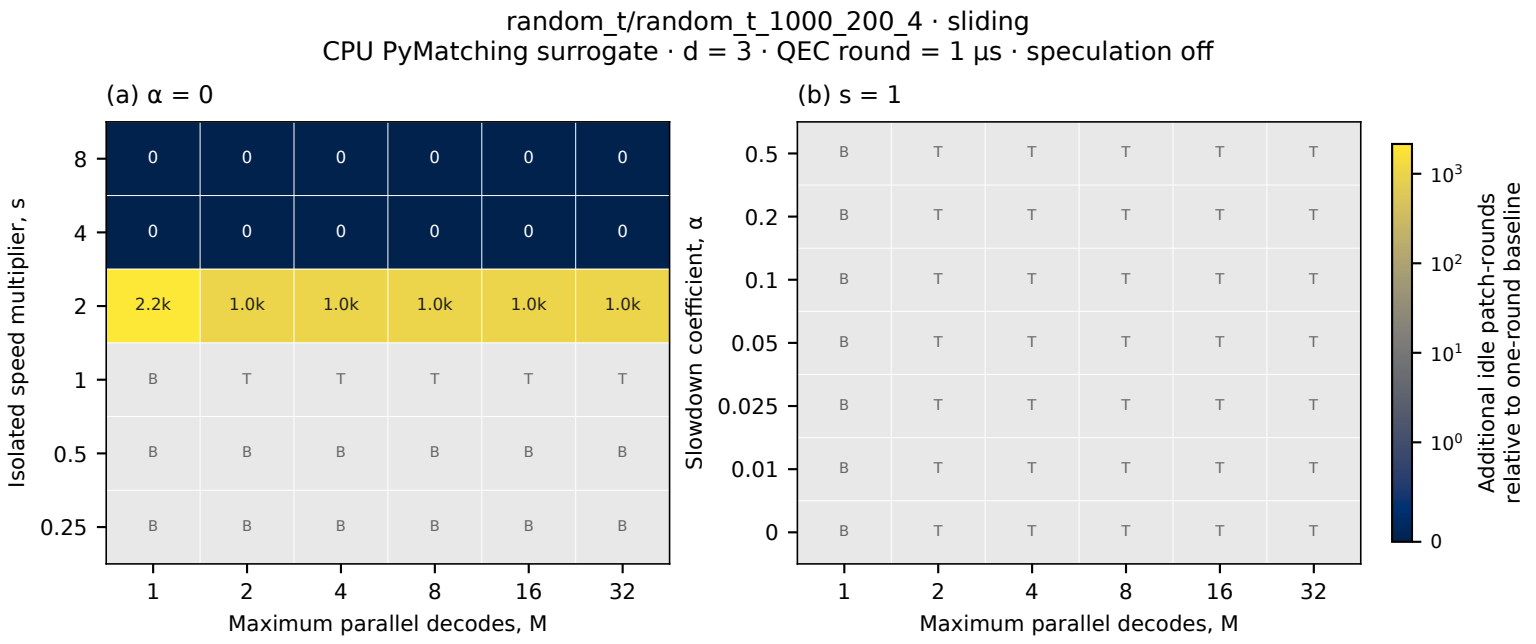

random_t/random_t_1000_200_3 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
3
0
0
0
0
0
1.1k
686
686
686
686
686
4.3k
2.6k
2.6k
2.6k
2.6k
2.6k
19k
7.5k
7.4k
7.4k
7.4k
7.4k
B
37k
18k
18k
18k
18k
B
B
42k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
19k
11k
11k
11k
11k
11k
19k
8.4k
8.4k
8.4k
8.4k
8.4k
19k
8.3k
8.3k
8.3k
8.3k
8.3k
19k
8.1k
8.1k
8.1k
8.1k
8.1k
19k
7.6k
7.6k
7.6k
7.6k
7.6k
19k
7.5k
7.4k
7.4k
7.4k
7.4k
19k
7.5k
7.4k
7.4k
7.4k
7.4k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

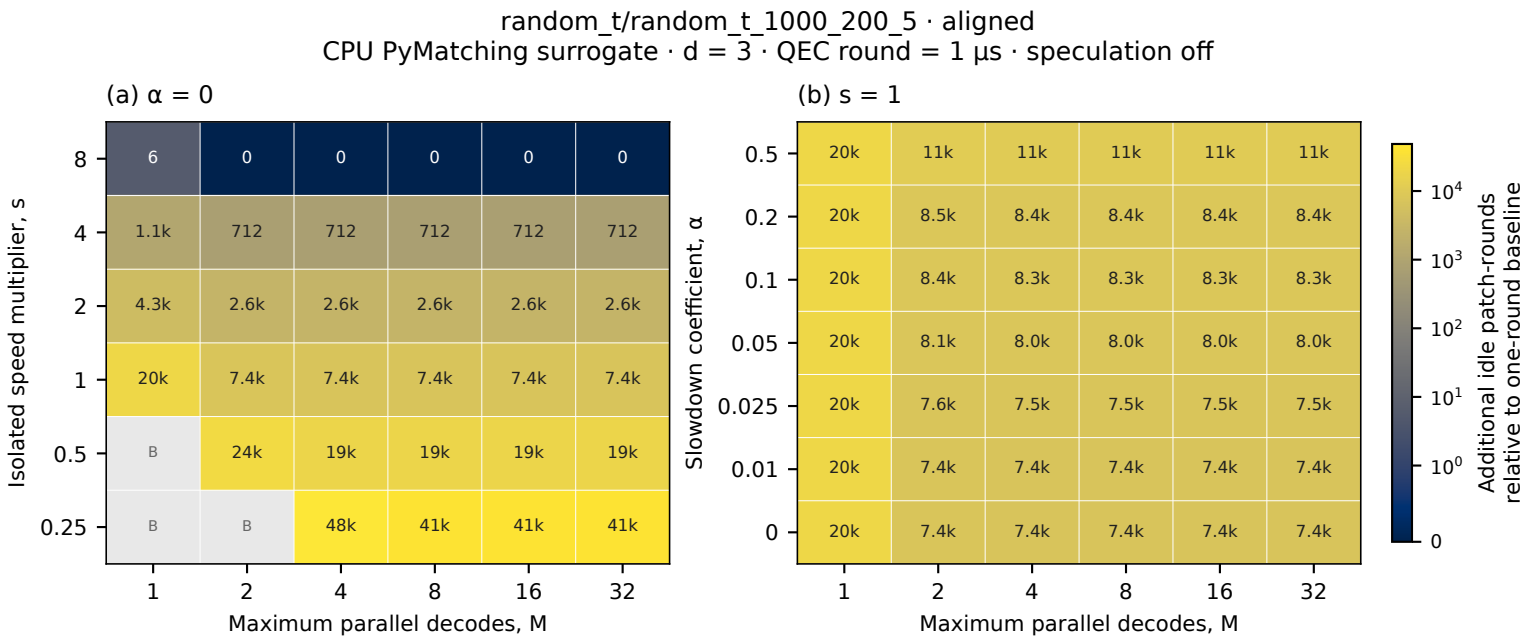

random_t/random_t_1000_200_3 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
5
0
0
0
0
0
1.6k
1.6k
1.6k
1.6k
1.6k
1.6k
3.4k
3.4k
3.4k
3.4k
3.4k
3.4k
13k
7.7k
7.7k
7.7k
7.7k
7.7k
B
20k
17k
17k
17k
17k
B
B
36k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
13k
8.4k
8.4k
8.4k
8.4k
8.4k
13k
8.2k
8.2k
8.2k
8.2k
8.2k
13k
8.1k
8.1k
8.1k
8.1k
8.1k
13k
7.7k
7.7k
7.7k
7.7k
7.7k
13k
7.7k
7.7k
7.7k
7.7k
7.7k
13k
7.7k
7.7k
7.7k
7.7k
7.7k
13k
7.7k
7.7k
7.7k
7.7k
7.7k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

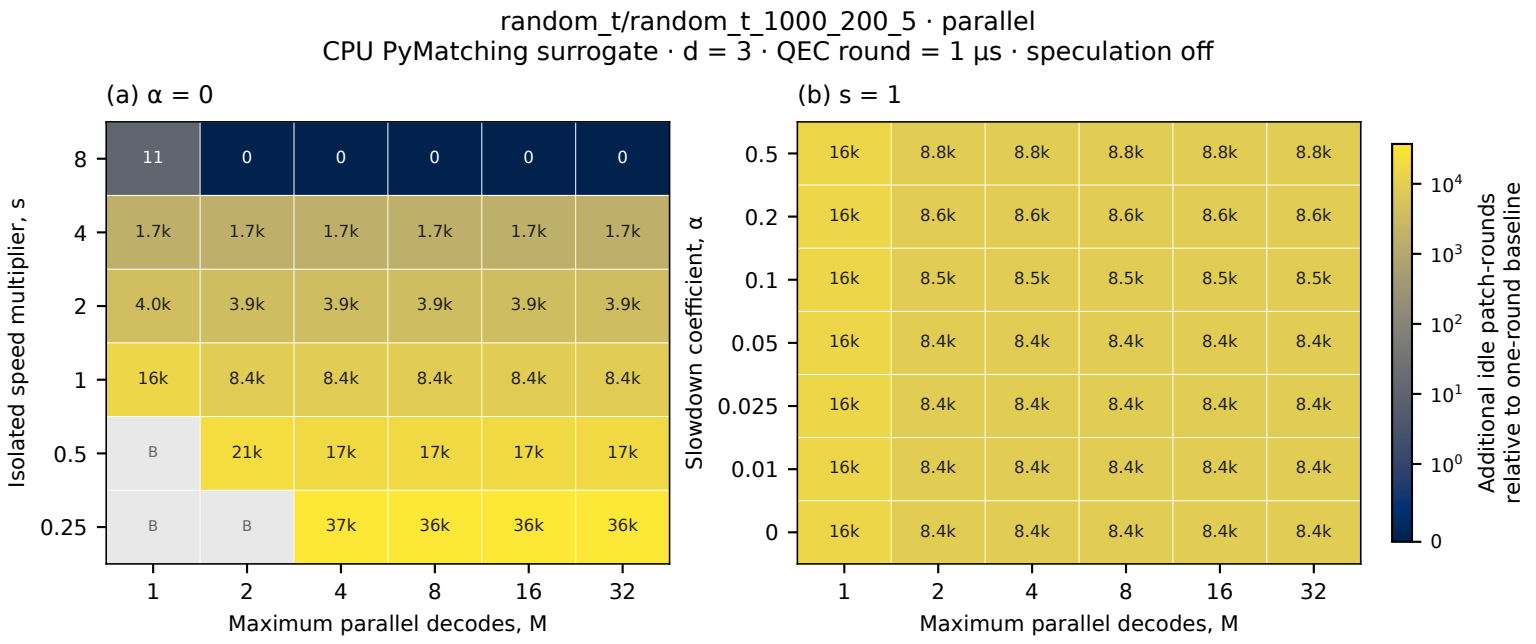

random_t/random_t_1000_200_3 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

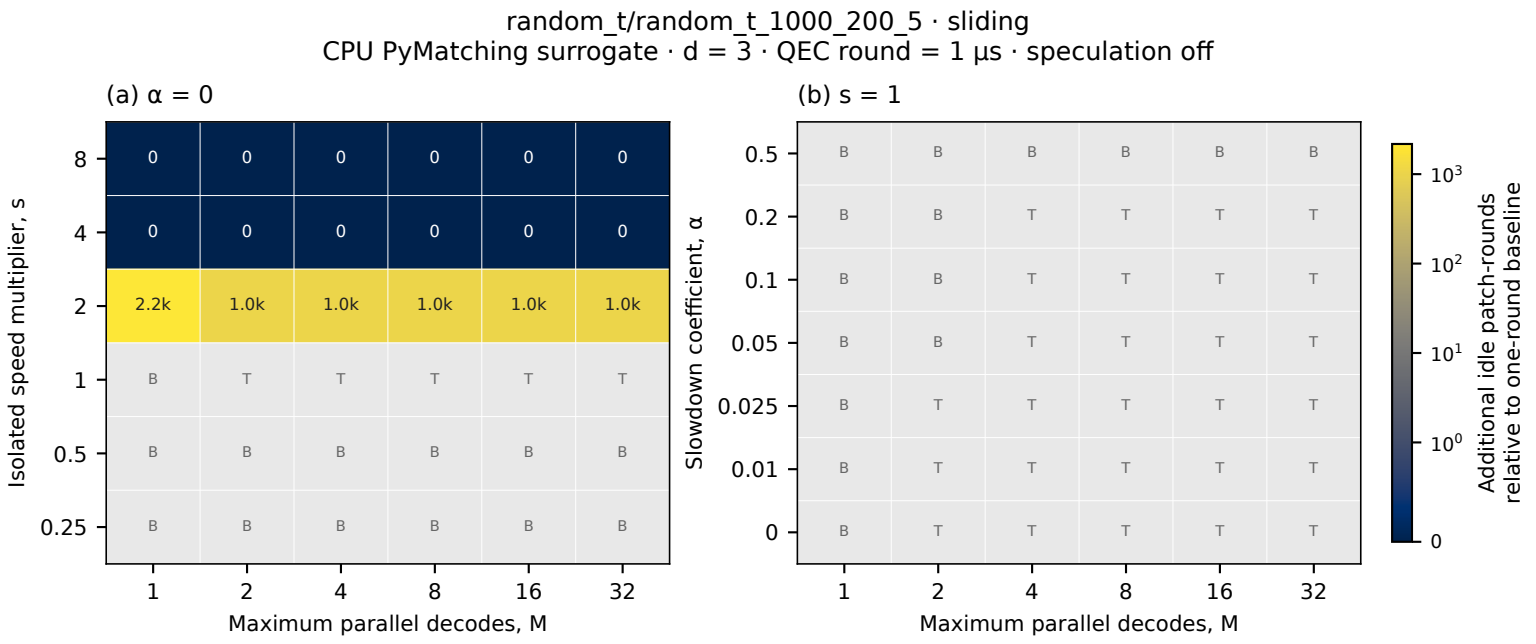

random_t/random_t_1000_200_4 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
3
0
0
0
0
0
1.1k
679
679
679
679
679
4.3k
2.6k
2.6k
2.6k
2.6k
2.6k
19k
7.5k
7.5k
7.5k
7.5k
7.5k
B
23k
19k
19k
19k
19k
B
B
42k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
19k
11k
11k
11k
11k
11k
19k
8.4k
8.4k
8.4k
8.4k
8.4k
19k
8.4k
8.4k
8.4k
8.4k
8.4k
19k
8.2k
8.1k
8.1k
8.1k
8.1k
19k
7.6k
7.6k
7.6k
7.6k
7.6k
19k
7.5k
7.5k
7.5k
7.5k
7.5k
19k
7.5k
7.5k
7.5k
7.5k
7.5k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0


random_t/random_t_1000_200_4 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
7
0
0
0
0
0
1.8k
1.8k
1.8k
1.8k
1.8k
1.8k
3.8k
3.7k
3.7k
3.7k
3.7k
3.7k
13k
7.9k
7.8k
7.8k
7.8k
7.8k
B
20k
17k
17k
17k
17k
B
B
37k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
13k
8.4k
8.4k
8.4k
8.4k
8.4k
13k
8.2k
8.2k
8.2k
8.2k
8.2k
13k
8.0k
8.0k
8.0k
8.0k
8.0k
13k
7.9k
7.9k
7.9k
7.9k
7.9k
13k
7.9k
7.8k
7.8k
7.8k
7.8k
13k
7.9k
7.8k
7.8k
7.8k
7.8k
13k
7.9k
7.8k
7.8k
7.8k
7.8k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

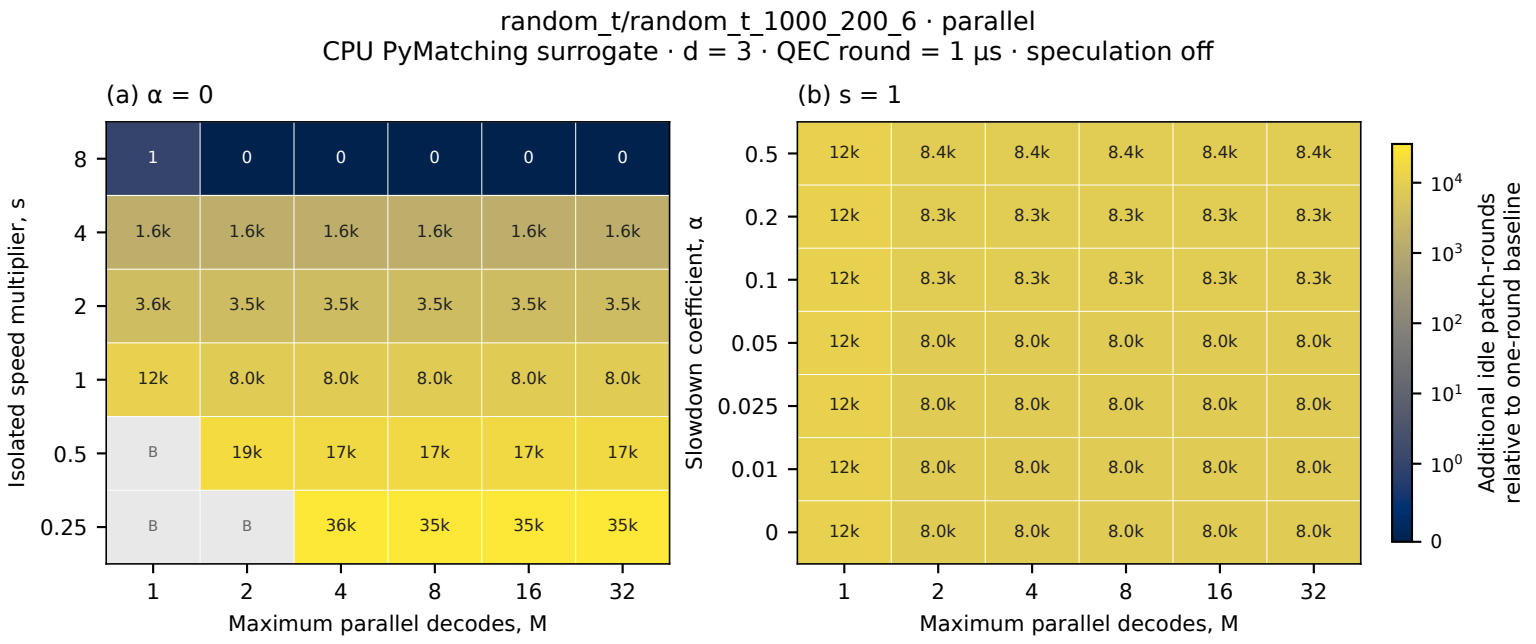

random_t/random_t_1000_200_4 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.2k
1.0k
B
T
10^3
10^2
10^1
10^0

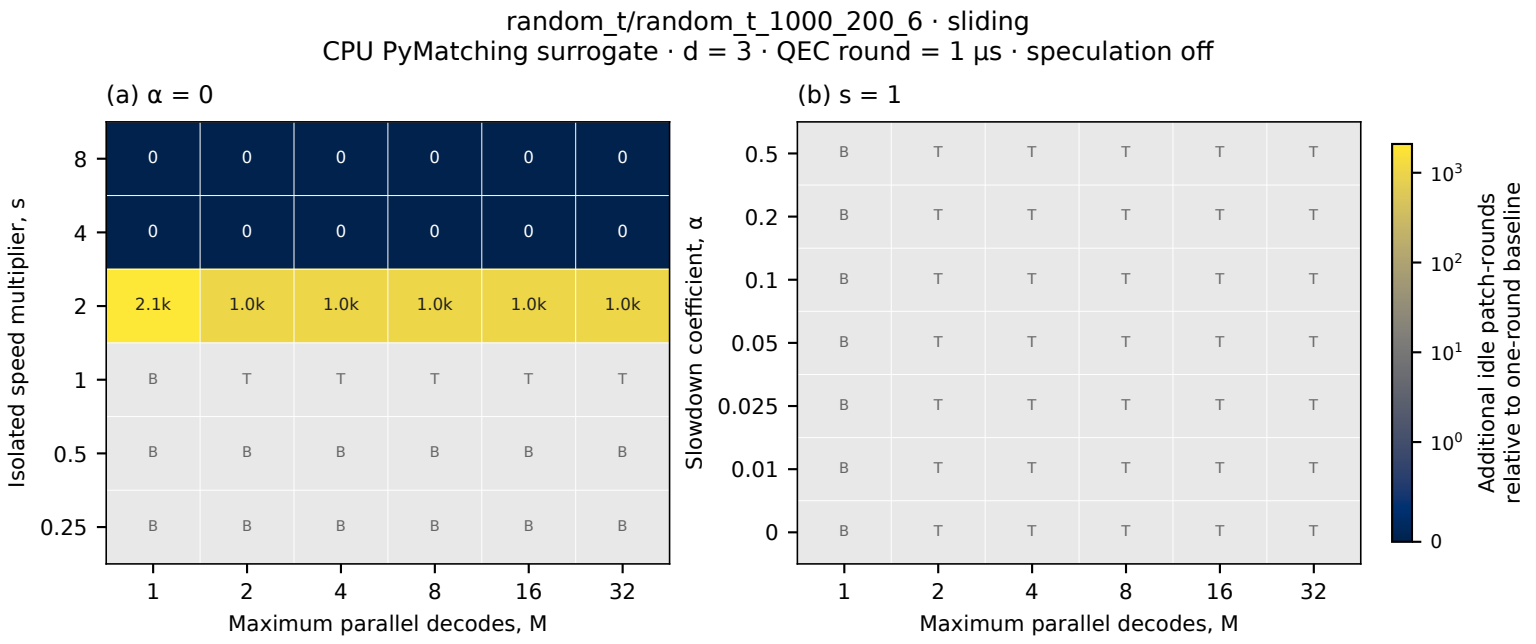

random_t/random_t_1000_200_5 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
6
0
0
0
0
0
1.1k
712
712
712
712
712
4.3k
2.6k
2.6k
2.6k
2.6k
2.6k
20k
7.4k
7.4k
7.4k
7.4k
7.4k
B
24k
19k
19k
19k
19k
B
B
48k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
20k
11k
11k
11k
11k
11k
20k
8.5k
8.4k
8.4k
8.4k
8.4k
20k
8.4k
8.3k
8.3k
8.3k
8.3k
20k
8.1k
8.0k
8.0k
8.0k
8.0k
20k
7.6k
7.5k
7.5k
7.5k
7.5k
20k
7.4k
7.4k
7.4k
7.4k
7.4k
20k
7.4k
7.4k
7.4k
7.4k
7.4k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

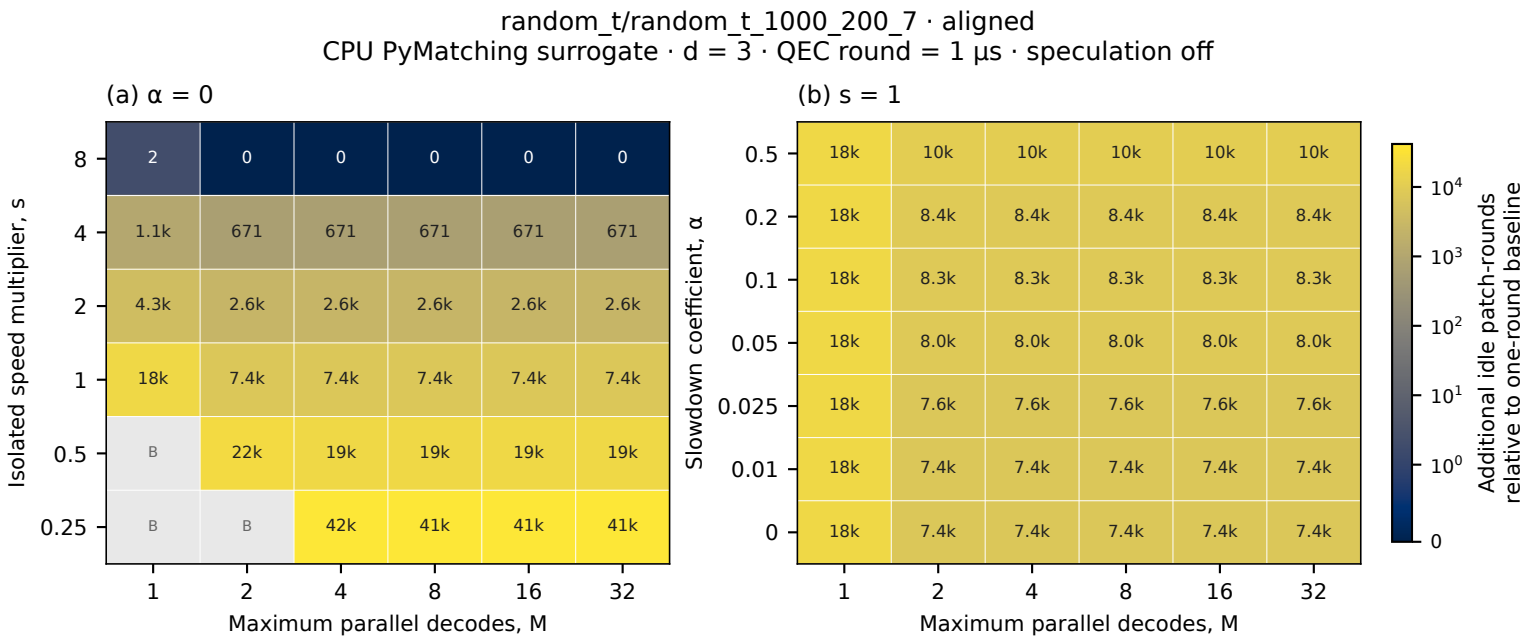

random_t/random_t_1000_200_5 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
11
0
0
0
0
0
1.7k
1.7k
1.7k
1.7k
1.7k
1.7k
4.0k
3.9k
3.9k
3.9k
3.9k
3.9k
16k
8.4k
8.4k
8.4k
8.4k
8.4k
B
21k
17k
17k
17k
17k
B
B
37k
36k
36k
36k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
16k
8.8k
8.8k
8.8k
8.8k
8.8k
16k
8.6k
8.6k
8.6k
8.6k
8.6k
16k
8.5k
8.5k
8.5k
8.5k
8.5k
16k
8.4k
8.4k
8.4k
8.4k
8.4k
16k
8.4k
8.4k
8.4k
8.4k
8.4k
16k
8.4k
8.4k
8.4k
8.4k
8.4k
16k
8.4k
8.4k
8.4k
8.4k
8.4k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

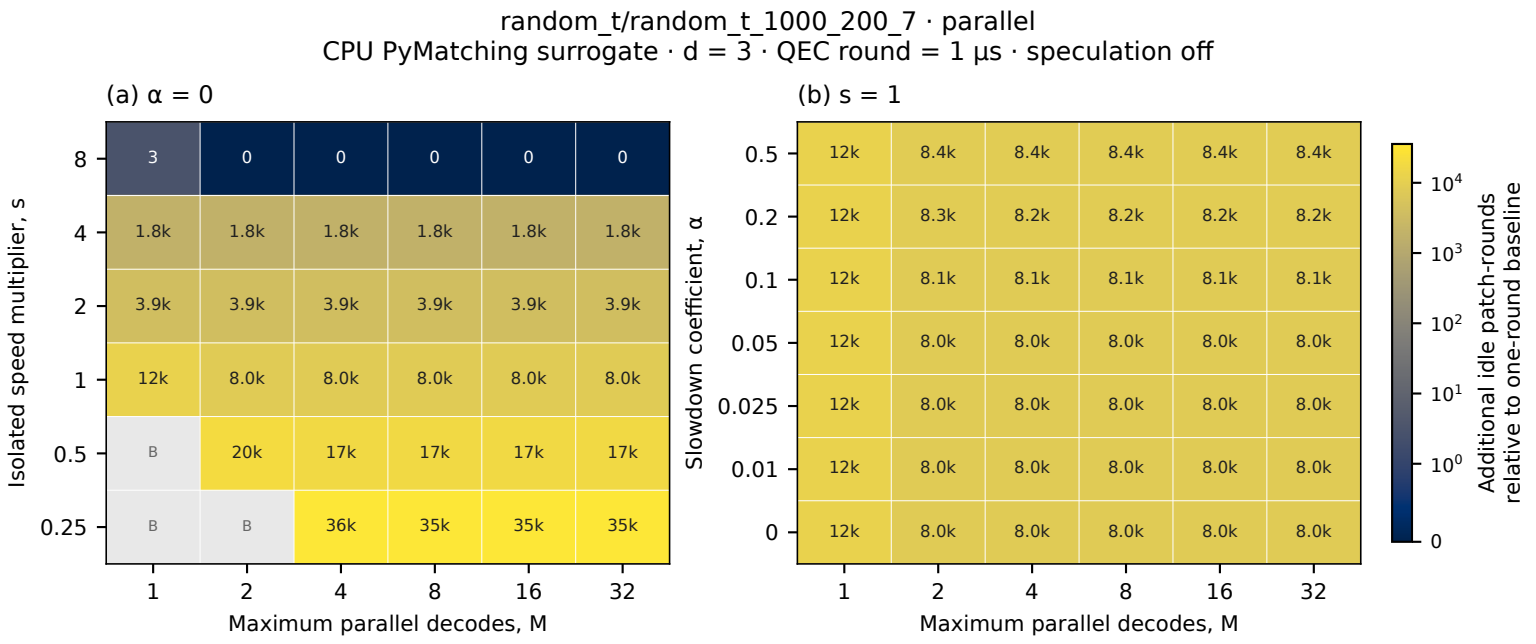

random_t/random_t_1000_200_5 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.2k
1.0k
B
T
10^3
10^2
10^1
10^0

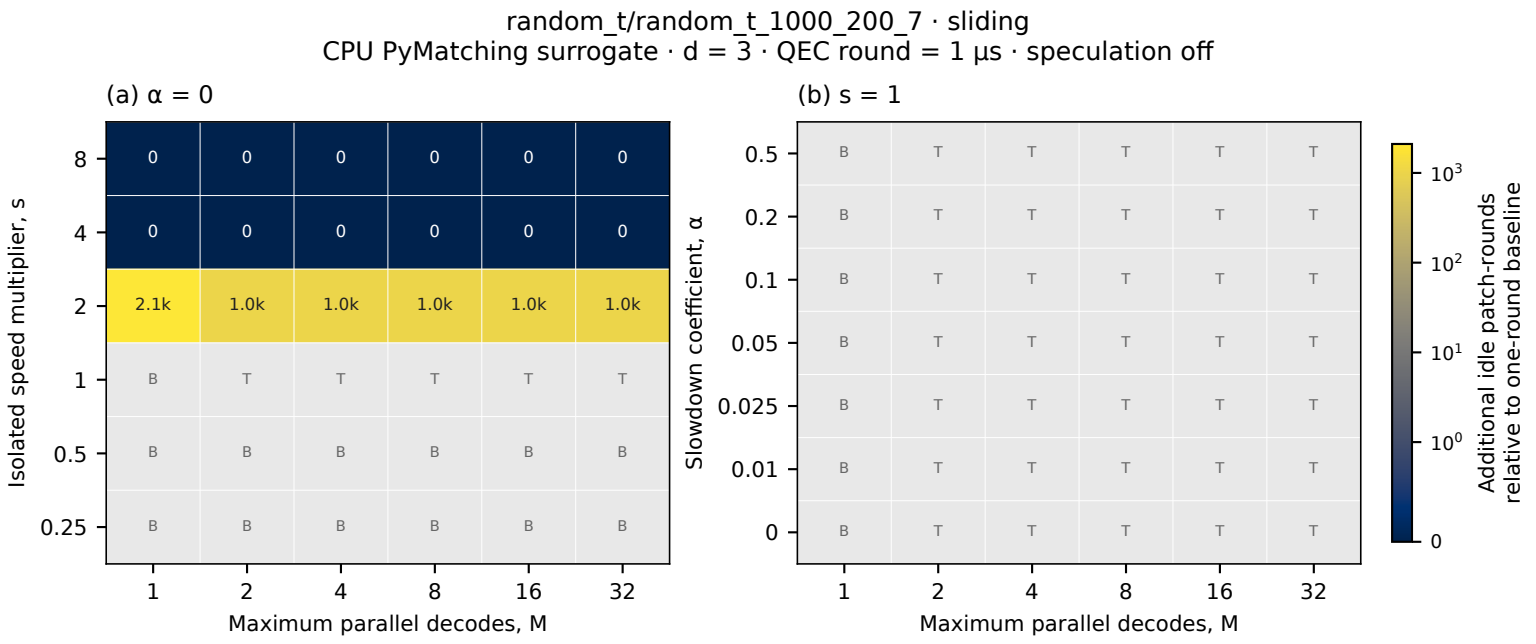

random_t/random_t_1000_200_6 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
1
0
0
0
0
0
1.1k
683
683
683
683
683
4.3k
2.7k
2.7k
2.7k
2.7k
2.7k
18k
7.4k
7.4k
7.4k
7.4k
7.4k
B
22k
18k
18k
18k
18k
B
B
41k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
18k
11k
10k
10k
10k
10k
18k
8.6k
8.5k
8.5k
8.5k
8.5k
18k
8.5k
8.4k
8.4k
8.4k
8.4k
18k
8.2k
8.2k
8.2k
8.2k
8.2k
18k
7.6k
7.6k
7.6k
7.6k
7.6k
18k
7.4k
7.4k
7.4k
7.4k
7.4k
18k
7.4k
7.4k
7.4k
7.4k
7.4k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

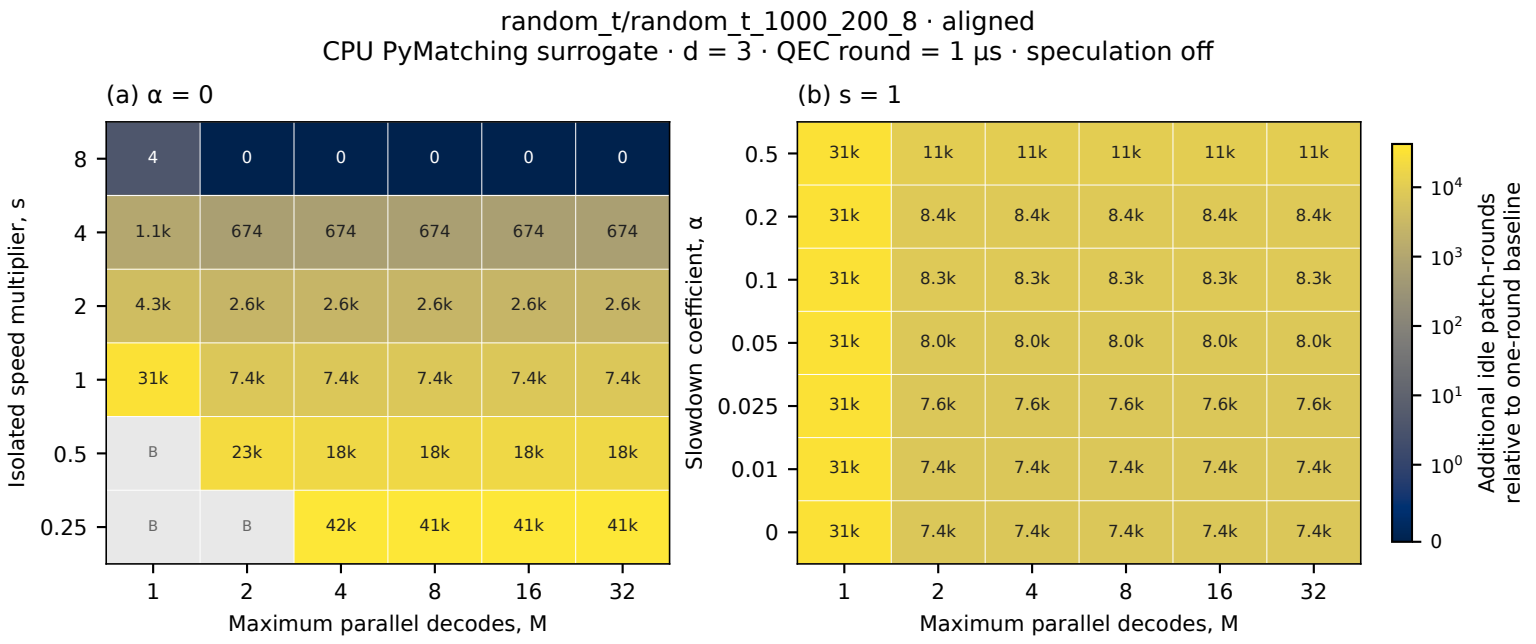

random_t/random_t_1000_200_6 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
1
0
0
0
0
0
1.6k
1.6k
1.6k
1.6k
1.6k
1.6k
3.6k
3.5k
3.5k
3.5k
3.5k
3.5k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
B
19k
17k
17k
17k
17k
B
B
36k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
12k
8.4k
8.4k
8.4k
8.4k
8.4k
12k
8.3k
8.3k
8.3k
8.3k
8.3k
12k
8.3k
8.3k
8.3k
8.3k
8.3k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

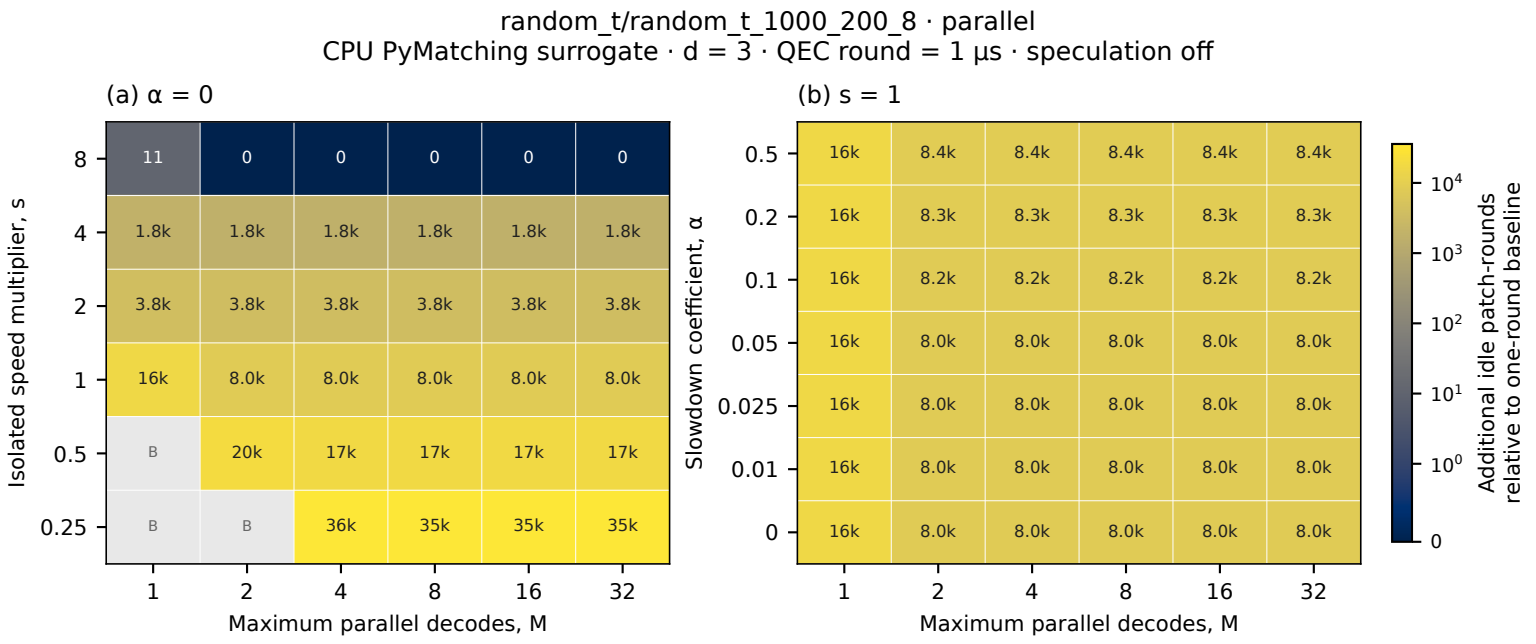

random_t/random_t_1000_200_6 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

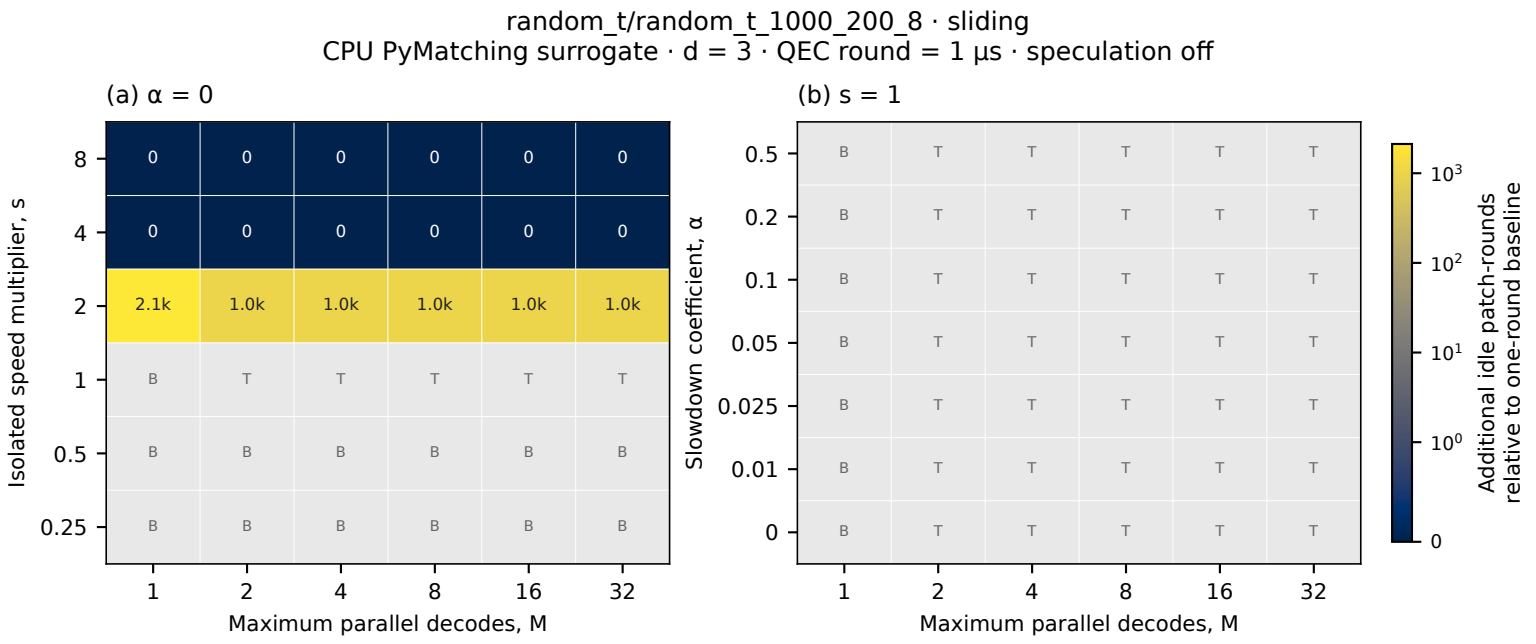

random_t/random_t_1000_200_7 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
2
0
0
0
0
0
1.1k
671
671
671
671
671
4.3k
2.6k
2.6k
2.6k
2.6k
2.6k
18k
7.4k
7.4k
7.4k
7.4k
7.4k
B
22k
19k
19k
19k
19k
B
B
42k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
18k
10k
10k
10k
10k
10k
18k
8.4k
8.4k
8.4k
8.4k
8.4k
18k
8.3k
8.3k
8.3k
8.3k
8.3k
18k
8.0k
8.0k
8.0k
8.0k
8.0k
18k
7.6k
7.6k
7.6k
7.6k
7.6k
18k
7.4k
7.4k
7.4k
7.4k
7.4k
18k
7.4k
7.4k
7.4k
7.4k
7.4k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

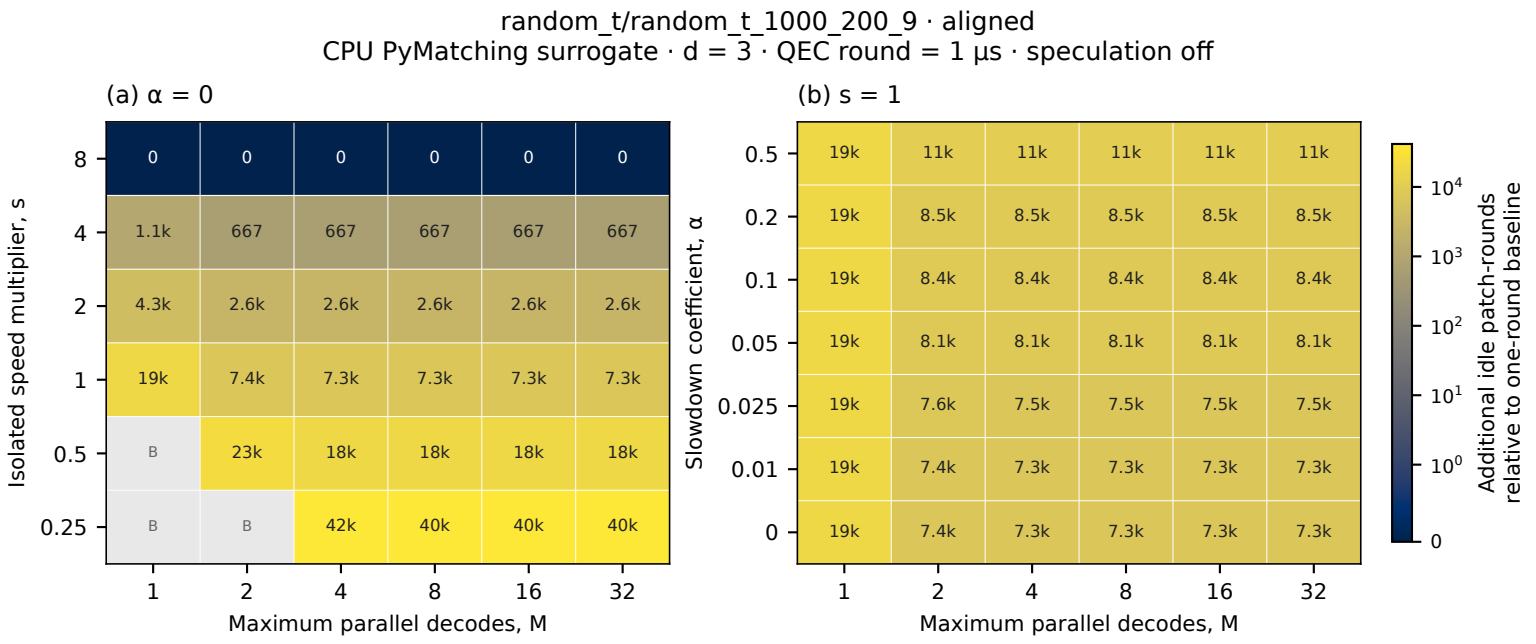

random_t/random_t_1000_200_7 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
3
0
0
0
0
0
1.8k
1.8k
1.8k
1.8k
1.8k
1.8k
3.9k
3.9k
3.9k
3.9k
3.9k
3.9k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
B
20k
17k
17k
17k
17k
B
B
36k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
12k
8.4k
8.4k
8.4k
8.4k
8.4k
12k
8.3k
8.2k
8.2k
8.2k
8.2k
12k
8.1k
8.1k
8.1k
8.1k
8.1k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
12k
8.0k
8.0k
8.0k
8.0k
8.0k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

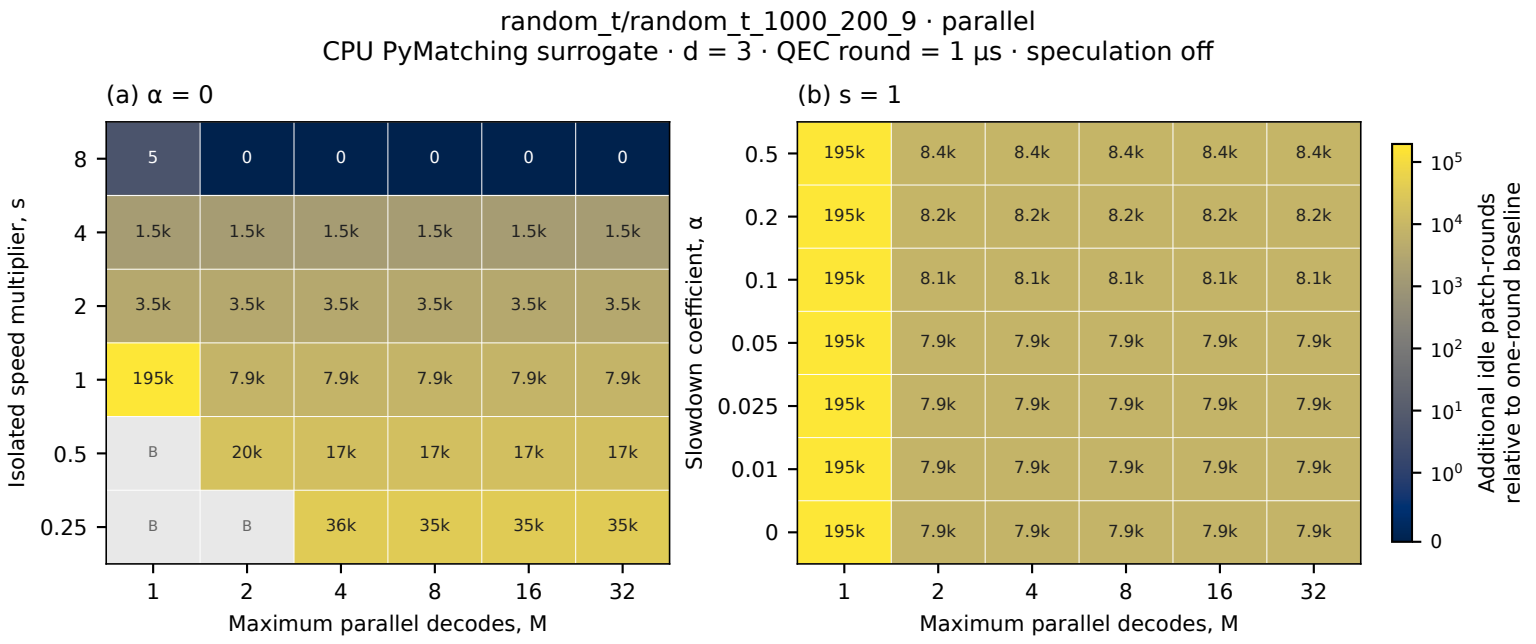

random_t/random_t_1000_200_7 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

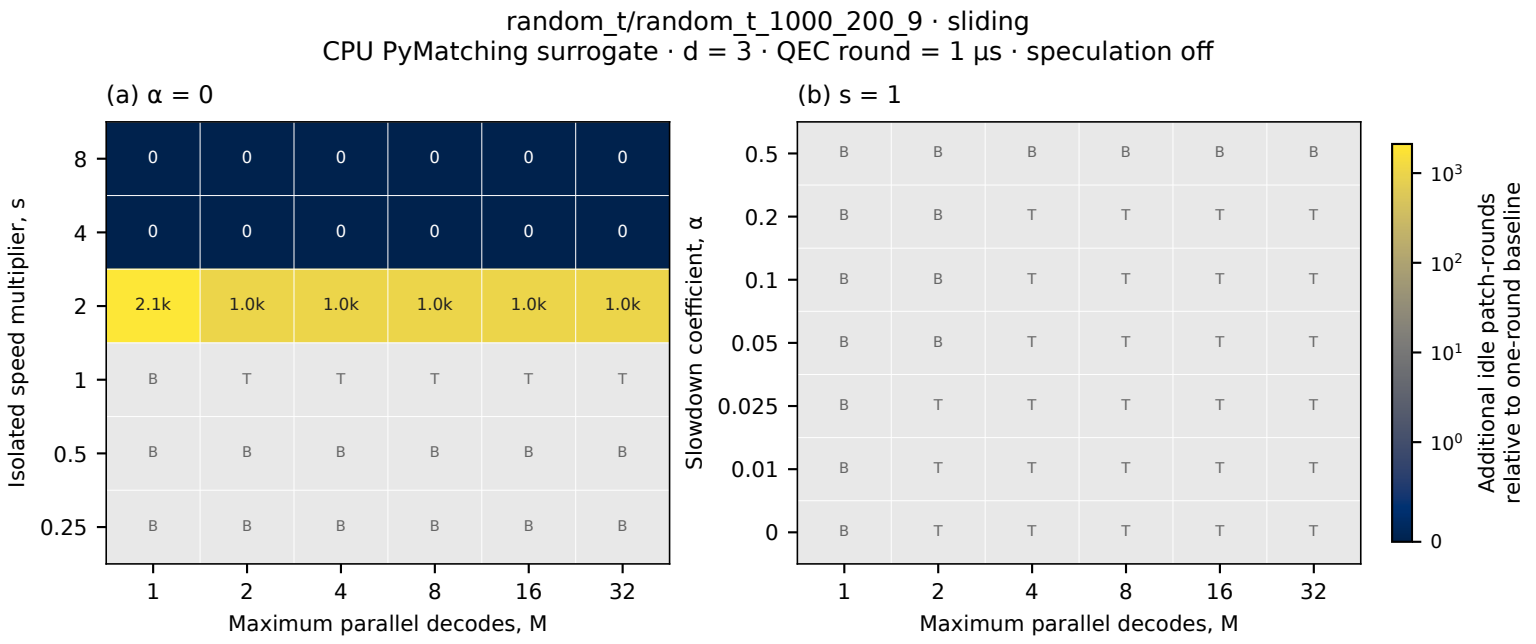

random_t/random_t_1000_200_8 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
4
0
0
0
0
0
1.1k
674
674
674
674
674
4.3k
2.6k
2.6k
2.6k
2.6k
2.6k
31k
7.4k
7.4k
7.4k
7.4k
7.4k
B
23k
18k
18k
18k
18k
B
B
42k
41k
41k
41k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
31k
11k
11k
11k
11k
11k
31k
8.4k
8.4k
8.4k
8.4k
8.4k
31k
8.3k
8.3k
8.3k
8.3k
8.3k
31k
8.0k
8.0k
8.0k
8.0k
8.0k
31k
7.6k
7.6k
7.6k
7.6k
7.6k
31k
7.4k
7.4k
7.4k
7.4k
7.4k
31k
7.4k
7.4k
7.4k
7.4k
7.4k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

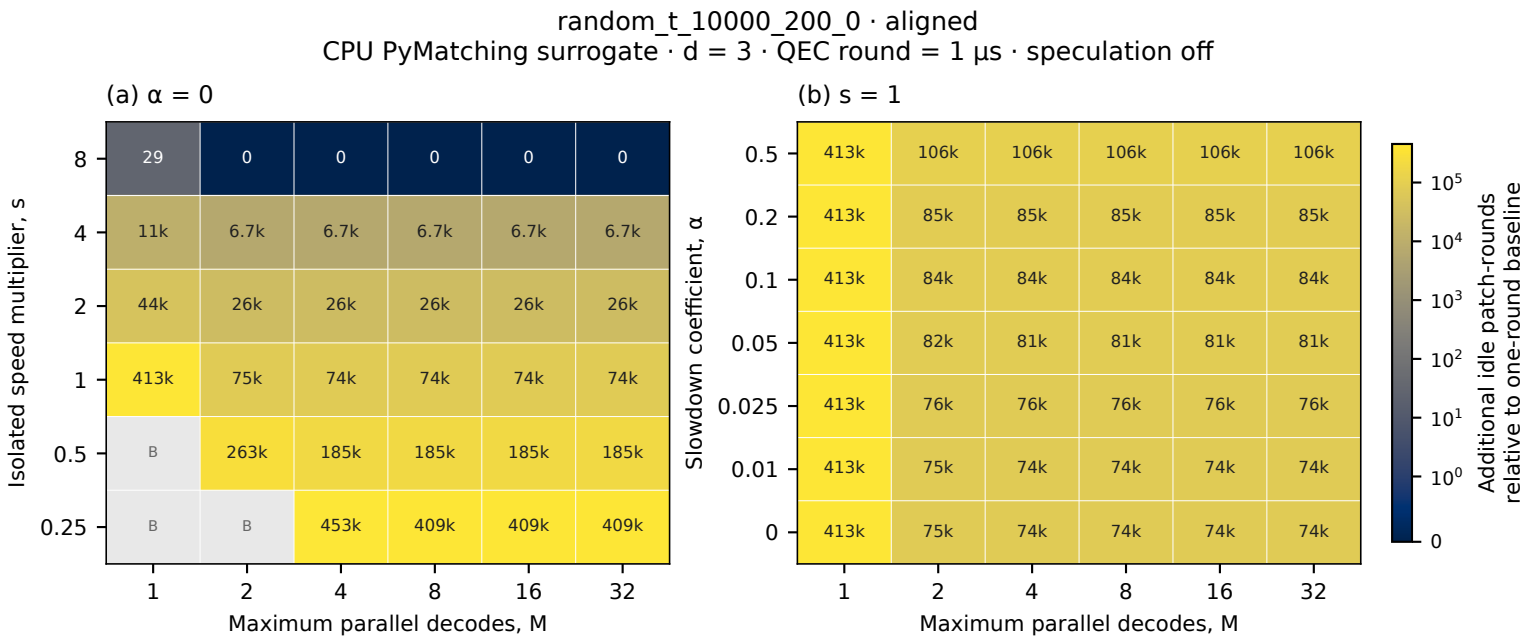

random_t/random_t_1000_200_8 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
11
0
0
0
0
0
1.8k
1.8k
1.8k
1.8k
1.8k
1.8k
3.8k
3.8k
3.8k
3.8k
3.8k
3.8k
16k
8.0k
8.0k
8.0k
8.0k
8.0k
B
20k
17k
17k
17k
17k
B
B
36k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
16k
8.4k
8.4k
8.4k
8.4k
8.4k
16k
8.3k
8.3k
8.3k
8.3k
8.3k
16k
8.2k
8.2k
8.2k
8.2k
8.2k
16k
8.0k
8.0k
8.0k
8.0k
8.0k
16k
8.0k
8.0k
8.0k
8.0k
8.0k
16k
8.0k
8.0k
8.0k
8.0k
8.0k
16k
8.0k
8.0k
8.0k
8.0k
8.0k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

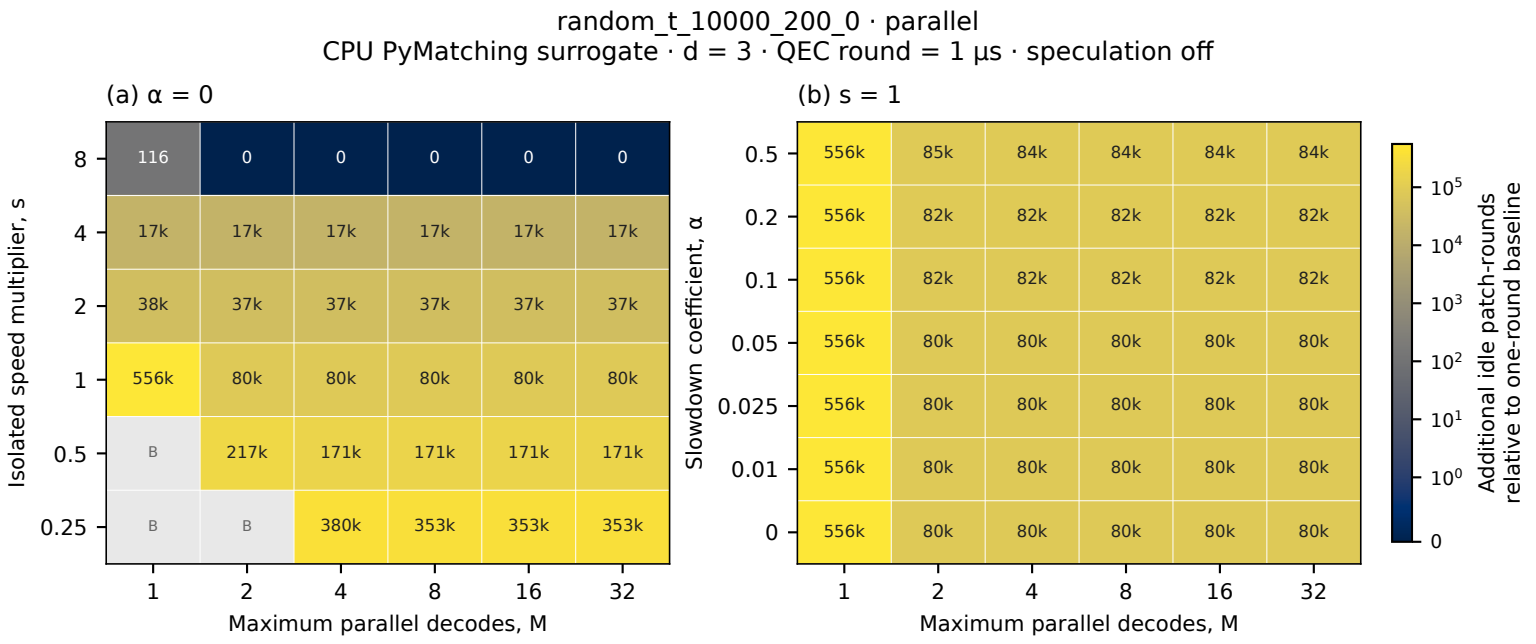

random_t/random_t_1000_200_8 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

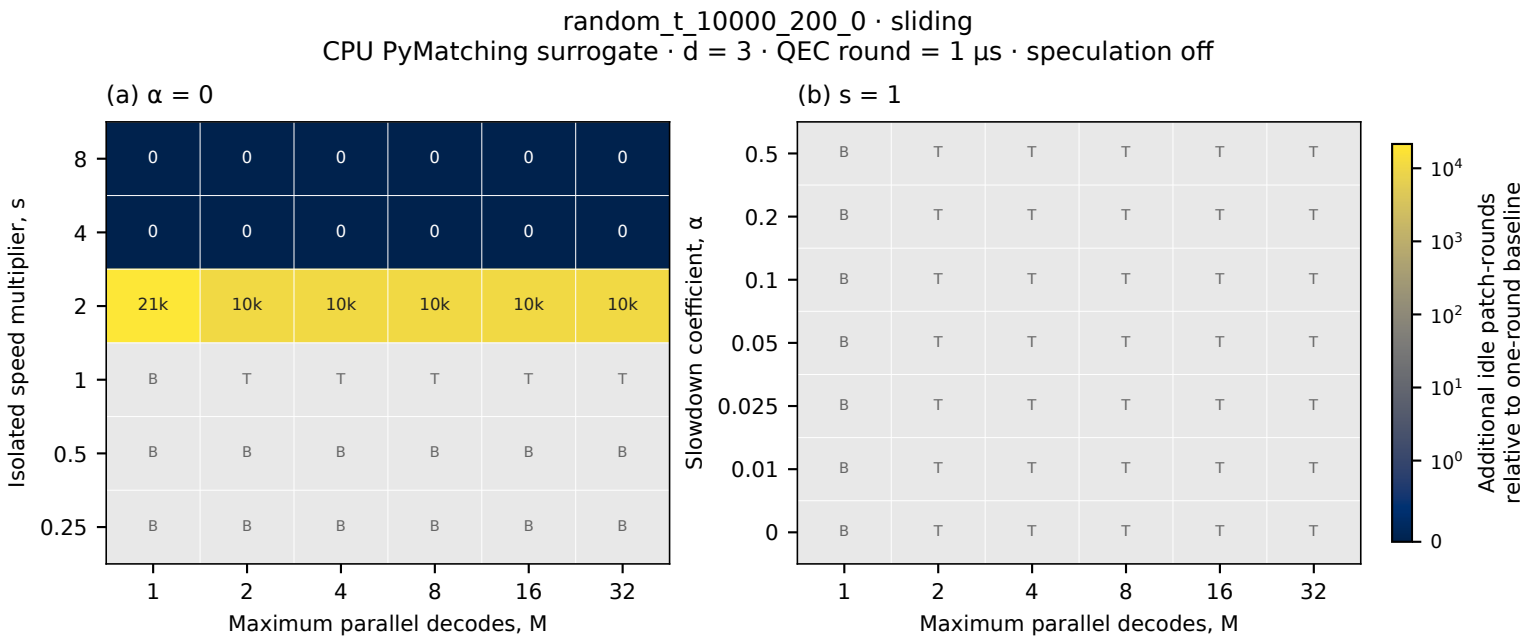

random_t/random_t_1000_200_9 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
0
1.1k
667
4.3k
2.6k
19k
7.4k
7.3k
B
23k
18k
42k
40k
Maximum parallel decodes, M
1
2
4
8
16
32
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
19k
11k
8.5k
8.4k
8.1k
7.6k
7.5k
7.4k
7.3k
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^4
10^3
10^2
10^1
10^0
0

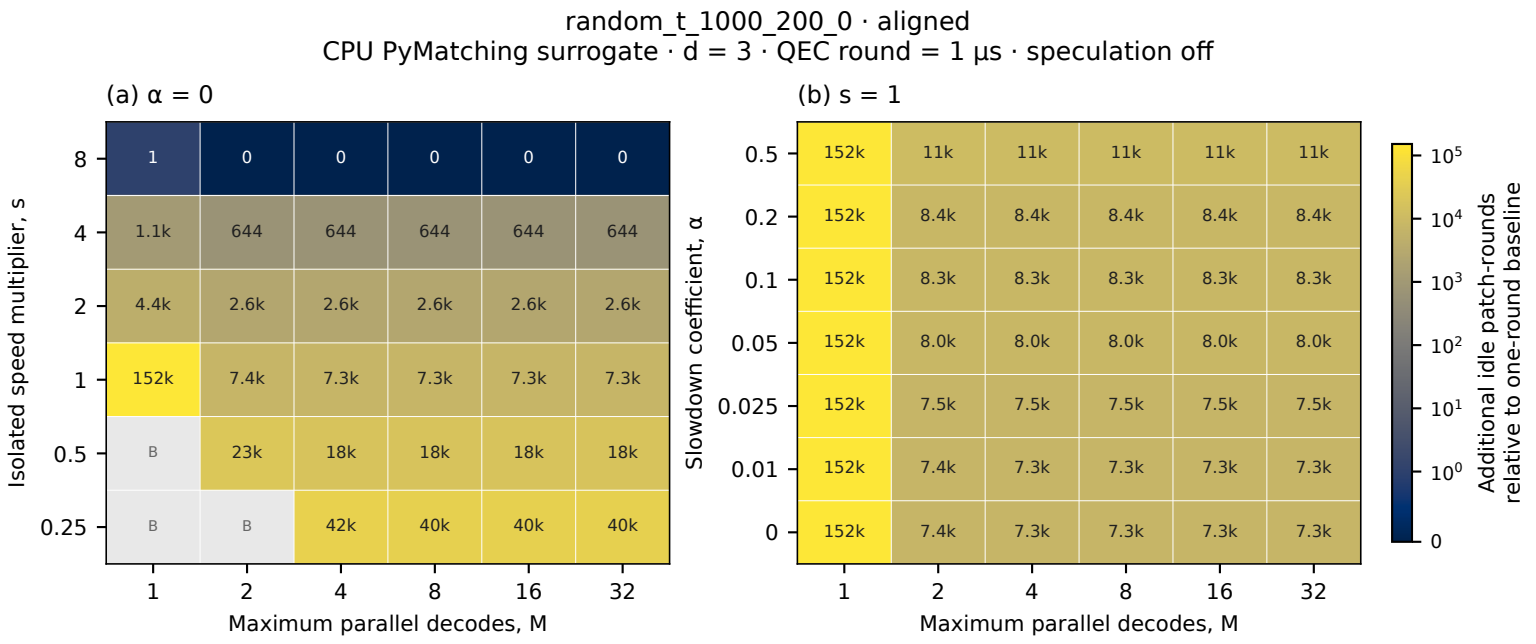

random_t/random_t_1000_200_9 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
5
0
0
0
0
0
1.5k
1.5k
1.5k
1.5k
1.5k
1.5k
3.5k
3.5k
3.5k
3.5k
3.5k
3.5k
195k
7.9k
7.9k
7.9k
7.9k
7.9k
B
20k
17k
17k
17k
17k
B
B
36k
35k
35k
35k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
195k
8.4k
8.4k
8.4k
8.4k
8.4k
195k
8.2k
8.2k
8.2k
8.2k
8.2k
195k
8.1k
8.1k
8.1k
8.1k
8.1k
195k
7.9k
7.9k
7.9k
7.9k
7.9k
195k
7.9k
7.9k
7.9k
7.9k
7.9k
195k
7.9k
7.9k
7.9k
7.9k
7.9k
195k
7.9k
7.9k
7.9k
7.9k
7.9k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^5
10^4
10^3
10^2
10^1
10^0
0

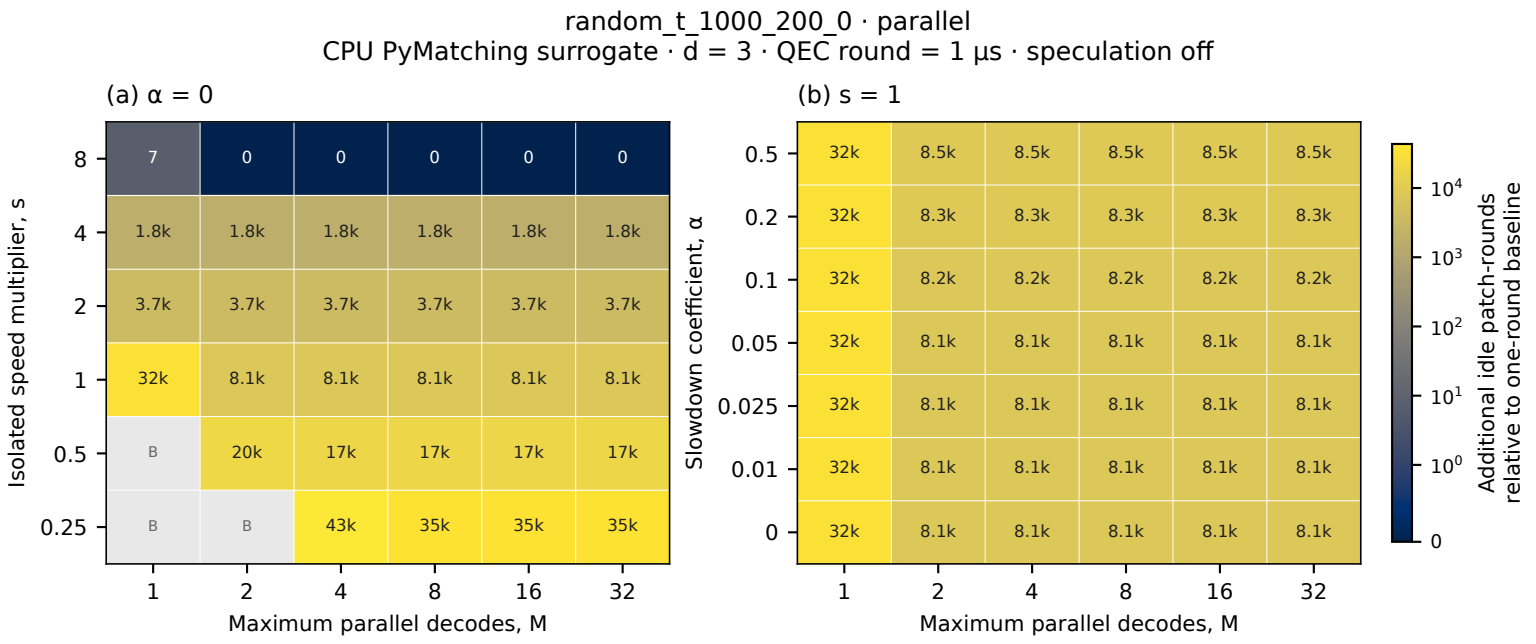

random_t/random_t_1000_200_9 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.1k
1.0k
B
T
10^3
10^2
10^1
10^0

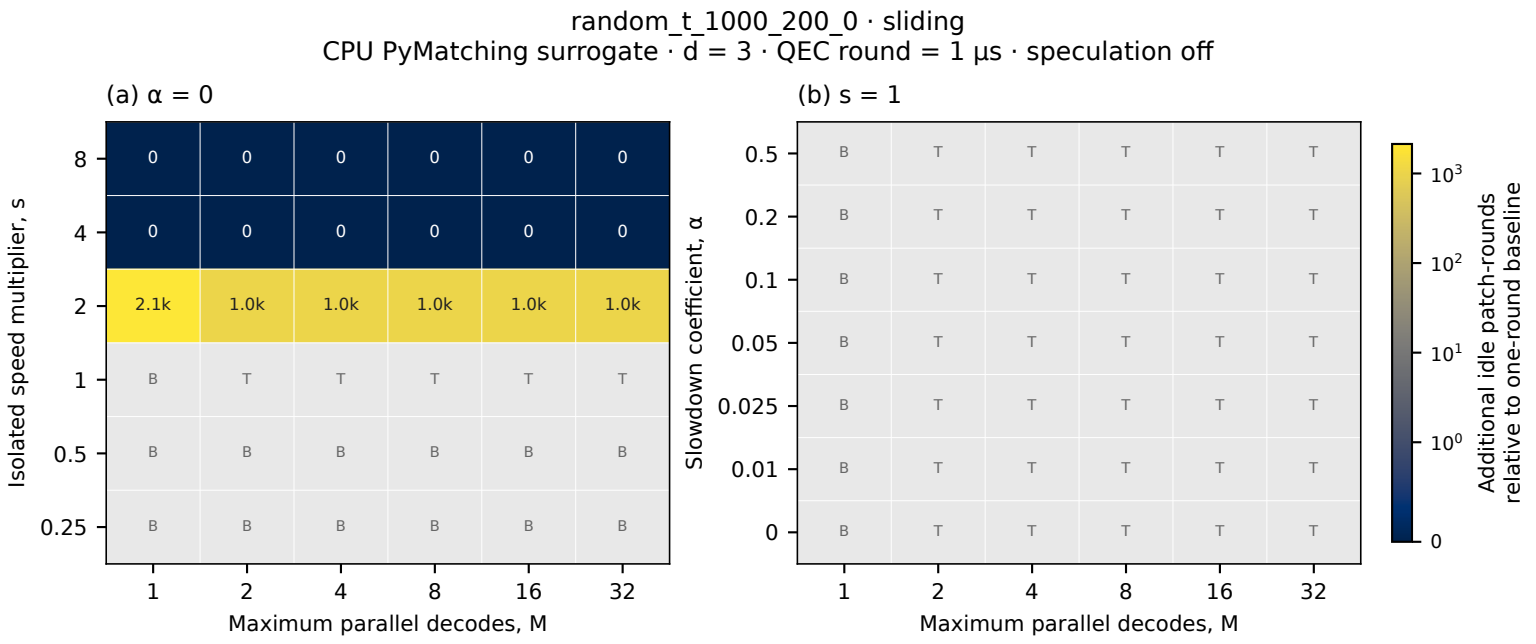

random_t_10000_200_0 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
29
0
0
0
0
0
11k
6.7k
6.7k
6.7k
6.7k
6.7k
44k
26k
26k
26k
26k
26k
413k
75k
74k
74k
74k
74k
B
263k
185k
185k
185k
185k
B
B
453k
409k
409k
409k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
413k
106k
106k
106k
106k
106k
413k
85k
85k
85k
85k
85k
413k
84k
84k
84k
84k
84k
413k
82k
81k
81k
81k
81k
413k
76k
76k
76k
76k
76k
413k
75k
74k
74k
74k
74k
413k
75k
74k
74k
74k
74k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^5
10^4
10^3
10^2
10^1
10^0
0

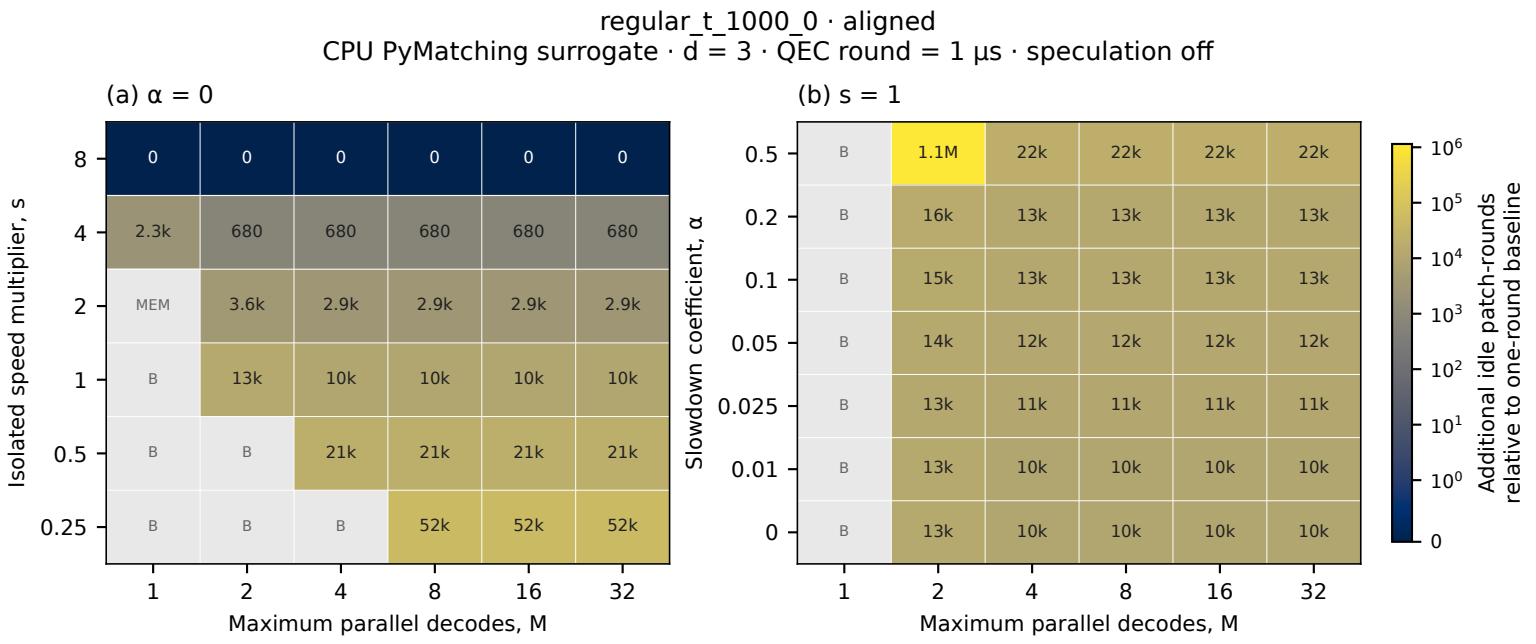

random_t_10000_200_0 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
116
0
0
0
0
0
17k
17k
17k
17k
17k
17k
38k
37k
37k
37k
37k
37k
556k
80k
80k
80k
80k
80k
B
217k
171k
171k
171k
171k
B
B
380k
353k
353k
353k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
556k
85k
84k
84k
84k
84k
556k
82k
82k
82k
82k
82k
556k
82k
82k
82k
82k
82k
556k
80k
80k
80k
80k
80k
556k
80k
80k
80k
80k
80k
556k
80k
80k
80k
80k
80k
556k
80k
80k
80k
80k
80k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^5
10^4
10^3
10^2
10^1
10^0
0

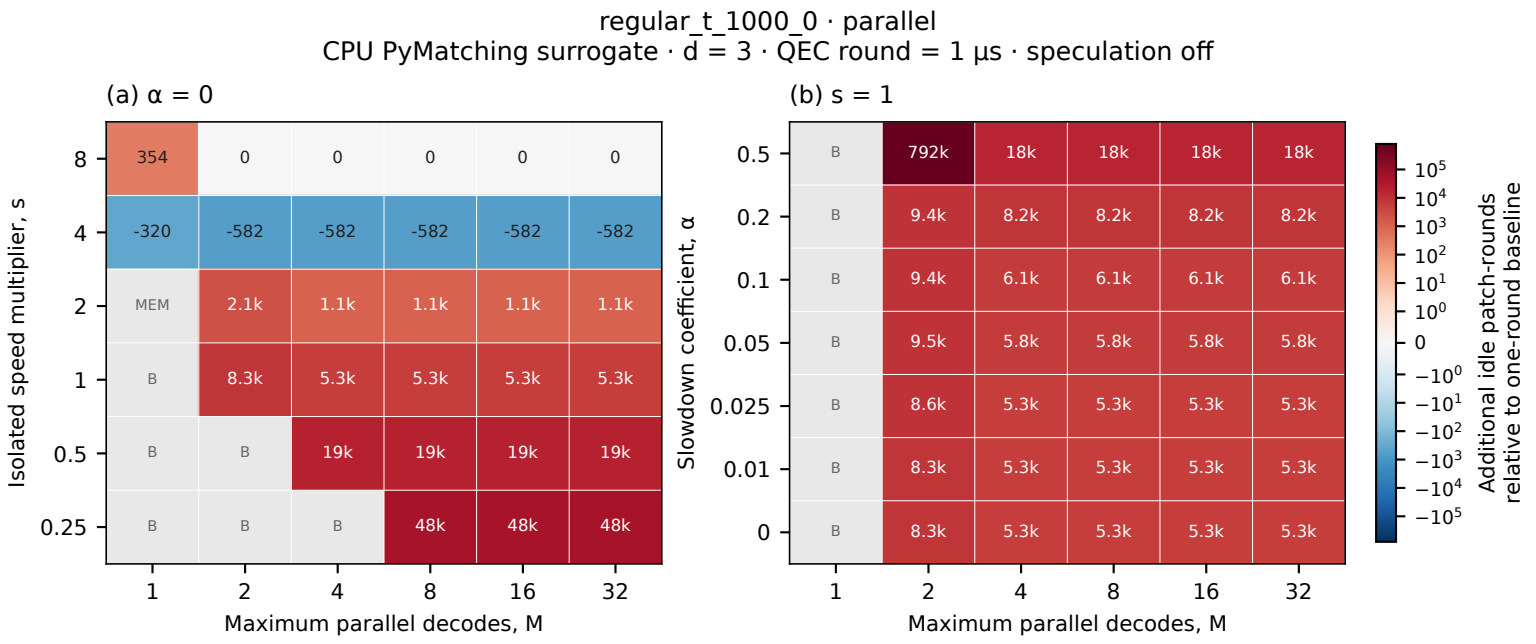

random_t_10000_200_0 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
21k
10k
B
T
10^4
10^3
10^2
10^1
10^0

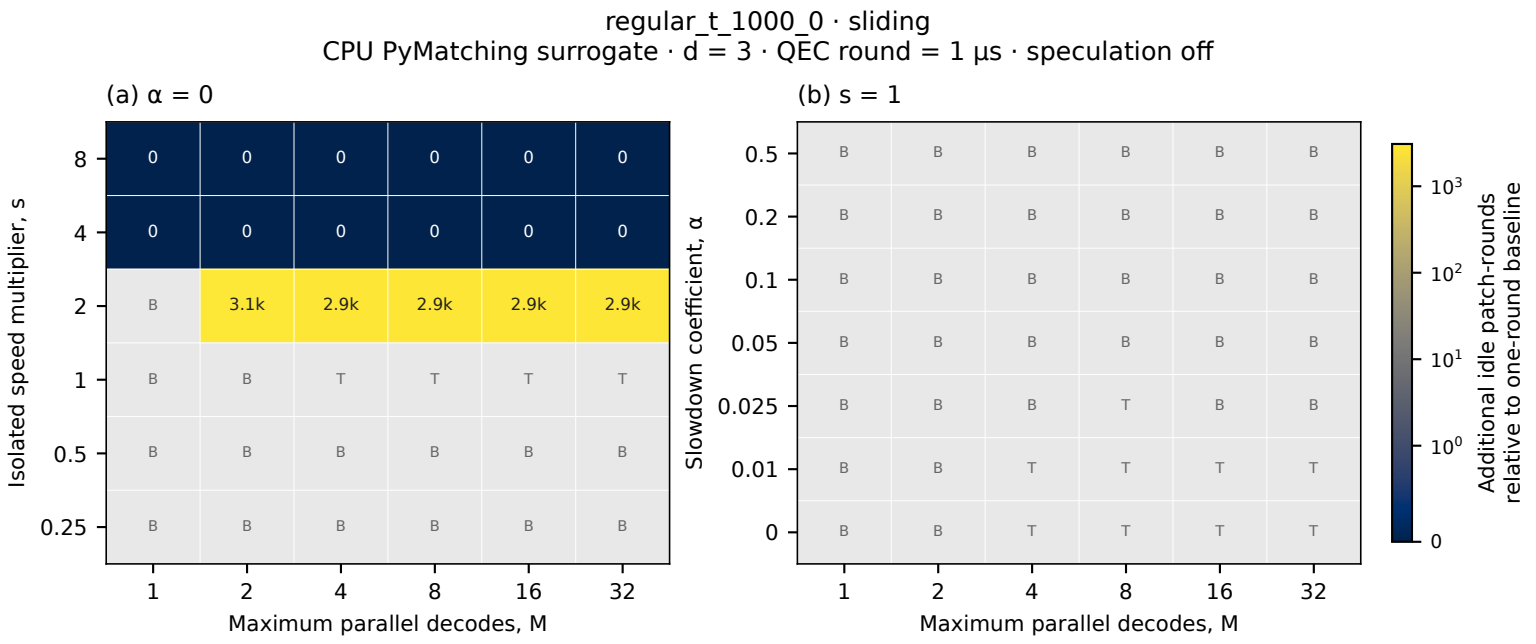

random_t_1000_200_0 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
1
0
0
0
0
0
1.1k
644
644
644
644
644
4.4k
2.6k
2.6k
2.6k
2.6k
2.6k
152k
7.4k
7.3k
7.3k
7.3k
7.3k
B
23k
18k
18k
18k
18k
B
B
42k
40k
40k
40k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
152k
11k
11k
11k
11k
11k
152k
8.4k
8.4k
8.4k
8.4k
8.4k
152k
8.3k
8.3k
8.3k
8.3k
8.3k
152k
8.0k
8.0k
8.0k
8.0k
8.0k
152k
7.5k
7.5k
7.5k
7.5k
7.5k
152k
7.4k
7.3k
7.3k
7.3k
7.3k
152k
7.4k
7.3k
7.3k
7.3k
7.3k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^5
10^4
10^3
10^2
10^1
10^0
0

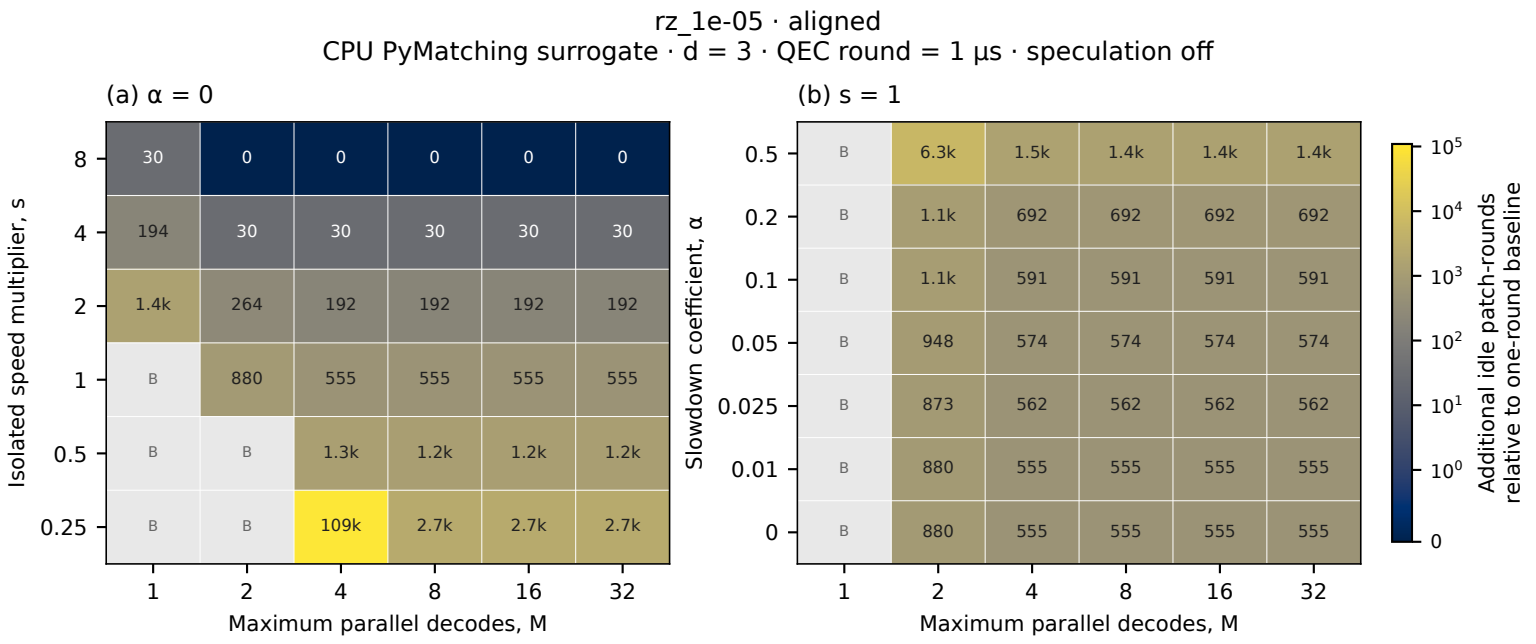

random_t_1000_200_0 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
7
0
1.8k
3.7k
32k
8.1k
B
20k
17k
43k
35k
8.5k
8.3k
8.2k

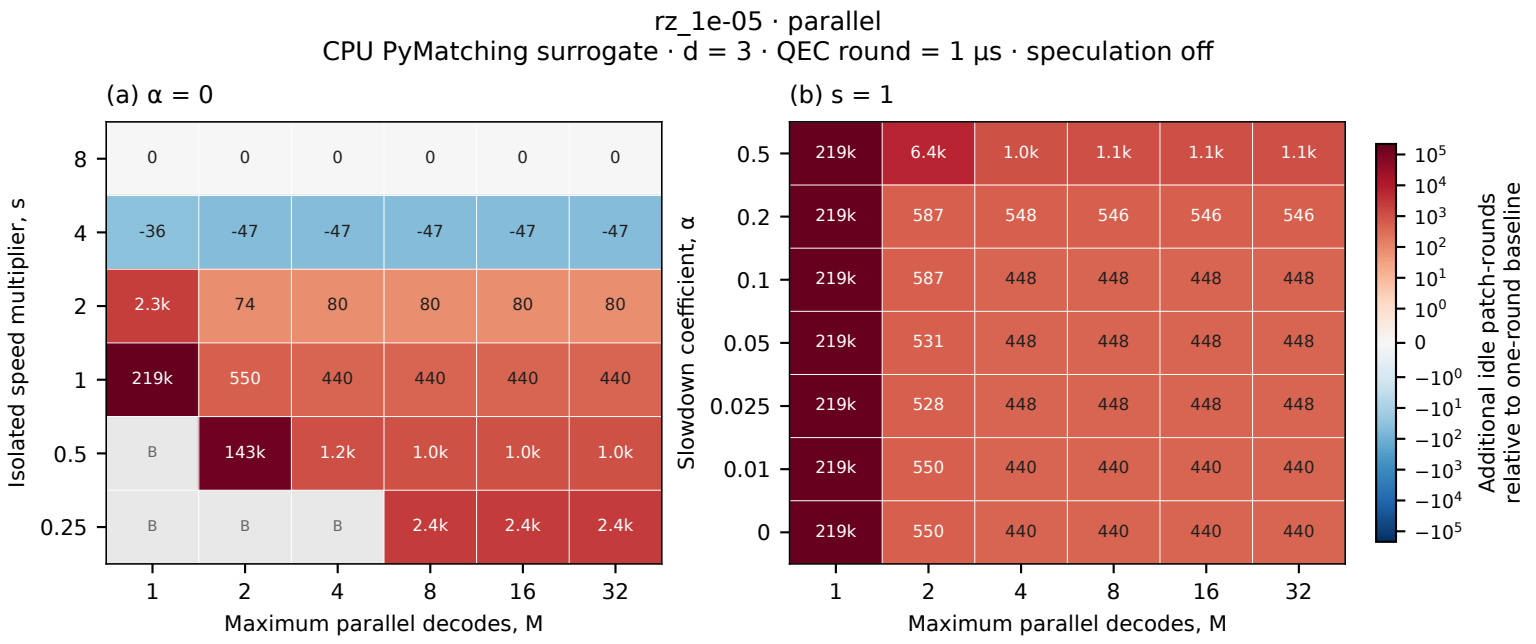

random_t_1000_200_0 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
2.1k
1.0k
B
T
1
2
4
8
16
32
10^3
10^2
10^1
10^0

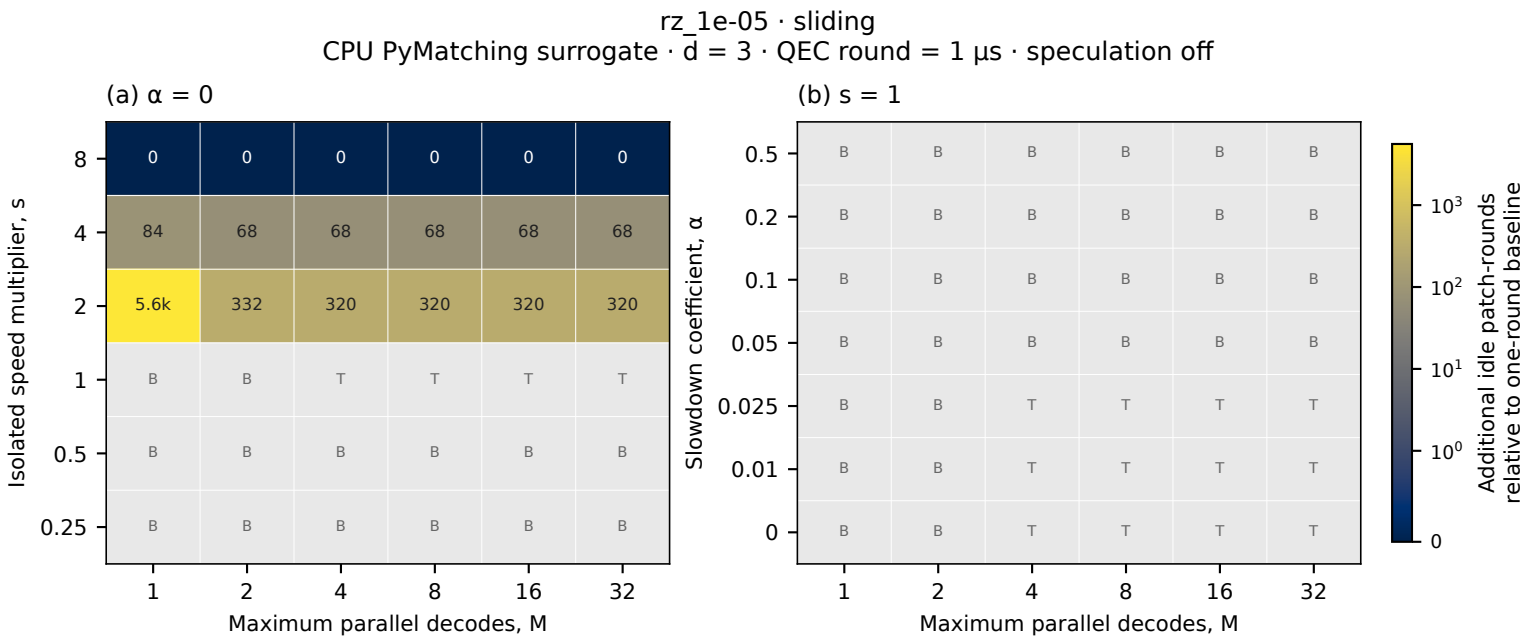

regular_t_1000_0 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
2.3k
680
MEM
3.6k
2.9k
B
13k
10k
21k
52k
1.1M
22k
16k
15k
14k
12k
11k
10
10^0
10^1
10^2
10^3
10^4
10^5
10^6

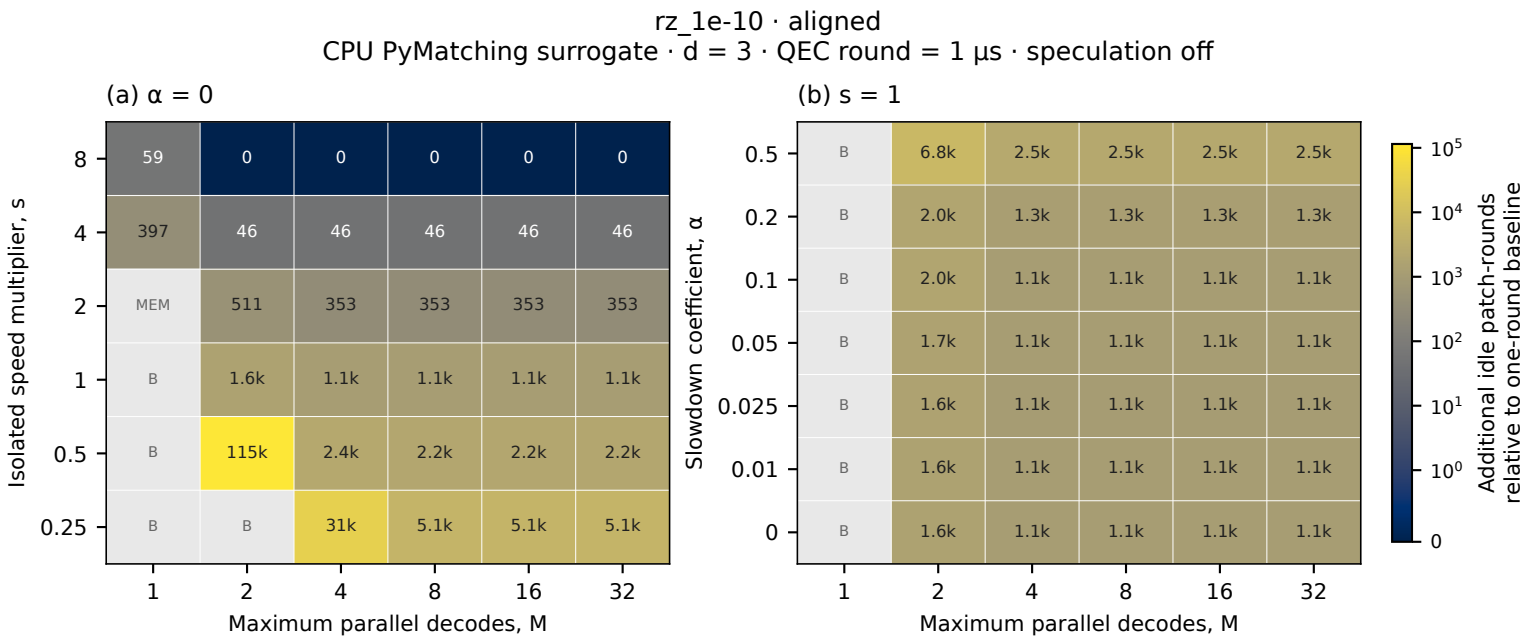

regular_t_1000_0 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
354
0
0
0
0
0
-320
-582
-582
-582
-582
-582
MEM
2.1k
1.1k
1.1k
1.1k
1.1k
B
8.3k
5.3k
5.3k
5.3k
5.3k
B
B
19k
19k
19k
19k
B
B
B
48k
48k
48k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
B
792k
18k
18k
18k
18k
B
9.4k
8.2k
8.2k
8.2k
8.2k
B
9.4k
6.1k
6.1k
6.1k
6.1k
B
9.5k
5.8k
5.8k
5.8k
5.8k
B
8.6k
5.3k
5.3k
5.3k
5.3k
B
8.3k
5.3k
5.3k
5.3k
5.3k
B
8.3k
5.3k
5.3k
5.3k
5.3k
1
2
4
8
16
32
Maximum parallel decodes, M
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
0
-10^{0}
-10^{1}
-10^{2}
-10^{3}
-10^{4}
-10^{5}
Additional idle patch-rounds
relative to one-round baseline

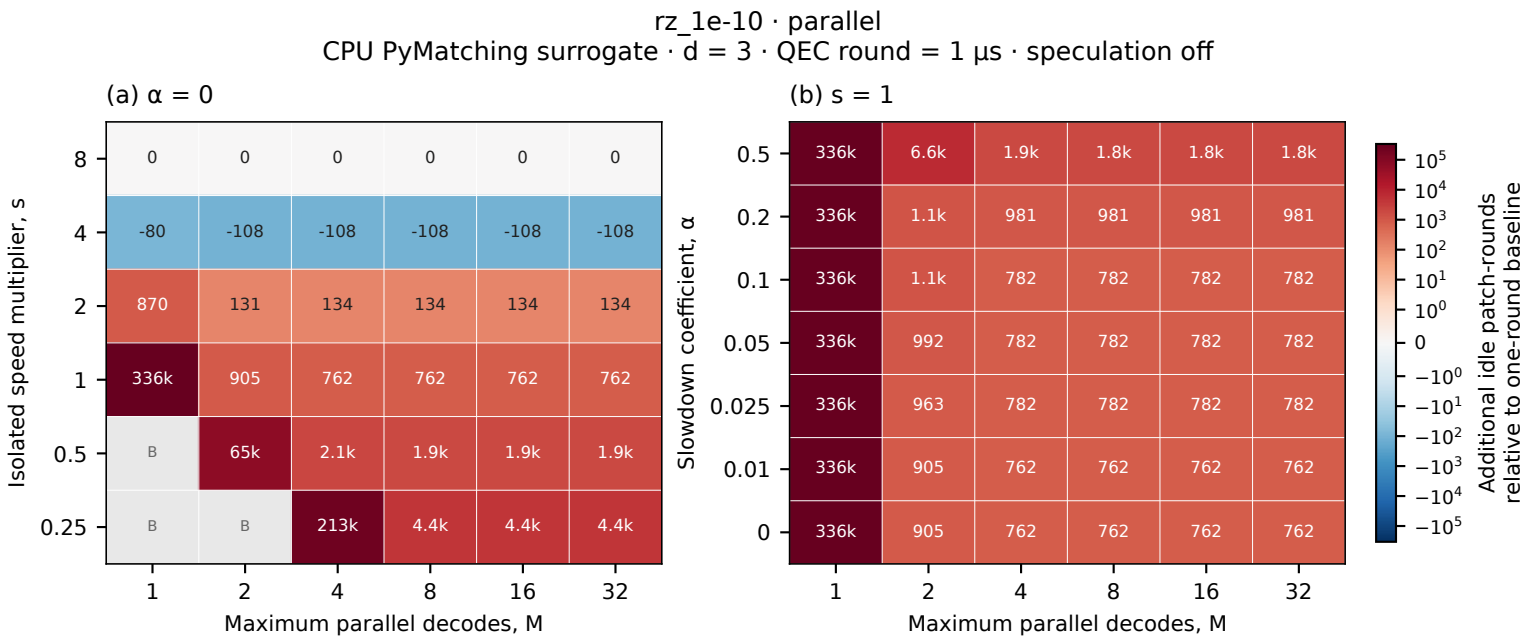

regular_t_1000_0 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
3.1k
2.9k
B
T
10^3
10^2
10^1
10^0

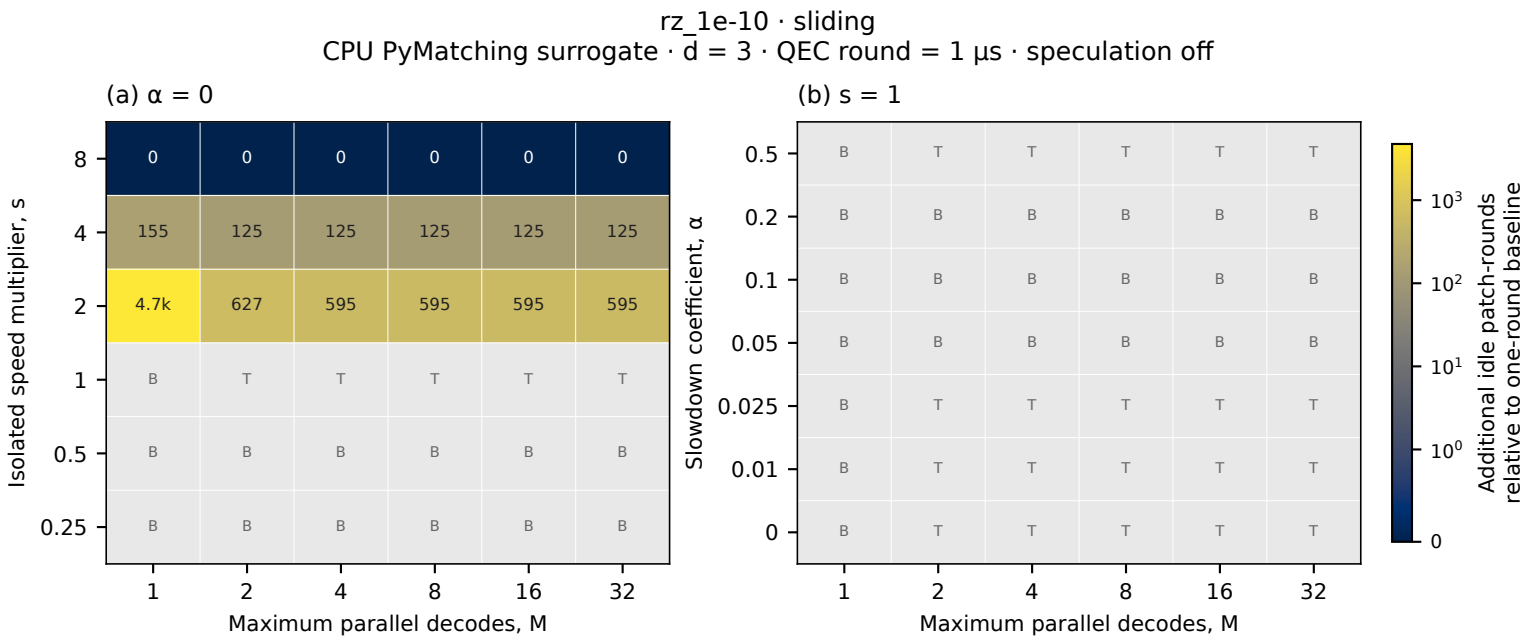

rz_1e-05 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
30 0 0 0 0 0
194 30 30 30 30 30
1.4k 264 192 192 192 192
B 880 555 555 555 555
B B 1.3k 1.2k 1.2k 1.2k
B B 109k 2.7k 2.7k 2.7k
1 2 4 8 16 32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
B 6.3k 1.5k 1.4k 1.4k 1.4k
B 1.1k 692 692 692 692
B 1.1k 591 591 591 591
B 948 574 574 574 574
B 873 562 562 562 562
B 880 555 555 555 555
B 880 555 555 555 555
1 2 4 8 16 32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^5
10^4
10^3
10^2
10^1
10^0
0

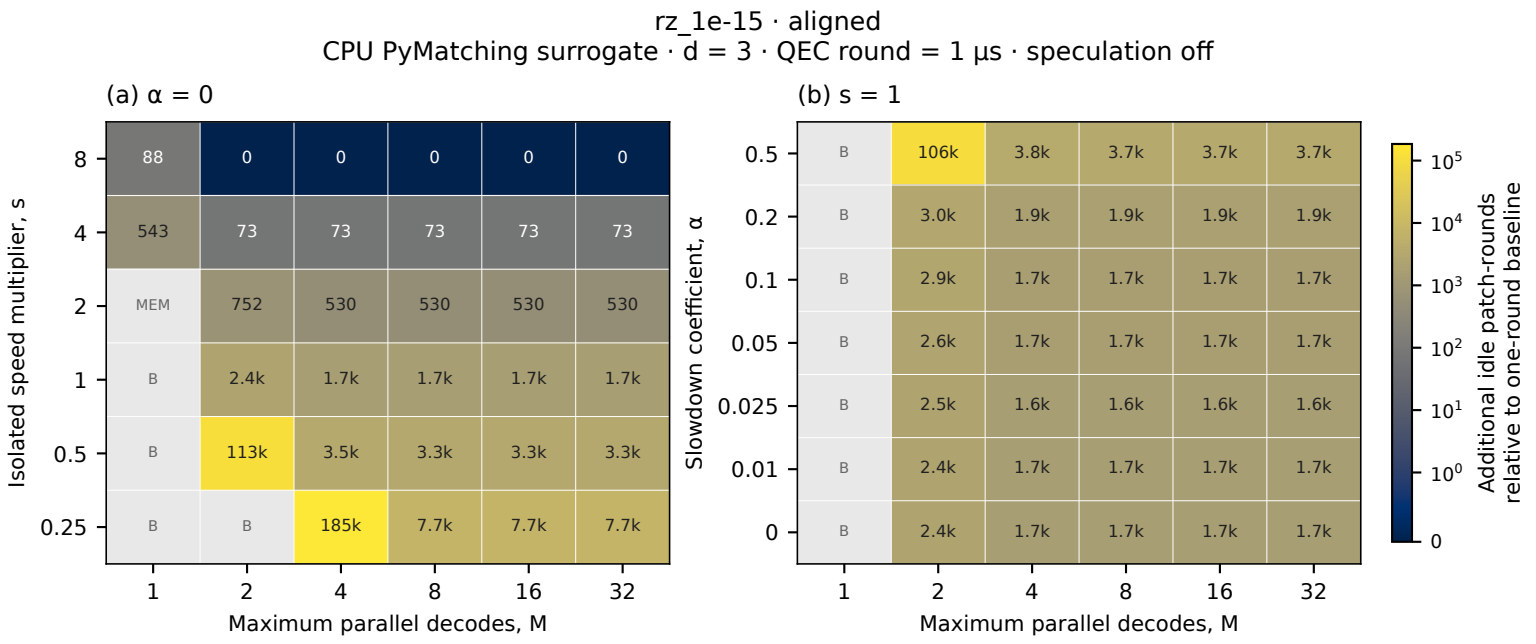

rz_1e-05 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0 0 0 0 0 0
-36 -47 -47 -47 -47 -47
2.3k 74 80 80 80 80
219k 550 440 440 440 440
B 143k 1.2k 1.0k 1.0k 1.0k
B B B 2.4k 2.4k 2.4k
219k 6.4k 1.0k 1.1k 1.1k 1.1k
219k 587 548 546 546 546
219k 587 448 448 448 448
219k 531 448 448 448 448
219k 528 448 448 448 448
219k 550 440 440 440 440
219k 550 440 440 440 440
10^5
10^4
10^3
10^2
10^1
10^0
0
-10^0
-10^1
-10^2
-10^3
-10^4
-10^5

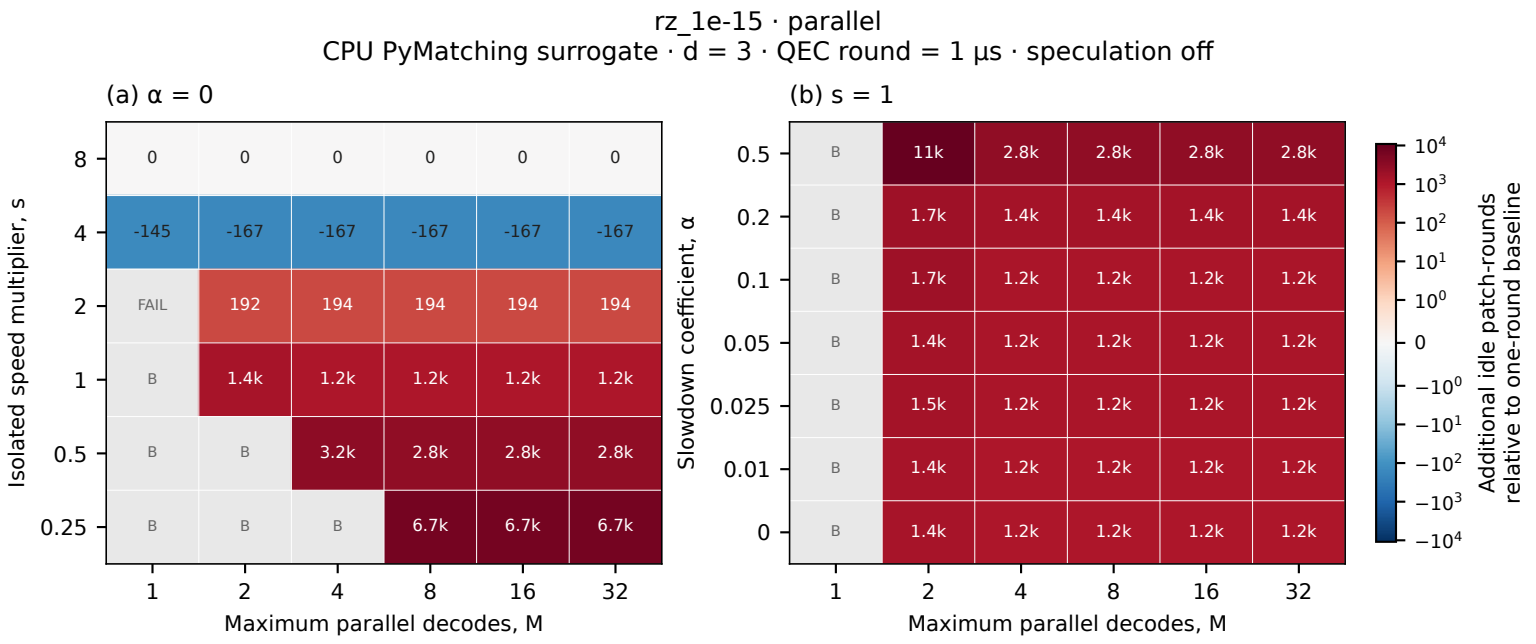

rz_1e-05 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
84
68
5.6k
332
320
B
T
10^3
10^2
10^1
10^0

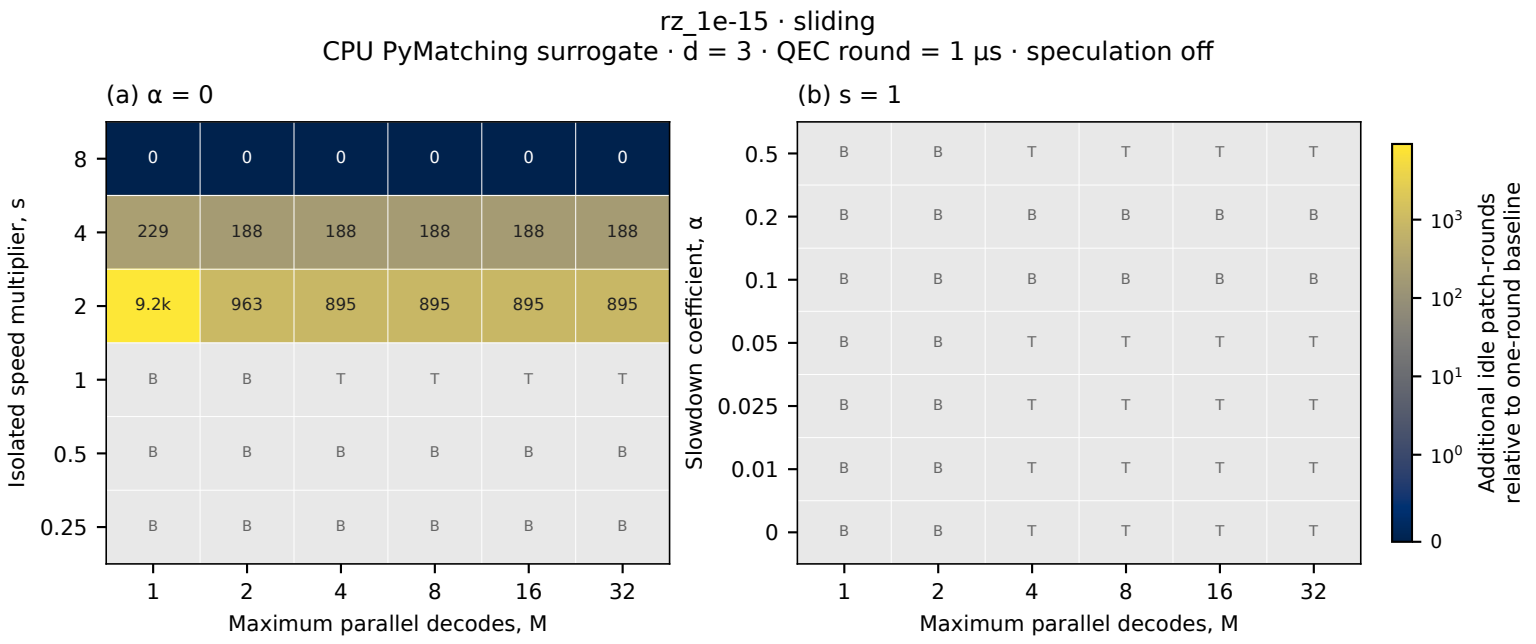

rz_1e-10 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
59 0 0 0 0 0
397 46 46 46 46 46
MEM 511 353 353 353 353
B 1.6k 1.1k 1.1k 1.1k 1.1k
B 115k 2.4k 2.2k 2.2k 2.2k
B B 31k 5.1k 5.1k 5.1k
B 6.8k 2.5k 2.5k 2.5k 2.5k
B 2.0k 1.3k 1.3k 1.3k 1.3k
B 2.0k 1.1k 1.1k 1.1k 1.1k
B 1.7k 1.1k 1.1k 1.1k 1.1k
B 1.6k 1.1k 1.1k 1.1k 1.1k
B 1.6k 1.1k 1.1k 1.1k 1.1k
B 1.6k 1.1k 1.1k 1.1k 1.1k
10^5
10^4
10^3
10^2
10^1
10^0
0

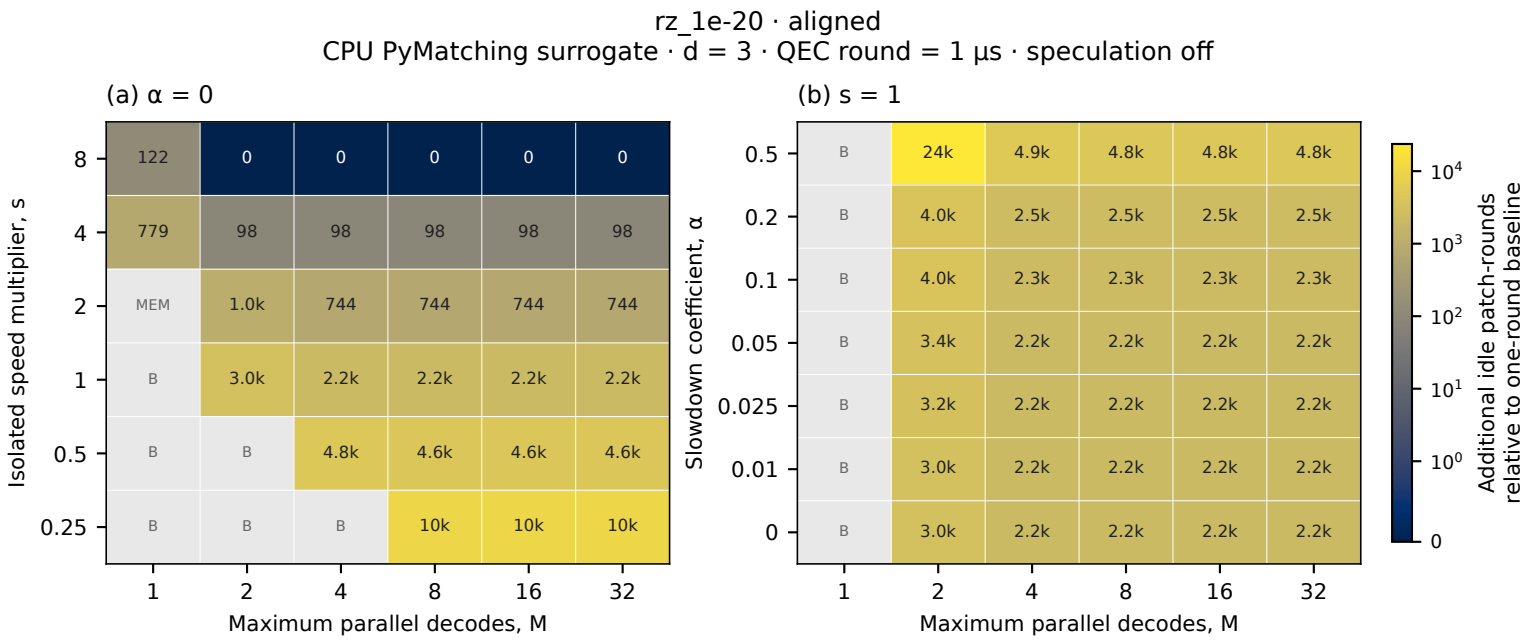

rz_1e-10 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0 0 0 0 0 0
-80 -108 -108 -108 -108 -108
870 131 134 134 134 134
336k 905 762 762 762 762
B 65k 2.1k 1.9k 1.9k 1.9k
B B 213k 4.4k 4.4k 4.4k
336k 6.6k 1.9k 1.8k 1.8k 1.8k
336k 1.1k 981 981 981 981
336k 1.1k 782 782 782 782
336k 992 782 782 782 782
336k 963 782 782 782 782
336k 905 762 762 762 762
336k 905 762 762 762 762
10^5
10^4
10^3
10^2
10^1
10^0
0
-10^0
-10^1
-10^2
-10^3
-10^4
-10^5

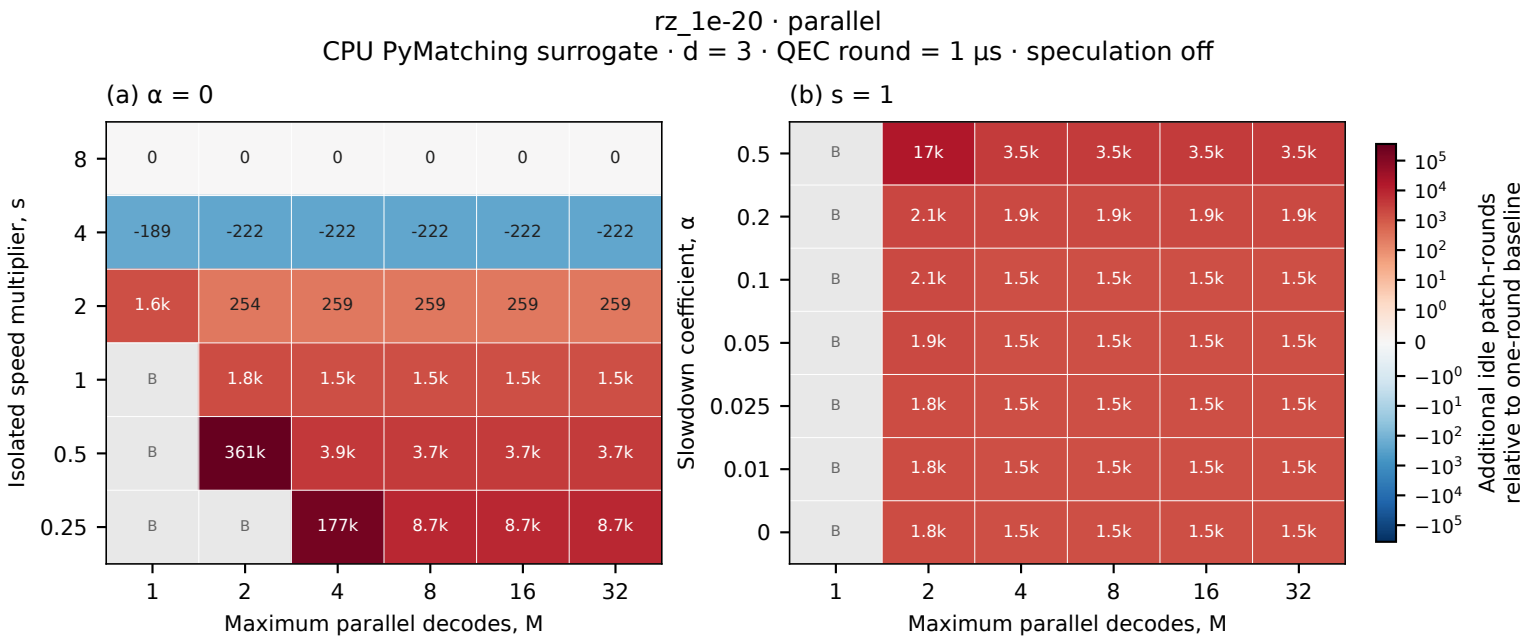

rz_1e-10 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
155
125
4.7k
627
595
B
T
10^3
10^2
10^1
10^0

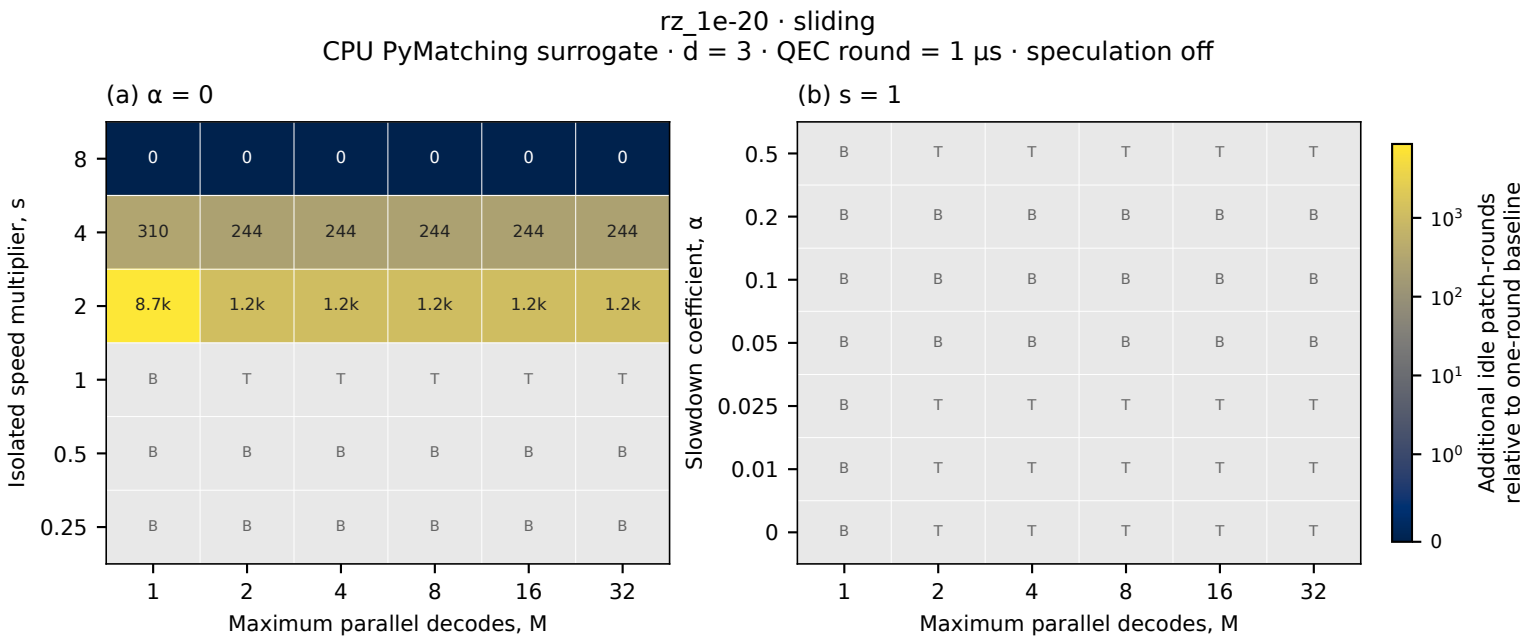

rz_1e-15 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
88 0 0 0 0 0
543 73 73 73 73 73
MEM 752 530 530 530 530
B 2.4k 1.7k 1.7k 1.7k 1.7k
B 113k 3.5k 3.3k 3.3k 3.3k
B B 185k 7.7k 7.7k 7.7k
B 106k 3.8k 3.7k 3.7k 3.7k
B 3.0k 1.9k 1.9k 1.9k 1.9k
B 2.9k 1.7k 1.7k 1.7k 1.7k
B 2.6k 1.7k 1.7k 1.7k 1.7k
B 2.5k 1.6k 1.6k 1.6k 1.6k
B 2.4k 1.7k 1.7k 1.7k 1.7k
B 2.4k 1.7k 1.7k 1.7k 1.7k
10^5
10^4
10^3
10^2
10^1
10^0
0

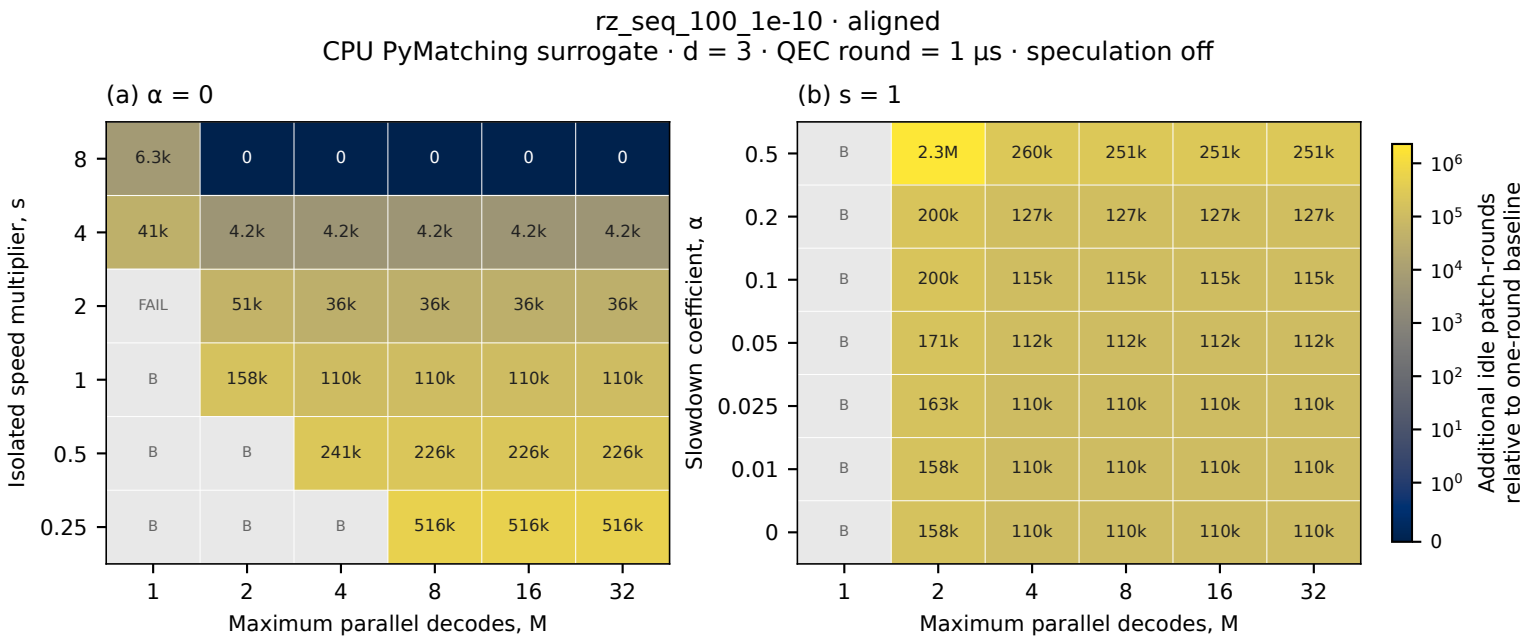

rz_1e-15 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0 0 0 0 0 0
-145 -167 -167 -167 -167 -167
FAIL 192 194 194 194 194
B 1.4k 1.2k 1.2k 1.2k 1.2k
B B 3.2k 2.8k 2.8k 2.8k
B B B 6.7k 6.7k 6.7k
B 11k 2.8k 2.8k 2.8k 2.8k
B 1.7k 1.4k 1.4k 1.4k 1.4k
B 1.7k 1.2k 1.2k 1.2k 1.2k
B 1.4k 1.2k 1.2k 1.2k 1.2k
B 1.5k 1.2k 1.2k 1.2k 1.2k
B 1.4k 1.2k 1.2k 1.2k 1.2k
B 1.4k 1.2k 1.2k 1.2k 1.2k
10^4
10^3
10^2
10^1
10^0
0
-10^0
-10^1
-10^2
-10^3
-10^4

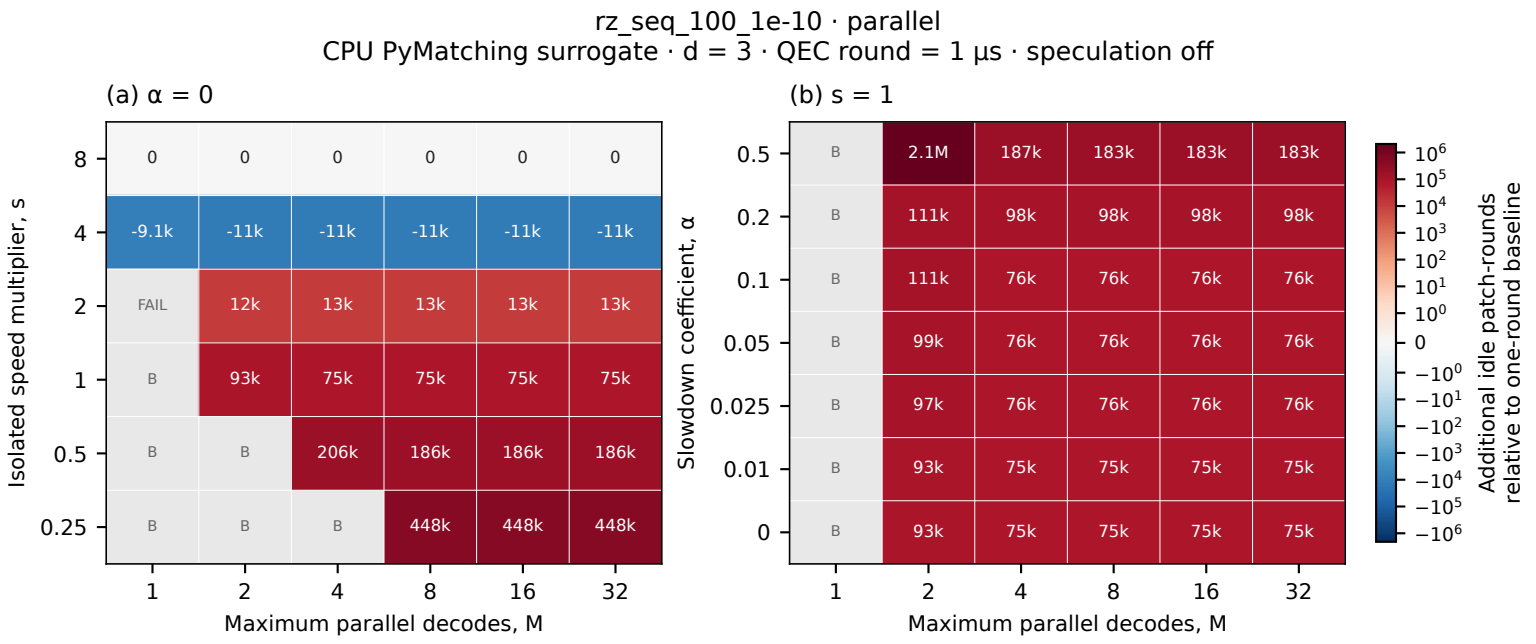

rz_1e-15 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
229
188
9.2k
963
895
B
T
10^3
10^2
10^1
10^0

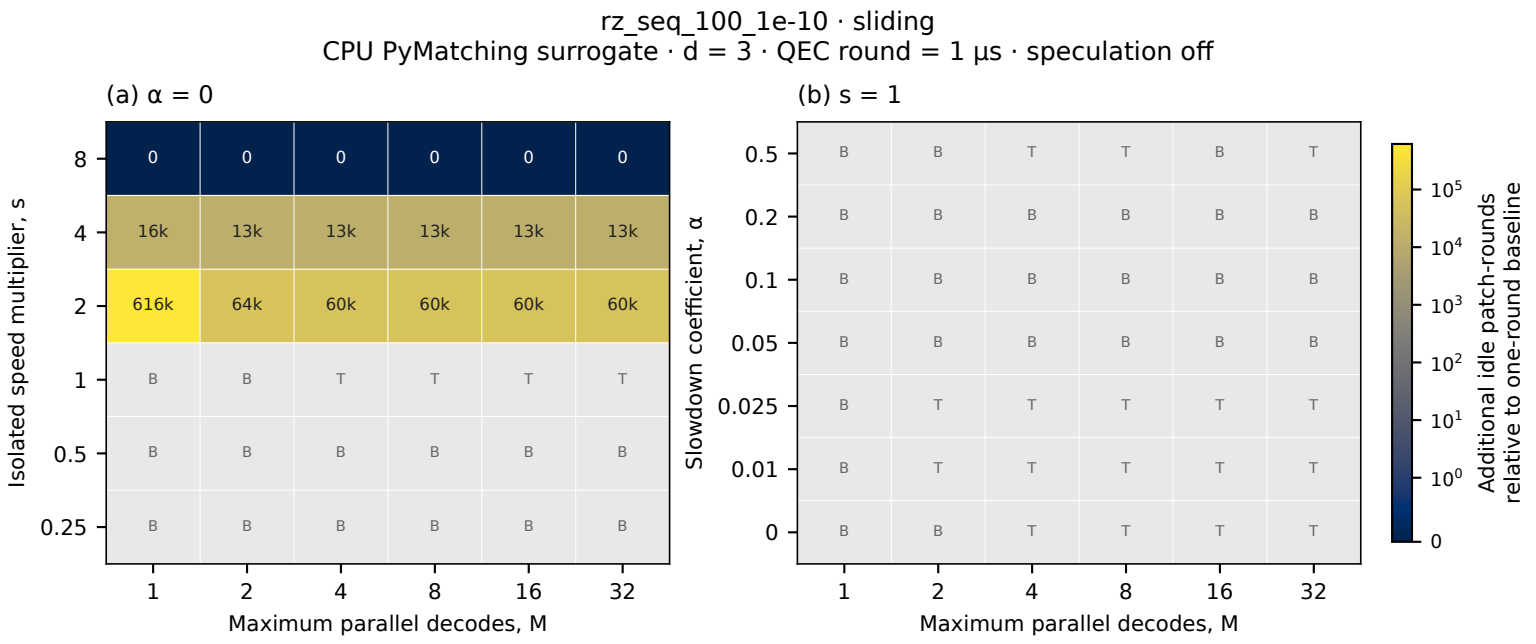

rz_1e-20 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
122 0 0 0 0 0
779 98 98 98 98 98
MEM 1.0k 744 744 744 744
B 3.0k 2.2k 2.2k 2.2k 2.2k
B B 4.8k 4.6k 4.6k 4.6k
B B B 10k 10k 10k
B 24k 4.9k 4.8k 4.8k 4.8k
B 4.0k 2.5k 2.5k 2.5k 2.5k
B 4.0k 2.3k 2.3k 2.3k 2.3k
B 3.4k 2.2k 2.2k 2.2k 2.2k
B 3.2k 2.2k 2.2k 2.2k 2.2k
B 3.0k 2.2k 2.2k 2.2k 2.2k
B 3.0k 2.2k 2.2k 2.2k 2.2k
10^4
10^3
10^2
10^1
10^0
0

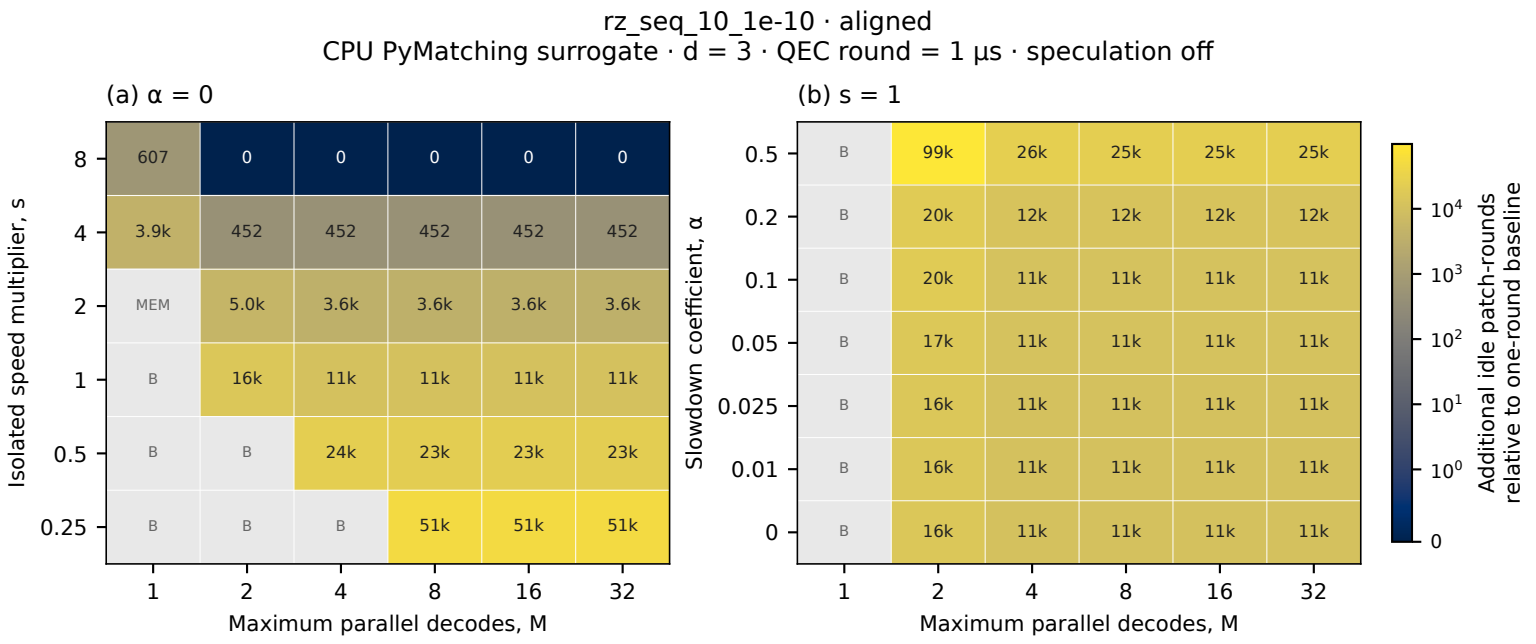

rz_1e-20 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
B
0
-189
-222
1.6k
254
259
1.8k
1.5k
361k
3.9k
3.7k
177k
8.7k
17k
3.5k
2.1k
1.9k
1.8k
10^5
10^4
10^3
10^2
10^1
10^0
-10^0
-10^1
-10^2
-10^3
-10^4
-10^5

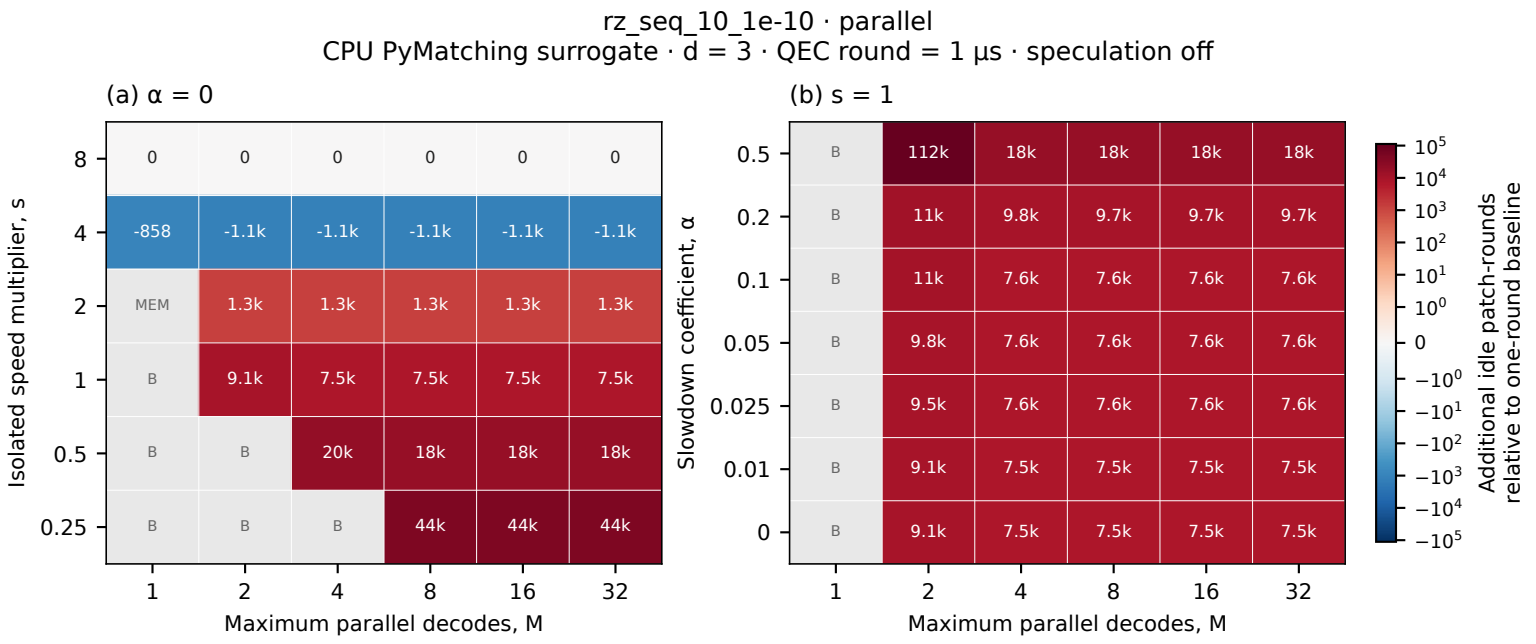

rz_1e-20 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
310
244
8.7k
1.2k
B
T
10^3
10^2
10^1
10^0

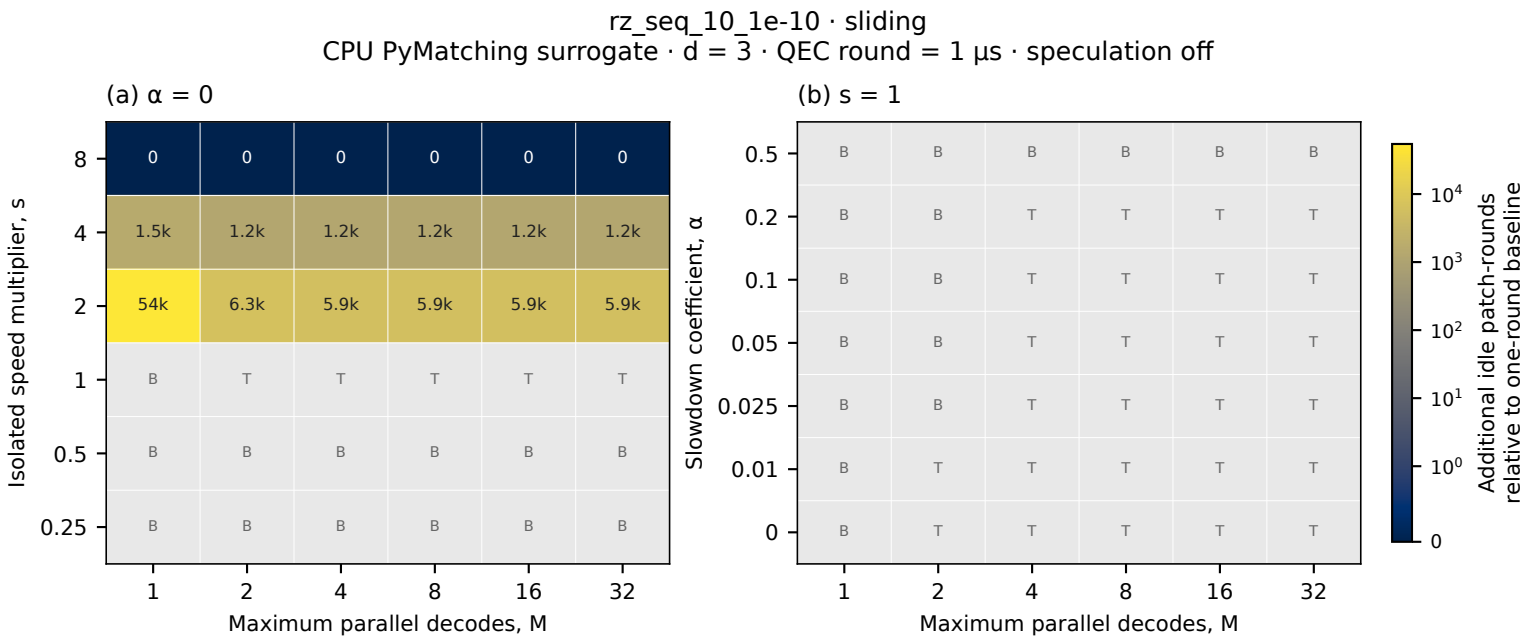

rz_seq_100_1e-10 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
6.3k
0
0
0
0
0
41k
4.2k
4.2k
4.2k
4.2k
4.2k
FAIL
51k
36k
36k
36k
36k
B
158k
110k
110k
110k
110k
B
B
241k
226k
226k
226k
B
B
B
516k
516k
516k
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
B
2.3M
260k
251k
251k
251k
B
200k
127k
127k
127k
127k
B
200k
115k
115k
115k
115k
B
171k
112k
112k
112k
112k
B
163k
110k
110k
110k
110k
B
158k
110k
110k
110k
110k
B
158k
110k
110k
110k
110k
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^6
10^5
10^4
10^3
10^2
10^1
10^0
0

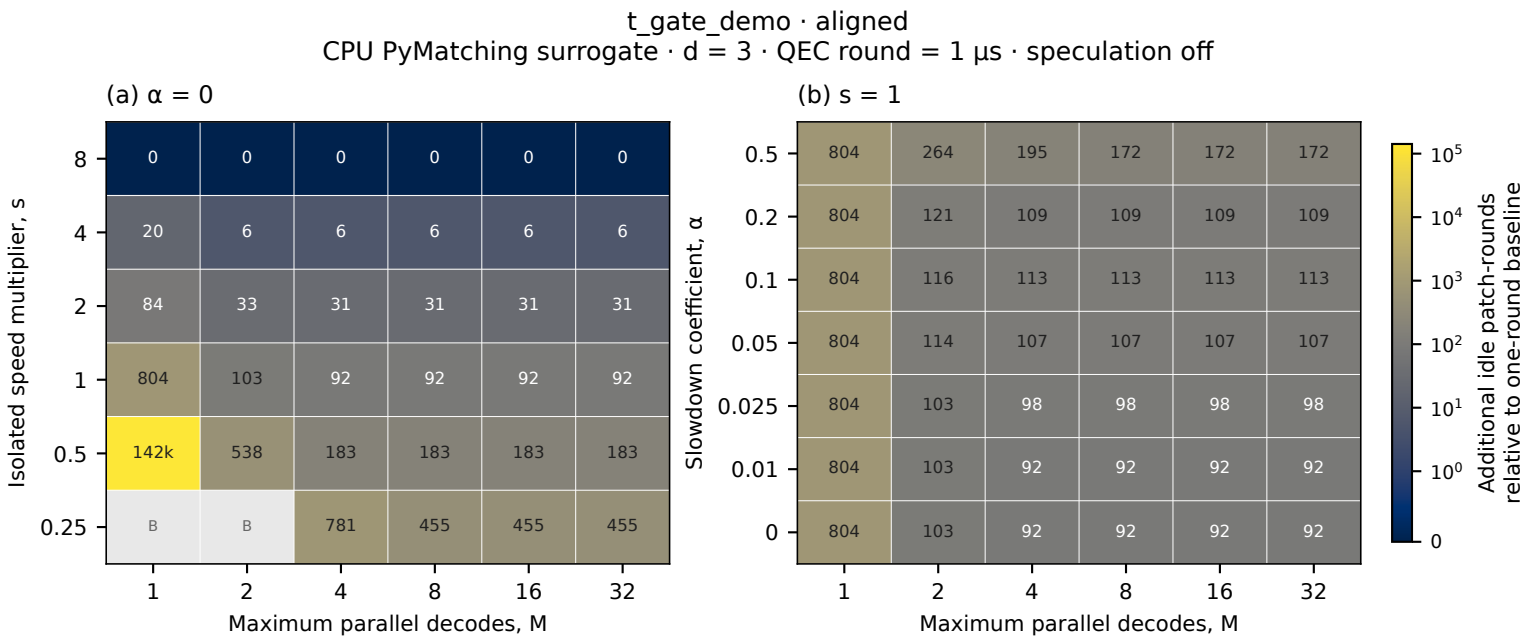

rz_seq_100_1e-10 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
0 0 0 0 0 0
-9.1k -11k -11k -11k -11k -11k
FAIL 12k 13k 13k 13k 13k
B 93k 75k 75k 75k 75k
B B 206k 186k 186k 186k
B B B 448k 448k 448k
1 2 4 8 16 32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
B 2.1M 187k 183k 183k 183k
B 111k 98k 98k 98k 98k
B 111k 76k 76k 76k 76k
B 99k 76k 76k 76k 76k
B 97k 76k 76k 76k 76k
B 93k 75k 75k 75k 75k
B 93k 75k 75k 75k 75k
1 2 4 8 16 32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^{6}
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
0
-10^{0}
-10^{1}
-10^{2}
-10^{3}
-10^{4}
-10^{5}
-10^{6}

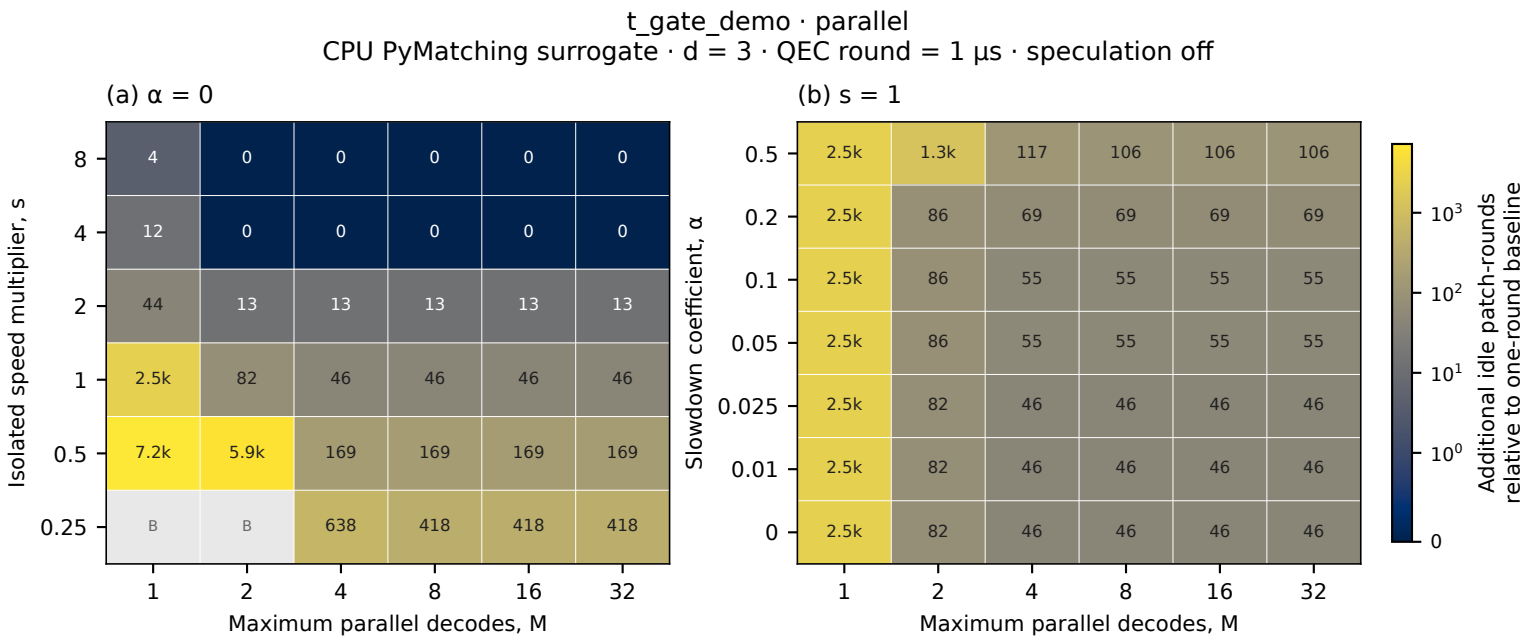

rz_seq_100_1e-10 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
16k
13k
616k
64k
60k
B
T
10^5
10^4
10^3
10^2
10^1
10^0

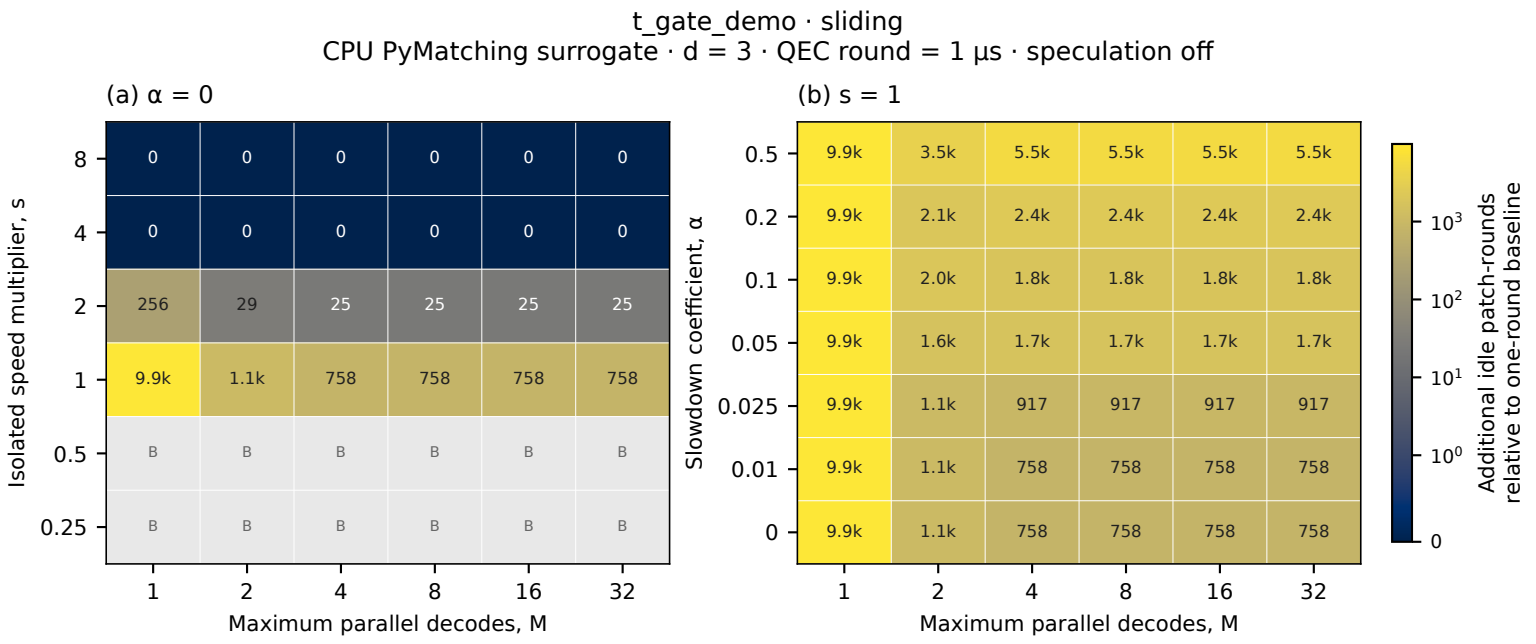

rz_seq_10_1e-10 · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
607
0
0
0
0
0
3.9k
452
452
452
452
452
MEM
5.0k
3.6k
3.6k
3.6k
3.6k
B
16k
11k
11k
11k
11k
B
B
24k
23k
23k
23k
B
B
B
51k
51k
51k
B
99k
26k
25k
25k
25k
B
20k
12k
12k
12k
12k
B
20k
11k
11k
11k
11k
B
17k
11k
11k
11k
11k
B
16k
11k
11k
11k
11k
B
16k
11k
11k
11k
11k
B
16k
11k
11k
11k
11k
10^4
10^3
10^2
10^1
10^0
0

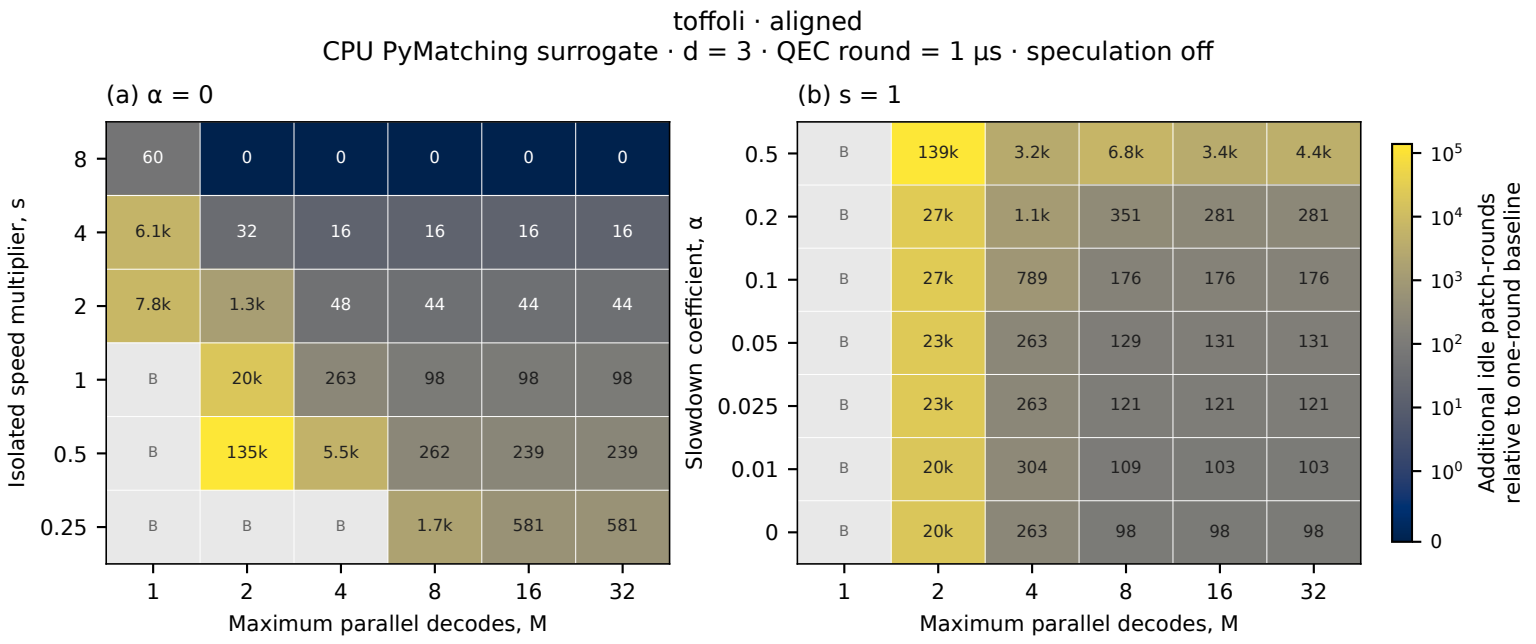

rz_seq_10_1e-10 · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
MEM
B
-858
-1.1k
1.3k
9.1k
7.5k
20k
18k
44k
112k
11k
9.8k
7.6k
9.5k
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
-10^{0}
-10^{1}
-10^{2}
-10^{3}
-10^{4}
-10^{5}

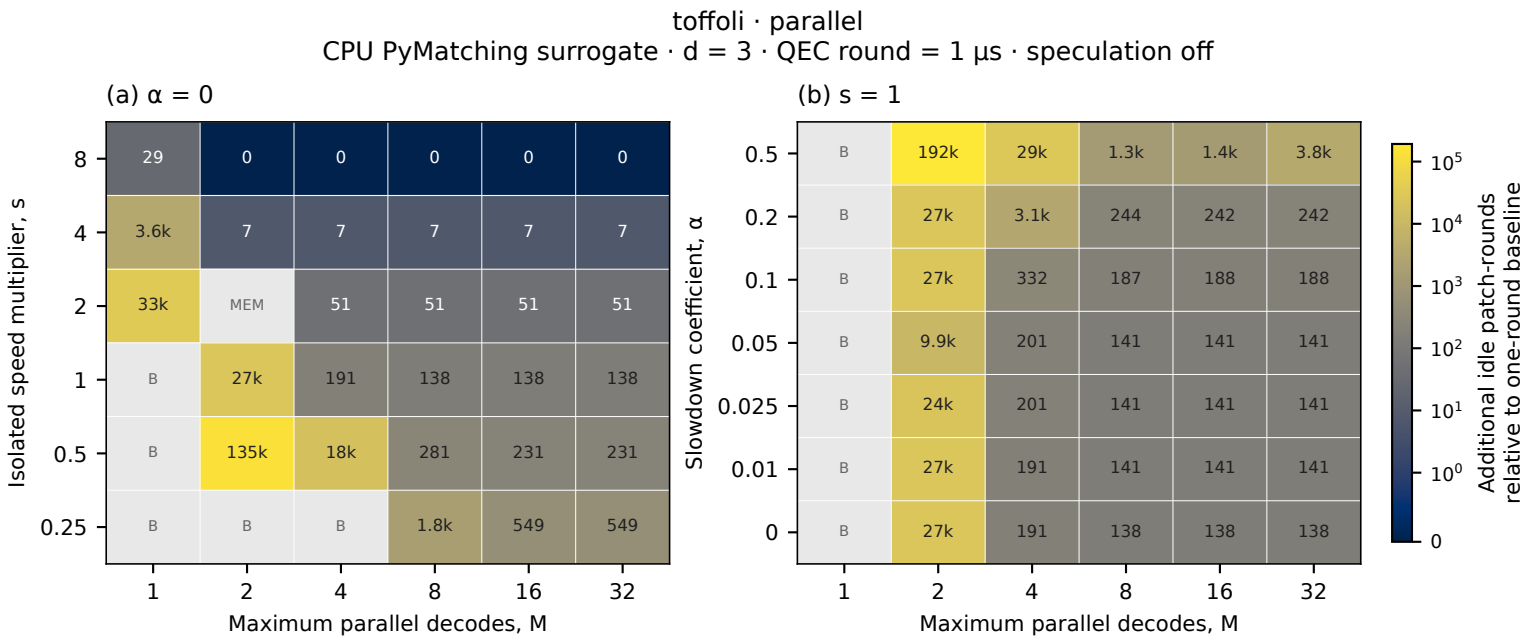

rz_seq_10_1e-10 · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
0
1.5k
1.2k
54k
6.3k
5.9k
B
T
10^4
10^3
10^2
10^1
10^0

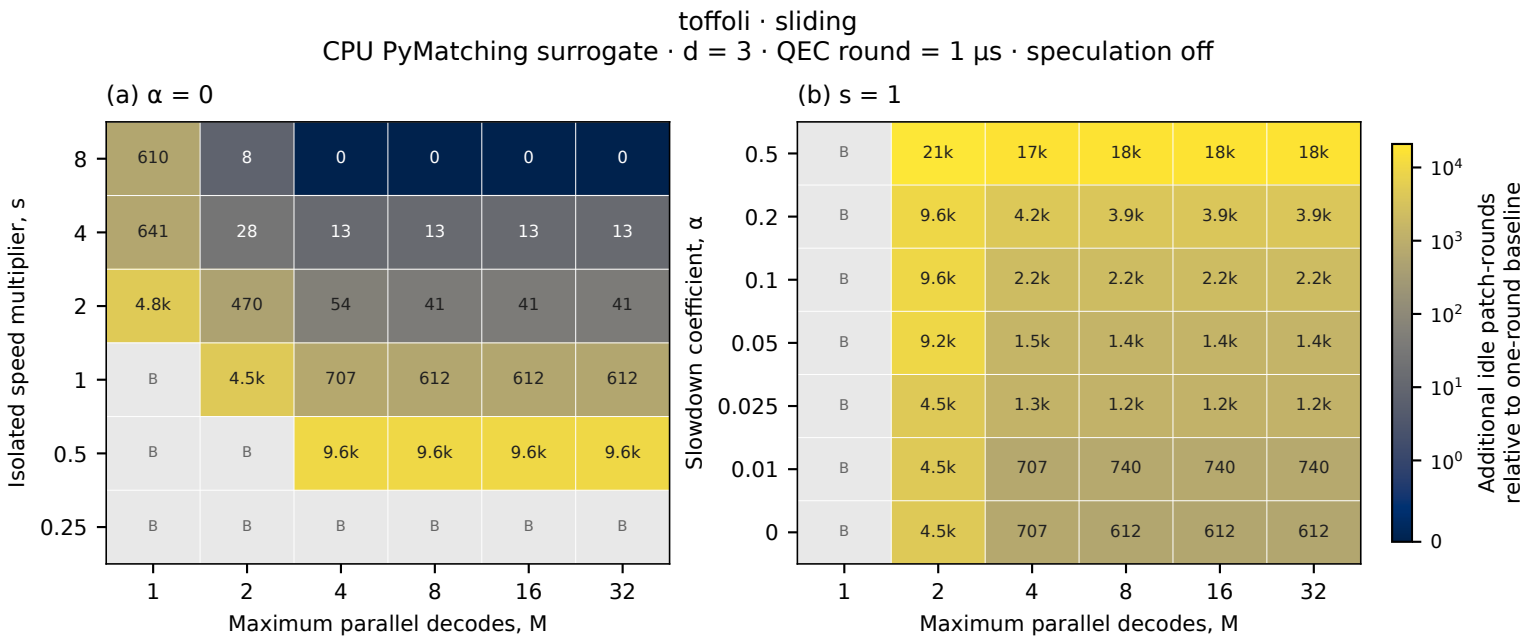

t_gate_demo · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
0 0 0 0 0 0
20 6 6 6 6 6
84 33 31 31 31 31
804 103 92 92 92 92
142k 538 183 183 183 183
B B 781 455 455 455
1 2 4 8 16 32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
804 264 195 172 172 172
804 121 109 109 109 109
804 116 113 113 113 113
804 114 107 107 107 107
804 103 98 98 98 98
804 103 92 92 92 92
804 103 92 92 92 92
1 2 4 8 16 32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^5
10^4
10^3
10^2
10^1
10^0
0


t_gate_demo · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
4 0 0 0 0 0
12 0 0 0 0 0
44 13 13 13 13 13
2.5k 82 46 46 46 46
7.2k 5.9k 169 169 169 169
B B 638 418 418 418
2.5k 1.3k 117 106 106 106
2.5k 86 69 69 69 69
2.5k 86 55 55 55 55
2.5k 86 55 55 55 55
2.5k 82 46 46 46 46
2.5k 82 46 46 46 46
2.5k 82 46 46 46 46
10^3
10^2
10^1
10^0
0


t_gate_demo · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
B
9.9k
3.5k
5.5k
2.1k
2.4k
2.0k
1.8k
1.6k
1.7k
1.1k
917
758
256
29
25
10^3
10^2
10^1
10^0


toffoli · aligned
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
Isolated speed multiplier, s
8
4
2
1
0.5
0.25
60
0
0
0
0
0
6.1k
32
16
16
16
16
7.8k
1.3k
48
44
44
44
B
20k
263
98
98
98
B
135k
5.5k
262
239
239
B
B
B
1.7k
581
581
1
2
4
8
16
32
Maximum parallel decodes, M
(b) s = 1
Slowdown coefficient, α
0.5
0.2
0.1
0.05
0.025
0.01
0
B
139k
3.2k
6.8k
3.4k
4.4k
B
27k
1.1k
351
281
281
B
27k
789
176
176
176
B
23k
263
129
131
131
B
23k
263
121
121
121
B
20k
304
109
103
103
B
20k
263
98
98
98
1
2
4
8
16
32
Maximum parallel decodes, M
Additional idle patch-rounds
relative to one-round baseline
10^5
10^4
10^3
10^2
10^1
10^0
0


toffoli · parallel
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
29
0
0
0
0
0
3.6k
7
7
7
7
7
33k
MEM
51
51
51
51
B
27k
191
138
138
138
B
135k
18k
281
231
231
B
B
B
1.8k
549
549
B
192k
29k
1.3k
1.4k
3.8k
B
27k
3.1k
244
242
242
B
27k
332
187
188
188
B
9.9k
201
141
141
141
B
24k
201
141
141
141
B
27k
191
141
141
141
B
27k
191
138
138
138
10^5
10^4
10^3
10^2
10^1
10^0
0


toffoli · sliding
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off
(a) α = 0
(b) s = 1
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline
8
4
2
1
0.5
0.25
0.5
0.2
0.1
0.05
0.025
0.01
0
1
2
4
8
16
32
610 8 0 0 0 0
641 28 13 13 13 13
4.8k 470 54 41 41 41
B 4.5k 707 612 612 612
B B 9.6k 9.6k 9.6k 9.6k
B B B B B B
B 21k 17k 18k 18k 18k
B 9.6k 4.2k 3.9k 3.9k 3.9k
B 9.6k 2.2k 2.2k 2.2k 2.2k
B 9.2k 1.5k 1.4k 1.4k 1.4k
B 4.5k 1.3k 1.2k 1.2k 1.2k
B 4.5k 707 740 740 740
B 4.5k 707 612 612 612
10^4
10^3
10^2
10^1
10^0
0